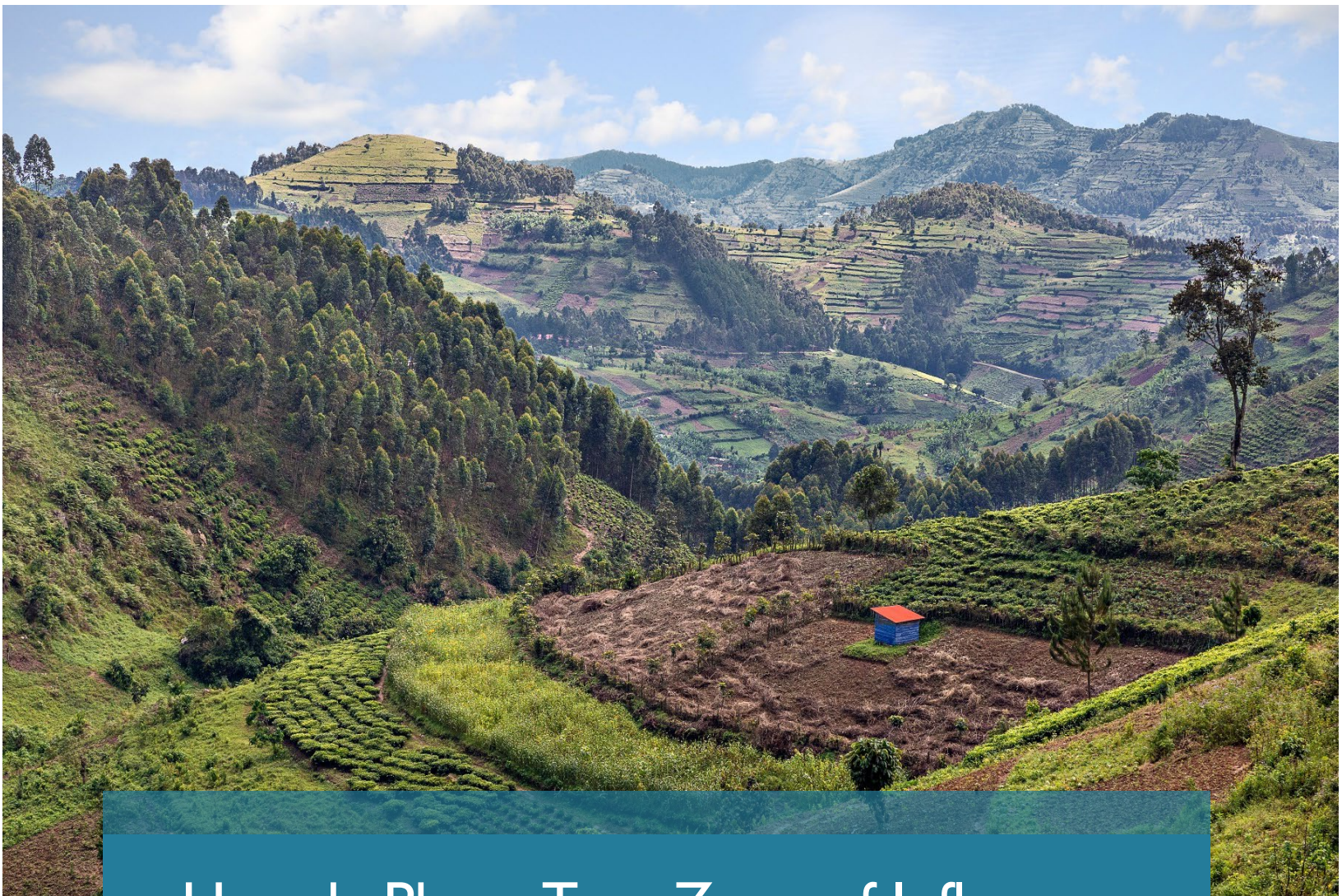




FEED THE FUTURE

The U.S. Government's Global Hunger & Food Security Initiative



Uganda Phase Two Zone of Influence and Resilience Focus Zone Midline Indicator Assessment—2022

FINAL REPORT

December 2024



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ABBREVIATIONS

5DE	Five Domains of Empowerment
A-WEAI	Abbreviated Women’s Empowerment in Agriculture Index
ARSSI	Ability to Recover from Shocks and Stresses Index
BMI	body mass index
CAPI	computer-assisted personal interviewing
CDCS	Country Development Cooperation Strategy
CPI	consumer price index
CWI	Comparative Wealth Index
EA	enumeration area
EBF	exclusive breastfeeding
FAO	Food and Agriculture Organization of the United Nations
FIES	Food Insecurity Experience Scale
HAZ	height-for-age z-score
HWISE-4	Brief Household Water Insecurity Experiences
ICDM	In-Country Data Manager
MAD	minimum acceptable diet
MDD	minimum dietary diversity
MDD-W	minimum dietary diversity—women
P2	phase two
PPP	purchasing power parity
REFS	Bureau for Resilience, Environment, and Food Security
RFZ	Resilience Focus Zone
SEI	shock exposure index
SOW	scope of work
USAID	United States Agency for International Development
WASH	water, sanitation, and hygiene
WHO	World Health Organization
WHZ	weight-for-height z-score
ZOI	Zone of Influence

EXECUTIVE SUMMARY

Feed the Future seeks to reduce poverty, hunger, and malnutrition among women and children; strengthen agriculture and food systems; increase income, resilience, women's empowerment, dietary diversity, and appropriate feeding practices; and improve hygienic environments. Program efforts are designed to impact populations in the Zone of Influence (ZOI) in Feed the Future target countries, including Uganda. The phase two ZOI (P2-ZOI) is the targeted subnational regions and districts where the program, guided by the Global Food Security Strategy 2017–2021, intends to achieve the greatest household- and individual-level impacts on poverty, hunger, and malnutrition (USAID, 2016). In areas of recurrent crises, the United States Agency for International Development (USAID) also aims to curb the need for humanitarian assistance by building capacity to mitigate, adapt to, and recover from shocks and stresses. In resilience focus countries, such as Uganda, program efforts are designed to strengthen household and community resilience among targeted populations in the Resilience Focus Zone (RFZ). Progress in achieving high-level Feed the Future objectives in the ZOI and resilience objectives in the RFZ is tracked using population-based indicators computed at baseline and periodically thereafter.

This is the final report for the 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Indicator Assessment (hereafter referred to as the “assessment”). The purpose of the assessment is to provide the USAID Bureau for Resilience, Environment, and Food Security, USAID Uganda, U.S. Government interagency partners, the Government of Uganda, and development partners with information to monitor short-term progress across a set of population-based indicators in the Uganda P2-ZOI and RFZ since the baseline values were established in 2019. The assessment is not designed to support conclusions of causality or program attribution.

In Uganda, the P2-ZOI comprises 44 districts¹ located across 4 regions:² East, North, Karamoja, and Southwest. Within the P2-ZOI, the RFZ covers 13 districts across the North, Karamoja, and Southwest regions. Results for the assessment are derived from the 2022 Uganda P2-ZOI and RFZ Midline Survey and the 2019 Uganda P2-ZOI and RFZ Baseline Survey. This report presents indicator estimates for the P2-ZOI and RFZ at midline, including analysis of changes since baseline. The 15 Feed the Future P2-ZOI indicators computed for this assessment for both the P2-ZOI and RFZ are:³

- Prevalence of poverty: Percent of people living on less than \$1.90/day 2011 purchasing power parity (PPP) [EG-c]⁴
- Depth of poverty of the poor: Mean shortfall of the poor relative to the \$1.90/day 2011 PPP poverty line [EG-h]
- Percent of people who are ‘near-poor,’ living on 100 percent to less than 125 percent of the \$1.90 2011 PPP poverty line [FTF Context-9]

¹ At baseline, the P2-ZOI included 41 districts. Since then, 3 districts have been subdivided into new districts, thereby increasing the total number of districts in the Uganda ZOI to 44 districts.

² Three of the regions—East, North, and Karamoja—align with Country Development Cooperation Strategy (CDCS) regional corridors in Uganda. The fourth region, Southwest, combines the Southwestern and Western CDCS regional corridors due to the small number of ZOI districts in the Western CDCS regional corridor.

³ With the exception of the Five Domains of Empowerment and Brief Household Water Insecurity Experiences Scale indicators, the Feed the Future indicators are defined in the 2019 version of the Feed the Future Indicator Handbook (USAID, 2018). The Five Domains of Empowerment indicator is defined in the 2023 version of the Feed the Future Indicator Handbook (USAID, 2023). The Brief Household Water Insecurity Experiences Scale indicator is not an official Feed the Future indicator but is collected and reported in P2-ZOI midline indicator assessments.

⁴ Text in brackets indicates the Feed the Future indicator number (USAID, 2018b; USAID, 2023).

- Percent of households below the comparative threshold for the poorest quintile of the asset-based Comparative Wealth Index (CWI) [EG-g]
- Prevalence of moderate and severe food insecurity in the population, based on the Food Insecurity Experience Scale (FIES) [EG-e]
- Ability to recover from shocks and stresses index [RESIL-a]
- Index of social capital at the household level [RESIL-b]
- Percent of households that believe local government will respond effectively to future shocks and stresses [RESIL-c]
- Percent of households participating in group-based savings, micro-finance, or lending programs [EG.4.2-a]
- Prevalence of women of reproductive age consuming a diet of minimum diversity [HL.9.1-d]
- Prevalence of exclusive breastfeeding of children under 6 months of age [HL.9.1-b]
- Percent of children 6-23 months of age consuming a minimum acceptable diet [HL.9.1-a]
- Percent of households with access to a basic sanitation service [HL.8.2-a]
- Percent of households with soap and water at a handwashing station on premise [HL.8.2-b]
- Prevalence of water insecurity, based on the Brief Household Water Insecurity Experiences Scale (HWISE-4)⁵
- Five Domains of Empowerment (5DE) score for women [EG.3-i]

In addition, the following expanded set of resilience indicators was computed for the RFZ:

- Index of Shock Exposure
- Absorptive capacity index
- Adaptive capacity index
- Transformative capacity index
- Household resilience capacity index
- Reduced Coping Strategies Index
- Index of Community Resilience

Furthermore, questions on coverage of promoted interventions were added to the 2022 Uganda P2-ZOI and RFZ Midline Survey questionnaire to capture data on households' participation in or access to the kinds of interventions that Feed the Future supports in Uganda. Descriptive statistics on agricultural cross-market functions, COVID-19, climate adaptation, and coverage of promoted interventions related to agriculture, resilience, women's nutrition and health, children's nutrition and health, and water, sanitation, and hygiene are provided in the Appendix 7.

A summary of Feed the Future P2-ZOI and RFZ indicator estimates—in total and by key disaggregates—are presented in Table ESI.

⁵ The HWISE-4 scale is a brief version of the Household Water Insecurity Experiences (HWISE) scale, designed to assess water insecurity at the household level. It consists of four questions that capture the frequency and severity of experiences related to water access and availability (Northwestern University Institute for Policy Research, n.d.).

Table ES1. Feed the Future ZOI and RFZ Indicator Estimates, in Total and by Key Disaggregates: Uganda 2019–2022

Indicator	Baseline (2019)			Midline (2022)			Diff.	P- value ^b	Sig. ^c
	Est.	95% CI	n ^a	Est.	95% CI	n ^a			
Prevalence of poverty: Percent of people living on less than \$1.90/day (2011 PPP) ^d									
All individuals (ZOI) ^e	41.1	37.1, 45.3	2,530	37.1	34.5, 39.7	2,995	-4.0		uc
All individuals (RFZ) ^e	37.9	25.8, 51.7	435	40.0	37.1, 42.9	1,653	2.1		uc
Depth of poverty of the poor: Mean percent shortfall of the poor relative to the \$1.90/day (2011 PPP) poverty line ^d									
All individuals (ZOI) ^e	33.8	32.2, 35.4	1,105	35.6		1,108	1.8		na
All individuals (RFZ) ^e	41.3	36.1, 46.5	213	41.6		610	0.3		na
Percent of people who are near-poor, living on 100 percent to less than 125 percent of the \$1.90/day (2011 PPP) poverty line ^d									
All individuals (ZOI) ^e	13.7	12.1, 15.3	2,530	11.8		2,995	-1.9		na
All individuals (RFZ) ^e	8.8	6.3, 12.2	435	12.5		1,653	3.7		na
Percent of households below the comparative threshold for the poorest quintile of the asset-based Comparative Wealth Index									
All households (ZOI)	39.8	36.2, 43.4	2,540	38.1	33.9, 42.6	2,995	-1.7	0.57	n/s
Gendered household type (ZOI)									
Male and female adults	39.5	35.6, 43.5	1,836	34.2	30.3, 38.3	2,401	-5.3	0.07	n/s
Female adults only	42.5	35.9, 49.4	450	53.7	46.9, 60.4	399	11.2	0.02	*
Male adults only	36.9	27.8, 47.0	253	57.5	42.9, 70.9	194	20.6	0.02	*
Children only	^	^		^	^		na	na	na
All households (RFZ)	43.6	28.5, 60.1	436	61.1	55.8, 66.3	1,653	17.5	0.05	*
Prevalence of moderate and severe food insecurity in the population, based on the Food Insecurity Experience Scale									
All households (ZOI)	70.7	63.5, 77.9	2,533	84.6	80.6, 88.5	2,981	13.9		s
Gendered household type (ZOI)									
Male and female adults	71.1	64.5, 77.7	1,829	84.2	80.1, 88.3	2,390	13.1		s
Female adults only	73.1	63.2, 83.0	450	87.4	83.5, 91.3	397	14.3		s
Male adults only	63.3	53.6, 73.0	253	82.3	73.0, 91.6	193	19.0		s
Children only	^	^		^	^		na		na
Severity (ZOI)									
Moderate	42.3	39.2, 45.4	2,533	47.9	45.3, 50.6	2,981	5.6		s

Indicator	Baseline (2019)			Midline (2022)			Diff.	p-value ^b	Sig. ^c
	Est.	95% CI	n ^a	Est.	95% CI	n ^a			
Severe	28.4	22.4, 34.3	2,533	36.6	31.1, 42.2	2,981	8.2		s
All households (RFZ)	70.6	54.9, 86.3	434	87.6	77.4, 97.8	1,645	17.0		s
Severity (RFZ)									
Moderate	41.0	36.6, 45.4	434	43.2	34.5, 51.9	1,645	2.2		n/s
Severe	29.7	17.3, 42.1	434	44.4	26.0, 62.8	1,645	14.7		n/s
Ability to Recover from Shocks and Stresses Index (Mean score)									
All households (ZOI)	4.4	4.3, 4.5	2,455	3.9	3.7, 4.0	2,972	-0.5	<0.001	***
Gendered household type (ZOI)									
Male and female adults	4.5	4.4, 4.6	1,794	3.9	3.7, 4.0	2,385	-0.6	<0.001	***
Female adults only	4.2	4.0, 4.4	431	3.7	3.5, 4.0	397	-0.5	0.003	**
Male adults only	4.4	4.2, 4.6	229	3.9	3.6, 4.2	189	-0.5	0.002	**
Children only	^	^		^	^		na	na	na
All households (RFZ)	4.7	4.5, 5.0	430	3.6	3.5, 3.8	1,638	-1.1	<0.001	***
Index of social capital at the household level (Mean score)									
Overall index									
All households (ZOI)	61.9	59.4, 64.5	2,529	60.1	57.9, 62.3	2,992	-1.8	0.28	n/s
Gendered household type (ZOI)									
Male and female adults	63.7	61.0, 66.4	1,827	61.0	58.6, 63.3	2,398	-2.7	0.14	n/s
Female adults only	56.2	51.7, 60.8	448	56.0	50.8, 61.2	399	-0.2	0.95	n/s
Male adults only	59.5	53.0, 66.1	253	56.7	50.1, 63.3	194	-2.8	0.55	n/s
Children only	^	^		^	^		na	na	na
All households (RFZ)	58.6	50.9, 66.3	435	56.3	52.8, 59.9	1,650	-2.3	0.60	n/s
Bonding sub-index									
All households (ZOI)	65.1	62.5, 67.7	2,529	60.6	58.4, 62.7	2,992	-4.5	0.009	**
Gendered household type (ZOI)									
Male and female adults	67.3	64.5, 70.1	1,827	61.5	59.2, 63.9	2,398	-5.8	0.002	**
Female adults only	59.5	54.5, 64.4	448	56.0	50.3, 61.8	399	-3.5	0.37	n/s
Male adults only	59.8	53.1, 66.5	253	57.8	51.7, 63.9	194	-2.0	0.66	n/s

Indicator	Baseline (2019)			Midline (2022)			Diff.	p-value ^b	Sig. ^c
	Est.	95% CI	n ^a	Est.	95% CI	n ^a			
Children only	^	^		^	^		na	na	na
All households (RFZ)	60.6	55.2, 66.1	435	57.7	54.2, 61.2	1,650	-2.9	0.37	n/s
Bridging sub-index									
All households (ZOI)	58.8	55.9, 61.6	2,529	59.6	56.7, 62.4	2,992	0.8	0.69	n/s
Gendered household type (ZOI)									
Male and female adults	60.1	57.2, 63.1	1,827	60.4	57.4, 63.5	2,398	0.3	0.89	n/s
Female adults only	53.0	48.1, 57.9	448	56.0	49.9, 62.1	399	3.0	0.46	n/s
Male adults only	59.3	51.3, 67.3	253	55.6	46.8, 64.4	194	-3.7	0.54	n/s
Children only	^	^		^	^		na	na	na
All households (RFZ)	56.6	45.5, 67.6	435	55.0	50.6, 59.4	1,650	-1.6	0.79	n/s
Percent of households that believe local government will respond effectively to future shocks and stresses									
All households (ZOI)	61.1	56.7, 65.3	2,294	59.0	53.9, 64.0	2,794	-2.1	0.55	n/s
Gendered household type (ZOI)									
Male and female adults	60.0	55.3, 64.6	1,656	58.1	52.8, 63.3	2,239	-1.9	0.59	n/s
Female adults only	63.8	56.9, 70.2	405	65.5	56.6, 73.4	373	1.7	0.76	n/s
Male adults only	63.6	55.1, 71.4	232	55.8	44.9, 66.2	181	-7.8	0.26	n/s
Children only	^	^		^	^		na	na	na
All households (RFZ)	64.7	49.3, 77.6	405	61.7	55.5, 67.5	1,523	-3.0	0.70	n/s
Percent of households participating in group-based savings, micro-finance, or lending programs									
All households (ZOI)	59.8	56.1, 63.3	2,178	62.9	59.3, 66.4	2,624	3.1	0.22	n/s
Gendered household type (ZOI)									
Male and female adults	60.8	56.8, 64.6	1,743	65.2	61.3, 68.9	2,231	4.4	0.12	n/s
Female adults only	55.8	49.4, 62.0	444	49.9	43.3, 56.4	393	-5.9	0.20	n/s
Male adults only									
Children only									
All households (RFZ)	64.3	56.1, 71.7	384	53.9	48.9, 58.9	1,459	-10.4	0.03	*
Five Domains of Empowerment score for women									
All women (ZOI)	0.92	0.91, 0.93	1,938	0.90	0.88, 0.92	2,335	-0.02	0.07	n/s

Indicator	Baseline (2019)			Midline (2022)			Diff.	p- value ^b	Sig. ^c
	Est.	95% CI	n ^a	Est.	95% CI	n ^a			
Women's age (ZOI)									
18–29 years	0.88	0.86, 0.90	537	0.85	0.81, 0.88	640	-0.03	0.05	n/s
30 years or older	0.93	0.93, 0.94	1,401	0.92	0.90, 0.94	1,695	-0.01	0.16	n/s
All women (RFZ)	0.93	0.91, 0.96	342	0.88	0.85, 0.93	1,277	-0.05	0.02	*
Percent of women who are empowered									
All women (ZOI)	75.8	72.9, 78.5	1,938	72.2	68.0, 76.1	2,335	-3.6	0.16	n/s
Women's age (ZOI)									
18–29 years	65.6	60.3, 70.5	537	60.6	54.0, 66.9	640	-5.0	0.24	n/s
30 years or older	79.5	76.5, 82.2	1,401	76.3	72.0, 80.2	1,695	-3.2	0.21	n/s
All women (RFZ)	79.8	72.6, 85.5	342	67.1	60.7, 72.9	1,277	-12.7	0.005	**
Average adequacy score of disempowered women									
All women (ZOI)	0.67	0.66, 0.68	533	0.64	0.63, 0.66	711	-0.03	0.002	**
Women's age (ZOI)									
18–29 years	0.65	0.64, 0.67	198	0.61	0.58, 0.63	258	-0.04	0.002	**
30 years or older	0.68	0.67, 0.69	335	0.66	0.65, 0.68	453	-0.02	0.046	*
All women (RFZ)	0.67	0.65, 0.69	82	0.63	0.61, 0.65	422	-0.04	0.013	*
Percent of disempowered women who are adequate in each A-WEAI indicator									
Input in productive decisions									
All women (ZOI)	91.3	88.7, 93.3	533	96.5	94.0, 98.0	711	5.2	<0.001	***
Women's age (ZOI)									
18–29 years	89.2	84.5, 92.6	198	92.8	86.8, 96.2	258	3.6	0.23	n/s
30 years or older	92.5	88.2, 95.3	335	98.7	96.6, 99.5	453	6.2	<0.001	**
Ownership of assets									
All women (ZOI)	96.3	94.0, 97.7	533	85.8	81.1, 89.5	711	-10.5	<0.001	***
Women's age (ZOI)									
18–29 years	94.9	89.9, 97.5	198	75.8	67.5, 82.5	258	-19.1	<0.001	***
30 years or older	97.1	94.9, 98.4	335	91.7	87.6, 94.5	453	-5.4	0.005	**
Access to and decisions on credit									

Indicator	Baseline (2019)			Midline (2022)			Diff.	p-value ^b	Sig. ^c
	Est.	95% CI	n ^a	Est.	95% CI	n ^a			
All women (ZOI)	12.1	9.2, 15.7	533	17.1	12.9, 22.5	711	5.0	0.09	n/s
Women's age (ZOI)									
18–29 years	13.0	9.3, 17.9	198	21.0	14.6, 29.3	258	8.0	0.07	n/s
30 years or older	11.5	8.1, 16.1	335	14.9	10.5, 20.7	453	3.4	0.31	n/s
Control over income									
All women (ZOI)	89.4	86.1, 92.0	533	95.6	92.0, 97.6	711	6.2	0.002	**
Women's age (ZOI)									
18–29 years	87.2	81.3, 91.4	198	90.8	82.3, 95.5	258	3.6	0.36	n/s
30 years or older	90.8	86.8, 93.7	335	98.3	96.3, 99.2	453	7.5	<0.001	***
Group membership									
All women (ZOI)	44.6	39.7, 49.6	533	34.8	29.0, 41.2	711	-9.8	0.02	*
Women's age (ZOI)									
18–29 years	39.5	33.0, 46.4	198	30.2	21.8, 40.2	258	-9.3	0.01	n/s
30 years or older	47.7	41.5, 54.0	335	37.5	29.8, 46.0	453	-10.2	0.053	n/s
Workload									
All women (ZOI)	42.6	37.0, 48.4	533	31.9	26.6, 37.7	711	-10.7	0.008	**
Women's age (ZOI)									
18–29 years	43.6	35.6, 52.0	198	32.4	24.7, 41.1	258	-11.2	0.06	n/s
30 years or older	42.0	36.3, 47.9	335	31.6	24.7, 39.5	453	-10.4	0.03	*
Percent of women of reproductive age consuming a diet of minimum diversity^f									
All women (ZOI)	29.2	26.5, 32.2	2,318	47.1	43.3, 51.0	3,085	17.9	<0.001	***
Women's age (ZOI)									
15–19 years	32.5	27.2, 38.2	486	47.9	41.9, 54.0	672	15.4	<0.001	***
20–49 years	28.4	26.0, 30.9	1,832	46.9	44.0, 49.7	2,413	18.5	<0.001	***
All women (RFZ)	24.5	18.7, 31.5	402	41.7	35.5, 48.2	1,739	17.1	<0.001	***
Prevalence of exclusive breastfeeding among children under 6 months of age^f									
All children (ZOI)	49.2	39.0, 59.4	161	41.9	31.4, 53.3	188	-7.3	0.35	n/s
Children's sex (ZOI)									

Indicator	Baseline (2019)			Midline (2022)			Diff.	p-value ^b	Sig. ^c
	Est.	95% CI	n ^a	Est.	95% CI	n ^a			
Male	38.8	26.7, 52.4	87	32.9	20.2, 48.6	94	-5.9	0.55	n/s
Female	60.3	47.7, 71.6	74	51.1	39.0, 63.1	94	-9.2	0.30	n/s
All children (RFZ)	^	^	28	76.0	64.7, 84.6	120	na	na	na
Percent of children 6–23 months of age consuming a minimum acceptable diet^f									
All children (ZOI)	11.1	8.4, 14.5	555	27.2	21.7, 33.4	669	16.1	<0.001	***
Children's sex (ZOI)									
Male	10.0	6.7, 14.8	285	32.7	25.2, 41.3	353	22.7	<0.001	***
Female	12.2	8.3, 17.6	270	20.2	13.8, 28.7	316	8.0	0.07	n/s
All children (RFZ)	12.5	7.3, 20.5	107	16.0	12.2, 20.8	376	3.5	0.37	n/s
Percent of households with access to basic sanitation service									
All households (ZOI)	23.2	20.3, 26.4	2,540	23.1	20.2, 26.3	2,995	-0.1	0.93	n/s
Gendered household type (ZOI)									
Male and female adults	25.1	21.8, 28.6	1,836	24.9	21.8, 28.3	2,401	-0.2	0.88	n/s
Female adults only	18.5	14.0, 24.1	450	15.3	10.4, 22.1	399	-3.2	0.22	n/s
Male adults only	18.4	13.1, 25.2	253	16.3	9.0, 27.8	194	-2.1	0.51	n/s
Children only	^	^	1	^	^	1	na	na	na
Residence (ZOI)									
Urban	27.4	21.0, 34.8	627	25.7	18.3, 34.8	552	-1.7	0.25	n/s
Rural	22.1	19.0, 25.6	1,913	22.5	19.4, 25.8	2,443	0.4	0.75	n/s
All households (RFZ)	13.8	8.9, 20.9	436	15.1	12.5, 18.2	1,653	1.3	0.47	n/s
Percent of households with soap and water at handwashing station on premises									
All households (ZOI)	31.7	28.3, 35.2	2,287	19.3	15.0, 24.4	2,823	-12.4	<0.001	***
Gendered household type (ZOI)									
Male and female adults	31.3	27.5, 35.3	1,654	20.6	15.8, 26.3	2,258	-10.7	0.001	**
Female adults only	29.9	23.8, 36.9	398	15.1	9.8, 22.5	379	-14.8	0.002	**
Male adults only	36.8	29.3, 45.1	234	11.6	6.3, 20.6	185	-25.2	<0.001	***
Children only	^	^	1	^	^	1	na	na	na
Residence (ZOI)									

Indicator	Baseline (2019)			Midline (2022)			Diff.	p-value ^b	Sig. ^c
	Est.	95% CI	n ^a	Est.	95% CI	n ^a			
Urban	45.9	40.3, 51.7	545	24.1	12.4, 41.5	518	-21.8	0.008	**
Rural	28.2	24.5, 32.2	1,742	18.1	13.9, 23.2	2,305	-10.1	0.001	**
All households (RFZ)	30.8	19.6, 44.8	366	12.8	10.3, 15.7	1,545	-18.0	0.008	**
Prevalence of household water insecurity, based on the Brief Household Water Insecurity Experiences Scale (HWISE-4)									
All households (ZOI)				37.2	32.8, 41.7	2,995			
Gendered household type (ZOI)									
Male and female adults				37.2	32.7, 41.9	2,401			
Female adults only				36.8	29.7, 44.6	399			
Male adults only				37.9	26.7, 50.5	194			
Children only				^	^	1			
Residence (ZOI)									
Urban				38.7	30.8, 47.2	552			
Rural				36.8	31.7, 42.2	2,443			
All households (RFZ)				47.4	43.3, 51.6	1,653			
Shock exposure index (Mean score)									
All households (RFZ)	18.3	16.9, 19.7	433	27.1	25.8, 28.5	1,649	8.8	<0.001	***
Household resilience capacity index (Mean score)									
All households (RFZ)	54.8	50.3, 59.3	435	39.3	37.2, 41.4	1,650	-15.5	na	na
Absorptive capacity index (Mean score)									
All households (RFZ)	59.4	55.8, 62.9	435	40.4	38.2, 42.6	1,650	-18.9	na	na
Adaptive capacity index (Mean score)									
All households (RFZ)	46.5	42.7, 50.3	435	41.0	39.3, 42.7	1,650	-5.5	na	na
Transformative capacity index (Mean score)									
All households (RFZ)	50.1	43.3, 57.0	435	43.0	39.7, 46.4	1,650	-7.1	na	na
Index of community resilience (Mean score)									
All households (RFZ)	32.4	28.2, 36.5	435	27.2	25.0, 29.3	1,650	-5.2	na	na
Reduced coping strategies index (Mean score)									
All households (RFZ)	8.5	6.7, 10.3	435	16.1	14.7, 17.4	1,629	7.6	<0.001	***

^a Results not statistically reliable, n<30; — Results not available, n=0

A-WEAI=Abbreviated Women's Empowerment in Agriculture Index, CI=confidence interval, Est.=estimate, PPP=purchasing power parity, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, disaggregate sample sizes may not total to the aggregated sample size.

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Significance tests could not be performed for the depth of poverty of the poor or the percent of people who are near-poor due to the methodology used to calculate midline estimates.

^c For the prevalence of poverty, statistical tests of difference could not be conducted. An asterisk (*) indicates that the baseline and midline estimate 95% confidence intervals (CIs) are not overlapping, while “uc” indicates that the baseline and midline 95% CIs overlap. For the prevalence of moderate and severe food insecurity, midline and baseline estimate differences greater than the midline margin of error are considered statistically significant (s=significant; n/s=not significant). For all other indicators, midline and baseline estimate differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^d Indicator estimates imputed using survey-to-survey imputation at the midline. The model was developed using 2019 Feed the Future Uganda P2-ZOI Baseline Survey data.

^e For the poverty indicators, the n's presented are the number of households—not the number of household members—even though the indicator estimates are calculated using a household member sample weight.

^f Estimates are based on de facto household members.

Notes:

Estimates are sample-weighted; numbers are unweighted.

Individual-level estimates are based on de jure household members, except where noted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

KEY FINDINGS

About the 2022 Uganda P2-ZOI and RFZ Midline Indicator Assessment

As the U.S. Government's global hunger and food security initiative, Feed the Future seeks to reduce poverty, hunger, and undernutrition among women and children; strengthen agriculture and food systems; increase income, resilience, women's empowerment, dietary diversity and appropriate feeding practices; and improve hygienic environments. Program efforts are designed to impact populations in the Zone of Influence (ZOI) in Feed the Future target countries, including Uganda. In areas of recurrent crises, the United States Agency for International Development (USAID) also aims to curb the need for humanitarian assistance by building capacity to mitigate, adapt to, and recover from shocks and stresses. In resilience focus countries, such as Uganda, program efforts are designed to strengthen household and community resilience among targeted populations in the Resilience Focus Zone (RFZ).

Uganda is both a Feed the Future target and resilience focus country. One of the main tools to track progress in achieving high-level Feed the Future objectives in the ZOI and resilience objectives in the RFZ is a set of population-based indicators calculated at baseline and then periodically thereafter. The purpose of the 2022 Feed the Future Uganda Phase Two ZOI (P2-ZOI) and RFZ Midline Indicator Assessment (hereafter referred to as the "assessment") is to provide the USAID Bureau for Resilience, Environment, and Food Security (REFS), USAID Uganda, U.S. Government interagency partners, the Government of Uganda, and development partners with information about short-term changes across a set of population-based indicators in the Uganda P2-ZOI and RFZ. This is the first assessment of the indicators in Uganda since the baseline values were established in 2019. Midline data were collected in the P2-ZOI on poverty, food security, agricultural cross-market functions, women's empowerment in agriculture, resilience, women's and children's nutrition, water and sanitation practices, climate adaptation, COVID-19, and coverage of promoted interventions. Additional midline data were collected in the RFZ on resilience capacities and coping strategies.

This report presents the results from the assessment, including changes between baseline and midline indicator estimates.

Phase Two Zone of Influence and Resilience Focus Zone

The P2-ZOI in Uganda covers 44 districts⁶ located across 4 regions:⁷ East, North, Karamoja, and Southwest (Figure 1). Within the P2-ZOI, the RFZ covers 13 districts across the North, Karamoja, and Southwest regions.^{8,9}

Survey methods

⁶ At baseline, the P2-ZOI included 41 districts. Since then, 3 districts have been subdivided into new districts, thereby increasing the total number of districts in the Uganda ZOI to 44 districts.

⁷ Three of the regions—East, North, and Karamoja—align with Country Development Cooperation Strategy (CDCS) regional corridors in Uganda. The fourth region, Southwest, combines the Southwestern and Western CDCS regional corridors due to the small number of ZOI districts in the Western CDCS regional corridor.

⁸ The 44 districts are as follows, with the 13 RFZ districts denoted by an asterisk: Abim*, Amudat*, Amuru, Bududa, Buhweju, Bulambuli, Bushenyi, Gulu, Ibanda, Isingiro*, Kaabong*, Kabale, Kamwenge*, Kanungu, Kapchorwa, Karengs*, Kasese, Kisoro, Kitagwenda*, Kole, Kotido*, Lamwo*, Lira, Manafwa, Mbale, Mbarara, Mitooma, Moroto*, Nabilatuk*, Nakapiripirit*, Namisindwa, Napak*, Ntungamo, Nwoya, Omoro, Oyam, Rubanda, Rubirizi, Rukiga*, Rukungiri, Rwampara*, Sheema, Sironko, and Tororo.

⁹ Kitagwenda was retained as an RFZ district at midline to align with RFZ districts at baseline.

Midline data were collected through the implementation of the 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey (hereafter referred to as the P2-ZOI Midline Survey), which used a multi-stage random sample representative of the population in the Uganda P2-ZOI and RFZ. A total of 3,200 households were sampled from 128 enumeration areas (EAs) in the P2-ZOI, and interviews were completed in 2,995 households (a 98% response rate among eligible households) (Table 2.2.1). The sample included 1,775 households in the Uganda RFZ, and among eligible households, 1,653 completed interviews (a 99% response rate among eligible households).

Midline data collection was carried out from October 28 to December 14, 2022.

Respondents to individual-level questionnaire modules in the P2-ZOI Midline Survey included primary adult female decision-makers (the female adult who makes the more important decisions in the household), women of reproductive age (15–49 years of age), caregivers of children under 5 years of age, and farmers 15 years of age or older responsible for cultivating crops.

Baseline results presented in this report were generated from the 2019 Feed the Future Uganda P2-ZOI Baseline Survey (hereafter referred to as the P2-ZOI Baseline Survey). Although the P2-ZOI Baseline Survey was not designed to include a representative sample of the RFZ, 436 households in the RFZ completed interviews, and these data were used to generate RFZ estimates. P2-ZOI Baseline Survey data collection was carried out from September 7 to November 25, 2019. More information on survey methods used to conduct the P2-ZOI Baseline Survey can be found in the P2-ZOI Baseline Report (Feed the Future, 2022).

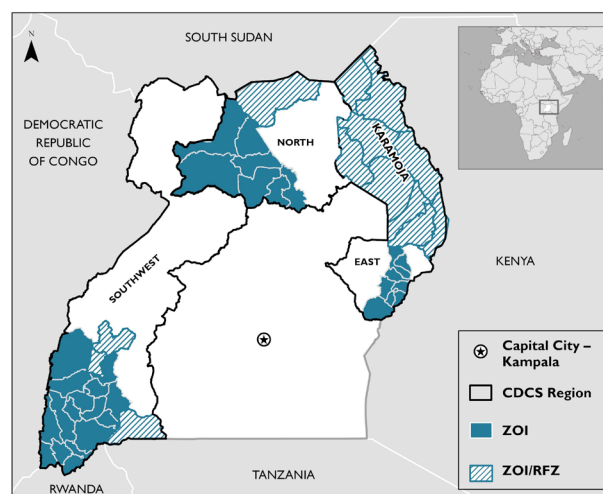
Tests of differences between baseline and midline indicator estimates were conducted to detect statistically significant changes over time, where feasible. Statistically significant changes are noted throughout this report.¹⁰

Demographic characteristics

Population estimates

The total estimated population in the Uganda P2-ZOI at midline is approximately 12.2 million people, and the estimated number of households in the P2-ZOI at midline is approximately 2.2 million (Tables

Figure 1. Map of the Uganda P2-ZOI and RFZ



¹⁰ The significance level used for results in this report is $p < 0.05$, except for the poverty and food insecurity indicators, for which traditional tests of significance could not be performed. For prevalence of poverty, the change between baseline and midline is considered statistically significant if the baseline and midline estimate 95 percent confidence intervals do not overlap. There was no way to determine statistical significance of change over time for depth of poverty and prevalence of near-poor indicators. For the food insecurity indicator, the change between baseline and midline is considered statistically significant if the difference between the baseline and midline estimates is greater than the margin of error for the midline estimate.

1.2.1 and 1.2.2). The estimated population in the RFZ at midline is approximately 2.0 million, and the estimated number of households in the RFZ at midline is approximately 358,000 (Tables 1.2.1 and 1.2.2).

Household composition

Between baseline and midline, there was a significant increase in household size in the Uganda P2-ZOI, from 5.1 to 5.6 members per household (Table 3.1.1). The mean number of children under 5 years of age (baseline: 0.8, midline: 0.8) and children 5–17 years of age (baseline: 2.0, midline: 2.1) did not change significantly, but the mean number of male adults (baseline: 1.1, midline: 1.3) and female adults (baseline: 1.2, midline: 1.4) per household increased significantly (Table 3.1.1).

Similar trends in household composition were evident in the RFZ. Between baseline and midline, there was a significant increase in household size, from 5.1 to 5.8 members per household (Table 3.1.1). The mean number of children under 5 years of age (baseline: 0.8, midline: 0.9) and children 5–17 years of age (baseline: 2.1, midline: 2.3) did not change significantly, but the mean number of male adults (baseline: 1.0, midline: 1.2) and female adults (baseline: 1.2, midline: 1.3) per household increased significantly (Table 3.1.1).

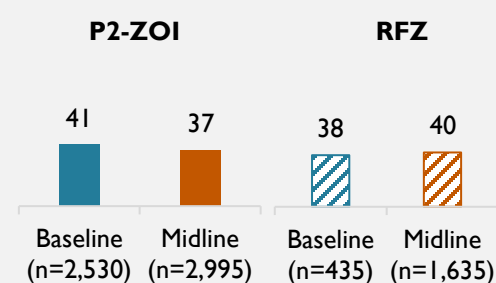
Poverty

This section presents baseline and midline P2-ZOI and RFZ results for poverty indicators, including three consumption-based indicators and one asset-based indicator.

Prevalence of poverty

An individual is considered poor if their daily per capita consumption expenditures fall below a global threshold of USD 1.90 at 2011 purchasing power parity (PPP) prices.¹¹ The percent of individuals in the P2-ZOI living below the USD 1.90 poverty threshold was 41% at baseline and 37% at midline. In the RFZ, the percent was 38% at baseline and 40% at midline (Figure 2).

Figure 2. Prevalence of poverty (USD 1.90 poverty threshold) in the P2-ZOI and RFZ



Depth of poverty of the poor

The depth of poverty of the poor represents the average percent consumption shortfall of the poor, relative to the global poverty line of USD 1.90 at 2011 PPP. The depth of poverty of the poor (mean percent shortfall of the poor relative to the USD 1.90 threshold) in the P2-ZOI was similar at baseline (34%) and midline (36%) (Table 4.1.1). In other words, poor individuals in the P2-ZOI, on average, lived on USD 1.25 per person per day at baseline and USD 1.22 per person per day at midline (adjusted for inflation). In the RFZ, the depth of poverty of the poor was 41% at baseline and 42% at midline (Table 4.1.1). In other words, poor individuals in the RFZ,

¹¹ In September 2022, the World Bank adjusted the global poverty threshold to USD 2.15 at 2017 PPP. This report presents the poverty at the previous threshold of USD 1.90 at 2011 PPP, in accordance with the 2019 *Feed the Future Indicator Handbook* (USAID, 2018).

on average, lived on USD 1.12 per person per day at baseline and USD 1.10 per person per day at midline (adjusted for inflation). No tests of significance were performed.¹²

Percent of people who are near-poor

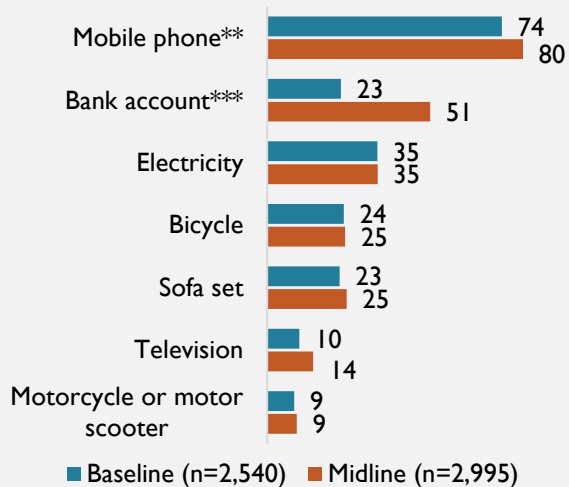
Individuals who are near-poor are those living at 100% to less than 125% of the global poverty threshold, or between USD 1.90 and 2.38 per person per day at 2011 PPP. Those who are near-poor are more likely to be pushed into poverty when faced with difficult times. The percent of individuals in the P2-ZOI who were near-poor was similar at baseline (14%) and midline (12%) (Table 4.1.1). In the RFZ, the percent of near-poor individuals was 9% at baseline and 13% at midline (Table 4.1.1). No tests of significance were performed.

Asset-based Comparative Wealth Index

Asset ownership can be used to predict a household's long-term welfare—its capacity to earn income and withstand shocks in the future. The number and type of assets that a household owns are associated with household resilience across national contexts. Asset-based wealth indices have increasingly been used as alternatives to income and consumption expenditure-based measures.

Whereas other asset-based wealth indices reflect a relative measure of wealth relevant to the situation in the country at the time of the survey, the Comparative Wealth Index (CWI) is comparable across countries and time, allowing for direct comparison of economic status. Using the CWI, households are categorized into standardized quintiles based on level of asset ownership.

Figure 3. Percent of households in the P2-ZOI that have or own selected assets



Significance levels: *p<0.05, **p<0.01, ***p<0.001

¹² Due to the survey-to-survey imputation method used to calculate a consumption aggregate for each household, reliable standard error estimates are not available for the depth of poverty of the poor; therefore, statistical tests of difference were not performed, and only differences between baseline and midline are presented.

As shown in Figure 3, the percent of households in the P2-ZOI that had a mobile phone (baseline: 74%, midline: 80%) and bank account (baseline: 23%, midline: 51%) increased significantly. However, for most selected assets, the percent of households that had or owned them did not change significantly between baseline and midline.¹³

Figure 4 shows that the percent of households in the P2-ZOI in the poorest CWI quintile did not change significantly between baseline and midline (baseline: 40%, midline: 38%). However, the percent of households in the RFZ in the poorest CWI quintile increased significantly between baseline and midline (baseline: 44%, midline: 61%).

Food insecurity

Food insecurity exists when people do not have access to enough quality food for an active and healthy life. The Food Insecurity Experience Scale (FIES) measures the percent of households that experienced moderate or severe food insecurity¹⁴ due to a lack of money or other resources in the 12 months before the survey.

Figure 5 shows that between baseline and midline, the prevalence of moderate and severe food insecurity in the P2-ZOI increased significantly (baseline: 71%, midline: 85%). In addition, both the prevalence of moderate food insecurity (baseline: 42%, midline: 48%) and the prevalence of severe food insecurity (baseline: 28%, midline: 37%) increased significantly.

Similar to the P2-ZOI results, the prevalence of moderate and severe food insecurity in the RFZ also increased significantly (baseline: 71%, midline: 88%). However, in the RFZ, neither the prevalence of moderate food insecurity (baseline: 41%, midline: 43%) nor the prevalence of severe food insecurity (baseline: 30%, midline: 44%) changed significantly (Figure 5).

Figure 4. Percent of households in the P2-ZOI and RFZ in the lowest CWI quintile

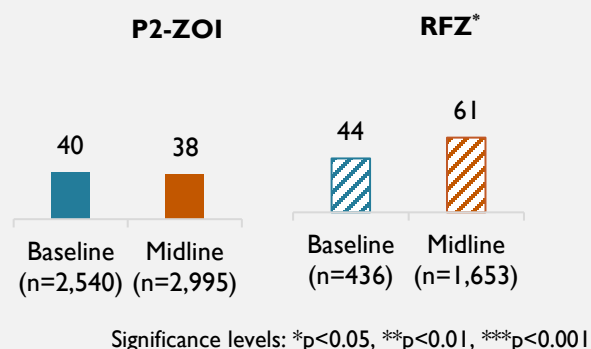
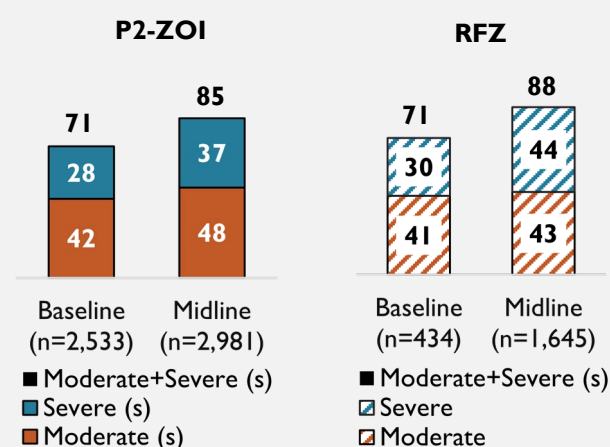


Figure 5. Percent of households in the P2-ZOI and RFZ experiencing moderate and severe food insecurity



(s): significant using a margin of error test

Note: Moderate food insecurity and severe food insecurity may not sum to moderate and severe food insecurity due to rounding.

¹³ These results were calculated for the RFZ.

¹⁴ Moderate food insecurity is associated with the inability to regularly eat healthy, nutritious diets. Severe food insecurity is more related to insufficient quantity of food in general.

Resilience

Household resilience, as defined by USAID, reflects a household's ability to mitigate, adapt to, and recover from shocks and stresses in a manner that reduces chronic vulnerability and facilitates inclusive growth. Shocks and stresses are events and trends with the potential to negatively impact a household's well-being (e.g., food security) at the time of the shock or stress and in the future. This section presents baseline and midline P2-ZOI and RFZ results on shock and stress exposure and perceived recovery; COVID-19; participation in group-based savings, micro-finance, and loan programs; perceptions of local government responsiveness; and social capital.

Exposure to shocks and stresses and perceived ability to recover

The mean Ability to Recover from Shocks and Stresses Index (ARSSI) score for households in the P2-ZOI decreased significantly from 4.4 at baseline to 3.9 at midline (Table 5.3.1). Figure 6 shows that, on average, households in the P2-ZOI experienced 9 shocks or stresses during the 12 months preceding the P2-ZOI Midline Survey, a significant increase from 6 shocks or stresses experienced during the 12 months preceding the P2-ZOI Baseline Survey.

There were significant increases in the percent of households in the P2-ZOI that experienced 17 of the 19 shocks and stresses included in the surveys (Figure 7). At baseline and midline, the two shocks or stresses households in the P2-ZOI most commonly experienced were an increase in food prices (baseline: 74%, midline: 95%) and too little rain (baseline: 70%, midline: 79%). The two shocks with the largest percentage point increases between baseline and midline were unemployment of youth (baseline: 23%, midline: 55%) and delays in food assistance (baseline: 11%, midline: 37%).

In the RFZ, the mean ARSSI score also decreased significantly from 4.7 at baseline to 3.6 at midline (Table 5.3.1), and the mean number of shocks or stresses experienced by households in the 12 months preceding the survey also increased significantly (baseline: 6, midline: 9) (Figure 6). There were significant increases in the percent of households in the RFZ that experienced 10 of the 19 shocks and stresses included in the surveys (Figure 7). As in the P2-ZOI, at baseline and midline, the two shocks or stresses households most commonly experienced were an increase in food prices (baseline: 75%, midline: 96%) and too little rain (baseline: 77%, midline: 85%). The two shocks or stresses with the largest percentage point increases between baseline and midline in the RFZ were delays in food assistance (baseline: 5%, midline: 41%) and unemployment of youth (baseline: 20%, midline: 49%).

Figure 6. Mean number of shocks and stresses experienced by households in the P2-ZOI and RFZ in the 12 months preceding the survey

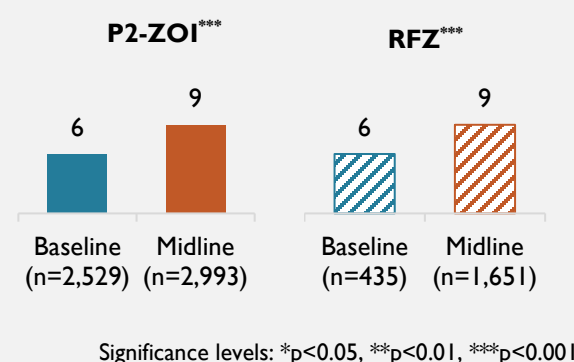
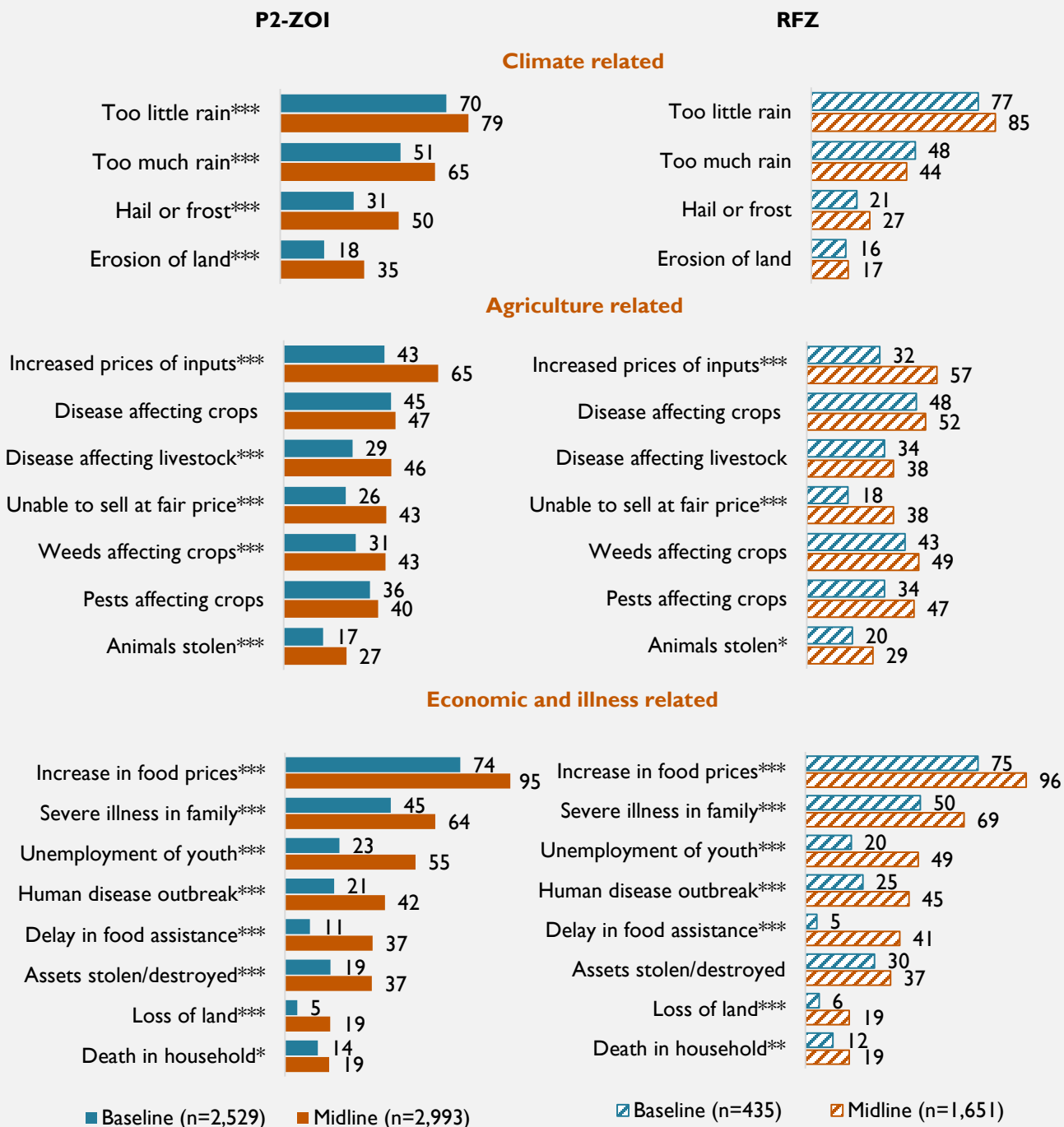


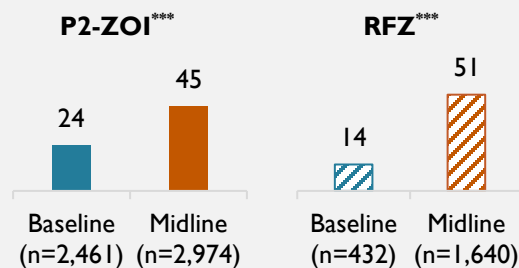
Figure 7. Percent of households in the P2-ZOI and RFZ that experienced each shock or stress in the 12 months before the survey



Significance levels: *p<0.05, **p<0.01, ***p<0.001

At midline, 45% of households in the P2-ZOI that experienced at least 1 shock or stress in the 12 months preceding the survey believed their ability to meet their food needs at the time of the survey was worse than before the shock exposure, a significant increase since baseline (24%) (Figure 8). A similar pattern is seen in households' perceived ability to meet their food needs during the upcoming year. At midline, 30% of households that experienced at least 1 shock or stress in the 12 months preceding the survey believed their ability to meet their food needs in the year after the survey would be worse than before the shock or stress, a significant increase from baseline (18%) (Table 5.3.3).

Figure 8. Percent of households in the P2-ZOI and RFZ that perceived their ability to meet their food needs at the time of survey as worse than before the shock exposure



Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

At midline, 51% of households in the RFZ that experienced at least 1 shock or stress in the 12 months preceding the survey believed their ability to meet their food needs at the time of the survey was worse than before they experienced the shock or stress, a significant increase since baseline (14%) (Figure 8). A similar pattern is seen in households' perceived ability to meet their food needs during the upcoming year. At midline, 38% of households that experienced at least 1 shock or stress in the 12 months preceding the survey believed their ability to meet their food needs in the year after the survey would be worse than before the shock or stress, a significant increase from baseline (13%) (Table 5.3.3).

COVID-19

The global spread of COVID-19, beginning in 2020, represented an unprecedented shock globally. As such, the P2-ZOI Midline Survey sought to collect information on household-level impacts of COVID-19.

At midline, nearly all households in the P2-ZOI (99%) said that they were affected by the COVID-19 pandemic (Table 5.2.4). Households reported experiencing worry or concern (96%), difficulty moving freely or socializing (94%), difficulty purchasing food or goods (88%), loss of income or inability to work (73%), and the need to spend savings or sell assets (69%) due to COVID-19 (Table 5.2.4).

Participation in group-based savings, micro-finance, or lending programs

Between baseline and midline, the percent of households in the P2-ZOI that participated in group-based savings, micro-finance, or lending programs did not change significantly (baseline: 60%, midline: 63%) (Table 5.4.5). However, the percent of households in the RFZ that participated in group-based savings, micro-finance, or lending programs decreased significantly between the two time points (baseline: 64%, midline: 54%) (Table 5.4.5).

Perceptions of local government responsiveness

Between baseline and midline, the percent of households in the P2-ZOI that believed their local government would help the community cope with future shocks or stresses did not change significantly (baseline: 61%, midline: 59%) (Table 5.4.4). Estimates for the RFZ were similar to the P2-ZOI, with no significant changes between the two time points (baseline: 65%, midline: 62%) (Table 5.4.4).

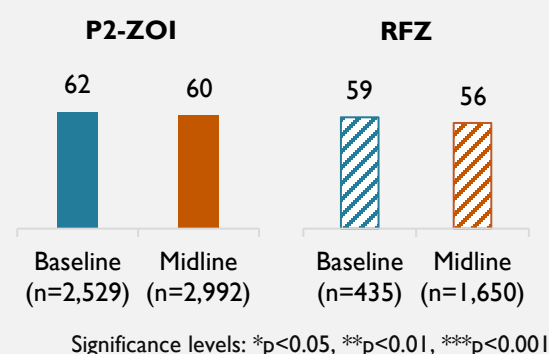
Social capital

Social capital has been shown to be an important source of resilience across different shocks and stresses, geographies, and populations. There are two sub-indices that comprise the social capital index: the bonding sub-index and the bridging sub-index. The bonding sub-index measures a household's ability to give and receive support *within* its community, and the bridging sub-index measures a household's ability to give and receive support from *outside* its community.

Figure 9 shows that the mean social capital index score in the P2-ZOI did not change significantly between baseline and midline (baseline: 62, midline: 60). While the mean bonding sub-index score decreased significantly (baseline: 65, midline: 61), the mean bridging sub-index score in the P2-ZOI did not change significantly (baseline: 59, midline: 60) (Tables 5.4.2 and 5.4.3).

In the RFZ, there were no significant changes in the mean social capital index score between the two points in time (baseline: 59, midline: 56) (Figure 9). There were also no significant changes in the mean bonding sub-index score (baseline: 61, midline: 58) and mean bridging sub-index score (baseline: 57, midline: 55) (Tables 5.4.2 and 5.4.3).

Figure 9. Mean social capital index score in the P2-ZOI and RFZ



Expanded resilience indicators (RFZ only)

In this section, baseline and midline results for an expanded set of resilience indicators are presented for the RFZ, including the shock exposure index (SEI), the resilience capacity indices (absorptive, adaptive, and transformative), the index of community resilience, and the reduced coping strategies index (TANGO International, 2018 and Maxwell & Caldwell, 2008). The data to calculate these indicators were collected only in the RFZ where USAID program efforts are specifically designed to build capacities to mitigate, adapt to, and recover from shocks and stresses.

Shock exposure index

The SEI measures the overall degree of context-specific shock and stress exposure at the household level by accounting for the number of shocks and stresses that households experienced and the perceived severity of those shocks and stresses on household food consumption. For this assessment, the SEI is measured on a scale from 0 to 76.¹⁵ Between baseline and midline, the average SEI for households in the RFZ increased significantly (baseline: 18, midline: 27) (Table 5.7.3).

Resilience capacity indices

Resilience capacity represents the potential for taking proactive measures to cope with shocks or stresses. Resilience capacity is measured as a set of indices, one for each of the three dimensions of resilience capacity: absorptive capacity, adaptive capacity, and transformative capacity. These are all

¹⁵ The scale is dependent on the number of shocks and stresses included in the calculation. For this analysis, because the surveys included 19 shocks and stresses, each of which could have a perceived severity between 1 and 4, the scale is 0–76.

combined into a composite index: the household resilience capacity index.¹⁶ The four indices are measured on a scale from 0 to 100. Statistical tests of difference were not performed comparing the baseline and midline estimates for the four indices, but tests were performed for each of the indices' component indicators. The household resilience capacity index score for the RFZ was 55 at baseline and 39 at midline (Table 5.7.4).

The absorptive capacity index measures a household's ability to mitigate exposure to shocks or stresses through preventative measures and appropriate coping strategies to avoid permanent, negative impacts. The mean absorptive capacity index score in the RFZ was 59 at baseline and 40 at midline (Figure 10). Of the seven component indicators that comprise the absorptive capacity index, four significantly decreased, with the largest decreases in access to cash savings (baseline: 73, midline: 48) and availability of informal safety nets (baseline: 68, midline: 50).

Figure 10. Absorptive capacity index and component indicator scores in the RFZ



Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

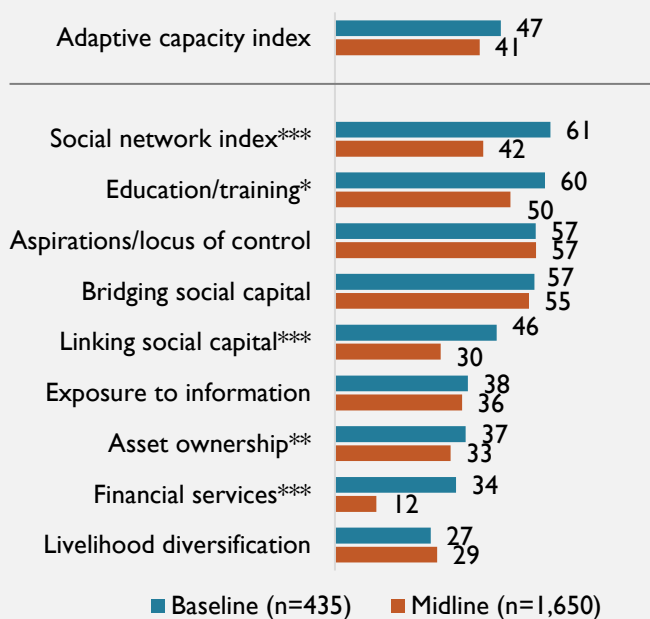
Note: Test of difference not performed on absorptive capacity index.

¹⁶ A test of difference between baseline and midline was not performed for the household resilience, absorptive, adaptive, or transformative capacity indices due to concerns about the comparability of baseline and midline index estimates created using factor analysis with large differences in sample size and important differences in the sampling design approaches.

The adaptive capacity index measures a household's ability to make proactive and informed choices about alternative livelihood strategies based on an understanding of changing conditions. The mean adaptive capacity index score for households in the RFZ was 47 at baseline and 41 at midline (Figure 11). Of the nine component indicators that comprise the adaptive capacity index, five decreased significantly. Those with the largest decreases are the social network index (baseline: 61, midline: 42), linking social capital (baseline: 46, midline: 30), and the availability of financial services (baseline: 34, midline: 12).

The transformative capacity index measures whether there is an enabling environment to facilitate a household's recovery from shocks or stresses. Transformative capacity involves the governance mechanisms, policies/regulations, infrastructure, community networks, and formal and informal social protection mechanisms that constitute the enabling environment for systemic change. The mean transformative capacity score for households in the RFZ was 50 at baseline and 43 at midline (Figure 12). The four highest scoring transformative capacity index component indicators at baseline did not change significantly between baseline and midline; however, there were significant decreases in five of the remaining seven component indicators. The largest decreases among the five component indicators were in linking social capital (baseline: 46, midline: 30) and the availability of or access to basic services (baseline: 45, midline: 31).

Figure 11. Adaptive capacity index and component indicator scores in the RFZ



Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
 Note: Test of difference not performed on adaptive capacity index.

Figure 12. Transformative capacity index and component indicator scores in the RFZ



Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
 Note: Test of difference not performed on transformative capacity index.

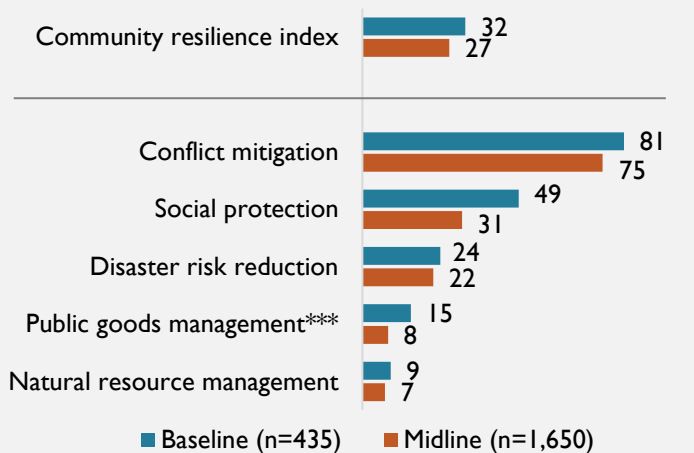
Index of community resilience

Community resilience reflects the ability of a community to plan for and successfully adapt to actual or potential adverse events in a timely manner. The index of community resilience is calculated on a scale from 0 to 100 using five component indicators. The mean index of community resilience score for households in the RFZ was 32 at baseline and 27 at midline (Figure 13).¹⁷ One component indicator, public goods management (baseline: 15, midline: 8), decreased significantly.

Reduced coping strategies index

The reduced coping strategies index is an indicator of household food insecurity that accounts for the frequency and severity of five food consumption behaviors—eating less preferred foods, borrowing food or money from friends and relatives, limiting portions at mealtimes, limiting adult intake, and reducing the number of meals per day—that households have used in the seven days prior to the survey. It is calculated on a scale from 0 to 56, with a higher score indicating greater household food insecurity. Between baseline and midline, the reduced coping strategies index increased significantly (baseline: 9, midline: 16) in the RFZ (Table 5.7.7). The percent of households that used each of the five coping strategies also increased significantly between baseline and midline, with the largest percentage point increases in households that limited portions at mealtime (baseline: 33%, midline: 78%) and reduced the number of meals per day (baseline: 27%, midline: 75%) (Figure 14).

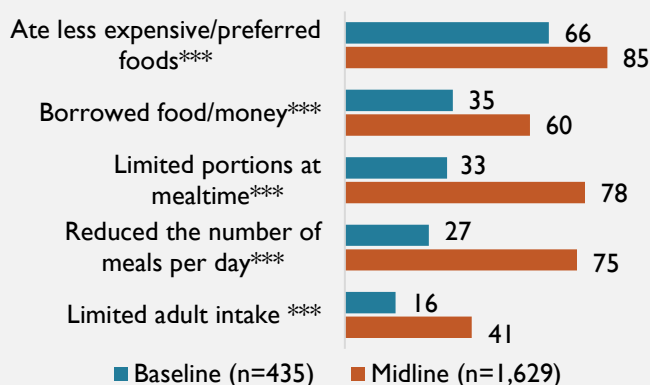
Figure 13. Index of community resilience and component indicator scores in the RFZ



Significance levels: *p<0.05, **p<0.01, ***p<0.001

Note: Test of difference not performed on the community resilience index or on the social protection component indicator.

Figure 14. Percent of households in the RFZ that used each coping strategy during the 7 days preceding the survey



Significance levels: *p<0.05, **p<0.01, ***p<0.001

¹⁷ A test of difference between baseline and midline was not performed for the index of community resilience due to concerns about the comparability of baseline and midline index estimates created using factor analysis with large differences in sample size and important differences in the sampling design approaches.

Climate adaptation

Addressing the immediate and long-term impacts of climate change underpins the success of Feed the Future and the ability of communities to sustain development outcomes by anticipating, preparing for, and adapting to current and future climate impacts. Feed the Future aims to help people, communities, and countries anticipate, prepare for, and adapt to current and future climate impacts.

Results presented in this section are used to help understand how individuals and communities are responding to the challenges posed by climate change. Descriptive results in this section focus on household climate change knowledge and concern, application of climate adaptive agricultural technologies and practices, and access to insurance and other services designed to help manage the impacts of weather-related problems on livelihoods. Climate information was collected only in the P2-ZOI Midline Survey to generate descriptive statistics for the P2-ZOI; thus, results are presented only for the P2-ZOI at midline.

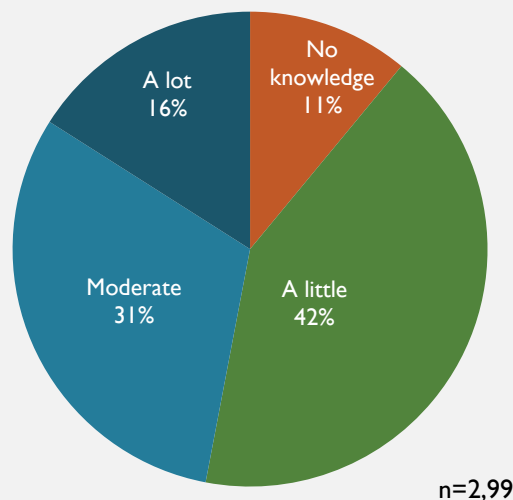
Climate change knowledge and concern

Figure 15 shows that approximately 1 in 10 households reported having no knowledge about climate change; about 4 in 10 said that they knew a little, and almost half said that they knew a moderate amount or a lot.

Only 27% of households reported receiving information to help prepare for or manage weather-related problems, and among those that received information, the main mechanisms of receiving information were radio (65%) and word of mouth (39%) (Table 5.5.2).

Approximately one-third of households (31%) were very concerned about the impacts of climate change on day-to-day life and income, and the majority of households (64%) were somewhat or moderately concerned; only 5% were not concerned (Table 5.5.1).

Figure 15. Percent distribution of households in the P2-ZOI by their level of climate change knowledge



Application of climate adaptive agricultural practices

Feed the Future promotes the use of climate adaptive agricultural technologies and practices to reduce risk and minimize the severity of the impacts of climate change on livelihoods. Examples of these practices include using flood- or drought-tolerant seed, adjusting planting times and fertilizer use, rotating crops, and utilizing drip irrigation methods.

Among households that cultivated crops in the P2-ZOI, 25% reported adopting at least 1 climate adaptive practice related to crop production in the 12 months before the survey (Table 5.5.3). Among households that raised livestock or other farm animals, 12% adopted a climate adaptive practice related to animal production (Table 5.5.4).

About one-quarter (24%) of households that cultivated crops or raised livestock or other farm animals faced barriers to adopting agricultural practices to help manage

weather-related problems related to their crop or animal production (Table 5.5.5). Figure 16 shows the percent of households that cultivated crops or raised livestock or other farm animals that faced specific barriers in the 12 months before the survey. Among those households that experienced any barriers, the two most common barriers were money for inputs (53%) and access to land (45%).

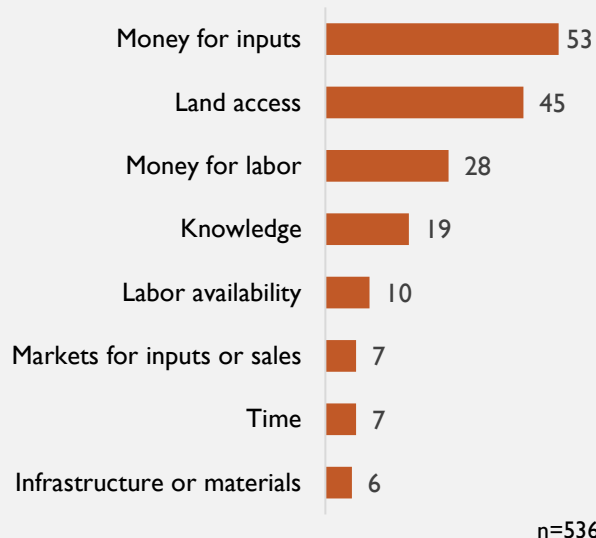
Access to insurance, income diversification activities, and participation in committees or groups to manage weather-related problems

Only 1% of households in the P2-ZOI obtained any type of insurance in the 12 months preceding the survey (Table 5.5.7). Fewer than 1% of agricultural households in the P2-ZOI received payments for weather-related problems that affected their crop or livestock production (Table 5.5.8).

One in 10 households in the P2-ZOI participated in training or received support to help diversify their sources of income during the 12 months preceding the survey (Table 5.5.10). Among these households, the most common topics of training or support received were about new crops or products (32%), petty trade (28%), and raising livestock (25%) (Table 5.5.10).

Finally, 8% of households in the P2-ZOI participated in a group or committee working to strengthen the community's ability to manage weather-related problems during the 12 months preceding the survey (Table 5.5.6), and 3% of households participated in a group or committee that monitors or influences land use (Tables 5.5.9).

Figure 16. Percent of households in the P2-ZOI that experienced each barrier to adopting practices to help manage weather-related problems



Note: Includes only households that cultivated crops or raised livestock that experienced any barriers

Nutrition

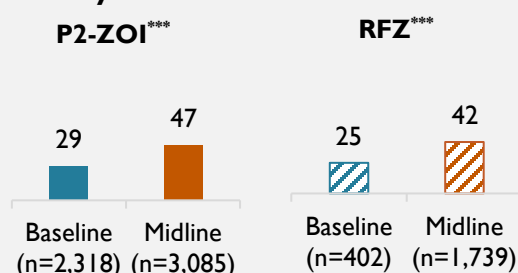
Good nutrition is essential to the health and well-being of individuals, families, communities, and entire nations, and it is core to the Feed the Future objectives. Women and girls have unique nutritional requirements throughout their life cycle—especially during adolescence, before and during pregnancy, and while breastfeeding. Nutritional vulnerability of children is also greatest during the 1,000 days between conception and a child’s second birthday because this is a critical period of growth and development. This section presents baseline and midline P2-ZOI and RFZ results for women’s and children’s dietary intake indicators.¹⁸

Women’s minimum dietary diversity

Women’s minimum dietary diversity (MDD-W) is achieved when women of reproductive age (15–49 years of age) consume foods from 5 of 10 food categories during the day and night before the survey. As shown in Figure 17, at midline, 47% of the women in the P2-ZOI consumed a diet of minimum diversity, a significant increase from baseline (29%).

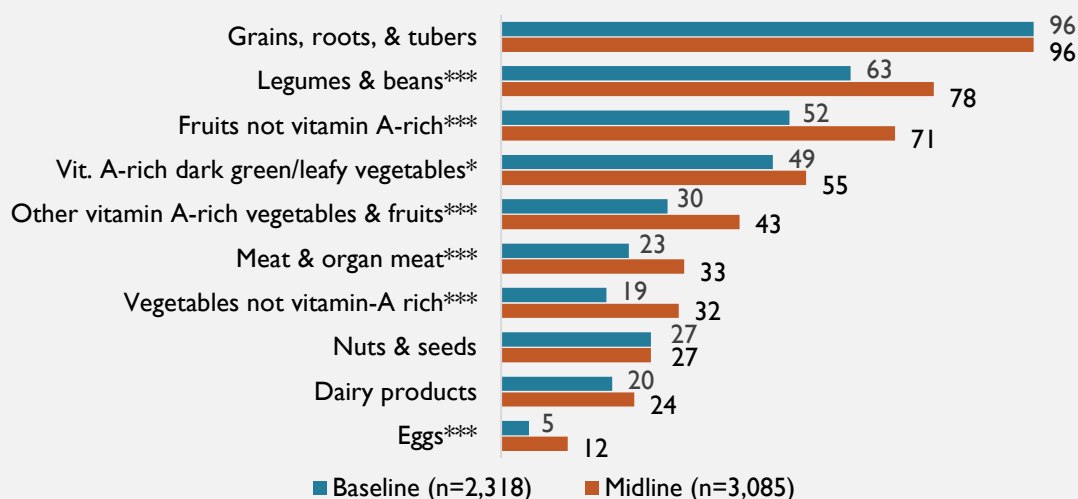
Between baseline and midline, there was also a significant increase in the percent of women of reproductive age in the RFZ who consumed a diet of minimum diversity (baseline: 25%, midline: 42%) (Figure 17).

Figure 17. Percent of women of reproductive age in the P2-ZOI and RFZ who consumed a diet of minimum diversity



Significance levels: *p<0.05, **p<0.01, ***p<0.001

Figure 18. Percent of women of reproductive age in the P2-ZOI who consumed foods in each food category



Significance levels: *p<0.05, **p<0.01, ***p<0.001

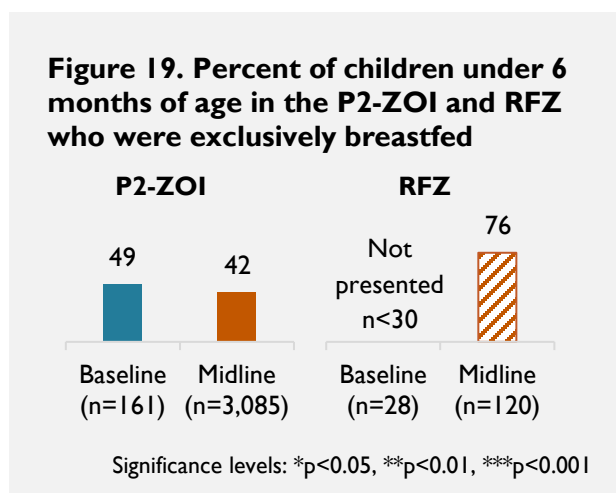
¹⁸ Women’s and children’s nutritional status indicators were not calculated as part of the assessment because Demographic and Health Survey data collected in 2022 or 2023 were not available for Uganda, and the collection of anthropometry data is not included in any Feed the Future P2-ZOI Midline Surveys.

Figure 18 shows that the percent of women in the P2-ZOI consuming foods in each of 7 of the 10 food groups increased significantly between the two time points, with the largest percentage point increase in fruits not rich in vitamin A (baseline: 52%, midline: 71%).

Exclusive breastfeeding

Children under 6 months of age who are exclusively breastfed tend to have better nutritional status and lower rates of morbidity and mortality in the short and long term. The World Health Organization (WHO) and UNICEF recommend that children initiate breastfeeding within the first hour of birth and be exclusively breastfed for the first 6 months of life—meaning that no other foods or liquids, including water, are provided.

As Figure 19 shows, in the P2-ZOI, the prevalence of exclusive breastfeeding (EBF) did not change significantly between baseline and midline (baseline: 49%, midline: 42%).

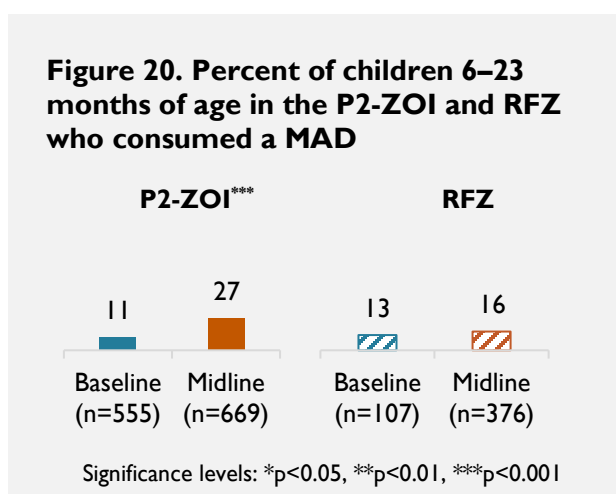


In the RFZ, the prevalence of EBF at midline was 76% (Figure 19). Because the sample of children 0–5 months of age in the RFZ at baseline was not large enough to generate a baseline estimate, a comparison of baseline and midline estimates for the RFZ could not be conducted for the indicator.

Children's minimum acceptable diet

Children 6–23 months of age consume a minimum acceptable diet (MAD) when they achieve minimum dietary diversity (MDD) (i.e., consume foods from at least 5 of 8 food categories), meet minimum meal frequency requirements for their age and breastfeeding status, and meet the minimum milk feeding frequency (for non-breastfed children).

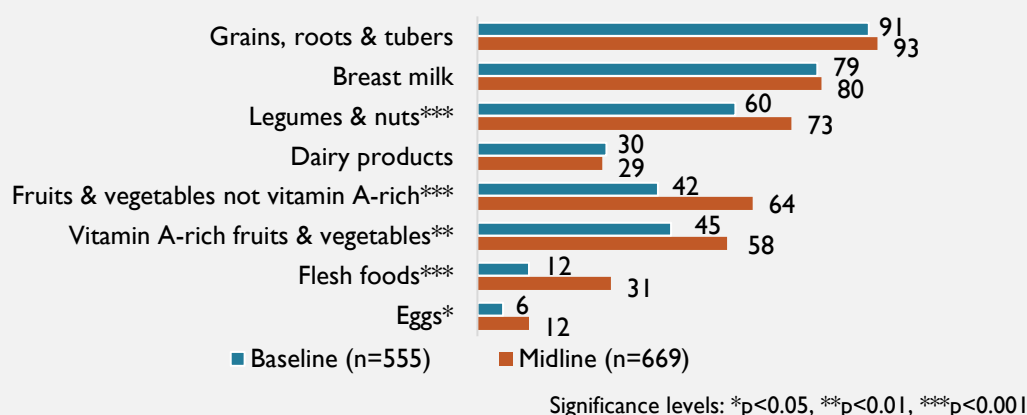
At midline, 27% of children in the P2-ZOI consumed a MAD, a significant increase from baseline (11%) (Figure 20).



In the RFZ, the percent of children who consumed a MAD did not change significantly (baseline: 13%, midline: 16%) (Figure 20).

Figure 21 shows that the percent of children in the P2-ZOI who consumed foods in each of five of the eight food categories increased significantly between baseline and midline. The percent of children consuming non-vitamin A-rich fruits and vegetables (baseline: 42%, midline: 64%) and the percent of children consuming flesh foods (baseline: 12%, midline: 31%) showed the largest percentage point increases.

Figure 21. Percent of children 6–23 months in the P2-ZOI consuming foods from each food category



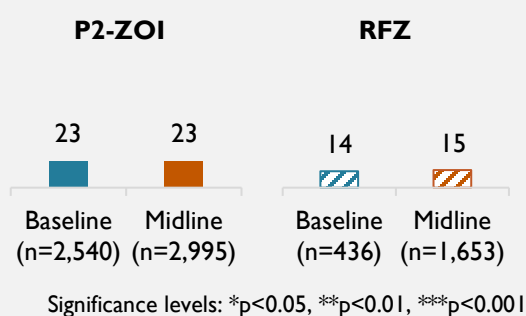
Water, sanitation, and hygiene

Reliable access to safe water, sanitation, and hygiene saves lives, improves livelihoods, and makes communities more resilient. This section presents baseline and midline P2-ZOI and RFZ results for water, sanitation, and hygiene indicators. It also includes results for a new indicator on water insecurity for the Uganda P2-ZOI and RFZ at midline only.

Access to a basic sanitation service

A basic sanitation service, as defined by the WHO/UNICEF Joint Monitoring Programme for Water Supply, Sanitation, and Hygiene, consists of a sanitation facility that: (1) hygienically separates human excreta from human contact (i.e., an improved sanitation facility) and (2) is not shared with other households. A limited sanitation service meets the first requirement but not the second—that is, the sanitation facility is improved but shared with other households. Unimproved sanitation does not meet the first requirement and may or may not meet the second.

Figure 22. Percent of households in the P2-ZOI and RFZ that have a basic sanitation service



The percent of households in the P2-ZOI with a basic sanitation service remained unchanged between baseline and midline (baseline: 23%, midline: 23%) (Figure 22). Furthermore, the percent of households in the P2-ZOI with a limited sanitation service (baseline: 15%, midline: 15%), with unimproved sanitation including open defecation (baseline: 62%, midline: 62%), and practicing open defecation (baseline: 8% and midline: 7%) did not change significantly between the two time points (Table 3.3.4).

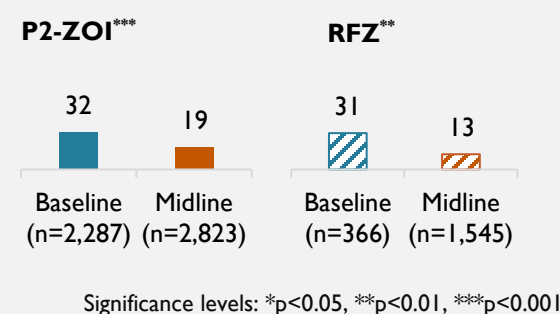
In the RFZ, the results also did not change significantly between the two time points: basic sanitation service (baseline: 14%, midline: 15%), limited sanitation service (baseline: 19%, midline: 13%), unimproved sanitation including open defecation (baseline: 67%, midline: 72%), and open defecation (baseline: 25%, midline: 28%) (Table 3.3.4).

Soap and water at the handwashing station on the premises

Having soap and water at handwashing stations helps prevent the transmission of certain illnesses, especially diarrheal illnesses, which can lead to malnutrition. Between baseline and midline, the percent of households in the P2-ZOI with soap and water at handwashing stations on the premises decreased significantly (baseline: 32%, midline: 19%) (Figure 23).

The percent of households in the RFZ with soap and water at handwashing stations on the premises also decreased significantly (baseline: 31%, midline: 13%) (Figure 23).

Figure 23. Percent of households in the P2-ZOI and RFZ that have soap and water at a handwashing station on their premises



Household water insecurity

At midline, household water insecurity was measured for the first time using the Brief Household Water Insecurity Experiences (HWISE-4) Scale. The HWISE-4 Scale measures the percent of households that experienced water availability, accessibility, and reliability challenges in the four weeks before the survey. As measured by the HWISE-4 Scale, 37% of households were water insecure in the P2-ZOI (Table 3.4.1). Household water insecurity in the RFZ was 47% at midline (Table 3.4.1).

Women's empowerment in agriculture

The Abbreviated Women's Empowerment in Agriculture Index (A-WEAI) is an aggregate measure of women's empowerment and gender parity in the agricultural domain among primary adult male and female decision-makers. The A-WEAI comprises two sub-indices: the Five Domains of Empowerment (5DE) and the Gender Parity Index. Although the P2-ZOI Baseline Survey collected data on the Gender Parity Index and 5DE to construct the A-WEAI, the P2-ZOI Midline Survey only collected data to produce the 5DE score for primary adult female decision-makers.

The 5DE score assesses the degree to which women in agriculture are empowered in five domains: production, resources, income, leadership, and time. Women are considered empowered when they achieve adequacy in four of these five domains. At both baseline and midline, about three-quarters of women in the P2-ZOI were considered empowered (baseline: 76%, midline: 72%) (Tables 6.2.1). In the RFZ, the percent of women who were empowered decreased significantly between baseline and midline (baseline: 80%, midline: 67%) (Table 6.2.2).

The 5DE results for the P2-ZOI are included in Table 6.2.1, but the components for each of the five domains, comparing baseline and midline results, are presented in this section.

Production

The production domain is assessed by participation in economic activities and input into decision-making regarding those activities. At both baseline and midline, nearly all women in the P2-ZOI participated in at least one economic activity (baseline: 96%, midline: 95%) (Table 6.4.1).

Figure 24 shows that the majority of women participated in food crop farming at both baseline and midline (baseline: 90%, midline: 89%). Among the six activities, only the percent of women participating in livestock raising increased significantly (baseline: 52%, midline: 61%).

Among women who participated in each economic activity, 90% or more participated in at least some decisions related to the activity at baseline and midline (Table 6.4.2).

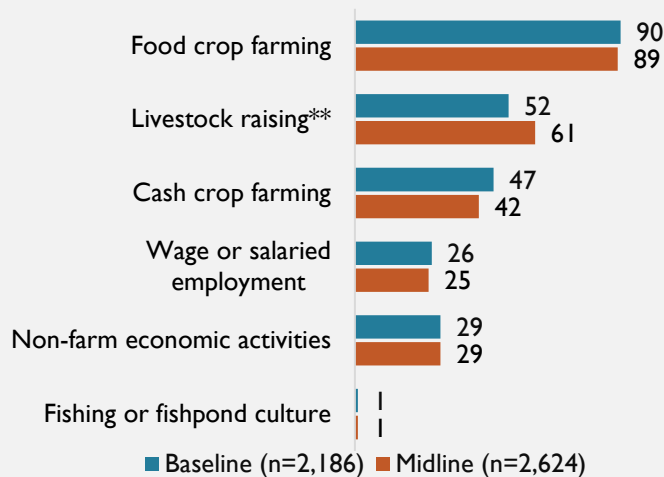
Resources

The resources domain is assessed based on asset ownership (either solely or jointly owned by the woman), access to credit at the household level, and input into decision-making about credit.

Women's ownership of productive assets largely remained the same or decreased significantly between baseline and midline (Figure 25). At both baseline and midline, hand tools for farming were the most commonly owned assets (baseline: 89%, midline: 85%). About half of women reported owning small consumer durables (baseline: 56%, midline: 56%) and poultry (baseline: 51%, midline: 47%). As Figure 25 shows, the largest significant percentage point decreases at midline were in the ownership of agricultural land (baseline: 79%, midline: 58%), cell phones (baseline: 62%, midline: 50%), and means of transportation (baseline: 25%, midline: 11%).

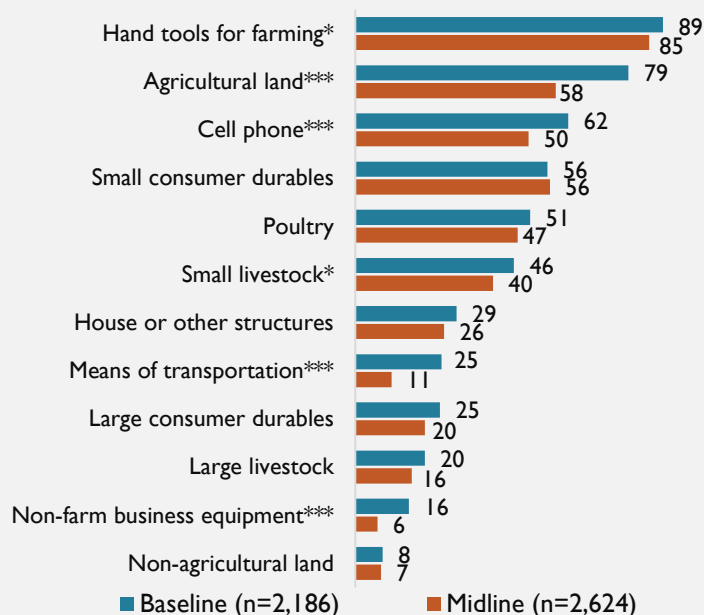
At midline, 69% of women reported that their household accessed credit in the 12 months preceding the survey, a significant increase from baseline (62%) (Table 6.4.4). Of the women whose households accessed credit, approximately 9 in 10 participated in decisions about whether to borrow or how to use the credit at both baseline and midline (Table 6.4.4).

Figure 24. Percent of women in the P2-ZOI participating in each economic activity



Significance levels: *p<0.05, **p<0.01, ***p<0.001

Figure 25. Percent of women in the P2-ZOI who solely or jointly own selected productive assets



Significance levels: * p< 0.05, ** p<0.01, p<0.001

Figure 26 shows sources of credit accessed by those households in the P2-ZOI in which women reported that their household accessed credit. The most common source of credit accessed at baseline and midline was group-based micro-finance (baseline: 33%, midline: 37%), followed by friends or relatives, which increased significantly (baseline: 20%, midline: 34%).

Income

The income domain is assessed based on the level of input into decision-making about the use of income from economic activities. At both baseline and midline, nearly all women (baseline: 98%, midline: 99%) in the P2-ZOI provided input into decisions about the use of income from economic activities or about household expenditures¹⁹ (Table 6.4.5).

Leadership

The leadership domain is assessed based on active membership in one or more types of community groups. Figure 27 shows that membership in credit or micro-finance groups was the most common (baseline: 49%, midline: 50%). Membership decreased significantly in religious groups (baseline: 40%, midline: 34%) and other women's groups (baseline: 39%, midline: 9%).

Time

The time domain is assessed based on the amount of time women spend performing work activities. At baseline and midline, nearly all women participated in work activities the day preceding the survey (baseline: 98%, midline: 99%), and the amount of time women spent on work activities increased significantly between baseline

Figure 26. Percent of women in the P2-ZOI who reported their household accessed credit, by source

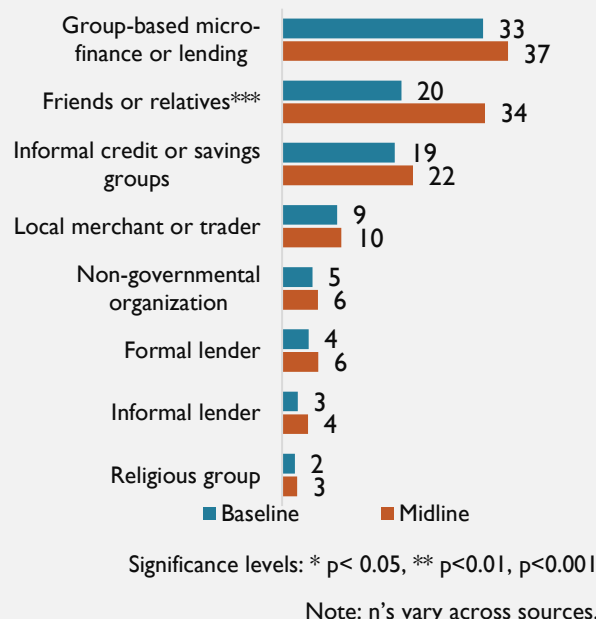
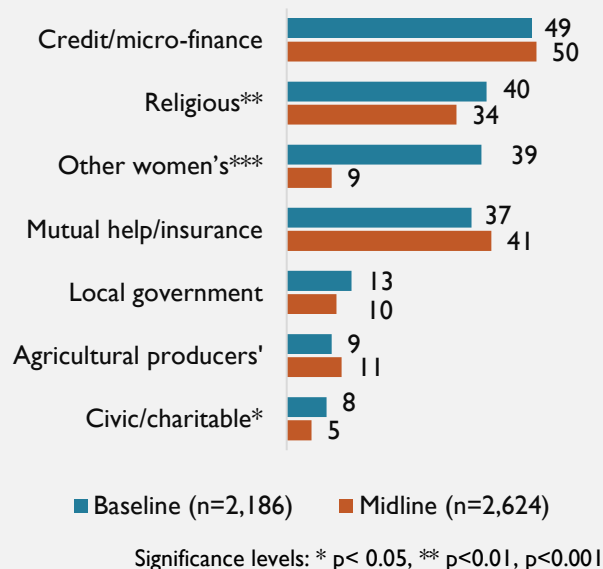


Figure 27. Percent of women in the P2-ZOI participating in each type of group



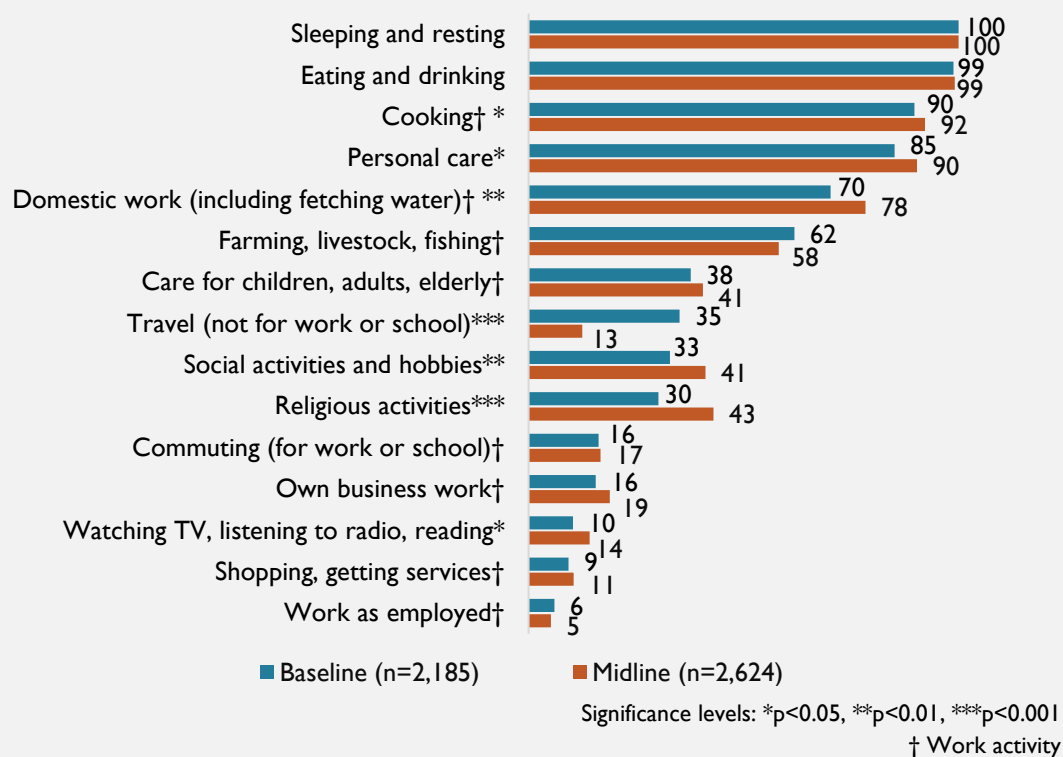
¹⁹ The six economic activities are food crop farming, cash crop farming, livestock raising, fishing or fishpond culture, non-farm economic activities, and wage or salaried employment. Two types of household expenditures are included: major and minor.

(8 hours 46 minutes) and midline (9 hours 35 minutes) (Table 6.4.7).²⁰

Figure 28 shows the percent of women participating in specific work and non-work activities. The largest significant increases between baseline and midline were in the percent of women who performed domestic work (baseline: 70%, midline: 78%), participated in social activities and hobbies (baseline: 33%, midline: 41%), and participated in religious activities (baseline: 30%, midline: 43%). The percent of women who traveled decreased significantly (baseline: 35%, midline: 13%).

Looking across work activities, there were significant increases between baseline and midline in the mean amount of time women spent on cooking (baseline: 2 hours 29 minutes, midline: 2 hours 56 minutes) and performing domestic work (baseline: 1 hour 21 minutes, midline: 1 hour 52 minutes) (Table 6.4.7).

Figure 28. Percent of women in the P2-ZOI who spent time on each activity the day preceding the survey



Agricultural cross-market functions

One of the high-level objectives of Feed the Future is inclusive and sustainable agricultural-led economic growth (USAID, 2016). To align with this objective, Feed the Future in Uganda focuses on creating efficiencies in cross-market functions, applying a market systems approach to improving rules, roles, and relationships among market actors for sustainable development and improvements across the agricultural sector. As such, data collected in the P2-ZOI Baseline and Midline Surveys include

²⁰ Work activities included the following: school; working as employed; own business work; food crop farming; cash crop farming; livestock raising; fishing or fishpond farming; commuting to/from work; shopping or accessing services; weaving, sewing, or textile care; cooking; domestic work; fetching water; and caring for children or caring for adults.

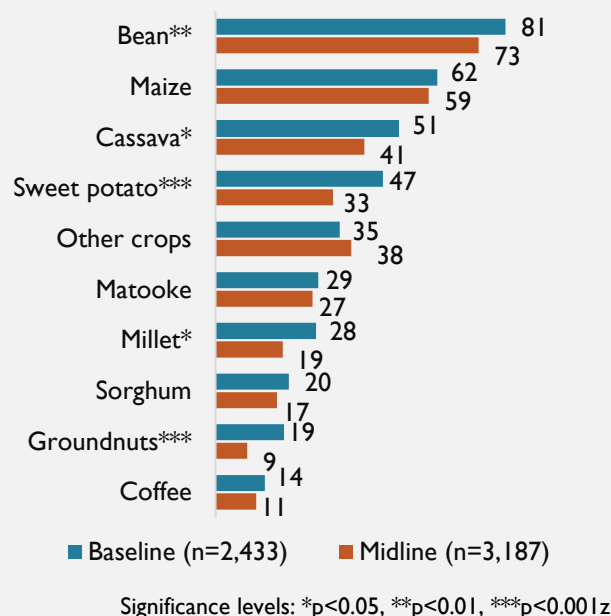
information on crops cultivated, sources of seeds, fertilizer use, pesticide use, agricultural equipment use, information received to plant and sell crops, methods to collect and transport harvested crops from plots, methods to dry and store harvested crops, selling crops, agricultural credit, farmers' groups, and income diversification. The results in this section are only presented for the P2-ZOI.

Crops cultivated

Figure 29 shows the percent of farmers who cultivated specific crops in the P2-ZOI at baseline and midline. Beans remained the crop most commonly cultivated at midline (73%), despite a significant decrease from baseline (81%). The percent of farmers who cultivated cassava, sweet potato, millet, and groundnuts also decreased significantly between baseline and midline.

The reasons for cultivating crops were similar at baseline and midline. Only 2% of farmers cultivated their crops solely for sale, about 2 in 5 farmers cultivated crops only for home consumption (baseline: 43%, midline: 42%), and more than half cultivated crops for both home consumption and for sale (baseline: 55%, midline: 56%) (Table 7.7a).

Figure 29. Percent of farmers in the P2-ZOI who cultivated specific crops



Sources of seeds

Farmers' sources of seeds did not change significantly between baseline and midline in the P2-ZOI; about half of farmers bought all their seeds (baseline: 52%, midline: 57%), about one-quarter of farmers did not buy any seeds (baseline: 24%, midline: 20%), and about another one-quarter of farmers bought some and received some (baseline: 24%, midline: 23%) (Table 7.2a).

The most common sources of purchased seeds were markets (non-agriculture dealers) (baseline: 42%, midline: 41%), followed by retailers or agriculture dealers with cash (baseline: 29%, midline: 36%), with the latter being a significant increase (Table 7.2a). About three in four farmers reported that seeds were available in their communities at midline (73%); however, this was a significant decrease from baseline (81%) (Table 7.2a).

Among farmers who received some or all of their seeds, the most common reason for not purchasing seeds was that they had their own saved seed or received seed from a friend or relative (baseline: 92%, midline: 89%). The second most common reason for not purchasing seeds was due to the high cost (baseline: 8%, midline: 14%), a reason that increased significantly between baseline and midline (Table 7.2b).

Fertilizer use

Fertilizer use in the P2-ZOI was similar at baseline and midline. About one in five crop farmers *applied* fertilizer to their crops (baseline: 18%, midline: 21%) (Table 7.3a). Of those who applied fertilizer, the majority reported that it was available in their community (baseline: 74%, midline: 72%) and that they *purchased* the fertilizer (baseline: 74%, midline: 65%) (Table 7.3a). The most common source of purchased fertilizer at baseline and midline was a retailer or agricultural dealer. The percent of crop farmers who purchased fertilizer from a retailer or agricultural dealer increased significantly between baseline and midline (baseline: 38%, midline: 53%) (Table 7.3a).

At baseline and midline, a similar percent of farmers reported the use of only inorganic fertilizer (baseline: 37%, midline: 43%) and only organic fertilizer (baseline: 42%, midline: 45%) (Table 7.3a). A smaller percent of farmers reported using both types (baseline: 17%, midline: 12%) (Table 7.3a).

Among farmers who did not *apply* fertilizer, the most common reason was the high cost (baseline: 72%, midline: 78%). The percent of farmers who reported not using fertilizer due to lack of knowledge about the added value increased significantly (baseline: 8%, midline: 13%) (Table 7.3b). Among farmers who did not *purchase* the fertilizer they used, the most common reasons for not purchasing were high cost (baseline: 43%, midline: 30%) and a preference to use compost or manure from their own animals (Table 7.3b). The percent of farmers with a preference to use compost or manure increased significantly (baseline: 42%, midline: 63%) (Table 7.3b).

Pesticide use

Pesticide use in the P2-ZOI was similar at baseline and midline (baseline: 33%, midline: 36%) (Table 7.4a). Among farmers who bought pesticides, the percent who bought pesticides from a retailer or agriculture dealer (baseline: 39%, midline: 60%) and from a wholesaler (baseline: 5%, midline: 10%) increased significantly, whereas the percent who bought pesticides in a market (baseline: 17%, midline: 27%) decreased significantly (Table 7.4a).

Among farmers who did not apply pesticides, the most common reasons were the same at baseline and midline: high cost (baseline: 66%, midline: 70%) and not needed (baseline: 21%, midline: 19%) (Table 7.4b). Similar to fertilizer use, the percent of farmers who did not use pesticides due to a lack of knowledge about its added value increased significantly (baseline: 8%, midline: 13%) (Table 7.4b).

Agricultural equipment use

At both baseline and midline, the use of hoes was nearly universal among farmers in the P2-ZOI (baseline: 99%, midline: 98%) (Table 7.5). At midline, plows were used by about one in five farmers (21%); however, this was a significant decrease from baseline (29%) (Table 7.5).

Although uncommon, tractor use increased significantly between baseline and midline (baseline: 1%, midline: 4%) (Table 7.5). Few farmers (1% or less) used drip irrigation, mechanized dryers, power tillers, rotators for irrigation, solar dryers or drying racks, and walking tractors (Table 7.5).

Information sources for planting and selling crops

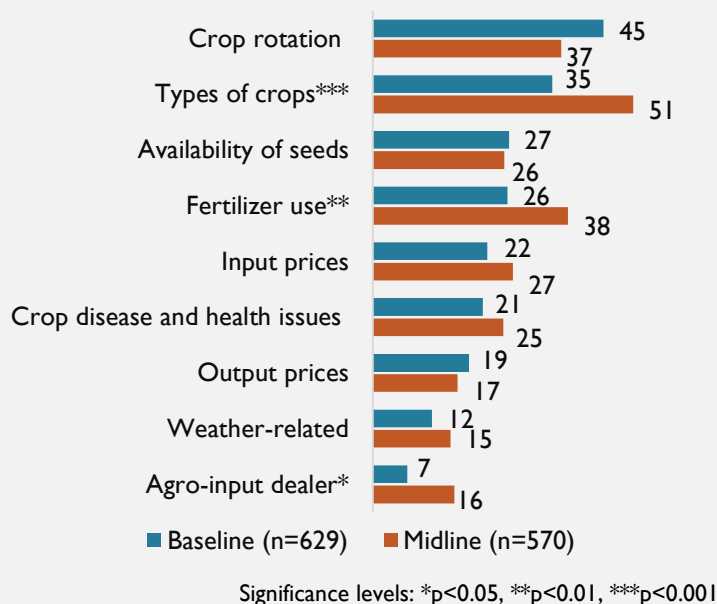
Overall, the percent of farmers who received information on planting and selling crops decreased significantly (baseline: 26%, midline: 17%) (Table 7.6a).

At both baseline and midline, the most common sources of information were friends and neighbors (baseline: 23%, midline: 20%), farmers' groups or associations (baseline: 11%, midline: 9%), and other farmers (baseline: 10%, midline: 10%), with no significant change detected between the two time points (Table 7.6a).

The most common method of receiving information was through radio programs (baseline: 48%, midline: 52%), but there were significant increases in the percent of farmers who received information in face-to-face meetings (baseline: 19%, midline: 32%) and who received information by phone (baseline: 1%, midline: 7%) (Table 7.6b).

Figure 30 shows the types of information farmers used to plant and sell their crops in the P2-ZOI at baseline and midline. There were significant increases in the percent of farmers who used information on types of crops to grow (baseline: 35%, midline: 51%), fertilizer use (baseline: 26%, midline: 38%), and agro-input dealers (baseline: 7%, midline: 16%).

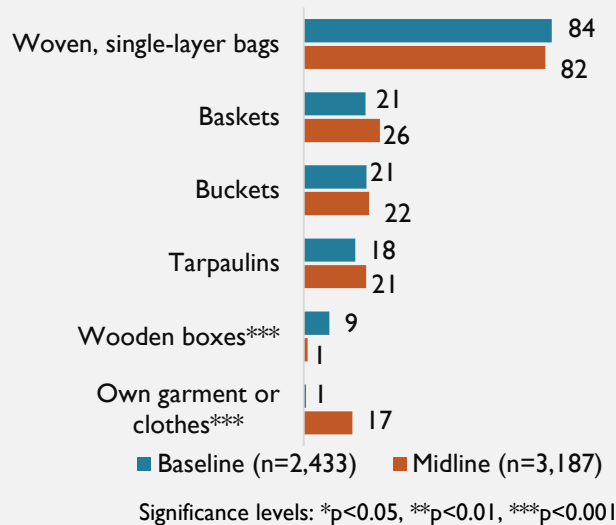
Figure 30. Types of information used, among farmers in the P2-ZOI who received any information



Containers to collect and transport harvested crops from plots

As shown in Figure 31, woven, single-layer bags were the most common containers used by farmers to collect harvested crops from their plots (baseline: 84%, midline: 82%). About one in five farmers used baskets (baseline: 21%, midline: 26%), buckets (baseline: 21%, midline: 22%), or tarpaulins (baseline: 18%, midline: 21%) at both baseline and midline. There was a significant decrease in the use of wooden boxes (baseline: 9%, midline: 1%) and a significant increase in the use of farmers' own garments or clothes (baseline: 1%, midline: 17%).

Figure 31. Containers farmers in the P2-ZOI used to collect crops



Methods used to transport harvested crops from the plot to a drying or storage location did not change significantly between baseline and midline in the P2-ZOI. Walking or

carrying crops on farmers' heads remained the main method of transport (baseline: 88%, midline: 86%), followed by bicycle (baseline: 17%, midline: 19%), and motorbike or boda boda (baseline: 10%, midline: 13%) (Table 7.7b).

Methods of drying and storing harvested crops

Although the majority of farmers dried their harvested crops, there was a small but significant decrease between baseline and midline (baseline: 97%, midline: 94%) (Table 7.8a). Among farmers who dried their crops, the most common methods used were drying on the bare ground (baseline: 58%, midline: 62%) and tarpaulins (baseline: 48%, midline: 43%) (Table 7.8a).

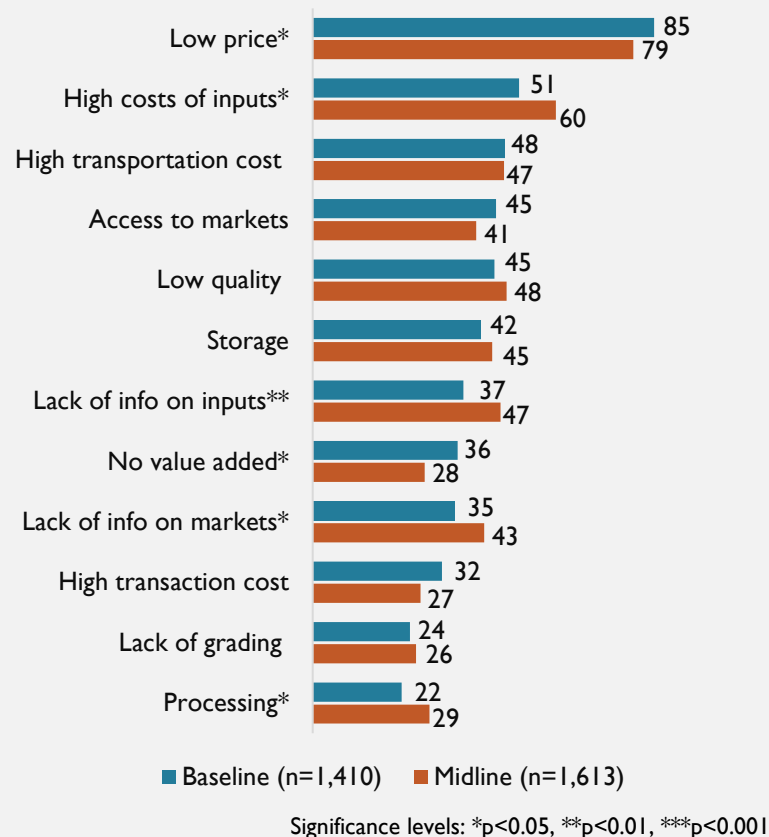
Between baseline and midline, there was a significant increase in the percent of farmers who stored their harvested crops (baseline: 78%, midline: 90%) (Table 7.8b). Among farmers who stored their harvested crops, woven, single-layer bags were most commonly used (baseline: 84% midline: 83%); however, there were significant changes in the use of other storage containers or materials (Table 7.8b). The use of woven, two- or three-layer bags increased significantly (baseline: 6%, midline: 13%), and the use of farmers' own garments or clothes also increased significantly (baseline: 2%, midline: 9%); the use of hermetic bags decreased significantly (baseline: 16%, midline: 10%) (Table 7.8b). Harvested crops were most commonly stored in residential houses at both baseline and midline (baseline: 86%, midline: 86%) (Table 7.8b).

Points of sale, means of transport to market, and major problems faced

At baseline and midline, about 3 in 5 farmers sold crops they cultivated in the 12 months preceding the survey (baseline: 57%, midline: 58%) (Table 7.9a). The most common points of sale were local markets (baseline: 48%, midline: 42%) and traders (baseline: 41%, midline: 45%) (Table 7.9a). The primary means of transport to the points of sale included walking or carrying crops on head (baseline: 45%, midline: 40%), by motorbike or boda boda (baseline: 37%, midline: 33%), and by bicycle (baseline: 25%, midline: 21%) (Table 7.9a). There were no significant changes between baseline and midline in the modes of transport to points of sale, with the exception of truck transport, which decreased significantly (baseline: 5%, midline: 2%) (Table 7.9a).

Farmers who sold their crops reported encountering several substantial challenges when selling their crops. Although selling for a low price was the most common problem faced, there was a significant decrease in respondents reporting this issue between the two time points (baseline: 85%, midline: 79%) (Figure 32). There were significant increases in the percent of farmers who reported facing problems with the high cost of inputs (baseline: 51%, midline: 60%), lack of information about agricultural inputs (baseline: 37%, midline: 47%), lack of information about agricultural markets (baseline: 35%, midline: 43%), and processing (baseline: 22%, midline: 29%). The percent of farmers who reported no value added as a major issue decreased significantly between baseline and midline (baseline: 36%, midline: 28%) (Figure 32).

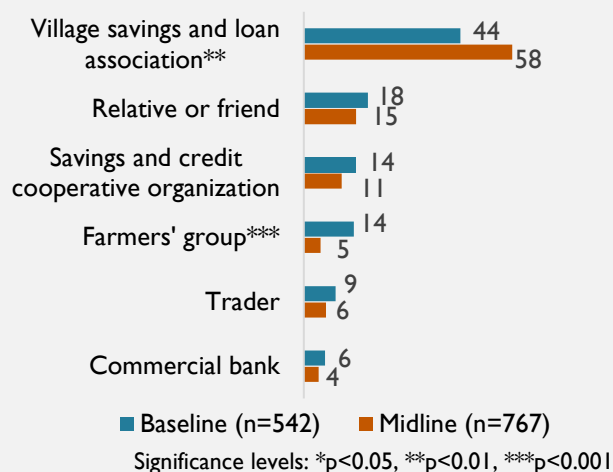
Figure 32. Major problems farmers in the P2-ZOI faced selling their crops, among farmers who sold their crops



Use of agricultural credit and membership in a farmers' group

The percent of farmers who accessed agricultural credit in the P2-ZOI did not change significantly between baseline (24%) and midline (22%) (Table 7.10a). As Figure 33 shows, village savings and loan associations were the most common sources of credit, with a significant increase between the two time points (baseline: 44%, midline: 58%). The next most common sources of credit were relatives or friends (baseline: 18%, midline: 15%) and savings and credit cooperative organizations (baseline: 14%, midline: 11%). The percent of farmers who accessed credit through a farmers' group decreased significantly (baseline: 14%, midline: 5%).

Figure 33. Sources of agricultural credit among farmers in the P2-ZOI who took credit



Farmers most commonly took agricultural credit to purchase agricultural inputs (baseline: 61%, midline: 56%) and to hire labor (baseline: 48%, midline: 46%) (Table 7.10b). Expanding or purchasing land, though less common, was also a reason given for accessing credit (baseline: 14%, midline: 16%) (Table 7.10b).

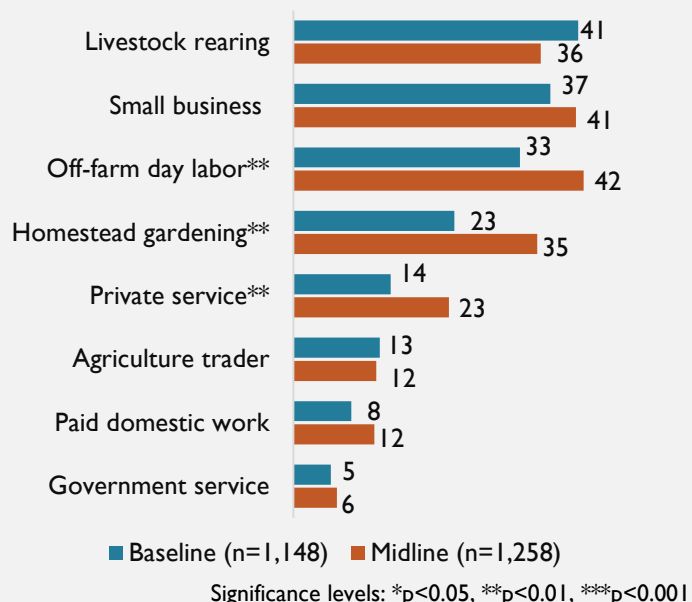
Among farmers who did not take agricultural credit, the most common reasons were that credit was not needed (baseline: 56%, midline: 54%), terms and conditions were not favorable (baseline: 23%, midline: 30%, a significant increase), and collateral was unavailable (baseline: 22%, midline: 20%) (Table 7.10c). The percent of farmers who were not aware of credit services was small but increased significantly (baseline: 6%, midline: 10%) (Table 7.10c).

Participation in additional activities for cash or food

There was a small but significant decrease in the percent of farmers who participated in additional activities for cash or food (baseline: 46%, midline: 42%) (Table 7.11).

Figure 34 shows that the most common activities farmers undertook were livestock rearing, small business work, and off-farm day labor, and it also shows significant increases in the percent of farmers who participated in off-farm day labor, homestead gardening, and private service work.

Figure 34. Additional activities farmers in the P2-ZOI participated in for cash or food, among farmers who took on additional activities



Coverage of promoted interventions

The P2-ZOI Midline Survey was used to collect descriptive information on the types of Feed the Future promoted interventions that reached households or individuals in the P2-ZOI in the 12 months preceding the survey.²¹ Interventions included activities related to agriculture, resilience, women's health and nutrition, children's health and nutrition, and water, sanitation, and hygiene. This descriptive information was only collected in the P2-ZOI Midline Survey; thus, results are presented only for the P2-ZOI and RFZ at midline.

Agriculture

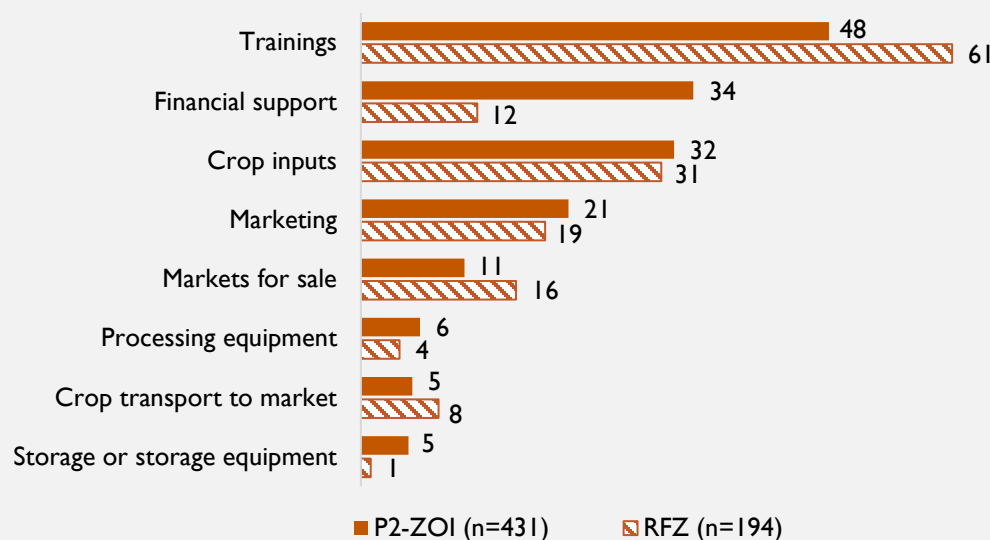
This section presents information on participation in promoted agriculture-related interventions, including crop and livestock trainings attended, services accessed through farmers' groups, and receipt of veterinary services for livestock.

²¹ For women's and children's health and nutrition interventions, the timeframe asked varied and was often shorter than 12 months.

Of those crop farmers who attended training, they mainly participated in trainings on growing green-leafy vegetables (18%) and pest or disease management (11%) (Table 7.12.1).

Figure 35 shows the types of services provided by farmers' groups. Training was the most common service provided through the groups in the P2-ZOI (48%) and in the RFZ (61%). The percent of farmers who reported that financial support was available through farmers' groups was 34% in the P2-ZOI and 12% in the RFZ.

Figure 35. Services provided by farmers' groups, according to crop farmers who were active farmers' group members in the P2-ZOI and RFZ

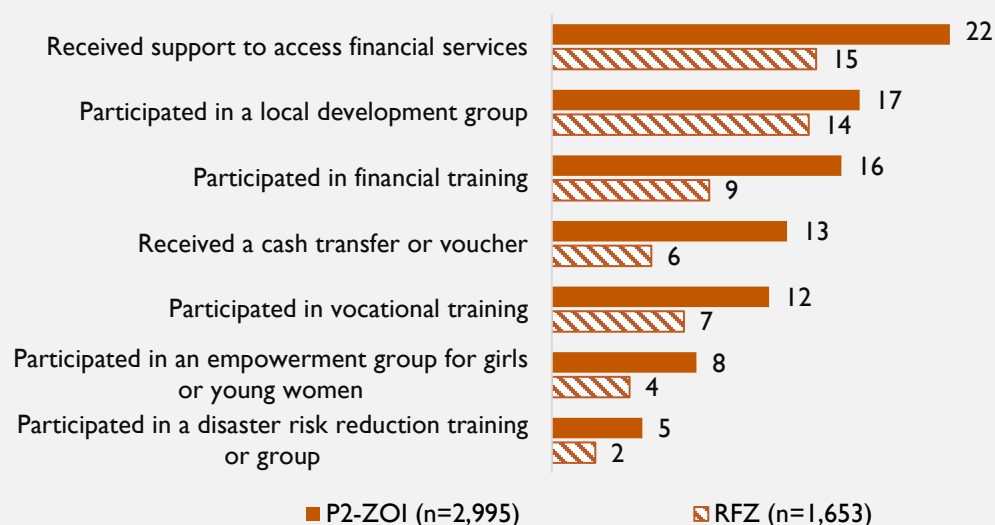


Participation in livestock-related training was low; only 10% of households in the P2-ZOI and RFZ that raised livestock or other farm animals participated in training on livestock farming, and a smaller percent of households participated in training on improving livestock health and productivity (7% in the P2-ZOI and 8% in the RFZ) and training on livestock commercialization (1% in both the P2-ZOI and RFZ) (Table 7.12.1). Receipt of veterinary services from a trained veterinary officer (3% in the P2-ZOI and 4% in the RFZ), and support to access livestock vaccines and medication (1% in both the P2-ZOI and RFZ) were also very low at midline (Table 7.12.3).

Resilience

This section presents information on participation in promoted resilience interventions, including support to access financial services, community development groups and committees, and training. Figure 36 shows that in both the P2-ZOI and RFZ, support to access financial services (22% in the P2-ZOI and 15% in the RFZ), participation in a local development group (17% in the P2-ZOI and 14% in the RFZ), and participation in financial training (16% in the P2-ZOI and 9% in the RFZ) were the most common resilience-related interventions accessed by households.

Figure 36. Percent of households participating in promoted resilience-related interventions in the P2-ZOI and RFZ



Women's health and nutrition

This section presents information on participation of women of reproductive age in promoted women's health and nutrition interventions, including counseling on maternal nutrition, timing and spacing of pregnancies and receipt of antenatal checks and services during pregnancy. At midline, 22% of women 15–49 years of age in the P2-ZOI and 27% of women 15–49 years of age in the RFZ received counseling on maternal nutrition from a health worker in the 4 months preceding the survey, and 35% of women 15–49 years of age in the P2-ZOI and 33% of women 15–49 years of age in the RFZ received counseling on healthy timing and spacing of pregnancies from a health worker during the same time period (Table 8.2.1).

The majority of women 15–49 years of age who were pregnant in the 12 months preceding the survey received an antenatal check-up from a health worker (84% in the P2-ZOI and 89% in the RFZ), and among those who gave birth, the mean number of check-ups was 5 in both the P2-ZOI and RFZ. Furthermore, most women 15–49 years of age who gave birth received iron and folic acid while pregnant (93% in the P2-ZOI and 97% in the RFZ) (Table 8.2.1).

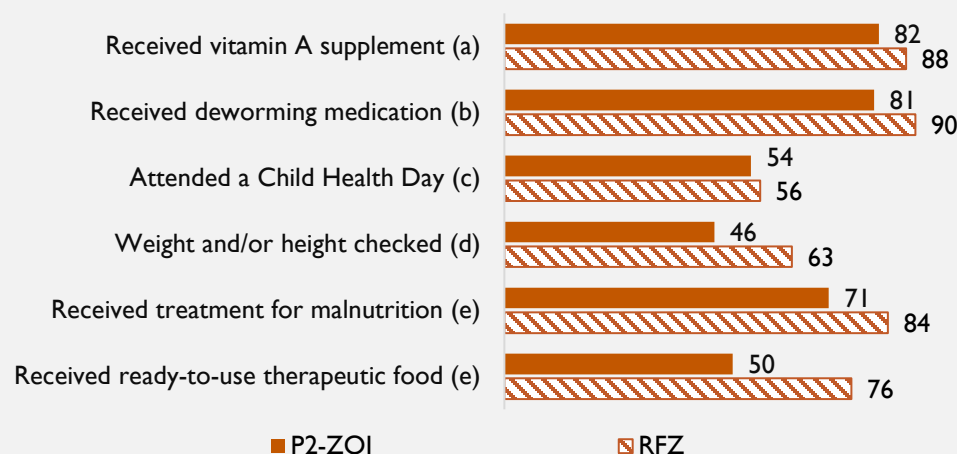
Children's health and nutrition

This section presents information on participation in promoted children's health and nutrition interventions and services received by caregivers of children under 5 years of age or by the children themselves. In the P2-ZOI, 43% of children under 2 years of age had caregivers who were counseled on breastfeeding, and the same percent (43%) had caregivers who were counseled on infant and young child feeding (Table 8.4.1). Caregivers of 37% of children under 5 years of age in the P2-ZOI were counseled on the benefits of growth monitoring (Table 8.4.1).

In the RFZ, 58% of children under 2 years of age had caregivers who were counseled on breastfeeding, and 49% had caregivers who were counseled on infant and young child feeding (Table 8.4.1). Caregivers of 49% of children under 5 years of age in the RFZ were counseled on the benefits of growth monitoring (Table 8.4.1).

As shown in Figure 37, the percent of children who had their height or weight checked by a health worker was 46% in the P2-ZOI and 63% in the RFZ. The majority of children diagnosed as malnourished received treatment in both the P2-ZOI (71%) and in the RFZ (84%). Of those children diagnosed as malnourished in the P2-ZOI, half (50%) received ready-to-use therapeutic food. In the RFZ, 76% of children diagnosed as malnourished received ready-to-use therapeutic food.

Figure 37. Percent of children who received promoted health and nutrition interventions in the P2-ZOI and RFZ



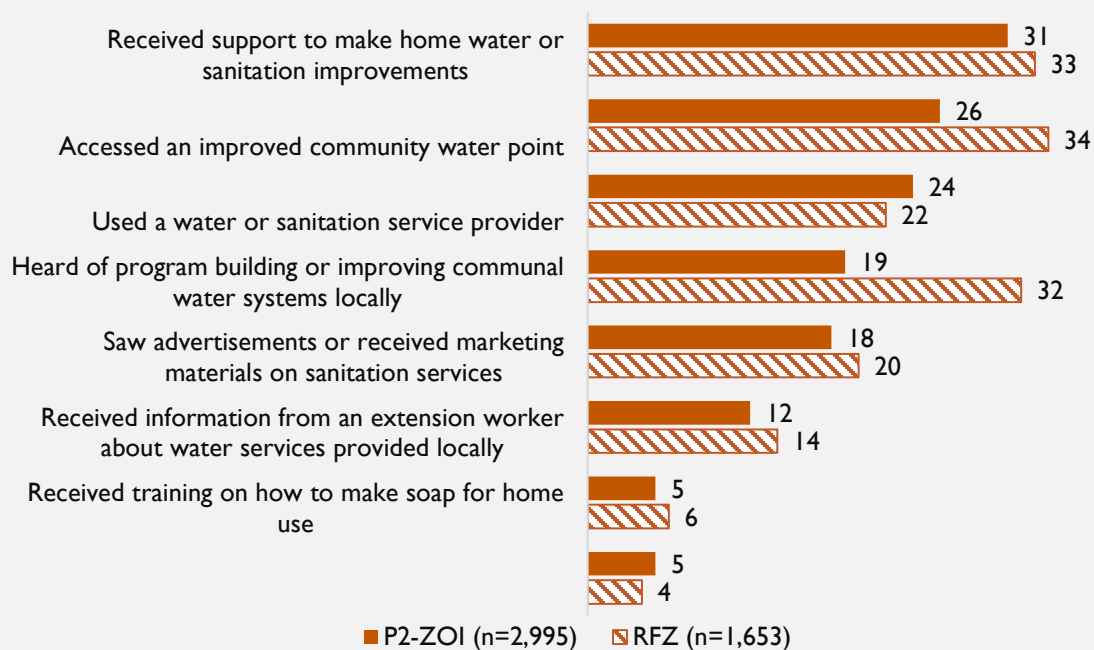
Notes:

- (a) Children 6–59 months, past 6 months (P2-ZOI n=2,250, RFZ n=1,342)
- (b) Children 12–59 months, past 6 months (P2-ZOI n=2,045 RFZ n=1,228)
- (c) Children 0–59 months, past 6 months (P2-ZOI n=2,444, RFZ n=1,465)
- (d) Children 0–59 months, past 4 months (P2-ZOI n=2,444, RFZ n=1,465)
- (e) Children 0–59 months, past 4 months (P2-ZOI n=258, RFZ n=182)

Water, sanitation, and hygiene

This section presents information on participation in promoted water, sanitation, and hygiene interventions, including training or counseling on hygiene practices, support to make home water or sanitation improvements, and access to improved community water points and water or sanitation service providers. As shown in Figure 38, participation in training or counseling on hygiene practices was the most common intervention accessed (31% in the P2-ZOI and 33% in the RFZ), followed by receipt of support to make home water or sanitation improvements (26% in the P2-ZOI and 34% in the RFZ) and access to an improved community water point (24% in the P2-ZOI and 22% in the RFZ). Finally, 19% of households in the P2-ZOI and 32% of households in the RFZ utilized a water or sanitation service provider.

Figure 38. Percent of households that participated in promoted water, sanitation, and hygiene interventions



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APPENDICES

The appendices present background information and the methodology used to obtain indicator estimates. There are seven appendices:

- A1. Feed the Future and resilience programming in Uganda
- A2. Data sources
- A3. Sampling and weighting
- A4. Survey design, implementation, and data analysis
- A5. Limitations, corrections, updates, and additional considerations
- A6. Measures and reporting conventions
- A7. Results tables

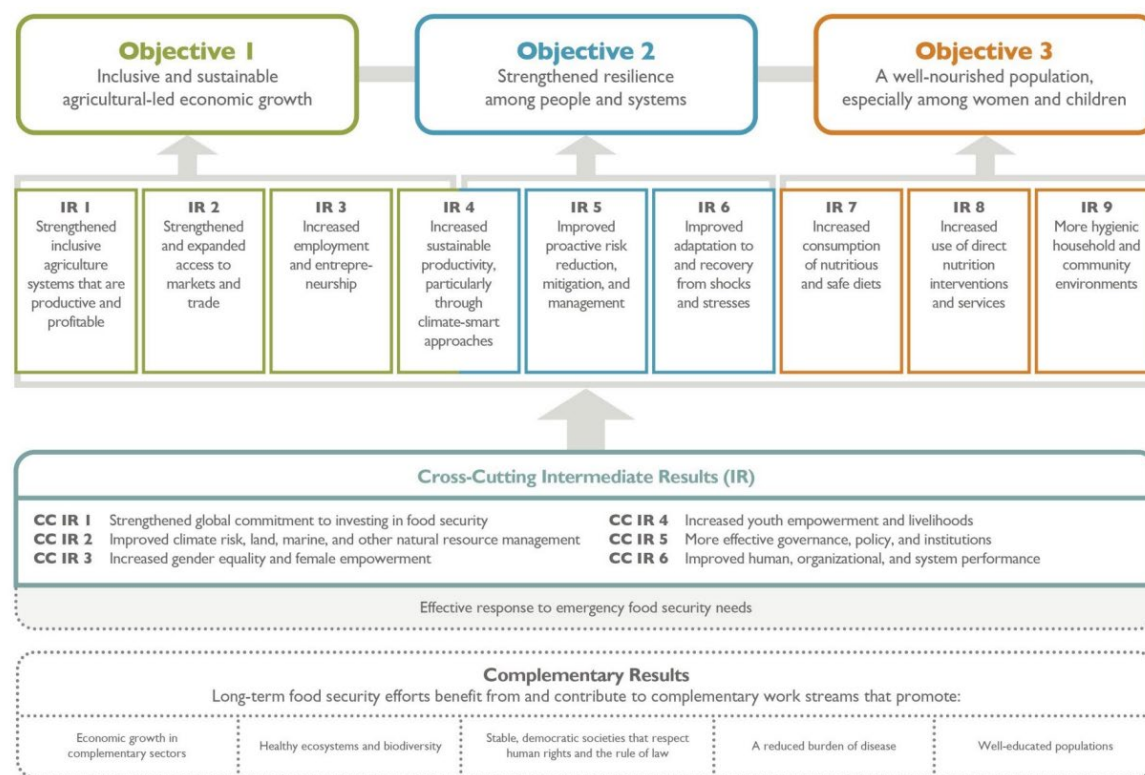
The information in these appendices primarily focuses on methods used to conduct the P2-ZOI Midline Survey and to obtain indicator estimates at midline, but also includes some information on baseline methods. More detailed information on the P2-ZOI Baseline Survey methodology can be found in the P2-ZOI Survey Baseline Report (Feed the Future, 2022). Additional information on standard methods and protocols followed during survey planning and implementation and data analysis can be found in the *Feed the Future ZOI Survey Methods Toolkit—Midline* (hereafter referred to as “the Midline Toolkit”) (USAID, 2021) and *Feed the Future ZOI Survey Methods Toolkit—Baseline* (hereafter referred to as “the Baseline Toolkit”) (USAID, 2018a).²²

A1. Feed the Future and resilience programming overview in Uganda

Feed the Future’s overarching goal is to sustainably reduce global hunger, malnutrition, and poverty. Reaching this goal requires the achievement of three main objectives as laid out in the Global Food Security Strategy 2017-2021 Results Framework (Figure A1.1) (USAID, 2016). The first objective is to achieve inclusive and sustainable agricultural-led economic growth. The second objective is to strengthen resilience among people and systems. The third objective is to have a well-nourished population, especially among women and children.

²² The materials included in the Baseline Toolkit and Midline Toolkit are referred to as “core” materials. These core materials include technical guides, templates, and tools to standardize data collection processes and ensure data quality across Feed the Future ZOI Surveys.

Figure A1.1. U.S. Government Global Food Security Strategy 2017–2021 Results Framework



Feed the Future in Uganda plans to achieve these objectives by employing a food systems, market systems, and resilience approach. Feed the Future Uganda targets a range of actors involved in market activities, including women, youth, the ultra-poor, landless, and other marginalized groups that face varied and multiple constraints to engaging with the agricultural markets to develop their capacity so that they can productively engage in market systems as their livelihoods. Feed the Future Uganda also invests in initiatives capable of catalyzing significant positive changes for a broader spectrum of beneficiaries, thereby maximizing the impact of U.S. Government resources through collaborative efforts with other partners (Feed the Future, 2018).

In addition, USAID aims to build sources of resilience to contribute to a sustainable reduction in vulnerability and more inclusive growth. USAID envisions that these results will contribute to reduced humanitarian need in the long term. In alignment with that vision, USAID Uganda designated an RFZ where resilience building efforts would be focused. The districts that comprise the RFZ historically and persistently receive high levels of humanitarian assistance due to repeat, large-scale humanitarian emergencies and shocks and stresses that are recurrent or protracted crises. RFZ districts have high rates of chronic poverty, persistently high acute malnutrition, chronic health burdens, and conflict risk. In addition, they have an enabling environment that includes political will, an acceptable level of security and physical access, and access to social services. Finally, they have a comparative advantage; that is—they have existing or planned USAID humanitarian and development programs, and other U.S. Government agencies and donors have programs, presence, and capabilities in the districts. USAID

Uganda works to sequence, layer, and integrate programming within the RFZ in collaboration with partners. Resilience building requires a multisectoral approach, building on existing investments whenever possible to improve efficiency and impact (USAID, 2022).

A2. Data sources

Data for the P2-ZOI and RFZ indicator estimates presented in this assessment report are drawn from two sources: the P2-ZOI Baseline Survey and the P2-ZOI Midline Survey.

The P2-ZOI Baseline Survey was implemented by ICF and the Uganda Bureau of Statistics. Data collection took place from September 7 to November 25, 2019. ICF also led the implementation of this assessment, including the P2-ZOI Midline Survey, with the Uganda-based International Research Consortium (IRC). Data collection took place from October 28 to December 14, 2022. Data collection for both surveys fell within the second harvest season (FEWS NET, n.d.).

The baseline and midline questionnaires were customized for the Uganda context and specific data requests required by USAID Uganda. The P2-ZOI Baseline Survey included one main questionnaire, and the P2-ZOI Midline Survey included a main questionnaire and a parallel questionnaire. At midline, the main questionnaire was administered to collect data used to calculate P2-ZOI and RFZ indicators and to collect additional information about the effects of COVID-19 on households, coverage of the types of interventions promoted by Feed the Future in the P2-ZOI, and climate knowledge and adaptation services. The parallel questionnaire was administered to collect consumption expenditure data from a smaller number of households to validate the prediction model used to compute the poverty indicator estimates using a survey-to-survey imputation methodology (Dang et al., 2024).

Survey modules for the P2-ZOI Baseline and Midline Surveys are outlined in Table A2.1. Eligibility criteria varied across modules, based on their purpose and content. Additional information about the P2-ZOI Baseline Survey questionnaire can be found in the P2-ZOI Baseline Survey Report (Feed the Future, 2022), and additional information about the P2-ZOI Midline Survey questionnaires can be found in **Appendix 4.2**.

Table A2.1. Uganda P2-ZOI Baseline and Midline Survey Modules

Module	Baseline	Midline—Main	Midline—Parallel ^a
1	Household roster and demographics	Household roster and demographics	Household roster and demographics
2	Dwelling characteristics and household assets Includes WASH	Dwelling characteristics and household assets Includes expenditures on utilities, WASH, water insecurity, and coverage of promoted livestock and WASH interventions	Dwelling characteristics and household assets
2A	n/a	Climate adaptation	n/a
3	Food security	Food security	n/a
4	Women's nutrition and anthropometry	Women's nutrition (excluding anthropometry)	n/a

Module	Baseline	Midline—Main	Midline—Parallel ^a
		Includes coverage of promoted women's health and nutrition interventions	
5	Children's nutrition and anthropometry	Children's nutrition (excluding anthropometry) Includes coverage of promoted children's health and nutrition interventions	n/a
6	Women's empowerment in agriculture (primary adult female and male decision-makers ^b)	Women's empowerment in agriculture (primary adult female decision-makers only ^b)	n/a
7	Cross-market functions	Cross-market functions Includes coverage of promoted interventions related to crop cultivation	n/a
8	Household consumption expenditures	n/a	Household consumption expenditures
R	Resilience	Resilience Includes COVID-19 and coverage of promoted resilience interventions	n/a

n/a=not applicable, WASH=water, sanitation, and hygiene

^a The parallel survey was used to collect data to validate the survey-to-survey imputation model used to compute the consumption aggregate, which was used to estimate the three Feed the Future P2-ZOI poverty indicators at midline.

^b Primary adult decision-makers are de jure household members 18 years of age or older who are identified as the primary adult male or primary adult female responsible for both social and economic decision-making in the household. When both exist in a single household, primary adult female and male decision-makers are typically, but not necessarily, husband and wife.

A3. Sampling and weighting

A3.1 Sample size determination

The main aim of the Midline Indicator Assessment is to provide sample-weighted estimates of the P2-ZOI and RFZ population-based indicators (including their standard errors and confidence intervals) to track changes between the P2-ZOI Baseline and Midline Surveys. This requires a descriptive population-based survey with a sample size that is adequate to enable the calculation of reliable indicator estimates at a single point in time (i.e., midline). Although the P2-ZOI Midline Survey is not designed to detect statistically significant differences in indicator estimates since the P2-ZOI Baseline Survey, it is still possible to detect statistically significant differences if the difference between baseline and midline estimates is large enough.

The main survey sample size was determined following the guidance developed by REFS for P2-ZOI Midline Surveys (USAID, 2021). The parallel survey sample size (a minimum of 300 households) was

determined following unpublished survey-to-survey imputation guidelines from the World Bank, provided to ICF prior to data collection (Carletto et al., 2022).

As per the guidance for the main survey, the required sample sizes for the following P2-ZOI indicators were determined:²³

- Prevalence of moderate and severe food insecurity in the population, based on the FIES
- Percent of households below the comparative threshold for the poorest quintile of the asset-based CWI
- Percent of households participating in group-based savings, micro-finance, or lending programs
- Percent of households with access to a basic sanitation service
- Percent of households with soap and water at a handwashing station on premises
- Percent of children 6–23 months consuming a MAD
- Prevalence of EBF of children under 6 months of age
- Percent of women of reproductive age consuming a diet of minimum diversity
- ARSSI
- Index of social capital at the household level
- Percent of households that believe local government will respond effectively to future shocks and stresses

Calculation of the initial main survey sample size

The initial required sample sizes were computed for the indicators listed above based on a 5 percent margin of error for the estimated level and precision of each indicator.²⁴

Table A3.1 presents the calculation of the initial sample size for the main survey based on the input parameters for the indicators, as given in the table using the formula that follows.

Table A3.1. Calculation of the Initial Sample Size for the P2-ZOI Midline Survey Indicators

Indicator type	Indicator	P_{est}	M	$Z_{1-\frac{\alpha}{2}}$	$\sigma_{x_{est}}^*$	D	$n_{initial}$
Percentage/ proportion	Percent households (FIES)	73.4	5%	1.96	n/a	4.1	1,237
	Percent households (CWI) [†]	39.8	5%	1.96	n/a	3.7	1,363
	Percent households (savings programs) [†]	67.9	5%	1.96	n/a	3.1	1,039
	Percent households (basic sanitation)	23.2	5%	1.96	n/a	3.5	959
	Percent households (handwashing)	31.7	5%	1.96	n/a	3.3	1,098

²³ The indicator *Prevalence of water insecurity*, based on HWISE-4, was not part of the sample size calculations because it is a new indicator that was added at midline.

²⁴ The estimated level and precision of each indicator were based on indicator estimates, means, and standard deviations from the P2-ZOI Baseline Survey.

Indicator type	Indicator	P_{est}	M	$Z_{1-\frac{\alpha}{2}}$	$\sigma_{X_{est}}^*$	D	$n_{initial}$
	Percent households (local government)†	61.1	5%	1.96	n/a	4.8	1,754
	Percent children (MAD)	11.1	5%	1.96	n/a	1.4	213
	Percent women (dietary diversity)	29.2	5%	1.96	n/a	2.5	795
Mean	Mean households (ARSSI)†	4.4	5%	1.96	1.3	3.1	416
	Mean households (social capital)†	62.7	5%	1.96	34.1	3.7	1,682

* Estimated standard deviation for mean indicators

† Indicators corresponding to the Global Food Security Strategy Strategic Objective 2: Strengthened resilience among people and systems.

$$n_{initial} = D \times P_{est}(1 - P_{est}) \left(\frac{Z_{1-\frac{\alpha}{2}}}{M} \right)^2 \quad (1)^{25}$$

Where,

$n_{initial}$ = estimated initial sample size required for the survey based on the indicator.

D = the estimated design effect for the indicator in this survey. The estimate is based on the value from the P2-ZOI Baseline Survey for all indicators.

P_{est} = the estimated prevalence or mean of the indicator at the midline. This value is derived from the baseline for this indicator.

$Z_{1-\frac{\alpha}{2}}$ = the critical value for the normal probability distribution. The significance level is set at $\alpha=0.05$, giving a value of $Z_{0.975}=1.96$.

M = the margin of error. The margin of error used for the P2-ZOI Midline Survey is 5 percent (i.e., $M=0.05$).

Based on the initial sample size calculation, the *Percent of households that believe local government will respond effectively to future shocks and stresses* indicator, noted as “the local government indicator” in Table A3.1, had the largest initial sample size required across all indicators (1,754 households).²⁶

Calculation of the final main survey sample size

²⁵ For estimation of mean indicators, the term $P_{est1}-P_{est}$ in formula (1) is replaced by X_{est2} , the estimated variance for the distribution of the mean indicator, X .

²⁶ Note for the *Prevalence of exclusive breastfeeding of children under 6 months of age* indicator, the REFS guidance indicates using a formula to estimate the minimum required household sample size for the survey to include at least 70 children under 6 months of age (see Equation 3). The household sample size computed for this indicator will be compared with the largest final household sample size required across the other indicators to determine the final required household sample size.

To derive the final required household sample size, the initial sample sizes were adjusted to account for the number of households that need to be contacted to obtain the required number of individuals for the given sub-population for individual level-indicators and anticipated household non-response. Therefore, the final sample size, denoted by n_{final} , is the product of the initial sample size and the adjustment for the number of households to contact for the given individual sub-population for individual-level indicators (i.e., adj_1) and the adjustment for anticipated household non-response (i.e., adj_2) for each indicator:

$$n_{final} = n_{initial} \times adj_1 \times adj_2 \quad (2)$$

Where,

- n_{final} = estimated final sample size required for the survey based on the indicator.
- adj_1 = adjustment for the number of households to contact to obtain the required number of individuals for the given sub-population for an individual-level indicator.²⁷
- adj_2 = adjustment for anticipated household non-response, which is equal to the inverse of the expected response rate.

Table A3.2 illustrates the computation of the final required sample size for each indicator.²⁸ An expected household response rate of 99 percent was assumed for the indicators based on the achieved household response rate from the P2-ZOI Baseline Survey.

Table A3.2. Calculation of the Final Sample Size for the P2-ZOI Midline Survey Indicators

Indicator type	Indicator	$n_{initial}$	Proportion of the population in the age group underlying the indicator	adj_1	Expected response rate	adj_2	n_{final}
Percentage/ proportion	Percent households (FIES)	1,237	n/a	n/a	99%	1.01	1,243
	Percent households (CWI)*	1,363	n/a	n/a	99%	1.01	1,377
	Percent households (savings programs)*	1,039	n/a	n/a	99%	1.01	1,049
	Percent households (basic sanitation)	959	n/a	n/a	99%	1.01	969
	Percent households (handwashing)	1,098	n/a	n/a	99%	1.01	1,109
	Percent households	1,754	n/a	n/a	99%	1.01	1,772

²⁷ This adjustment is determined based on the Stukel-Deitchler Inflation using the P2-ZOI Baseline Survey data on the average household size and proportion of the population made up of the given sub-population for each indicator. Please refer to Appendix A of the *Feed the Future Population-based Survey Sampling Guide* (Stukel, 2018) for a more detailed discussion of this adjustment factor.

²⁸ For individual-level indicators, the final required sample size computation will include an additional adjustment to determine the number of households to contact for the given individual sub-population.

Indicator type	Indicator	$n_{initial}$	Proportion of the population in the age group underlying the indicator	adj_1	Expected response rate	adj_2	n_{final}
	(local government)*						
	Percent children (MAD)	213	0.04	5.23	99%	1.01	1,125
	Percent women (dietary diversity)	795	0.21	1.21	99%	1.01	1,035
Mean	Mean households (ARSSI)*	416	n/a	n/a	99%	1.01	420
	Mean households (social capital)*	1,682	n/a	n/a	99%	1.01	1,699

* Indicators corresponding to the Global Food Security Strategy Strategic Objective 2: Strengthened resilience among people and systems.

The minimum required household sample size for the *Prevalence of exclusive breastfeeding of children under 6 months of age indicator* (or “EBF indicator”) was determined using the following formula:

$$n_c = \frac{70}{(1 - \exp^{-PP \times HHSZ})} \times adj_{NR} \quad (3)$$

Where,

- n_c = minimum required household sample size for the EBF indicator.
- PP = estimated proportion of the P2-ZOI population that is under 6 months of age.
- $HHSZ$ = estimated average household size in the P2-ZOI.
- adj_{NR} = inverse of the expected household response rate.

Table A3.3 shows the parameter values used to calculate the minimum required household sample size for the EBF indicator.

Table A3.3. Minimum Sample Size Required for the EBF Indicator

Indicator	$HHSZ$	PP	Expected response rate	adj_{NR}	n_c
EBF	5.06	1.2	99%	1.01	1,201

The EBF indicator required a sample size of 1,201 households to enable at least an estimated 70 children under 6 months of age to be part of the P2-ZOI Midline Survey sample.

Final main survey sample size for the P2-ZOI and RFZ

Based on the sample size calculations in Tables A3.1 and A3.2, the indicator with the overall largest required final household sample size was the local government indicator, requiring 1,772 households. Because it met the sample size requirements of all other indicators, it was retained as the minimum

sample size needed to produce indicator estimates for the P2-ZOI. Rounding up to maintain a sample take of 25 households per cluster, the final required sample size was 1,775 households for the P2-ZOI.

REFS's guidance for determining the sample size in the RFZ is to use the largest sample size among the ZOI-level indicators that correspond to the Global Food Security Strategy Strategic Objective 2: Strengthened resilience among people and systems as the RFZ sample size. These indicators are marked with an asterisk (*) in Tables A3.1 and A3.2. For the Uganda RFZ, the local government indicator also had the largest required sample size of 1,772 households across all Strategic Objective 2 indicators. Rounding up to maintain the sample take of 25 households per EA, the final required sample size was 1,775 households for the RFZ.

Based on the sample size calculations, the P2-ZOI and the RFZ required a minimum sample size of 1,775 households each. However, because the RFZ sits entirely within the P2-ZOI, the final number of households was more than the required sample for a proportional allocation of the P2-ZOI sample within the overlapping area of the P2-ZOI and RFZ. The non-overlapping portion makes up 79.7% of the P2-ZOI population, and the overlapping area makes up 20.3% of the P2-ZOI population. Table A3.4 presents the percent distribution of the household populations across the overlapping and non-overlapping areas of the P2-ZOI and RFZ, based on the latest available household population projections from the Uganda Bureau of Statistics at the time of the sample design (Uganda Bureau of Statistics, 2023). The table also presents the proportional allocation for the minimum required sample size for the P2-ZOI and RFZ and the final sample allocation for each area.

Table A3.4. Final Sample Size and Allocation to Primary Strata for the P2-ZOI Midline Survey

Stratum	Area	Percent distribution of the estimated population	Proportional allocation for the minimum required sample size for the P2-ZOI and RFZ (n=1,772)	Final sample allocation*
Non-RFZ	P2-ZOI non-overlap	79.7%	1,412	1,425
RFZ	RFZ in overlap	100.0%	1,772	1,775
	P2-ZOI in overlap	20.3%	360	
Total (overall P2-ZOI)		100%	3,544	3,200

* The final sample allocations across the primary strata presented in this table have been rounded up slightly to maintain the sample take of 25 households per cluster for the survey.

Accounting for the overlap, the final total required sample was 3,200 households across both the P2-ZOI and the RFZ. This final sample size and sample allocation met and exceeded the minimum required sample sizes across all indicators for the P2-ZOI and RFZ as determined in the sample size calculations presented in Tables A3.1 and A3.2. With a sample take of 25 households per cluster for the main survey, a total of 128 clusters were sampled: 71 clusters in the RFZ stratum and 57 clusters in the non-RFZ stratum.

Calculation of the final parallel survey sample size

The parallel survey sample was selected as an additional sample of households in the same clusters as the main survey. To ensure that the minimum required sample size of 300 households for the parallel survey was met, a sample of 3 additional households was selected in each cluster sampled for the main survey, resulting in a 384-household sample size for the parallel survey. Therefore, 28 households were selected in each cluster, 25 for the main survey and 3 for the parallel survey.

A3.2 Sample design

The sample of households for the P2-ZOI Midline Survey followed a multi-stage stratified cluster sampling design. The EAs from the 2014 Uganda National Population and Housing Census that fell within the 44 districts making up the P2-ZOI comprised the first-stage sampling frame for the survey. Before sample selection, the frame was stratified to distinguish RFZ and non-RFZ districts, regions (East, Karamoja, North, and Southwest), and urban and rural residence to form a total of 12 strata to meet the survey objectives and improve precision of survey estimates. Table A3.5 presents the distribution of the number of EAs, households, and population across the sampling strata, together with the number of sampled EAs and households per stratum.

Table A3.5. Distribution of EAs, Households, and Total Population in Each Stratum

Stratum	Primary	Region	Urban/Rural Residence	Population information		Sample information	
				Number of EAs	Number of households	Number of sampled clusters	Number of sampled households
1	RFZ	Karamoja	Urban	299	23,419	4	100
2	RFZ	Karamoja	Rural	2,109	138,054	27	675
3	RFZ	Northern	Urban	63	4,452	2	50
4	RFZ	Northern	Rural	315	22,726	4	100
5	RFZ	South-Western	Urban	158	18,217	4	100
6	RFZ	South-Western	Rural	1,459	150,703	30	750
Total (RFZ Primary Stratum)				4,403	357,571	71	1,775
7	Non-RFZ	Eastern	Urban	1,056	80,375	3	75
8	Non-RFZ	Eastern	Rural	6,931	350,505	12	300
9	Non-RFZ	Northern	Urban	683	79,659	3	75
10	Non-RFZ	Northern	Rural	3,252	282,439	9	225
11	Non-RFZ	South-Western	Urban	2,390	249,351	8	200
12	Non-RFZ	South-Western	Rural	7,722	651,220	22	550
Total (Non-RFZ Primary Stratum)				22,034	1,693,549	57	1,425
Overall Total (P2-ZOI)				26,437	2,051,120	128	3,200

Note: The population information presented in this table is from the 2014 Uganda National Population and Housing Census.

In the first stage of sampling, 128 EAs were selected as primary sampling units from the sampling frame, based on the 2014 Uganda National Population and Housing Census EAs, across the 44 districts comprising the P2-ZOI using systematic probability proportional to size sampling. Additionally, during this first stage of sampling, 58 EAs were selected as part of a reserve sample using the two-phase sampling approach as described in Section 5.3 of the *Feed the Future PBS Sampling Guide* (Stukel, 2018). The reserve sample was selected in districts where there were potential security risks that could make certain EAs inaccessible during fieldwork. The number of reserve EAs selected per stratum was based on the proportion of EAs in that stratum that fall within districts that were identified to be high risk.

Based on the information in the sampling frame, the average EA size in the survey area was 80 households. Given the time that had passed since the census, and the expected increase in the average EA size over time, the threshold for EA segmentation was set at 300 households following the *Feed the Future PBS Sampling Guide* (Stukel, 2018). Based on this threshold, no EAs were found to be exceptionally large, thus requiring no segmentation. In the next stage, a 2-phase sampling approach was used to select a sample of 28 households in each cluster—25 households for the main survey and 3 households for the parallel survey—using systematic sampling from a comprehensive list of households generated during the household listing operation, which was fielded from August 1 to September 17, 2022, within the sampled clusters. In the final stage of sampling, eligible individuals within the selected households were sampled using a “take-all” approach (i.e., all eligible individuals found were included in the survey).

A3.3 Weighting

The weighting of survey data used information available from the 2014 Uganda National Household and Population Census EA frame (i.e., the first-stage sampling frame), as well as information collected during the listing and data collection processes. This information included the following: (1) EA measure of size used for selection of EAs (where the measure of size is in terms of the number of households in the EA), (2) number of households within the strata from which EAs are drawn, (3) number of households listed within the EA, and (4) response rates at household level and at individual level for each of the individual sampling groups of the survey. The sampling weights presented in Table A3.6 were calculated accounting for the stages of sampling (i.e., the design weights) and levels of non-response across these groups (i.e., the non-response adjustment per sampling group). Table A3.6 also presents the indicators or results to which each sampling weight was applied and the survey modules in which the data to calculate the indicators or results were collected.

Table A3.6. Sampling Weights, Relevant Indicators or Results, and Survey Modules

Sampling weight	Indicator or result	Module
Household ^a	Main survey household-level results	2, 2A, 3, R
	Parallel survey validation of the survey-to-survey imputation model used to compute the consumption aggregate (used to generate a household member weight)	8
Women 15–49 years of age	MDD-W indicator and coverage of promoted women’s health and nutrition interventions’ results	4
Children under 5 years of age	Coverage of promoted children’s health and nutrition interventions results	5

Sampling weight	Indicator or result	Module
Children under 2 years of age ^b	EBF and MAD indicators (baseline only)	5
Children 0–5 months of age	EBF indicator (midline only)	5
Children 6–23 months of age	MAD indicator (midline only)	5
Primary female decision-makers	Women’s empowerment in agriculture results	6
Crop farmers	Cross-market function results	7

^a Two household sampling weights were calculated—one for the main survey and one for the parallel survey. The household sampling weight was the only sampling weight calculated for the parallel survey.

^b Sampling weights at baseline could only be computed for children 0–23 months of age (i.e., under 2 years of age), as “age-in-months” information was not collected in the household roster to determine children who fell within the 0–5 months and 6–23-month age groups.

Design weights were calculated based on the separate sampling probabilities for each stage of sampling. The following notation is used to illustrate the sample weighting process:

f_{1hi} = first-stage sampling probability of the i -th EA from stratum h (EA selection).
 f_{2hij} = second-stage sampling probability of the j -th household, in sampled EA i , from stratum h (household selection).

As mentioned in the sample design section, a two-phase sampling approach was used during the first stage of sampling to select a reserve sample of EAs following the guidance as described in Section 5.3 of the *Feed the Future PBS Sampling Guide* (Stukel, 2018). Therefore, the probability of selecting the i -th EA from stratum h , in the P2-ZOI sample is calculated as follows:

$$f_{1hi} = \frac{m_h \times N_{hi}}{N_h} \times \frac{m_h^*}{m_h} = \frac{m_h^* \times N_{hi}}{N_h} \quad (4)$$

Where,

m_h = number of EAs selected at the first phase of the first stage from stratum h .
 m_h^* = number of EAs released for interviewing from stratum h .
 N_{hi} = total number of households in selected EA i from stratum h (as per the first-stage sampling frame).
 N_h = total number of households across all EAs from stratum h (as per the first-stage sampling frame).

The sampling of households at the second stage was also undertaken using a two-phase sampling approach to enable the selection of household samples for both the main and parallel surveys. Therefore, the second-stage probability of selecting household j , in sampled EA i , from stratum h for the main survey is calculated as follows:

$$f_{2hij} = \frac{n_{hi}}{N_{hi}^*} \times \frac{n_{hi}^m}{n_{hi}} = \frac{n_{hi}^m}{N_{hi}^*} = \frac{25}{N_{hi}^*} \quad (5)$$

And for the parallel survey, the second-stage probability of selecting household k , in sampled EA i , from stratum h is calculated as follows:

$$f_{2hik}^p = \frac{n_{hi}}{N_{hi}^*} \times \frac{n_{hi}^p}{n_{hi}} = \frac{n_{hi}^p}{N_{hi}^*} = \frac{3}{N_{hi}^*} \quad (6)$$

Where,

$n_{hi} = 28$ = number of households selected in sampled EA i , from stratum h during the first phase of the second stage of sampling. A fixed number of 28 households was selected from each cluster during the first phase of the second stage of sampling.

$n_{hi}^m = 25$ = number of households subsampled from n_{hi} for the main survey sample.

$n_{hi}^p = 3$ = number of households subsampled from n_{hi} for the parallel survey sample.

N_{hi}^* = total number of households listed in sampled EA i , from stratum h .

Therefore, the overall probability of selecting a household is the product of the selection probabilities across these two stages for the main and parallel surveys:

$$\text{Main survey: } f_{hij} = f_{1hi} \times f_{2hij} = \frac{m_h^* \times N_{hi}}{N_h} \times \frac{25}{N_{hi}^*} \approx \frac{m_h^* \times 25}{N_h}, \quad \text{if } N_{hi} \approx N_{hi}^* \quad (7)$$

$$\text{Parallel survey: } f_{hik}^p = f_{1hi} \times f_{2hik}^p = \frac{m_h^* \times N_{hi}}{N_h} \times \frac{3}{N_{hi}^*} \approx \frac{m_h^* \times 3}{N_h}, \quad \text{if } N_{hi} \approx N_{hi}^* \quad (8)$$

The design weight for each household is the inverse of its overall selection probability for the main and parallel surveys:

$$\text{Main survey: } W_{hij} = \frac{1}{f_{hij}} = \frac{N_h \times N_{hi}^*}{m_h^* \times N_{hi} \times 25} \approx \frac{N_h}{m_h^* \times 25}, \quad \text{if } N_{hi} \approx N_{hi}^* \quad (9)$$

$$\text{Parallel survey: } W_{hik}^p = \frac{1}{f_{hik}^p} = \frac{N_h \times N_{hi}^*}{m_h^* \times N_{hi} \times 3} \approx \frac{N_h}{m_h^* \times 3}, \quad \text{if } N_{hi} \approx N_{hi}^* \quad (10)$$

Note that the design weight at the individual level is the same as the design weight at the household level because there was no subsampling of individuals within the household for any of the individual sub-populations.

The final sampling weight for the main and parallel surveys was calculated with the design weight adjusted for non-response for each of the selected clusters. Response rates were calculated at the EA level as ratios of the number of interviewed units over the number of eligible units within adjustment classes, where units could be household or individual (e.g., woman of reproductive age, child, farmer, or primary adult female decision-maker).

A4. Survey design, implementation, and data analysis

This appendix describes the methodology used to conduct the P2-ZOI Midline Survey, including the household listing, questionnaire design, training and pretesting, ethics review and approval, fieldwork, data management, and data analysis.²⁹

A4.1 Household listing

A listing of all households was completed in the 128 selected clusters prior to the selection of the household sample. The household listing operation was conducted from August 1 to September 17, 2022. Each of the 20 listing teams comprised an experienced lister and a cartographer. Teams followed all recommended procedures in the *Feed the Future Computer-Assisted Personal Interviewing Household Listing Manual* (USAID, 2021), including mapping, numbering, and listing all structures, dwelling units, and households within the designated boundaries of the cluster. Field teams also recorded the name of a responsible adult or emancipated minor in each household. A total of 13,498 households were listed during the household listing operation.

Three clusters were replaced during the household listing operation.³⁰ Two clusters in the Karamoja region (one cluster in Nakapiripirit district and one cluster in Moroto district) were found to be sparsely populated during the listing operation and had an insufficient number of households from which to sample, so they were replaced by reserve clusters. Additionally, one cluster in the Mbale district was replaced due to access issues caused by a lack of support from the local authority.

Community sensitization with community leaders in each cluster was conducted during the household listing operation to explain the purpose of the survey and to request community cooperation. Listing teams provided community leaders with an official letter from USAID Uganda describing the survey and benefits that may accrue to the country and community from the survey findings.

A4.2 Questionnaire design

The P2-ZOI Midline Survey main and parallel questionnaires were developed to collect all information required according to the assessment scope of work (SOW), including meeting Uganda-specific data requirements. In doing so, comparability with the P2-ZOI Baseline Survey questionnaire and customization of the survey content for the Ugandan context were ensured.

The questionnaire used in the P2-ZOI Baseline Survey served as the starting point for developing the questionnaires for the P2-ZOI Midline Survey. The P2-ZOI Baseline Survey questionnaire, developed from the core baseline questionnaire in the Baseline Toolkit (USAID, 2018a), included an additional expanded resilience module to collect information on resilience capacities and coping strategies. A module on agriculture cross-market functions replaced the standard priority value chain commodity modules.

²⁹ More information on the P2-ZOI Baseline Survey methodology can be found in the P2-ZOI Survey Baseline Report (Feed the Future, 2022).

³⁰ It was determined that two of the five reserve clusters were required during fieldwork, as discussed in Appendix 4.5.

Some changes to the content of the core baseline questionnaire were implemented in the core midline questionnaires and thus in the P2-ZOI Midline Survey main questionnaire, including the following:

- Removed the anthropometry sections of Module 4, *Women's nutrition*, and Module 5, *Children's nutrition*, because anthropometry data would not be collected
- Removed Module 6, *Women's empowerment in agriculture*, for primary adult male decision-makers
- Removed Module 8, *Household consumption expenditure* (but included in the parallel survey questionnaire)
- Retained questions about certain assets and expenditures that were in Module 8 to enable the generation of poverty indicator estimates using a survey-to-survey imputation approach
- Removed questions about the perceived severity of shocks and stresses on household income from Module R
- Added a new module on climate adaptation
- Added questions to measure water insecurity based on the HWISE-4 Scale
- Added questions on interventions of the types promoted by Feed the Future in the P2-ZOI

In addition to applying the changes made to the core questionnaires to the P2-ZOI Midline Survey questionnaire, several components of Module R, *Resilience*, used to calculate resilience capacity indices were streamlined. Streamlining was undertaken to reduce respondent burden, including removing questions not used in the resilience capacity index calculations. Skips and filters were added in Modules 3 and R so that questions relevant only to the calculation of the expanded set of resilience indicators for the RFZ were asked only of households in the RFZ. Finally, questions about livelihood coping strategies were added to Module 3, *Food security*, to be asked only in the RFZ.

As noted in Appendix 5.3, the differences did not affect the comparability of baseline and midline results because they were accounted for during analysis.

The questionnaires were translated into six target languages—Luganda, Lumasaba/Lugisu, Lou, Runyankole/Rukiga, Runyoro/Tutoro, and Ngakarimojong—following the Feed the Future Translation Protocol.³¹ Further improvements to translations were made in response to issues identified in trainings and pretesting before fieldwork.

A4.3 Trainings and pretesting

Training and pretesting were conducted according to the 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Indicator Assessment SOW and the 2022 Feed the Future Uganda P2-ZOI and RFZ Survey Protocol, which were developed using the guidance and template in the Midline Toolkit at the time.³²

³¹ The Translation Protocol was updated in March 2023, but according to the protocol in place during the P2-ZOI Midline Survey preparation phase (the May 2021 version), each translation was orally back-translated into English, and any identified issues with the translations were resolved.

³² The P2-ZOI Midline Survey Protocol Template was updated in March 2023; an April 2021 version of the Survey Protocol Template was used for Uganda. In the newer version of the template, the paper questionnaire pretest was replaced by cognitive testing and the CAPI pretest was replaced by a survey pretest.

A paper pretest to assess questionnaire flow and translation was conducted in Wakiso, Mukono Mbale, Gulu, Nakapiripirit, Mbarara, and Kamwenge districts over five days (September 11–15, 2022).³³ The paper pretest included asking respondents a set of assessment questions about selected survey questions to determine their understanding of those items. Survey questions were revised, as needed, based on respondents' responses to these assessment questions. A total of 139 household interviews (111 for the main survey and 28 for the parallel survey) were conducted. Approximately 20 interviews were completed in each of the 6 target languages as well as in English.

The training of trainers was conducted over 12 days (August 30–September 12, 2022) and was followed by the computer-assisted personal interviewing (CAPI) pretest.³⁴ The CAPI pretest was conducted in Kampala and Wakosi districts over two days (September 15–16, 2022). A total of 54 household interviews (35 for the main survey and 19 for the parallel survey) were conducted. The number of household interviews by language ranged from 3 (Ngakarimojong) to 16 (English).

The main field staff training was held over 14 days (September 23–October 8, 2022) and was followed immediately by the pilot. The pilot was conducted in Lira, Kamwenge, Kiruhura, Mbale, and Napak districts over three days (October 15–17, 2022). A total of 290 household interviews (202 for the main survey and 88 for the parallel survey) were conducted. The number of household interviews by language ranged from 21 (Lumasaba/Lugisu) to 89 (Runyankole/Rukiga). Survey questionnaires and other materials were updated throughout training, pretesting, and the pilot, as necessary.

A4.4 Ethics review and approval

ICF Institutional Review Board approval was granted on July 26, 2022, and the Mildmay Uganda Research Ethics Committee approval was granted on August 31, 2022.

A4.5 Fieldwork

Fieldwork began on October 28, 2022, and ended on December 14, 2022. There were 12 field teams, each comprising 2 interviewing pairs and 1 Field Supervisor. Because of the gender-sensitive nature of some aspects of the main questionnaire, each Interviewer team had at least one female Interviewer who interviewed female respondents for those modules with the sensitive questions. Field Supervisors completed their duties as prescribed in the *Field Supervisor's Manual* and *Interviewers' Manual*, which were both customized for the P2-ZOI Midline Survey. Field teams were regularly visited by Quality Control and Support teams who carried out their responsibilities as described in the *Quality Control and Support*

³³ A paper questionnaire pretest, which included a set of cognitive probing questions to help understand issues with respondents' comprehension of selected questions, was conducted in Uganda. Although probing questions were asked, there was no formal training on cognitive probing or formal cognitive testing. The paper questionnaire pretest has since been replaced by cognitive testing, which includes formal training materials and a separate customizable interview guide.

³⁴ The CAPI pretest, which was meant to focus specifically on the functionality of the customized CAPI data collection application, became the survey pretest in April 2023, to reflect a broader focus. Although the CAPI pretest did include assessing questionnaire flow and translations and ensuring successful data transmission, the survey pretest emphasizes these aspects more and also highlights testing the broader CAPI system, including reviewing data on the server.

Team's Manual. All survey data were regularly sent to a secure server and downloaded and reviewed by the In-Country Data Manager (ICDM) following the procedures described in the *ICDM's Manual*.³⁵

During fieldwork, field check tables were produced, reviewed, and discussed on a weekly basis. Each table had a target or threshold based on the P2-ZOI Baseline Survey results or WHO life tables (for tables on age ratios). The field check tables were reviewed to determine whether targets were met and to identify those teams that did not meet targets or did not show improvements. Meetings were held each week to discuss any issues that specific teams were having and to co-create a follow-up plan.

During fieldwork, two clusters were replaced by EAs in the reserve sample. In Kaabong district, one cluster was replaced due to security issues, and in Omoro district, the community refused to allow the survey to be carried out. A total of five clusters were replaced in the P2-ZOI Midline Survey. Fieldwork in all other clusters was completed as planned.

A4.6 Post-fieldwork data processing

After fieldwork was completed, the ICDM reviewed the data for consistency and accuracy. The ICDM also reviewed the open-ended “other (specify)” responses, translated them into English, and suggested how to recode responses that fit into existing response categories. Data were then transferred to the Data Manager for further cleaning. Additional checks were completed at this stage to confirm eligibility of respondents for interview. The checks included confirming that respondents interviewed met the eligibility criteria for the module and had consented to be interviewed. Variables in which respondents could give multiple answers were also broken into bivariate variables for each response to a question. The required sampling weights (see Appendix 4.3) were computed and added to the data. Finally, household-level and individual-level data files were exported from the Census and Survey Processing System with all direct personally identifiable information removed and shared with the data analysis team.

A4.7 Data analysis

Analyses were performed using Stata and R software and documented in syntax files.³⁶ P2-ZOI indicators were calculated following guidance in the *Guide to Feed the Future Statistics for P2-ZOI Midline Surveys* and template syntax available in the Midline Toolkit (USAID, 2021). The resilience capacity indices for the RFZ were calculated following TANGO International's guidance (2018) with additional input from REFS; the reduced coping strategy index results for the RFZ were calculated following the World Food Programme's guidance (2023). The cross-market function results were computed following baseline methods to enable comparisons over time.

³⁵Core versions of the manuals referred to in this paragraph with standard protocols followed across P2-ZOI Midline Surveys can be found in the Midline Toolkit. The core versions of *Interviewer's Manual*, *Field Supervisor's Manual*, and *Quality Control and Support Team's Manual* were being finalized for the Midline Toolkit during the P2-ZOI Midline Survey preparation phase, so the content of the manuals used in Uganda may differ slightly from those available in the Midline Toolkit, but the standard protocols were the same. The content of the *Quality Control and Support Team's Manual* is standard and does not require adaptation across ZOI Surveys. The *ICDM's Manual* is not included in the Midline Toolkit, but the procedures in the core *ICDM's Manual* in the Baseline Toolkit were largely followed and adapted as needed for the P2-ZOI Midline Survey.

³⁶ Although the Humanitarian Food Assistance Averted and Food Consumption Score indicators were included in the assessment SOW, they are being calculated and reported separately.

All indicator estimates were generated to appropriately account for the complex sample survey design (i.e., stratification, cluster sampling, and sampling weights). Estimates for individual-level indicators may be reported for de jure or de facto populations, depending on the topic or indicator:

- Results related to cross-market functions, promoted interventions, and women's empowerment in agriculture are calculated for de jure household members—that is, usual household members regardless of whether they stayed in the household the night before the survey.
- Results related to women's and children's dietary intake are calculated for de facto household members—that is, individuals who stayed in the household the night before the survey, regardless of whether they are usual household members.

A5. Limitations, corrections, updates, and additional considerations

This appendix focuses on limitations, corrections, updates, and additional considerations related to the comparability of baseline and midline results and the interpretation of changes between the two points in time. Limitations related to sampling and weighting are presented first, followed by data capture limitations. The third section presents information related to analysis—corrections to baseline estimates, updates to calculation methodology, and additional considerations specific to this assessment. The final section presents external influences.

A5.1 Sampling and weighting

Two limitations related to sampling and weighting should be taken into account when comparing baseline and midline results and interpreting changes in indicator estimates over time: the stratification of the RFZ and sampling weights for children 0–5 months and 6–23 months of age.

Stratification of the RFZ

The P2-ZOI Midline Survey is representative of both the P2-ZOI and the RFZ; however, the P2-ZOI Baseline Survey is representative of only the P2-ZOI. Because sampling for the P2-ZOI Baseline Survey did not stratify between RFZ and non-RFZ areas, the number of households that completed interviews in the RFZ at midline (1,653) is much greater than at baseline (436). Furthermore, if the P2-ZOI Midline Survey had not been required to be representative of the RFZ, the sample size would have been 1,772 households, but because the P2-ZOI Midline Survey was required to be representative of the RFZ and because of the overlap of the RFZ and the P2-ZOI, the P2-ZOI sample was 3,200 households. The larger sample size allowed for estimates with greater precision than there otherwise would have been for indicator estimates generated from a descriptive survey.

Sampling weights for children under 2 years of age

At baseline, sampling weights specific to children 0–5 months and 6–23 months of age could not be calculated because children's age in month was not collected in the household roster. Therefore, in the baseline analyses, the sampling weight for children under 2 years of age was used for children 0–5 months and 6–23 months of age under the assumption that the non-response rate for children under 2 years of age was the same as (or a good approximation for) children 0–5 months and 6–23 months. In contrast, at midline, age in months for children under 1 year of age was collected in the household roster allowing for sampling weights specific to children 0–5 months of age and 6–23 months of age to

be calculated. As a result of these differences, the baseline and midline sampling weights for children may not have the same level of precision.

A5.2 Data capture

This section summarizes an issue with data collection that may affect the interpretation or limit the comparability of baseline and midline data.

Availability of/access to insurance

The availability of/access to insurance component indicator, which was used in the computation of the Absorptive Capacity Index and the Household Resilience Capacity Index in the baseline analysis, was not included in the midline analysis. A logic error in the P2-ZOI Midline Survey CAPI system resulted in data missing for 78 percent of households in the RFZ, so the component indicator was removed from both baseline and midline indicator estimate calculations in the midline analysis to ensure the baseline and midline results would be comparable.

A5.3 Analysis

This section summarizes corrections to baseline indicator calculations, updates to indicators and indicator calculations between baseline and midline, and additional analysis considerations. The baseline poverty indicator estimates were not affected, but the corrections and updates described in this section for other indicators resulted in differences in the baseline estimates reported in the P2-ZOI Baseline and Midline Reports. The limitations and additional considerations in this section should be taken into account when looking at trends between the two points in time.

Corrections to baseline estimates

Reduced coping strategies index

The calculation of results related to one baseline indicator was corrected during the midline analysis. During the midline analysis, an error was found in the calculation of the reduced coping strategy index score and its complementary results reported in the P2-ZOI Baseline Survey Report. Corrections made to baseline indicator calculations resulted in differences between indicator estimates reported in the P2-ZOI Baseline Report and those reported in this assessment.

The reduced coping strategy index was calculated to be 9.5 in the baseline analysis, but an error was found in the calculations of the “percent of households that used each of the five coping strategies during the 7 days preceding the survey.” When the error was corrected in the midline analysis, the reduced coping strategy index baseline estimate became 8.5. Furthermore, the percent of households that used three of the five coping strategies and the mean number of days each coping strategy was used changed as follows when the error was corrected:

Table A5.1. Updated Baseline Estimates for the Reduced Coping Strategies Index Results

Coping strategy	Percent of households that used each strategy		Mean number of days strategy was used, among households that used the strategy	
	Baseline report estimate (%)	Updated baseline estimate (%)	Baseline report estimate	Updated baseline estimate
Limited portions at mealtime	82.3	33.1	2.5	3.9
Reduced the number of meals per day	91.6	27.2	3.2	4.2
Limited adult intake	24.2	16.4	0.9	3.6
Ate less expensive or preferred foods	66.0	No change	2.5	3.9
Borrowed food or money from friends and relatives	34.9	No change	0.9	2.6

Updates to calculation methodology

This section summarizes changes between indicators and indicator calculations at baseline and midline. There is one indicator in the P2-ZOI Baseline Report that is not presented in this assessment:³⁷ the A-WEAI score, including the Gender Parity Index score. In addition, there are 10 indicators for which the calculation method was updated. With the exception of the poverty indicators, in the midline analysis, the updated methods were applied to the baseline data and new baseline indicator estimates were computed and compared to the midline indicator estimates to ensure comparability between the two points in time in this assessment. Therefore, the results reported in this assessment may not match the P2-ZOI Baseline Report results.³⁸ These indicators are as follows:

Poverty indicators³⁹

The three poverty indicator calculations were updated to use a survey-to-survey imputation method to generate midline indicator estimates.⁴⁰ The standard approach to estimating the poverty rate is based on detailed household consumption expenditure data collected in a population-based survey, which was

³⁷ See Section 1.3.1 of the *Guide to Feed the Future Statistics for P2-ZOI Midline Surveys* in the Midline Toolkit (USAID, 2021) for further details about the reasons for the exclusion of these indicators in the midline analyses.

³⁸ Further details on these indicators and differences in calculations between baseline and midline can be found in Section 1.3.2 of the *Guide to Feed the Future Statistics for P2-ZOI Midline Surveys* in the Midline Toolkit (USAID, 2021).

³⁹ The poverty indicators include: (1) Prevalence of poverty: Percent of people living on less than \$1.90/day (2011 PPP), (2) Depth of poverty of the poor: Mean percent shortfall of the poor relative to the \$1.90/day (2011 PPP) poverty line, (3) Percent of people who are “near-poor,” living on 100 percent to less than 125 percent of the \$1.90/day (2011 PPP) poverty line.

⁴⁰ To reduce the implementation burden and cost of P2-ZOI midline indicator assessments, comprehensive data on household consumption expenditure were not collected. Instead, a survey-to-survey imputation method developed by researchers at the World Bank was used (Dang, 2024). The survey-to-survey imputation methodology used to calculate the poverty indicators at midline requires a model to be developed from a previously implemented survey in the survey area, which in the case of P2-ZOI Midline Surveys is the P2-ZOI Baseline Survey implemented in that country.

done in P2-ZOI Baseline Surveys.⁴¹ The baseline estimates computed following the standard approach were retained for this assessment. Updates to the poverty indicator calculations apply only to midline estimates in this assessment.

The basic assumptions to implement the survey-to-survey imputation methodology are as follows: (1) the questions in the two surveys are asked in a consistent way and are comparable; (2) the variables from the base survey can be combined with the variables in the target survey; and (3) base survey data can be used to estimate the parameters of the predictive model and the distribution of their error terms to predict (impute) household consumption and obtain poverty estimates in the target survey.

The model includes utility expenditures, household and dwelling characteristics, improved water and sanitation, education, dependency, and asset ownership. In the Uganda analysis, per capita household utility expenditures were determined using the following data: expenditures on charcoal, charcoal paraffin or kerosene, candles, and matches in the seven days preceding the survey; expenditures on natural gas, electricity, and water utilities during the one month preceding the survey; and ownership of solar panels and generators at the time of the P2-ZOI Midline Survey. The utility expenditures were aggregated and divided equally among the number of usual household members. The questions used to collect these data in the P2-ZOI Midline Survey were identical to those in the P2-ZOI Baseline Survey.

The model was used to predict per capita consumption using P2-ZOI Baseline Survey data. The model's coefficients were then used with the model's predictor variable data from the P2-ZOI Midline Survey to impute per capita consumption for each household at midline. The imputed values were then used to estimate the prevalence of poverty, depth of poverty of the poor, and prevalence of near-poor at midline.

Due to the survey-to-survey imputation method used to calculate the consumption aggregate for each household, reliable standard error estimates are not available for the depth of poverty of the poor and prevalence of near-poor; therefore, statistical tests of difference were not performed for these two indicators.

Percent of households below the comparative threshold for the poorest quintile of the asset-based CWI

The reference country used to calculate the standardized quintile cutoffs and anchoring points was changed from the 2017 Senegal Demographic and Health Survey to the 2017–2018 Bangladesh Demographic and Health Survey, a country with a higher gross domestic product and greater variation in wealth distribution. Using a reference country with a greater variation in wealth distribution allows for better comparison with countries that also have varying distributions.

Prevalence of moderate and severe food insecurity in the population, based on the FIES

The calculation for this indicator was updated to include two calibration steps (one to calibrate midline to baseline and another to calibrate the baseline and midline to the global scale) based on guidance from

⁴¹ At baseline, per capita daily consumption was calculated by first determining the monetary value of daily household consumption expenditures per usual household member for specific categories of items (food consumption over the past 7 days; non-food expenditures over the past 7 days, 1 month, 3 months, and 12 months; housing; and durable goods) and then aggregating the values across all categories (USAID, 2018b). Per capita consumption values were then used to estimate the baseline prevalence of poverty values.

the Food and Agriculture Organization of the United Nations (FAO). The former was included to enable the comparison of midline to baseline estimates and the latter to use FAO's international reference scale to measure moderate and severe food insecurity. In the analysis at baseline, these calibration steps were not needed. In addition, the calculation for this indicator was updated at midline to use the household sampling weight rather than the household member sampling weight.

ARSSI

The calculation for the index was updated to include only data on the household's perceived severity of each shock or stress experienced on their food consumption. Previously, the calculation also included data on the household's perceived severity of each shock or stress experienced on their income. The latter was dropped from the calculation because the effects on a household's food consumption and income were found to be highly correlated.

Index of social capital at the household level

The calculation for the index was updated so reciprocity was better accounted for—that is, to include both the extent to which the household can lean on a certain group during difficult times (*i.e.*, relatives in their community, relatives outside their community, non-relatives in their community, and non-relatives outside their community) as well as the extent to which the household would give support to that group during difficult times. In the analysis at baseline, reciprocity was not fully accounted for in the indicator calculation methodology.

Percent of households participating in group-based savings, micro-finance, or lending programs

The calculation for this indicator was updated to use only data collected from primary adult female decision-makers. In the analysis at baseline, the calculation used data from both primary adult male and female decision-makers because Module 6, *Women's empowerment in agriculture*, was administered to both men and women. In the P2-ZOI Midline Survey, data from primary adult male decision-makers were not captured in Module 6.

Percent of children ages 6–23 months consuming a MAD

In alignment with updated WHO guidance (WHO & UNICEF, 2021), the criteria for consuming achieving MDD, a component of the children's MAD indicator, was updated to measure children 6–23 months of age who consumed at least five of eight food groups, including breastmilk. At baseline, MDD measured children 6–23 months of age who consumed four of seven food groups. In addition, the criteria for achieving minimum meal frequency for non-breastfed children, another component of the children's MAD indicator, was updated to require at least one solid, semi-solid, or soft foods feeding. In the analysis at baseline, minimum meal frequency for non-breastfed children was defined as four or more feedings of solid, semi-solid, soft food, or milk feeds, and at least two of these feedings had to be milk feeds.

Percent of households with soap and water at a handwashing station on premises

The denominator for this indicator was updated to include both those households where a place for handwashing was observed and those households where no handwashing facility exists in the denominator. The updated calculation aligns with the WHO's indicator definition and excludes households that do not provide permission to see the handwashing facility or where it is not seen for

some other reason (WHO & UNICEF, 2018). In the analysis at baseline, the calculation included all households in the denominator.

Additional considerations

The calculation of several Feed the Future P2-ZOI and RFZ indicators requires country-specific considerations. The poverty indicator calculations require country-specific values for PPP and consumer price indices (CPIs). The 5DE score calculations require adaptation to account for any country-specific credit sources, community groups, or daily activities. The ARSSI calculations require adaptation to include country-specific shocks and stresses. The resilience capacity index calculations require adaptation to reflect the country-customized questionnaire. These considerations for the Uganda analyses are described in this section. In addition to these analysis-related corrections and updates already highlighted, there are two items to highlight related to analysis: red palm data and secondary anthropometry data.

Poverty indicators

The poverty indicators were calculated using three global poverty thresholds in this assessment: \$1.25 per person per day (2005 PPP), \$1.90 per person per day (2011 PPP), and \$2.15 per person per day (2017 PPP). For all poverty indicator calculations, the November 2022 CPI for Uganda (138.48) was used because that was the month of fieldwork in which the majority of data were collected. In addition, the following Uganda-specific PPP conversion rates and CPIs were used:

To calculate the poverty indicators using the \$1.90 per person per day (2011 PPP) poverty line:

- 2011 PPP conversion factor: 946.890 Ugandan shillings per international dollar
- 2011 CPI: 85.10

To calculate the poverty indicators using the \$1.25 per person per day (2005 PPP) poverty line:

- 2005 PPP conversion factor: 744.618 Ugandan shillings per international dollar
- 2005 CPI: 45.22

To calculate the poverty indicators using the \$2.15 per person per day (2017 PPP) poverty line:

- 2017 PPP conversion factor: 1,221.088 Ugandan shillings per international dollar
- 2017 CPI: 112.10

SEI and ARSSI

There are 16 shocks and stresses included in the core P2-ZOI Baseline and Midline Survey questionnaires. During the P2-ZOI Baseline Survey questionnaire customization process, USAID Uganda chose to include 14 of these shocks and stresses, adapted and combined 2 of them, and added 4 more for a total of 19 shocks and stresses. The four shocks and stresses added were hail or frost, delay in food assistance, unemployment of youth, and human disease outbreak. The two shocks/stresses from the core questionnaires that were adapted and combined—unable to access crop inputs and unable to access livestock inputs—became “increased prices of agricultural or livestock inputs.” The same 19 shocks and stresses included in the P2-ZOI Baseline Survey questionnaire were included in the P2-ZOI Midline Survey main questionnaire to ensure comparability of the baseline and midline results. The SEI and ARSSI calculations were adjusted to account for the updates to the shocks and stresses in the questionnaires.

Resilience capacity indices

The resilience capacity indices calculations follow the guidance provided by and outlined in Appendix 2.3 of the P2-ZOI Baseline Survey Report (Feed the Future, 2022). However, two of the component indicator calculations were adjusted in the midline analysis due to the streamlining of the midline questionnaire (see Appendix 4.2).

- The number of information sources asked about in Module R7 was reduced from 19 to the 11 most relevant to the Ugandan context based on analysis of baseline data. The exposure to information component indicator estimates for both baseline and midline were, therefore, calculated using the 11 information topics included in the P2-ZOI Midline Survey questionnaire.
- The number of food and income sources asked about in Module R10 was reduced from 20 to the 14 most relevant to the Ugandan context based on analysis of baseline data. The livelihood diversification component indicator estimates for both baseline and midline were, therefore, calculated using the income and food sources included in the P2-ZOI Midline Survey questionnaire.

In addition, as noted in Appendix 5.2, the availability of/access to insurance component indicator was not included for either baseline or midline in the midline analysis because data were missing for the majority of households in the RFZ.

Red palm data

There are two indicators that estimate adequate dietary intake: percent of women of reproductive age 15–49 years consuming a diet of minimum diversity (MDD-W) and percent of children 6–23 months consuming a MAD. One component of MAD is MDD for children.

See Table A5.2 for the questions that capture the consumption of vitamin A-rich foods in the core questionnaires and the corresponding food groups to which they belong in the MDD calculations.

Table A5.2. Food Groups that Capture the Consumption of Vitamin A-rich Foods and the Relevant Questions in the P2-ZOI Baseline and Midline Survey Questionnaires

Food group		Question
Women		
7	Dark green leafy vegetables	• Any dark green leafy vegetables?
8	Other vitamin A-rich fruits and vegetables	• Pumpkin, carrots, squash, or sweet potatoes that are yellow or orange inside? • Ripe mangoes, ripe papayas? • Foods made with red palm oil, red palm nuts, or red palm nut pulp sauce?
Children		
6	Vitamin A-rich fruits and vegetables	• Pumpkin, carrots, squash, or sweet potatoes that are yellow or orange inside? • Any dark green leafy vegetables? • Ripe mangoes or papayas? • Foods made with red palm oil, red palm nuts, or red palm nut pulp sauce?

The question that captures the consumption of foods made with red palm oil, red palm nuts, or red palm nut pulp sauce deviates from MDD indicator calculation guidance (FAO, 2021; WHO & UNICEF, 2021), which states that red palm fruit and red palm pulp should be included in the questions about vitamin A-rich fruits but that red palm oil should be asked about in a separate, optional question because it is often included in mixed dishes and it is difficult to discern the amount consumed. Since red palm oil, red palm nuts, and red palm nut pulp sauce were grouped together in one question, red palm oil cannot be excluded unless red palm nuts and red palm nut pulp sauce are also excluded.

This distinction is important because palm oil that is pressed from the seed of the red palm is not rich in vitamin A, but palm oil that is derived from the oily pulp of the red palm fruit is very rich in vitamin A (WHO & UNICEF, 2021). By grouping red palm oil with red palm nut and red palm nut pulp sauce, there is a risk in over-reporting vitamin A in an individual's diet. For example, a respondent may report consuming foods made with red palm oil, red palm nut, or red palm nut pulp sauce because they consumed foods made with red palm oil. In this case, the dietary intake food score could indicate levels of vitamin A that are not representative. If palm oil is the only vitamin A rich food consumed, the individual will be counted as consuming an additional food group category. Because red palm products are grouped together as one category in the questionnaire, analysts are unable to tell if the respondent has consumed only red palm oil or a red palm product rich in vitamin A.

To test whether grouping red palm oil with red palm nut and red palm nut pulp sauce biases the respondent's dietary intake food score, a comparison of the women's and children's dietary intake food scores with the inclusion and exclusion of the red palm question across surveys was conducted (Tables 8.1.3 and T8.3.4). The analysis concluded that there is negligible impact on consuming an MDD for women or children whether the red palm question is included or excluded. Therefore, the red palm questions were retained in the calculation of women's and children's food group scores because this aligned with calculations for previous ZOI surveys, with the caveat that this approach may result in the (slight) overestimation of MDD for women and children.

Five Domains of Empowerment indicator

Compared to the core P2-ZOI Midline Survey main questionnaire and the calculation guidance in the *Guide to Feed the Future Statistics for P2-ZOI Midline Surveys*, there are two deviations:

- The question naming convention used at midline was the same as used at baseline even though the question naming convention was updated in the core P2-ZOI Midline Survey main questionnaire.
- USAID Uganda requested that three credit sources not included in the core questionnaire be added to Module 6.3B—religious groups, local merchants or traders, and other sources—at baseline, so these groups were also included at midline and included in the calculation of Indicator 2.2 when generating both baseline and midline indicator estimates.

Secondary anthropometry data

The assessment SOW included the computation of Feed the Future's four P2-ZOI anthropometry-based nutritional status indicators:

- HL.9-d: Prevalence of underweight (body mass index [BMI]<18.5) women of reproductive age

- HL.9-a: Prevalence of stunted (height-for-age z-score [HAZ]<-2) children under 5 (0–59 months)
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However, because the timing of a DHS Program-supported survey in Uganda did not align with the timing of the P2-ZOI Midline Survey, we were unable to compute the nutritional status indicators.

A5.4 External influences

Two external influences—COVID-19 and the 2020–2022 global economic crisis—should be considered when comparing baseline and midline indicator estimates and interpreting changes between the two points in time.

COVID-19

As a result of the global COVID-19 pandemic and the impact it has had on poverty and nutrition in low-income countries, some P2-ZOI and RFZ midline results may have been impacted atypically when compared to what may have been anticipated in a three-year time frame (Picchioni et al., 2022; Sánchez-Páramo et al., 2021).⁴² According to Sánchez-Páramo et al., 2021, “Globally, three to four years of progress toward ending extreme poverty are estimated to have been lost.” In some cases, stasis between baseline and midline may be considered a positive finding.

Global economic crisis

The COVID-19 pandemic and war in Ukraine, among other factors, triggered a global economic meltdown from 2020–2022. Global growth fell by about 5 percent in 2020 (International Monetary Fund, 2020). By 2022, global inflation had risen to almost 9 percent, almost 5 percent higher than in 2021 (World Bank Group, 2024). In Uganda, the pandemic and global economic crisis led to a decreased economic activity due to business closures, decreases in tourism and foreign investment, and reduced remittances sent from abroad (World Bank Group, 2021). The economic crisis that took place between the P2-ZOI Baseline and Midline Surveys may have contributed to some of the changes in P2-ZOI and RFZ indicator estimates.

A6. Measures and reporting conventions

A6.1 Indicator disaggregates

A standard set of indicator disaggregate variables is used in P2-ZOI Midline Survey results tables; these include gendered household type, wealth quintile, and shock exposure severity. In addition, sex and age categories are used for individual-level results, and residence is used for WASH results. For this assessment, RFZ/non-RFZ, region, and household education are also used as disaggregates. This section defines each of the disaggregate variables and when they are used.

⁴² Picchioni et al.’s systematic literature review (2022) found clear evidence in lower middle-income countries that decreased sources of income and reduced access, availability, and affordability of food due to the COVID-19 pandemic negatively affected individuals’ diet quality and nutrition, especially for women and lower socioeconomic groups.

Gendered household type

All standard household-level indicators for the P2-ZOI, with the exception of the poverty indicators, are disaggregated by gendered household type—that is: (1) households that include both male and female adults, 18 years of age or older; (2) households that include female adults, but no male adults; (3) households that include male adults, but no female adults; and (4) households with no adults (children only). This approach to conceptualizing household type is distinct from the standard “head of household” approach, which is embedded with presumptions about household gender dynamics, and may perpetuate existing social inequalities and prioritization of household responsibilities that may be detrimental to women. This variable is calculated using data on the age and sex of de jure household members, which are collected in the household roster.

Wealth quintile

Households are characterized into wealth quintiles according to their asset-based wealth index score, which considers several household and dwelling characteristics and household ownership of various assets, which are collected in Module 2, *Dwelling characteristics and household assets*. Wealth quintiles are survey-specific, and thus, as a disaggregate they allow comparison of indicator estimates based on relative, rather than absolute, asset-based wealth of the individuals in the sample. Wealth quintile is used as a disaggregate for many household-level indicators, as well as some individual-level nutrition indicators.

Shock exposure severity

The SEI assigns households a score depending on the number and severity of shocks and stresses the household experienced during the 12 months preceding the ZOI Survey. Respondents to Module R, *Resilience*, were asked whether their household experienced 19 shocks or stresses. For the shock exposure severity disaggregate, households are categorized into one of four categories based on their score: did not experience any shocks (SEI score of 0), low (SEI scores 1–21 for the midline; 1–13 for the baseline), moderate (SEI scores 22–32 for the midline, 14–22 for the baseline), and high (SEI scores 33–71 for the midline, 23–63 for the baseline). Households with an SEI score of 0 are assigned to the first category because they did not experience any shocks. The remaining three categories are meant to split households with an SEI score higher than 0 into roughly even categories. Thus, the shock exposure severity disaggregate allows comparison of indicator estimates based on relative, rather than absolute, shock exposure of the households in the sample. The SEI is used as a disaggregate for many household-level indicators, as well as some individual-level nutrition and agriculture indicators.

Residence

Residence is used primarily in tables presenting WASH results. Households are designated as being located in either urban or rural areas, as defined for Uganda. This EA-level designation is an attribute included in the survey sampling frame and applied to each household.

Resilience Focus Zone

This disaggregate is a binary variable indicating whether a household is located in the RFZ or in a non-RFZ area. The RFZ included the same geographic area at baseline and midline, but due to the creation of new districts between the two timepoints, the RFZ comprised 11 districts at baseline and 13 districts at

midline. The RFZ is located in the Karamoja, North, and Southwest regions; there are no RFZ districts in the East region.

Region

Region is a disaggregate used in several tables throughout the report. Three of the regions—East, Karamoja, and North—align with CDCS regional corridors in Uganda. The fourth region, Southwest, combines the southwestern and western CDCS regional corridors due to the small number of P2-ZOI districts in the western CDCS regional corridor (n=2).

Household education

Household educational attainment reflects the highest level of education attained by any member of the household, as reported in the household roster. This variable is used in tables that present household-level data, and comprises five categories: no education (households with no member who has received formal education); less than primary year 3 (households with at least one member who has received formal schooling, but with no member who has completed the third year of primary); completed primary year 3 (households with at least one member who has completed the third year of primary, but with no member who has completed the seventh year of primary); completed primary (households with at least one member who has completed the seventh year of primary, but with no member who has completed “A” level); and completed “A” level or higher (households with at least one member who has completed “A” level). Households are categorized in only one of the five categories.

Age in years

Data on household members’ age in years are collected in the household roster (Module 1). For women 15–49 years of age and primary adult female decision-makers, age in years is also captured in subsequent individual-level questionnaire modules to confirm eligibility. Age in years captured in individual-level modules is used whenever it is available for individuals 15 years of age or older. Categorical age disaggregates are created using age in years data, including:

- 18–29 years and 30 years or older: for primary adult decision-makers
- 15–19 years and 20–49 years: for women of reproductive age
- 15–29 and 30 years or older: for farmers and promoted intervention participants

Age in months

Date of birth is collected in Module 5, *Children’s nutrition*, rather than the roster so that children’s primary caregivers can be prompted to provide the most accurate age possible. Children’s date of birth is then converted into age in months. Children’s age in months is then used to create age categories for tables presenting children’s dietary intake indicators for children 6–11, 12–17, and 18–23 months of age and results related to participation in children’s health and nutrition interventions of the types promoted by Feed the Future in the ZOI for children less than 2 years (24 months) of age and 2 years (24 months) to less than 5 years (60 months) of age.

Sex

Sex—male or female—is a standard disaggregate in tables presenting children’s dietary indicators, cross-market function results, and results related to participation in interventions of the types promoted by

Feed the Future in the ZOI. The sex of household members is collected in the household roster, and children's sex is confirmed in Module 5, *Children's nutrition*.

A6.2 Reporting conventions

Conventions

This assessment is primarily comparative in nature. This section provides an overview of the conventions used in reporting the results.

- In the tables throughout this report, weighted point estimates and unweighted sample sizes are presented.
- Estimates are shown to one decimal place, except for the 5DE and the average adequacy score of the disempowered, which are shown to two decimal places, and unweighted sample sizes are shown as whole numbers.
- Estimates are suppressed when the unweighted sample size is insufficient to calculate a reliable point estimate ($n < 30$); this is denoted by the symbol “^” in the results table cell, and a table footnote defines the symbol.
- If there are no observations for a result or disaggregate category ($n = 0$), the symbol “-” is used in the results table to indicate that an estimate is not available.
- If there are observations for a result or disaggregate category ($n > 0$) but data are not available to calculate an estimate, N/A (not available) is entered for the estimate, 95 percent confidence interval, and number of observations.

Statistical tests of significance

Generally, three types of tests of significance were performed:

- To determine whether there is a statistically significant **difference** between the baseline and midline estimates (both proportions and means), a post-estimation Stata command was used to calculate confidence intervals and p-values for linear combinations of model coefficients.⁴³
- For results presented only at midline, to determine whether there is a statistically significant **association** between the outcome that is a proportion and a background characteristic (disaggregate variable), a Rao-Scott chi-squared test of association is performed.⁴⁴
- To determine whether there is a statistically significant **difference** in an outcome that is a mean by a background characteristic (disaggregate variable), a post-estimation command was used to calculate confidence intervals and p-values for linear combinations of model coefficients.⁴⁵

For most indicator estimates compared between baseline and midline in the results tables, the 95 percent confidence intervals for the baseline and midline estimates, the percentage point difference (proportions) or difference in means (means) between estimates, associated p-value, and level of significance are reported in results tables. The exceptions are prevalence of poverty, depth of poverty of

⁴³ The Stata “lincomest” post-estimation command was used to determine whether there was a statistically significant difference.

⁴⁴ The Stata “svy: tab var1 var2” command is used to perform a Rao-Scott chi-squared test of association.

⁴⁵ Again, the Stata “lincomest” post-estimation command was used to determine whether there was a statistically significant difference.

the poor, prevalence of near-poor, prevalence of moderate and severe food insecurity, and the resilience capacity indices. (See the exceptions explained below.)

P-values are presented according to the style used by the New England Journal of Medicine (* indicates $p < 0.05$, ** indicates $p < 0.01$, *** indicates $p < 0.001$, n/s indicates not significant, and na=not applicable): p-values larger than 0.01 are reported to two decimal places, and those between 0.01 and 0.001 to three decimal places; p-values smaller than 0.001 are reported as $p < 0.001$. However, p-values between 0.045 and 0.055 are presented to three decimal places to show whether the difference is statistically significant at the 0.05 level. If a test of difference could not be conducted, na (not applicable) is entered for the difference, p-value, and level of significance.

However, there are three exceptions where different procedures are followed:

- For the prevalence of poverty, statistical tests of difference could not be conducted. The asterisk (*) indicates that the baseline and midline estimate 95% confidence intervals are not overlapping, while “uc” indicates that the baseline and midline confidence intervals overlap.
- For the depth of poverty of the poor and percent of people who are near-poor, reliable standard error estimates are not available due to the survey-to-survey imputation method used to calculate a consumption aggregate for each household (see **Appendix 5.3** for more detail). Therefore, statistical tests of difference were not performed and only differences between baseline and midline are presented.
- For the prevalence of moderate and severe food insecurity, whether a statistically significant difference exists between the P2-ZOI Baseline and Midline Survey estimates was determined by comparing the difference of midline and baseline point estimates to the midline margin of error at the 95% level. If the difference of midline and baseline point estimates is greater than the midline margin of error, it is considered statistically significant. If not, it is considered not statistically significant. For food insecurity, differences found to be statistically significant are indicated by s=significant; n/s=not significant and na=not applicable.

A7. Results Tables

The results tables include the P2-ZOI Baseline and Midline Survey results for the indicators below and, in separate tables, the components that make up these indicators. The indicator tables are also presented by standard and country-specific disaggregates.

The 16 Feed the Future P2-ZOI indicators computed for this assessment for both the P2-ZOI and RFZ are:

- Prevalence of poverty: Percent of people living on less than \$1.90/day 2011 purchasing power parity (PPP) [EG-c]
- Depth of poverty of the poor: Mean shortfall of the poor relative to the \$1.90/day 2011 PPP poverty line [EG-h]
- Percent of people who are 'near-poor,' living on 100 percent to less than 125 percent of the \$1.90 2011 PPP poverty line [FTF Context-9]
- Percent of households below the comparative threshold for the poorest quintile of the asset-based Comparative Wealth Index (CWI) [EG-g]
- Prevalence of moderate and severe food insecurity in the population, based on the Food Insecurity Experience Scale (FIES) [EG-e]
- Ability to recover from shocks and stresses index [RESIL-a]
- Index of social capital at the household level [RESIL-b]
- Percent of households that believe local government will respond effectively to future shocks and stresses [RESIL-c]
- Percent of households participating in group-based savings, micro-finance, or lending programs [EG.4.2-a]
- Prevalence of women of reproductive age consuming a diet of minimum diversity [HL.9.1-d]
- Prevalence of exclusive breastfeeding of children under 6 months of age [HL.9.1-b]
- Percent of children 6-23 months of age consuming a minimum acceptable diet [HL.9.1-a]
- Percent of households with access to a basic sanitation service [HL.8.2-a]
- Percent of households with soap and water at a handwashing station on premise [HL.8.2-b]
- Prevalence of water insecurity, based on the Brief Household Water Insecurity Experiences Scale (HWISE-4)⁴⁶
- Five Domains of Empowerment (5DE) score for women [EG.3-i]

In addition, the following expanded set of resilience indicators was computed for the RFZ:

- Index of Shock Exposure
- Absorptive capacity index
- Adaptive capacity index
- Transformative capacity index

⁴⁶ The HWISE-4 scale is a brief version of the Household Water Insecurity Experiences (HWISE) scale, designed to assess water insecurity at the household level. It consists of four questions that capture the frequency and severity of experiences related to water access and availability (Northwestern University Institute for Policy Research, n.d.).

- Household resilience capacity index
- Reduced Coping Strategies Index
- Index of Community Resilience

Furthermore, questions on coverage of promoted interventions were added to the P2-ZOI Midline Survey questionnaire to capture data on households' participation in or access to the kinds of interventions that Feed the Future supports in Uganda. Descriptive statistics on agricultural cross-market functions, COVID-19, climate adaptation, and coverage of promoted interventions related to agriculture, resilience, women's nutrition and health, children's nutrition and health, and water, sanitation, and hygiene are provided in their respective tables.

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Reporting Conventions Summary Table

Condition	Action	Comment
Estimates (est) and confidence intervals (CI)		
If (obs=0)	Set est & CI to "—"	Results not available
If (obs>0 & obs<30)	Set est & CI to "^"	Results not statistically reliable
If (est=0 or est=100) & obs>=30	Set CI to "na"	na=not applicable
If a particular result is not available (e.g., data were not collected at BL due to a CAPI error or a change in response categories at midline)	Set est & CI to "N/A"	N/A=not available
If (indicator is five domains of empowerment or average adequacy score of the disempowered)	Report est & CI to two decimal places	
If (any other indicator or result)	Report est & CI to one decimal place	
Differences (Diff.), p-values, and levels of significance (Sig.)		
If (obs_BL=0) or (obs_ML=0) or (obs BL>0 & obs_BL<30) or (obs_ML>0 & obs_ML<30)	Set p-value, Diff., Sig. to "na"	na=not applicable
If (p-value>=0.05)	Set Sig. to "n/s"	n/s=not significant
If (p-value<0.05)	Set Sig. to "*"	Significant at 5% level
If (p-value<0.01)	Set Sig. to "**"	Significant at 1% level
If (p-value<0.001)	Set Sig. to "***"	Significant at 0.1% level
If (p-value<0.001)	Set p-value to "<0.001"	Report p-values as "<0.001"
If (p-value>0.01)	Report p-values to two decimal places	
If (p-value>=0.001 & p-value<0.010) or (p-value>=0.045 & p-value<0.055)	Report p-values to three decimal places	
The levels of significance presented for the prevalence of poverty (T4.1.1) and the prevalence of moderate and severe food insecurity (T5.1.1 and T5.1.2) follow different conventions; please see the tables notes for details. P-values are not reported in those tables.		
Standard deviation (SD) and standard error (SE)		
All estimates (i.e., 1 or 2 decimal places)	Report SD and SE to two decimal places	

Table A7.1: Feed the Future Phase Two ZOI Midline Survey Indicator Estimates, by Key Disaggregates: Uganda 2022

Indicator	Estimate	SD	SE	DEFF	Indicator incompletion rate ^a (%)	Unweighted number ^b (n)
Prevalence of poverty: Percent of people living on less than \$1.90/day (2011 PPP)^c						
All individuals (ZOI)^d	37.1		0.014		0.0	2,995
All individuals (RFZ)^d	40.0		0.015		0.0	1,653
Depth of poverty of the poor: Mean percent shortfall of the poor relative to the \$1.90/day (2011 PPP) poverty line^c						
All individuals (ZOI)^d	35.6				0.0	1,108
All individuals (RFZ)^d	41.6				0.0	610
Percent of people who are near-poor, living on 100 percent to less than 125 percent of the \$1.90/day (2011 PPP) poverty line^c						
All individuals (ZOI)^d	11.8				0.0	2,995
All individuals (RFZ)^d	12.5				0.0	1,653
Percent of households below the comparative threshold for the poorest quintile of the asset-based comparative wealth index						
All households (ZOI)	38.1		0.02	6.3	0.0	2,995
Gendered household type (ZOI)						
Male and female adults	34.2		0.02	4.5	0.0	2,401
Female adults only	53.7		0.04	2.0	0.0	399
Male adults only	57.5		0.07	3.6	0.0	194
Children only	^		^	^		1
All households (RFZ)	61.1		0.03	4.4	0.0	1,653
Prevalence of moderate and severe food insecurity in the population, based on the Food Insecurity Experience Scale						
All households (ZOI)	84.6		0.01	4.4	0.5	2,981
Gendered household type (ZOI)						
Male and female adults	84.2		0.02	4.0	0.5	2,390
Female adults only	87.4		0.02	1.0	0.5	397
Male adults only	82.3		0.03	1.1	0.5	193
Children only	^		^	^		1
Severity (ZOI)						
Moderate	47.9		0.01	1.9	0.5	2,981
Severe	36.6		0.03	4.8	0.5	2,981
All households (RFZ)	87.6		0.01	4.4	0.5	1,645
Severity (RFZ)						
Moderate	43.2		0.01	1.9	0.5	1,645
Severe	44.4		0.02	4.8	0.5	1,645
Ability to Recover from Shocks and Stresses Index (Mean score)						
All households (ZOI)	3.9	1.35		7.5	0.1	2,972
Gendered household type (ZOI)						
Male and female adults	3.9	1.35		6.5	0.1	2,385
Female adults only	3.7	1.30		3.2	0.0	397
Male adults only	3.9	1.42		2.0	0.0	189
Children only	^	^		^		1
All households (RFZ)	3.6	2.59		1.2	0.1	1,638
Index of social capital at the household level (Mean score)						
Overall index						
All households (ZOI)	60.1	35.48		2.9	0.1	2,992
Gendered household type (ZOI)						
Male and female adults	61.0	35.28		2.6	0.1	2,398
Female adults only	56.0	34.95		2.2	0.0	399
Male adults only	56.7	38.55		1.4	0.0	194
Children only	^	^		^		1
All households (RFZ)	56.3	69.79		1.0	0.2	1,650

Indicator	Estimate	SD	SE	DEFF	Indicator incompletion rate ^a (%)	Unweighted number ^b (n)
Bonding sub-index						
All households (ZOI)	60.6	39.01		2.3	0.1	2,992
Gendered household type (ZOI)						
Male and female adults	61.5	38.78		2.1	0.1	2,398
Female adults only	56.0	38.97		2.2	0.0	399
Male adults only	57.8	41.22		1.1	0.0	194
Children only	^	^		^		1
All households (RFZ)	57.7	76.64		0.8	0.2	1,650
Bridging sub-index						
All households (ZOI)	59.6	41.98		3.5	0.1	2,992
Gendered household type (ZOI)						
Male and female adults	60.4	41.70		3.1	0.1	2,398
Female adults only	56.0	41.73		2.1	0.0	399
Male adults only	55.6	45.70		1.8	0.0	194
Children only	^	^		^		1
All households (RFZ)	55.0	82.54		1.1	0.2	1,650
Percent of households that believe local government will respond effectively to future shocks and stresses						
All households (ZOI)	59.0		0.03	7.3	6.7	2,794
Gendered household type (ZOI)						
Male and female adults	58.1		0.03	6.3	6.7	2,239
Female adults only	65.5		0.04	3.0	6.5	373
Male adults only	55.8		0.05	1.9	6.7	181
Children only	^		^	^		1
All households (RFZ)	61.7		0.03	5.5	7.9	1,523
Percent of households participating in group-based savings, micro-finance, or lending programs						
All households (ZOI)	62.9		0.02	3.5	0.0	2,624
Gendered household type (ZOI)						
Male and female adults	65.2		0.02	3.6	0.0	2,231
Female adults only	49.9		0.03	1.7	0.0	393
Male adults only						
Children only						
All households (RFZ)	53.9		0.02	3.4	0.0	1,459
Five Domains of Empowerment score						
All women (ZOI)	0.90	0.47		3.9	12.4	2,335
Women's age (ZOI)						
18-29 years	0.85	0.41		2.4	14.9	640
30 years or older	0.92	0.37		3.5	9.6	1,695
All women (RFZ)	0.88	0.45		4.4	12.5	1,277
Percent of women who are empowered						
All women (ZOI)	72.2		0.02	5.8	12.4	2,335
Women's age (ZOI)						
18-29 years	60.6		0.03	2.7	14.9	640
30 years or older	76.3		0.02	4.1	9.6	1,695
All women (RFZ)	67.1		0.03	2.7	12.5	1,277
Average adequacy score of disempowered women						
All women (ZOI)	0.64	0.12		2.8	0.0	711
Women's age (ZOI)						
18-29 years	0.61	0.15		1.9	0.0	258
30 years or older	0.66	0.09		2.8	0.0	453
All women (RFZ)	0.63	0.19		2.7	0.0	422
Percent of disempowered women who are adequate in each A-WEAI indicator						
Input in productive decisions						
All women (ZOI)	96.5		0.01	1.8	0.0	711

Indicator	Estimate	SD	SE	DEFF	Indicator incompletion rate ^a (%)	Unweighted number ^b (n)
Women's age (ZOI)						
18-29 years	92.8		0.03	1.9	0.0	258
30 years or older	98.7		0.00	1.2	0.0	453
Ownership of assets						
All women (ZOI)	85.8		0.02	2.4	0.0	711
Women's age (ZOI)						
18-29 years	75.8		0.04	1.9	0.0	258
30 years or older	91.7		0.02	1.6	0.0	453
Access to and decisions on credit						
All women (ZOI)	17.1		0.02	2.7	0.0	711
Women's age (ZOI)						
18-29 years	21.0		0.04	2.0	0.0	258
30 years or older	14.9		0.02	2.2	0.0	453
Control over income						
All women (ZOI)	95.6		0.01	2.8	0.0	711
Women's age (ZOI)						
18-29 years	90.8		0.03	2.9	0.0	258
30 years or older	98.3		0.01	1.1	0.0	453
Group membership						
All women (ZOI)	34.8		0.03	2.8	0.0	711
Women's age (ZOI)						
18-29 years	30.2		0.05	2.5	0.0	258
30 years or older	37.5		0.04	3.0	0.0	453
Workload						
All women (ZOI)	31.9		0.03	2.3	0.0	711
Women's age (ZOI)						
18-29 years	32.4		0.04	1.4	0.0	258
30 years or older	31.6		0.04	2.7	0.0	453
Percent of women of reproductive age consuming a diet of minimum diversity^e						
All women (ZOI)	47.1		0.01	2.3	3.8	3,085
Women's age (ZOI)						
15-19 years	47.9		0.03	2.0	7.3	672
20-49 years	46.9		0.01	2.1	2.8	2,413
All women (RFZ)	41.7		0.02	1.8	4.1	1,739
Prevalence of exclusive breastfeeding among children under 6 months of age^e						
All children (ZOI)	41.9		0.05	1.7	4.1	188
Children's sex (ZOI)						
Male	32.9		0.07	1.9	2.1	94
Female	51.1		0.07	1.7	4.1	94
All children (RFZ)	76.0		0.04	0.9	2.4	120
Percent of children 6-23 months of age consuming a minimum acceptable diet^e						
All children (ZOI)	27.2		0.02	2.1	3.3	669
Children's sex (ZOI)						
Male	32.7		0.03	2.0	1.1	353
Female	20.2		0.03	2.0	5.4	316
All children (RFZ)	16.0		0.02	1.2	2.8	376
Percent of households with access to basic sanitation service						
All households (ZOI)	23.1		0.02	4.0	0.0	2,995
Gendered household type (ZOI)						
Male and female adults	24.9		0.02	3.5	0.0	2,401
Female adults only	15.3		0.03	2.7	0.0	399
Male adults only	16.3		0.05	2.7	0.0	194
Children only	^		^	^		1

Indicator	Estimate	SD	SE	DEFF	Indicator incompletion rate ^a (%)	Unweighted number ^b (n)
Residence (ZOI)						
Urban	25.7		0.04	5.0	0.0	552
Rural	22.5		0.02	3.6	0.0	2,443
All households (RFZ)	15.1		0.01	2.6	0.0	1,653
Percent of households with soap and water at handwashing station on premises						
All households (ZOI)	19.3		0.02	9.9	5.7	2,823
Gendered household type (ZOI)						
Male and female adults	20.6		0.03	9.5	6.0	2,258
Female adults only	15.1		0.03	2.9	5.0	379
Male adults only	11.6		0.03	2.0	4.6	185
Children only	^		^	^		1
Residence (ZOI)						
Urban	24.1		0.07	13.8	6.2	518
Rural	18.1		0.02	8.7	5.6	2,305
All households (RFZ)	12.8		0.01	2.4	6.5	1,545
Prevalence of household water insecurity, based on the Brief Household Water Insecurity Experiences Scale (HWISE-4)						
All households (ZOI)	37.2		0.02	6.5	0.0	2,995
Gendered household type (ZOI)						
Male and female adults	37.2		0.02	5.6	0.0	2,401
Female adults only	36.8		0.04	2.5	0.0	399
Male adults only	37.9		0.06	2.7	0.0	194
Children only	^		^	^		1
Residence (ZOI)						
Urban	38.7		0.04	3.6	0.0	552
Rural	36.8		0.03	7.3	0.0	2,443
All households (RFZ)	47.4		0.02	2.9	0.0	1,653
Shock exposure index (Mean score)						
All households (RFZ)	27.1	21.3		1.4	0.2	1,649
Household resilience capacity index (Mean score)						
All households (RFZ)	39.3	15.3		0.9	0.0	1,650
Absorptive capacity index (Mean score)						
All households (RFZ)	40.4	17.5		1.0	0.0	1,650
Adaptive capacity index (Mean score)						
All households (RFZ)	41.0	14.4		1.0	0.0	1,650
Transformative capacity index (Mean score)						
All households (RFZ)	43.0	19.4		0.9	0.0	1,650
Index of community resilience (Mean score)						
All households (RFZ)	27.2	16.8		0.8	0.0	1,650
Reduced coping strategies index (Mean score)						
All households (RFZ)	16.1	14.1		5.6	1.5	1,629

^a Results not statistically reliable, n<30; — Results not available, n=0

A-WEAI=Abbreviated Women's Empowerment in Agriculture Index, DEFF=design effect, PPP=purchasing power parity, RFZ=Resilience Focus Zone, SD=standard deviation, SE=standard error, ZOI=Zone of Influence

^a Indicator incompleteness rate is defined as: (Number of cases eligible for inclusion in indicator X estimate - Number of cases included in indicator X estimate) / Number of cases eligible for inclusion in indicator X estimate * 100

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, disaggregate sample sizes may not total to the aggregated sample size.

^c Indicator estimates imputed using survey-to-survey imputation at the midline. The model was developed using 2019 Feed the Future Uganda P2-ZOI Baseline Survey data.

^d For the poverty indicators, the n's presented are the number of households—not the number of household members—even though the indicator estimates are calculated using a household member sample weight.

^e Estimates are based on de facto household members.

Notes:

Estimates are sample-weighted; numbers of observations are unweighted.

Individual-level estimates are based on de jure household members, except where noted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 1.2.1: Population of Individuals in the ZOI, by Category, Uganda 2019-2022

Category of individuals	Baseline (2019)		Midline (2022)		Diff.
	N ^a	%	N ^a	%	
Total number of individuals	11,392,400	100.0	12,242,300	100.0	849,900
Total number of individuals, by key sub-population					
Children 0-5 months ^b	136,708	1.2	125,937	1.0	-10,771
Children 6-23 months ^b	501,266	4.4	512,850	4.2	11,584
Children 0-23 months ^b	597,962	5.2	652,597	5.3	54,635
Children 0-59 months ^b	1,645,976	14.4	1,737,439	14.2	91,463
Youth 15-29 years	2,905,062	25.5	3,465,767	28.3	560,705
Women of reproductive age (15-49 years)	2,437,974	21.4	2,830,651	23.1	392,677
Primary adult female decision-makers	2,005,062	17.6	2,029,503	16.6	24,441
Primary adult female decision-makers who participate in household agricultural production ^c	1,661,973	14.6	1,750,362	14.3	88,389
Crop farmers 15+ years	2,221,518	19.5	2,340,329	19.1	118,811
Total number of individuals, by residence^d					
Urban	2,118,986	18.6	2,363,431	19.3	244,445
Rural	9,273,414	81.4	9,878,869	80.7	605,455
Total number of individuals, by region					
East	2,540,505	22.3	2,868,716	23.4	328,211
Karamoja	717,721	6.3	1,091,326	8.9	373,605
North	2,403,796	21.1	1,912,752	15.6	-491,044
Southwest	5,730,377	50.3	6,369,506	52.0	639,129
Total number of individuals, by RFZ					
Non-RFZ	9,594,907	84.2	10,235,052	83.1	640,145
RFZ	1,797,493	15.8	2,007,248	16.9	209,755
Total number of individuals, by gendered household type					
Male and female adults	9,569,616	84.0	10,915,227	89.2	1,345,611
Female adults only	1,458,227	12.8	1,093,567	8.9	-364,660
Male adults only	364,557	3.2	233,265	1.9	-131,292
Children only	N/A	< 1.0	N/A	< 1.0	N/A
Children 0-5 months, by sex^b					
Male	72,046	0.6	60,919	0.5	-11,127
Female	67,102	0.6	65,018	0.5	-2,084
Children 6-23 months, by sex^b					
Male	250,633	2.2	277,925	2.3	27,292
Female	250,633	2.2	234,925	1.9	-15,708
Children 0-23 months, by sex^b					
Male	307,956	2.7	346,518	2.8	38,562
Female	290,006	2.5	306,078	2.5	16,072
Children 0-59 months, by sex^b					
Male	799,161	7.0	901,135	7.4	101,974
Female	846,815	7.4	836,305	6.8	-10,510

Category of individuals	Baseline (2019)		Midline (2022)		Diff.
	N ^a	%	N ^a	%	
Youth 15-29 years, by sex					
Male	1,435,442	12.6	1,676,159	13.7	240,717
Female	1,469,620	12.9	1,789,608	14.6	319,988
Primary adult female decision-makers, by age					
18-29 years	569,620	5.0	557,205	4.6	-12,415
30 years or older	1,435,442	12.6	1,469,868	12.0	34,426
Crop farmers' sex					
Male	900,000	7.9	908,016	7.4	8,016
Female	1,321,518	11.6	1,432,312	11.7	110,794
Crop farmers' age					
15-29 years	558,228	4.9	636,689	5.2	78,461
30 years or older	1,663,290	14.6	1,703,640	13.9	40,350

N/A=not available, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of individuals in the ZOI population

^b At baseline, the weighted percentage is determined using age-in-month information obtained in the children's nutrition module, so the weighted percentage does not account for eligible children who did not complete the module. At midline, required age-in-month information was obtained for all children under 1 year of age in the household roster, so the weighted percentage accounts for all children.

^c The weighted percentage is determined using information on participation in economic activities obtained in the Women's Empowerment in Agriculture module, so the weighted percentage does not account for eligible women who did not complete the module.

^d The urban/rural disaggregate uses the Uganda-specific definition of urban and rural reflected in the sampling frame at the time the sample was drawn.

Sources: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey; 2019 Feed the Future Uganda P2-ZOI Baseline Survey; Uganda Bureau of Statistics 2019 district-level population projection; Uganda Bureau of Statistics 2022 district-level population projection

Table 1.2.2: Number of Households in the ZOI, by Category, Uganda 2019-2022

Category of households	Baseline (2019)		Midline (2022)		Diff.
	N ^a	%	N ^a	%	
Total number of households	2,251,462	100.0	2,200,270	100.0	-51,192
Total number of households, by gendered household type					
Male and female adults	1,610,009	71.5	1,780,318	80.9	170,309
Female adults only	414,831	18.4	295,440	13.4	-119,391
Male adults only	225,576	10.0	124,269	5.6	-101,307
Children only	N/A	< 1.0	N/A	< 1.0	N/A
Total number of households, by residence^b					
Urban	463,532	20.6	454,203	20.6	-9,329
Rural	1,787,931	79.4	1,746,067	79.4	-41,864
Total number of households, by region					
East	498,547	22.1	520,076	23.6	21,529
Karamoja	121,647	5.4	174,245	7.9	52,598
North	451,221	20.0	377,730	17.2	-73,491
Southwest	1,180,048	52.4	1,128,219	51.3	-51,829
Total number of households, by RFZ					
Non-RFZ	1,900,031	84.3	1,841,934	84.3	-58,097
RFZ	351,431	15.7	358,336	15.7	6,905
Total number of households involved in agriculture^c	2,044,027	90.8	2,006,777	91.2	-37,250
Total number of households that cultivated any crops	1,952,938	86.7	1,927,545	87.6	-25,393
Total number of households, by livestock owned					
Milk cows or bulls	293,459	13.0	437,192	19.9	143,733
Other cattle	301,661	13.4	508,730	23.1	207,069
Horses, donkeys, or mules	N/A	< 1.0	NA	< 1.0	N/A
Goats	812,338	36.1	811,515	36.9	-823
Sheep	117,494	5.2	172,214	7.8	54,720
Chickens or other poultry	1,047,534	46.6	1,177,437	53.5	129,903
Fish	N/A	< 1.0	N/A	< 1.0	N/A
Pigs	387,181	17.2	515,121	23.4	127,940
Any other livestock	80,880	3.6	172,288	7.8	91,408

N/A=not available, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households in the ZOI. Number of households was calculated by dividing the estimated total ZOI population by household size.

^b The urban/rural disaggregate uses the Uganda-specific definition of urban and rural reflected in the sampling frame at the time the sample was drawn.

^c The term "Involved in agriculture" refers to ownership of the livestock presented in the table or cultivation of any crops.

Notes:

Percentages may not sum to the 100.0 percent, and numbers of households may not sum to the total due to rounding.

The number of households is not presented for any category comprising less than 1.0 percent of the sample; the percentage is too small to produce a valid ZOI-level estimate.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey; Uganda Bureau of Statistics 2019 district-level population projection; Uganda Bureau of Statistics 2022 district-level population projection

Table 2.2.1: Results of Household and Individual Interviews in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Respondent group	Total		Urban		Rural	
	Baseline (2019)	Midline (2022)	Baseline (2019)	Midline (2022)	Baseline (2019)	Midline (2022)
Households						
Number of households selected	2,675	3,200	675	600	2,000	2,600
Number of households occupied	2,567	3,043	638	564	1,929	2,479
Number of households interviewed	2,541	2,995	627	552	1,914	2,443
Household response rate (%) ^a	99.0	98.4	98.3	97.9	99.2	98.5
Children 0-5 months of age						
Number of eligible children		204		33		171
Number of eligible children whose caregiver was interviewed		196		31		165
Eligible children 0-5 months of age response rate (%) ^{b,c}		96.1		93.9		96.5
Children 6-23 months of age						
Number of eligible children		704		110		594
Number of eligible children whose caregiver was interviewed		692		107		585
Eligible children 6-23 months of age response rate (%) ^{b,c}		98.3		97.3		98.5
Children under 2 years of age						
Number of eligible children	724	908	171	143	553	765
Number of eligible children whose caregiver was interviewed	710	888	169	138	541	750
Eligible children under 2 years of age response rate (%) ^{b,c}	98.1	97.8	98.8	96.5	97.8	98.0
Children under 5 years of age						
Number of eligible children	2,007	2,495	444	375	1,563	2,120
Number of eligible children whose caregiver was interviewed	1,973	2,444	442	369	1,531	2,075
Eligible children under 5 years of age response rate (%) ^{b,c}	98.3	98.0	99.5	98.4	98.0	97.9
Women of reproductive age (15-49 years)						
Number of eligible women	2,805	3,772	680	720	2,125	3,052
Number of eligible women interviewed	2,422	3,208	592	620	1,830	2,588
Eligible women of reproductive age response rate (%) ^{b,c}	86.3	85.0	87.1	86.1	86.1	84.8
Primary adult female decision-makers (18+ years)						
Number of eligible women	2,265	2,752	527	483	1,738	2,269
Number of eligible women interviewed	2,189	2,625	516	459	1,673	2,166
Eligible primary adult female decision-maker response rate (%) ^{b,d}	96.6	95.4	97.9	95.0	96.3	95.5

Respondent group	Total		Urban		Rural	
	Baseline (2019)	Midline (2022)	Baseline (2019)	Midline (2022)	Baseline (2019)	Midline (2022)
Farmers age 15+ who cultivated crops^c						
Number of eligible farmers	2,505	3,279	470	527	2,035	2,752
Number of eligible farmers interviewed	2,436	3,207	451	510	1,985	2,697
Eligible farmer response rate (%) ^{b,d}	97.2	97.8	96.0	96.8	97.5	98.0

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a The household response rate is calculated based on the household result code on the household questionnaire cover page and is defined as the number of households completed divided by the number of households occupied. Households are considered completed if the household result code is completed—that is, both Modules 1 and 2 are completed. Households are considered to be occupied if the household result code is any of the following: completed, no household member at home, postponed/unavailable, refused, dwelling not found, household member too ill to respond/cognitively impaired, or partial complete. Households are considered to be unoccupied, and thus excluded from the household response rate calculations, if the household result code is any of the following: entire household absent for an extended period of time, dwelling vacant, address not a dwelling or dwelling destroyed. At baseline, the household response rates were calculated based on the result codes of Module 1 only.

^b Individual-level response rates are calculated based on the result codes in the relevant individual modules (i.e., Modules 4, 5, 6, and 7). These rates are defined as the number of eligible individuals interviewed divided by the number of eligible individuals. Eligibility determination for Modules 4, 5, and 6 is initiated in the household roster and confirmed in the respective module. If eligibility could not be confirmed (i.e., the individual did not complete the individual-level module), and the individual is eligible according to the information in the household roster, the individual was included in the relevant response rate calculation. (Note that for children under 2 years of age [Module 5], the primary caregivers of the children served as the respondents, not the children directly, but children—not primary caregivers—are included in response rate calculations.) Eligibility determination for Module 7 is determined in Module 2.

^c De facto household members only

^d De jure household members only

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.1.1: Comparison of Household Demographic Characteristics in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Characteristic	Region										Resilience Focus Zone				
	All households			East		Karamoja		North		Southwest		Non-RFZ		RFZ	
	Est.	95% CI	Sig. ^a	Est.	Sig. ^a	Est.	Sig. ^a	Est.	Sig. ^a	Est.	Sig. ^a	Est.	Sig. ^a	Est.	Sig. ^a
Mean household size															
Baseline	5.1	4.8, 5.3		5.1		5.9		5.3		4.9		5.0		5.1	
Midline	5.6	5.3, 5.8	**	5.5	n/s	6.3	n/s	5.0	n/s	5.7	**	5.5	*	5.8	*
Mean number of children under 2 years of age															
Baseline	0.3	0.3, 0.3		0.3		0.4		0.3		0.3		0.3		0.3	
Midline	0.3	0.3, 0.3	n/s	0.4	n/s	0.3	*	0.3	n/s	0.3	n/s	0.3	n/s	0.3	n/s
Mean number of children under 5 years of age															
Baseline	0.8	0.7, 0.8		0.8		1.1		0.8		0.7		0.8		0.8	
Midline	0.8	0.7, 0.8	n/s	0.8	n/s	1.1	n/s	0.8	n/s	0.7	n/s	0.8	n/s	0.9	n/s
Mean number of children 5 years of age or older (5-17 years)															
Baseline	2.0	1.9, 2.1		2.0		2.5		2.1		1.9		2.0		2.1	
Midline	2.1	2.0, 2.3	n/s	2.0	n/s	2.6	n/s	1.9	n/s	2.2	*	2.1	n/s	2.3	n/s
Mean number of youth (15-29 years)															
Baseline	1.3	1.2, 1.4		1.3		1.5		1.5		1.2		1.3		1.3	
Midline	1.6	1.5, 1.7	***	1.7	**	1.5	n/s	1.5	n/s	1.6	***	1.6	***	1.5	n/s
Mean number of women of reproductive age (15-49 years)															
Baseline	1.1	1.0, 1.1		1.0		1.2		1.1		1.1		1.1		1.1	
Midline	1.3	1.2, 1.4	***	1.3	***	1.3	n/s	1.1	n/s	1.3	***	1.3	***	1.3	n/s
Mean number of adult male household members ^b															
Baseline	1.1	1.0, 1.1		1.1		1.0		1.1		1.1		1.1		1.0	
Midline	1.3	1.2, 1.3	***	1.3	**	1.2	n/s	1.1	n/s	1.3	**	1.3	***	1.2	**
Mean number of adult female household members ^b															
Baseline	1.2	1.1, 1.2		1.1		1.3		1.3		1.2		1.2		1.2	
Midline	1.4	1.3, 1.4	***	1.4	***	1.4	n/s	1.2	n/s	1.4	***	1.4	***	1.3	*
Mean number of farmers 15+															
Baseline	1.0	0.9, 1.0		1.1		0.9		1.2		0.9		1.0		0.9	
Midline	1.1	1.0, 1.1	*	0.9	n/s	1.1	n/s	1.5	**	1.0	**	1.1	n/s	1.1	n/s
Household size (%)															
Small (1-5 members)															
Baseline	59.5	56.3, 62.7		58.7		47.6		55.8		62.6		59.8		58.4	
Midline	54.2	50.0, 58.3	*	56.7	n/s	38.4	n/s	62.0	n/s	52.8	*	55.3	n/s	48.2	*
Medium (6-10 members)															
Baseline	36.7	33.9, 39.5		38.3		43.6		39.4		34.2		36.6		37.3	
Midline	40.6	37.1, 44.2	n/s	36.7	n/s	56.3	n/s	34.8	n/s	41.9	*	39.4	n/s	46.6	*

Characteristic	Region										Resilience Focus Zone				
	All households			East		Karamoja		North		Southwest		Non-RFZ		RFZ	
	Est.	95% CI	Sig. ^a	Est.	Sig. ^a	Est.	Sig. ^a	Est.	Sig. ^a	Est.	Sig. ^a	Est.	Sig. ^a	Est.	Sig. ^a
Large (11 or more members)															
Baseline	3.8	2.9, 5.0		3.1		8.9		4.7		3.2		3.7		4.3	
Midline	5.2	4.1, 6.7	n/s	6.6	*	5.3	n/s	3.2	n/s	5.3	n/s	5.2	n/s	5.2	n/s
Household education(%)															
No education															
Baseline	3.0	2.3, 4.1		1.8		23.0		3.0		1.5		2.0		8.7	
Midline	3.3	2.5, 4.3	n/s	2.6	n/s	17.8	n/s	1.4	n/s	1.9	n/s	2.0	n/s	9.8	n/s
Less than primary year 3															
Baseline	4.1	3.1, 5.4		3.9		13.1		2.2		4.0		3.5		7.2	
Midline	3.7	2.8, 4.9	n/s	2.0	n/s	16.3	n/s	2.8	n/s	2.9	n/s	2.4	n/s	10.8	n/s
Completed primary year 3															
Baseline	28.9	26.6, 31.2		37.3		20.4		29.2		26.0		29.0		28.2	
Midline	26.1	22.5, 30.1	n/s	25.3	*	24.4	n/s	34.9	n/s	23.8	n/s	25.9	n/s	27.2	n/s
Completed primary															
Baseline	45.7	42.9, 48.6		44.3		28.0		44.0		48.8		46.6		40.8	
Midline	49.5	46.1, 52.9	n/s	53.7	n/s	32.0	n/s	47.3	n/s	51.1	n/s	51.7	n/s	38.4	n/s
Completed 'A' level or higher															
Baseline	18.3	15.9, 21.0		12.6		15.4		21.6		19.8		18.9		15.1	
Midline	17.4	14.5, 20.7	n/s	16.4	n/s	9.6	n/s	13.6	n/s	20.4	n/s	18.1	n/s	13.8	n/s
Number of households (n)															
Baseline	2,540			761		218		602		959		2,104		436	
Midline	2,995			345		744		416		1,490		1,342		1,653	

CI=confidence interval, Est.=estimate, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether an association exists between the outcome indicators and the disaggregate variables. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.001,

*** p<0.001; n/s=not significant; na=not applicable.

^b Feed the Future defines adult as an individual 18 years of age or older. Females and males 15-17 years of age are of reproductive age but are not considered adults by this definition.

Note: Estimates are based on de jure household members, except where noted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.1.2: Comparison of Characteristics of Primary Adult Female Decision-makers in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Characteristic	Baseline (2019)		Midline (2022)		Sig. ^a
	%	95% CI	%	95% CI	
Age					
18-24	14.2	12.6, 16.0	13.8	11.9, 15.8	n/s
25-29	13.8	12.1, 15.6	13.0	11.5, 14.8	n/s
30-34	12.8	11.5, 14.3	11.7	10.2, 13.3	n/s
35-39	11.9	10.6, 13.4	14.6	12.4, 17.1	n/s
40-44	10.2	8.9, 11.6	10.0	8.7, 11.5	n/s
45-49	9.8	8.3, 11.5	8.8	7.3, 10.4	n/s
50-54	7.0	5.9, 8.2	7.9	6.4, 9.9	n/s
55-59	5.3	4.4, 6.4	6.4	5.2, 7.8	n/s
60+	15.0	12.9, 17.4	13.8	11.9, 16.0	n/s
Marital status					
Married	67.7	64.6, 70.5	69.0	65.9, 72.0	n/s
Living in a consensual union	10.5	8.5, 12.9	8.8	6.6, 11.7	n/s
Widowed	15.5	13.5, 17.7	14.6	12.7, 16.6	n/s
Divorced or separated	1.0	0.6, 1.6	5.5	4.2, 7.1	***
Never married or in a union	5.4	4.3, 6.8	2.1	1.4, 3.2	***
Education level					
No education	23.6	21.1, 26.4	24.3	21.9, 26.9	n/s
Less than primary year 3	10.2	8.2, 12.6	8.6	6.8, 10.7	n/s
Completed primary year 3	35.8	33.4, 38.3	33.8	30.8, 36.9	n/s
Completed primary	24.5	21.8, 27.5	28.4	25.3, 31.9	n/s
Completed 'A' level or higher	5.8	4.4, 7.7	4.9	3.5, 6.8	n/s
Economic activity ^b					
Participates in some form of economic activity	96.1	94.9, 97.0	95.1	93.0, 96.6	n/s
Participation in economic activity by type ^c					
Farm	91.2	88.7, 93.1	91.3	88.1, 93.7	n/s
Non-farm	29.0	26.0, 32.2	28.7	25.7, 31.9	n/s
Wage/salaried	25.6	22.2, 29.3	25.1	21.6, 29.1	n/s
Number of primary female adult decision-makers (n) ^d	2,186		2,624		

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Both paid and unpaid types of economic activity are included. Domestic work, such as caring for children and the elderly or cooking and cleaning, is not included.

^c Farm work includes food crop farming, cash crop farming, livestock raising, or fishing/fishpond culture; non-farm work includes running small businesses or self-employment; and wage/salaried employment includes both agriculture and non-agriculture-based work that is salaried. Percentages do not add up to 100 percent because individuals can engage in more than one type of economic activity.

^d Number of primary adult female decision-makers who are de jure household members

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.2.1: Comparison of Households' Main Dwelling Construction Materials in the Uganda ZOI, in Total and by Residence, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Residence and construction material	Baseline (2019)		Midline (2022)		Sig. ^a
	%	95% CI	%	95% CI	
All households					
Roofing ^b					
Natural	20.3	17.3, 23.6	21.7	17.9, 25.9	n/s
Rudimentary	0.3	0.1, 0.8	0.3	0.1, 1.0	n/s
Finished	79.4	76.1, 82.4	77.6	73.3, 81.3	n/s
Other	0.0	na	0.5	0.3, 1.0	na
Exterior walls ^c					
Natural	6.8	5.4, 8.6	12.9	9.8, 16.7	**
Rudimentary	65.5	61.7, 69.1	54.2	49.6, 58.8	***
Finished	27.6	24.1, 31.5	30.5	26.1, 35.3	n/s
Other	0.1	0.0, 0.5	2.4	1.2, 4.7	**
Flooring ^d					
Natural	70.5	65.7, 75.0	67.6	61.6, 73.1	n/s
Rudimentary	0.1	0.0, 0.4	0.5	0.2, 1.0	*
Finished	29.4	25.0, 34.3	31.0	25.5, 37.0	n/s
Other	0.0	na	0.9	0.5, 1.6	na
Total number of households (n)	2,540		2,995		
Urban households					
Roofing ^b					
Natural	14.1	7.6, 24.8	17.1	7.6, 34.1	n/s
Rudimentary	0.5	0.1, 1.6	0.0	na	na
Finished	85.4	75.3, 91.8	82.7	65.8, 92.2	n/s
Other	0.0	na	0.2	0.0, 1.1	na
Exterior walls ^c					
Natural	3.0	1.5, 5.9	9.7	4.9, 18.5	n/s
Rudimentary	44.3	36.4, 52.4	46.3	35.2, 57.9	n/s
Finished	52.4	43.7, 60.8	40.1	27.9, 53.7	n/s
Other	0.3	0.0, 2.6	3.8	0.7, 18.1	n/s
Flooring ^d					
Natural	44.2	31.9, 57.3	56.0	39.0, 71.7	n/s
Rudimentary	0.0	na	0.7	0.2, 2.7	na
Finished	55.8	42.7, 68.1	42.9	27.1, 60.2	n/s
Other	0.0	na	0.4	0.1, 1.6	na
Total number of urban households (n)	627		552		
Rural households					
Roofing ^b					
Natural	21.8	18.7, 25.3	22.9	19.2, 27.0	n/s
Rudimentary	0.3	0.1, 0.9	0.3	0.1, 1.2	n/s
Finished	77.9	74.4, 81.0	76.3	72.2, 79.9	n/s
Other	0.0	na	0.6	0.3, 1.1	na

Residence and construction material	Baseline (2019)		Midline (2022)		Sig.
	%	95% CI	%	95% CI	
Exterior walls ^c					
Natural	7.8	6.1, 9.9	13.7	10.2, 18.1	**
Rudimentary	71.0	66.6, 75.0	56.3	51.3, 61.1	***
Finished	21.2	17.3, 25.7	28.0	23.4, 33.1	*
Other	0.0	na	2.0	1.1, 3.8	na
Flooring ^d					
Natural	77.3	71.7, 82.1	70.6	64.4, 76.2	n/s
Rudimentary	0.1	0.0, 0.5	0.4	0.2, 1.0	n/s
Finished	22.6	17.8, 28.2	27.9	22.3, 34.2	n/s
Other	0.0	na	1.1	0.6, 1.9	na
Total number of rural households (n)	1,913		2,443		

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; n/s=not significant; na=not applicable.

^b Natural roofs include no roof, thatch (palm leaf, straw, reed), and sod. Rudimentary roofs include bamboo, wood planks, and cardboard. Finished roofs include metal, wood, calamine/cement fiber, ceramic tiles, cement, and roofing shingles.

^c Natural walls include no walls, dirt, and cane/palm/tree trunks. Rudimentary walls include bamboo with mud, stone with mud, plywood, cardboard, reused wood, and unbaked bricks. Finished walls include wood planks/shingles, unbaked bricks covered with plaster, bricks, cement blocks, cement, and stone with lime/cement.

^d Natural floors include earth/sand, dung, and palm leaves. Rudimentary floors include wood planks and bamboo slats. Finished floors include parquet/polished wood, vinyl or asphalt strips, ceramic tiles, cement, and wall-to-wall carpet.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.2.2: Comparison of Household Dwelling Characteristics in the Uganda ZOI, in Total and by Residence, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Indicator and residence	Baseline (2019)			Midline (2022)			Sig. ^b
	Est.	95% CI	n ^a	Est.	95% CI	n ^a	
Percent using solid fuel for cooking (%) ^c							
All households	97.6	96.2, 98.5	2,540	85.8	79.9, 90.2	2,972	***
Residence							
Urban	95.3	88.5, 98.1	627	75.0	53.2, 88.8	542	*
Rural	98.2	96.9, 98.9	1,913	88.6	82.8, 92.5	2,430	***
Percent with access to electricity (%)							
All households	34.6	31.1, 38.3	2,540	34.7	30.5, 39.1	2,995	n/s
Residence							
Urban	49.0	37.4, 60.8	627	45.3	35.9, 55.1	552	n/s
Rural	30.8	27.4, 34.5	1,913	31.9	27.4, 36.8	2,443	n/s
Mean number of persons per sleeping room ^{d,e}							
All households	2.5	2.4, 2.6	2,540	2.7	2.6, 2.8	2,995	*
Residence							
Urban	2.4	2.3, 2.6	627	2.6	2.3, 2.9	552	*
Rural	2.6	2.5, 2.7	1,913	2.8	2.6, 2.9	2,443	*

CI=confidence interval, Est.=estimate, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^c Solid cooking fuels are not considered clean fuels and can have negative health impacts (Smith & Pillarisetti, 2017). Solid fuel is defined as charcoal, wood, animal dung, straw/shrubs/grass, and agriculture crop residue. Households in the "no food cooked in household category" are removed from the estimates.

^d The average number of persons per sleeping room is a common indicator of crowding (United Nations Development Group, 2003).

^e Estimates are based on de jure household members.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.3.1: Comparison of Household Drinking Water Characteristics in the Uganda ZOI, in Total and by Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Indicator and household characteristic	Baseline (2019)			Midline (2022)			Sig. ^b
	%	95% CI	n ^a	%	95% CI	n ^a	
Use a regularly available improved water source ^c							
All households	47.8	43.2, 52.5	2,540	46.3	41.4, 51.2	2,995	n/s
Residence							
Urban	49.1	40.4, 57.9	627	46.9	37.1, 57.0	552	n/s
Rural	47.5	42.2, 52.9	1,913	46.1	40.6, 51.6	2,443	n/s
Region							
East	58.1	50.0, 65.7	761	53.1	40.7, 65.2	345	n/s
Karamoja	46.2	34.1, 58.8	218	51.4	41.4, 61.4	744	n/s
North	52.7	42.7, 62.4	602	54.7	41.1, 67.6	416	n/s
Southwest	41.8	35.0, 49.0	959	39.5	33.9, 45.4	1,490	n/s
Resilience Focus Zone							
Non-RFZ	51.0	45.9, 56.1	2,104	47.4	41.9, 53.0	1,342	n/s
RFZ	30.7	23.6, 38.8	436	40.4	34.4, 46.7	1,653	n/s
Use appropriate water treatment practice or technology ^d							
All households	54.9	50.9, 58.9	2,540	48.9	42.3, 55.5	2,995	n/s
Residence							
Urban	67.0	58.6, 74.4	627	55.9	37.3, 73.1	552	n/s
Rural	51.8	47.4, 56.1	1,913	47.0	40.2, 53.9	2,443	n/s
Region							
East	50.3	44.9, 55.6	761	47.0	38.7, 55.4	345	n/s
Karamoja	17.1	9.2, 29.4	218	17.9	13.0, 24.3	744	n/s
North	10.0	7.5, 13.1	602	11.3	6.7, 18.4	416	n/s
Southwest	78.0	71.3, 83.4	959	67.1	54.6, 77.6	1,490	n/s
Resilience Focus Zone							
Non-RFZ	55.5	51.0, 59.9	2,104	50.5	42.8, 58.3	1,342	n/s
RFZ	51.8	38.1, 65.3	436	40.2	35.7, 44.8	1,653	n/s

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^c Improved water sources include piped water into the dwelling, piped water into the yard/plot, piped water to a neighbor, public tap/standpipe, tube well/borehole, protected dug well, protected spring, rainwater, bottled water, sachet water, tanker-truck, and cart with small tank (WHO & UNICEF, 2018). The indicator presented includes an indication of regularity in access to the water source—namely, that (a) water is available from this source all year round and (b) water from this source was available every day in the 2 weeks preceding the survey.

^d Appropriate water treatment practice or technology refers to methods that effectively kill or remove pathogens. This includes boiling the water, adding bleach or chlorine, using a water filter (ceramic, sand, composite), and solar disinfection (WHO, 2008). Practices such as straining through a cloth and letting it stand and settle are not considered effective approaches to water treatment. Other is also not considered an effective approach.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.3.2: Number of Trips Households Made to Collect Drinking Water from Their Main Drinking Water Source During the Week Preceding the Survey in the Uganda ZOI, in Total and by Household Characteristics, Feed the Future Phase Two ZOI and RFZ Midline Survey

Number of trips	All households (%)	Urban households (%)	Rural households (%)
No trips ^a	13.8	21.5	11.8
1-4 trips	7.3	5.4	7.8
5-9 trips	9.7	8.8	9.9
10 or more trips	69.3	64.4	70.6
Number of households	2,995	552	2,443

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a No trips includes households that use a main drinking water source located in their dwelling, yard, or plot or that use bottled water as their main drinking water source.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.3.3: Comparison of the Availability of Handwashing Stations with Soap and Water on Households' Premises and Basic Sanitation Services in the Uganda ZOI, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Indicator and household characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^b	Sig. ^c
	%	95% CI	n ^a	%	95% CI	n ^a			
Have soap and water at handwashing station on premises ^{d,e}									
All households	31.7	28.3, 35.2	2,287	19.3	15.0, 24.4	2,823	-12.4	<0.001	***
Gendered household type									
Male and female adults	31.3	27.5, 35.3	1,654	20.6	15.8, 26.3	2,258	-10.7	0.001	**
Female adults only	29.9	23.8, 36.9	398	15.1	9.8, 22.5	379	-14.8	0.002	**
Male adults only	36.8	29.3, 45.1	234	11.6	6.3, 20.6	185	-25.2	<0.001	***
Children only	^	^		^	^		na	na	na
Residence									
Urban	45.9	40.3, 51.7	545	24.1	12.4, 41.5	518	-21.8	0.008	**
Rural	28.2	24.5, 32.2	1,742	18.1	13.9, 23.2	2,305	-10.1	0.001	**
Resilience Focus Zone									
Non-RFZ	31.8	28.3, 35.5	1,921	20.5	15.6, 26.6	1,278	-11.3	<0.001	***
RFZ	30.8	19.6, 44.8	366	12.8	10.3, 15.7	1,545	-18.0	0.008	**
Region									
East	27.8	21.2, 35.5	749	19.5	13.5, 27.3	343	-8.3	0.10	n/s
Karamoja	4.5	1.4, 13.6	190	8.1	4.9, 13.3	723	3.6	0.27	n/s
North	13.4	8.6, 20.4	490	2.9	1.1, 7.7	411	-10.5	0.002	**
Southwest	42.5	37.7, 47.3	858	27.0	19.6, 36.0	1,346	-15.5	0.002	**
Have a basic sanitation service (improved sanitation, not shared) ^f									
All households	23.2	20.3, 26.4	2,540	23.1	20.2, 26.3	2,995	-0.1	0.93	n/s
Gendered household type									
Male and female adults	25.1	21.8, 28.6	1,836	24.9	21.8, 28.3	2,401	-0.2	0.88	n/s
Female adults only	18.5	14.0, 24.1	450	15.3	10.4, 22.1	399	-3.2	0.22	n/s
Male adults only	18.4	13.1, 25.2	253	16.3	9.0, 27.8	194	-2.1	0.51	n/s
Children only	^	^		^	^		na	na	na
Residence									
Urban	27.4	21.0, 34.8	627	25.7	18.3, 34.8	552	-1.7	0.25	n/s
Rural	22.1	19.0, 25.6	1,913	22.5	19.4, 25.8	2,443	0.4	0.75	n/s
Resilience Focus Zone									
Non-RFZ	24.9	21.7, 28.5	2,033	24.7	21.3, 28.4	1,342	-0.2	0.89	n/s
RFZ	13.8	8.9, 20.8	507	15.1	12.5, 18.2	1,653	1.3	0.47	n/s
Region									
East	10.2	7.4, 14.0	761	30.6	25.2, 36.7	345	20.4	<0.001	***
Karamoja	3.8	0.6, 20.6	218	6.7	4.4, 10.2	744	2.9	0.43	n/s
North	14.2	11.1, 18.1	602	24.6	18.6, 31.8	416	10.4	0.01	**
Southwest	34.1	29.2, 39.4	959	21.7	17.2, 27.0	1,490	-12.4	<0.001	***

^ Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^c Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^d A handwashing station is a location where people wash their hands. These can be fixed locations or movable devices that may be placed in a convenient spot for use. The soap may be in bar, powder, or liquid form. The soap must be at the handwashing station or reachable by hand when standing in front of it. The proportion of the population with a basic handwashing facility with soap and water available on premises is Sustainable Development Goal indicator 6.2.1b (United Nations Statistics Division, n.d.).

^e The number of households for which a handwashing station was observed includes those households that do not have a handwashing station on premises but excludes households that do not provide permission to see the handwashing facility or where it is not seen for some other reason.

^f Improved sanitation facilities are those that separate human excreta from human contact; they include the categories flush to piped sewer system, flush to septic tank, flush to pit latrine, flush to "don't know where," composting toilet, ventilated improved pit latrine, and pit latrine with a slab (WHO & UNICEF, 2018).

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.3.4: Comparison of Household Sanitation Characteristics in the Uganda ZOI, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Indicator and household characteristic	Baseline (2019)			Midline (2022)			Sig. ^b
	Est.	95% CI	n ^a	Est.	95% CI	n ^a	
Use improved sanitation, shared ^c (%)							
All households	14.7	11.8, 18.2	2,540	15.3	12.3, 18.8	2,995	n/s
Residence							
Urban	29.9	18.9, 43.8	627	24.8	16.6, 35.2	552	n/s
Rural	10.8	8.2, 14.0	1,913	12.8	9.8, 16.6	2,443	n/s
Region							
East	10.5	7.3, 14.9	761	27.9	18.7, 39.5	345	**
Karamoja	16.0	7.5, 31.2	218	13.3	8.9, 19.4	744	n/s
North	16.9	10.2, 26.5	602	20.9	14.5, 29.1	416	n/s
Southwest	15.6	11.3, 21.0	959	7.9	5.3, 11.5	1,490	**
Resilience Focus Zone							
Non-RFZ	13.9	11.1, 17.3	2,104	15.7	12.3, 19.8	1,342	n/s
RFZ	19.1	10.8, 31.6	436	13.2	10.2, 17.0	1,653	n/s
Use unimproved sanitation ^d (%)							
All households	62.1	58.5, 65.6	2,540	61.7	57.5, 65.9	2,995	n/s
Residence							
Urban	42.7	33.3, 52.7	627	50.2	39.5, 61.0	552	n/s
Rural	67.1	63.1, 70.9	1,913	64.7	60.1, 69.1	2,443	n/s
Region							
East	79.3	72.9, 84.4	761	41.5	31.9, 51.8	345	***
Karamoja	80.2	61.6, 91.1	218	80.0	74.0, 84.9	744	n/s
North	68.9	60.0, 76.6	602	54.6	45.2, 63.7	416	*
Southwest	50.3	45.2, 55.5	959	70.7	64.5, 76.2	1,490	***
Resilience Focus Zone							
Non-RFZ	61.2	57.4, 64.8	2,104	59.8	54.9, 64.6	1,342	n/s
RFZ	67.1	54.9, 77.4	436	71.7	66.7, 76.2	1,653	n/s
Practice open defecation ^e (%)							
All households	7.6	6.1, 9.5	2,540	6.9	5.6, 8.6	2,995	n/s
Residence							
Urban	3.8	1.7, 8.4	627	3.9	1.8, 8.3	552	n/s
Rural	8.6	6.9, 10.8	1,913	7.7	6.2, 9.6	2,443	n/s
Region							
East	8.1	5.1, 12.6	761	5.5	2.9, 10.1	345	n/s
Karamoja	68.6	47.4, 84.2	218	54.4	43.7, 64.7	744	n/s
North	7.7	5.6, 10.5	602	6.6	3.6, 11.8	416	n/s
Southwest	1.1	0.5, 2.4	959	0.4	0.1, 1.1	1,490	n/s

Indicator and household characteristic	Baseline (2019)			Midline (2022)			Sig. ^b
	Est.	95% CI	n ^a	Est.	95% CI	n ^a	
Resilience Focus Zone							
Non-RFZ	4.4	3.3, 5.8	2,104	2.9	1.9, 4.5	1,342	n/s
RFZ	25.3	14.4, 40.5	436	27.7	22.6, 33.4	1,653	n/s
Mean number of household members who regularly practice open defecation at home or at work							
All households				0.8	0.68, 0.98	2,928	
Residence							
Urban				0.5	0.31, 0.68	533	
Rural				0.9	0.73, 1.10	2,395	
Region							
East				1.1	0.56, 1.65	334	
Karamoja				3.5	2.84, 4.08	695	
North				0.7	0.48, 0.87	416	
Southwest				0.4	0.26, 0.49	1,483	
Resilience Focus Zone							
Non-RFZ				0.6	0.44, 0.78	1,327	
RFZ				2.0	1.70, 2.29	1,601	

CI=confidence interval, Est.=estimate, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^c Improved sanitation facilities are those that separate human excreta from human contact; they include the categories flush to piped sewer system, flush to septic tank, flush to pit latrine, flush to "don't know where," composting toilet, ventilated improved pit latrine, and pit latrine with a slab (WHO & UNICEF, 2018).

^d Unimproved sanitation facilities are those that do not adequately separate human excreta from human contact. Unimproved sanitation facilities include flush or pour flush toilets that empty into "somewhere else," open pits or pit latrines without a slab, bucket toilets, and hanging toilets. Households that report having no sanitation facility or report using the bush or field are considered to be using an unimproved sanitation facility. "Other" responses are also considered to be unimproved. (WHO & UNICEF, 2018).

^e Households that report having no sanitation facility or using the bush or field are considered as practicing open defecation.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.4.1: Prevalence of Household Water Insecurity in the Uganda ZOI, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Midline Survey

Household characteristic	%	Sig. ^a	Number of households (n) ^b
All households	37.2		2,995
Gendered household type		n/s	
Male and female adults	37.2		2,401
Female adults only	36.8		399
Male adults only	37.9		194
Children only	^		1
Wealth quintile		***	
Highest (wealthiest)	30.4		475
Fourth	30.6		442
Middle	36.2		486
Second	34.6		557
Lowest (poorest)	51.3		1,035
Shock exposure severity		*	
Did not experience any shocks	^		19
Low	31.1		1,019
Moderate	38.3		981
High	41.5		972
Number of trips to collect drinking water in past week		n/s	
No trips ^c	36.4		497
1-4 trips	37.7		171
5-9 trips	36.3		317
10 or more trips	37.4		2,010
Residence		n/s	
Urban	38.7		552
Rural	36.8		2,443
Region		***	
East	47.8		345
Karamoja	59.3		744
North	34.2		416
Southwest	29.8		1,490
Resilience Focus Zone		***	
Non-RFZ	35.2		1,342
RFZ	47.4		1,653

Household characteristic	%	Sig. ^a	Number of households (n) ^b
Household education		n/s	
No education	38.8		196
Less than primary year 3	48.5		204
Completed primary year 3	40.3		798
Completed primary	34.7		1,322
Completed 'A' level or higher	36.7		475

^a Results not statistically reliable, n<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether an association exists between the outcome indicator and the disaggregate variables. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c No trips includes households that use water piped to their dwelling, plot, or yard or that use bottled water as their main drinking water source.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.4.2: Percent of Households in the Uganda ZOI that Worried about Water Availability for Household Needs in the 4 Weeks Preceding the Survey, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Midline Survey

Household characteristic	Average score (range 0-3)	Sig. ^a	Never	Rarely (1-2 times)	Some- times (3-10 times)	Often/ always (>10 times)	Number of households (n) ^b
All households	0.7		50.7	29.2	17.7	2.4	2,995
Gendered household type		n/s					
Male and female adults	0.7		50.9	29.3	17.3	2.5	2,401
Female adults only	0.7		49.7	30.4	18.2	1.7	399
Male adults only	0.8		49.2	24.4	23.2	3.1	194
Children only	^		^	^	^	^	1
Wealth quintile		***					
Highest (wealthiest)	0.6		55.7	25.4	17.3	1.6	475
Fourth	0.6		61.0	21.4	15.3	2.2	442
Middle	0.7		53.6	26.5	17.4	2.6	486
Second	0.7		51.0	26.7	20.4	1.8	557
Lowest (poorest)	0.9		35.3	43.3	17.7	3.6	1,035
Shock exposure severity		***					
Did not experience any shocks	0.7		^	^	^	^	19
Low	0.7		54.8	31.7	11.9	1.6	1,019
Moderate	0.7		51.7	28.1	17.9	2.3	981
High	0.8		46.1	27.6	23.2	3.1	972
Number of trips to collect drinking water in past week		n/s					
No trips ^c	0.7		53.1	30.1	14.0	2.8	497
1-4 trips	0.7		48.7	30.5	18.4	2.4	171
5-9 trips	0.7		52.4	28.9	16.2	2.6	317
10 or more trips	0.7		50.2	28.9	18.6	2.3	2,010
Residence		n/s					
Urban	0.8		50.5	23.5	22.4	3.5	552
Rural	0.7		50.7	30.7	16.5	2.1	2,443
Region		***					
East	0.8		40.9	35.8	20.9	2.5	345
Karamoja	0.9		24.1	63.0	11.0	1.9	744
North	0.7		51.1	30.8	15.0	3.1	416
Southwest	0.6		59.2	20.4	18.2	2.2	1,490
Resilience Focus Zone		***					
Non-RFZ	0.7		53.0	26.7	18.1	2.3	1,342
RFZ	0.8		39.0	42.1	15.8	3.1	1,653
Household education		n/s					
No education	0.7		42.2	47.3	9.6	0.8	196
Less than primary year 3	0.9		42.6	34.1	17.8	5.4	204
Completed primary year 3	0.8		47.1	30.4	19.6	3.0	798
Completed primary	0.7		52.7	28.4	16.8	2.1	1,322
Completed 'A' level or higher	0.7		53.8	25.1	18.9	2.1	475

^ Results not statistically reliable, n<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether an association exists between the outcome indicator and the disaggregate variables. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c No trips includes households that use a main drinking water source located in their dwelling, yard, or plot or that use bottled water as their main drinking water source.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.4.3: Percent of Households in the Uganda ZOI that Had to Change Schedules or Plans Due to Their Water Situation in the Past 4 Weeks, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Midline Survey

Household characteristic	Average score (range 0-3)	Sig. ^a	Never	Rarely (1-2 times)	Some- times (3-10 times)	Often/ always (>10 times)	Number of households (n) ^b
All households	0.7		54.5	26.0	16.8	2.7	2,995
Gendered household type		n/s					
Male and female adults	0.7		54.0	27.0	16.5	2.5	2,401
Female adults only	0.7		56.2	22.6	17.8	3.4	399
Male adults only	0.7		57.7	19.1	19.2	4.0	194
Children only	^		^	^	^	^	1
Wealth quintile		**					
Highest (wealthiest)	0.6		59.0	24.2	15.5	1.2	475
Fourth	0.6		63.1	19.8	14.2	2.9	442
Middle	0.6		56.8	24.3	16.0	2.9	486
Second	0.7		54.3	26.0	17.0	2.6	557
Lowest (poorest)	0.9		41.9	33.8	20.5	3.8	1,035
Shock exposure severity		***					
Did not experience any shocks	^		^	^	^	^	19
Low	0.5		59.4	28.8	10.4	1.4	1,019
Moderate	0.7		54.8	24.7	16.9	3.5	981
High	0.8		49.6	24.5	22.7	3.1	972
Number of trips to collect drinking water in past week		n/s					
No trips ^c	0.6		58.2	24.8	15.9	1.1	497
1-4 trips	0.7		50.3	28.5	19.2	2.0	171
5-9 trips	0.6		55.9	26.7	14.3	3.1	317
10 or more trips	0.7		54.0	25.8	17.1	3.1	2,010
Residence		n/s					
Urban	0.7		53.8	22.0	20.8	3.4	552
Rural	0.7		54.6	27.0	15.8	2.6	2,443
Region		***					
East	0.9		40.5	33.1	23.7	2.7	345
Karamoja	0.8		32.9	53.9	12.7	0.5	744
North	0.7		55.9	24.9	14.2	5.0	416
Southwest	0.6		63.8	18.7	15.2	2.4	1,490
Resilience Focus Zone		n/s					
Non-RFZ	0.7		56.0	23.7	17.3	3.1	1,342
RFZ	0.7		46.6	37.8	14.5	1.1	1,653
Household education		n/s					
No education	0.7		48.8	35.9	12.4	2.8	196
Less than primary year 3	0.8		44.5	33.1	15.4	7.0	204
Completed primary year 3	0.8		51.1	25.9	19.2	3.9	798
Completed primary	0.6		57.4	25.6	14.8	2.2	1,322
Completed 'A' level or higher	0.7		54.3	23.8	20.3	1.6	475

^a Results not statistically reliable, n<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether an association exists between the outcome indicator and the disaggregate variables. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c No trips includes households that use a main drinking water source located in their dwelling, yard, or plot or that use bottled water as their main drinking water source.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.4.4: Percent of Households in the Uganda ZOI that Had to Go Without Handwashing Due to Water Problems in the Past 4 Weeks, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Midline Survey

Household characteristic	Average score (range 0-3)	Sig. ^a	Never	Rarely (1-2 times)	Some-times (3-10 times)	Often/always (>10 times)	Number of households (n) ^b
All households	0.8		52.8	24.4	16.5	6.3	2,995
Gendered household type		n/s					
Male and female adults	0.8		52.6	25.3	15.9	6.1	2,401
Female adults only	0.8		51.1	21.0	20.1	7.7	399
Male adults only	0.7		59.7	18.4	16.3	5.7	194
Children only	^		^	^	^	^	1
Wealth quintile		***					
Highest (wealthiest)	0.5		66.7	18.8	11.8	2.7	475
Fourth	0.7		57.5	22.1	13.3	7.1	442
Middle	0.7		55.5	21.6	17.2	5.6	486
Second	0.8		54.8	20.0	18.8	6.5	557
Lowest (poorest)	1.0		33.7	37.4	19.9	9.0	1,035
Shock exposure severity		n/s					
Did not experience any shocks	^		^	^	^	^	19
Low	0.8		51.2	27.6	14.4	6.8	1,019
Moderate	0.8		53.4	22.3	17.6	6.7	981
High	0.7		53.9	23.5	17.2	5.4	972
Number of trips to collect drinking water in past week		n/s					
No trips ^c	0.7		57.1	23.8	15.8	3.4	497
1-4 trips	0.6		57.6	27.1	12.0	3.3	171
5-9 trips	0.7		55.5	21.8	16.9	5.8	317
10 or more trips	0.8		51.1	24.6	17.0	7.3	2,010
Residence		n/s					
Urban	0.7		56.8	19.4	19.1	4.6	552
Rural	0.8		51.7	25.7	15.8	6.8	2,443
Region		***					
East	0.8		47.3	32.1	15.2	5.4	345
Karamoja	0.9		23.1	64.0	11.3	1.6	744
North	1.0		45.4	18.9	22.8	12.9	416
Southwest	0.6		62.4	16.5	15.8	5.3	1,490
Resilience Focus Zone		***					
Non-RFZ	0.7		56.2	21.2	16.2	6.4	1,342
RFZ	1.0		35.1	40.8	18.1	6.0	1,653
Household education		n/s					
No education	0.9		39.2	40.2	16.7	3.9	196
Less than primary year 3	0.9		39.8	34.1	18.8	7.2	204
Completed primary year 3	0.8		47.8	26.6	18.5	7.2	798
Completed primary	0.7		55.9	22.5	15.7	5.9	1,322
Completed 'A' level or higher	0.7		56.9	21.5	15.2	6.4	475

^a Results not statistically reliable, n<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether an association exists between the outcome indicator and the disaggregate variables. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c No trips includes households that use a main drinking water source located in their dwelling, yard, or plot or that use bottled water as their main drinking water source.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.4.5: Percent of Households in the Uganda ZOI that Experienced an Inadequate Amount of Drinking Water in the Household in the Past 4 Weeks, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Midline Survey

Household characteristic	Average score (range 0-3)	Sig. ^a	Never	Rarely (1-2 times)	Some- times (3-10 times)	Often/ always (>10 times)	Number of households (n) ^b
All households	0.7		54.8	25.8	16.5	2.9	2,995
Gendered household type		n/s					
Male and female adults	0.7		55.0	26.4	15.6	3.0	2,401
Female adults only	0.7		52.8	24.7	20.2	2.3	399
Male adults only	0.7		57.9	18.4	20.7	2.9	194
Children only	^		^	^	^	^	1
Wealth quintile		***					
Highest (wealthiest)	0.5		61.9	23.3	13.4	1.5	475
Fourth	0.6		60.9	20.6	16.3	2.2	442
Middle	0.7		56.8	21.2	19.6	2.4	486
Second	0.7		54.3	24.4	16.8	4.5	557
Lowest (poorest)	0.8		43.1	37.3	15.9	3.7	1,035
Shock exposure severity		*					
Did not experience any shocks	^		^	^	^	^	19
Low	0.6		59.3	26.5	11.3	2.8	1,019
Moderate	0.7		53.4	26.7	17.5	2.4	981
High	0.7		52.3	23.9	20.4	3.4	972
Number of trips to collect drinking water in past week		n/s					
No trips ^c	0.6		57.8	28.4	10.3	3.6	497
1-4 trips	0.6		55.3	28.3	13.8	2.6	171
5-9 trips	0.7		55.6	24.2	15.5	4.7	317
10 or more trips	0.7		54.1	25.2	18.1	2.6	2,010
Residence		n/s					
Urban	0.7		56.3	21.5	17.2	5.1	552
Rural	0.7		54.5	26.9	16.3	2.4	2,443
Region		***					
East	0.8		48.4	29.4	19.9	2.3	345
Karamoja	0.9		24.7	64.7	9.2	1.3	744
North	0.7		54.8	25.2	13.2	6.8	416
Southwest	0.6		62.4	18.3	17.1	2.2	1,490
Resilience Focus Zone		***					
Non-RFZ	0.7		57.3	22.5	17.0	3.2	1,342
RFZ	0.8		42.0	42.2	13.9	1.8	1,653
Household education		n/s					
No education	0.7		48.6	38.7	10.6	2.1	196
Less than primary year 3	0.8		46.8	29.3	18.6	5.4	204
Completed primary year 3	0.7		53.5	23.4	20.4	2.7	798
Completed primary	0.7		55.8	26.2	15.1	2.8	1,322
Completed 'A' level or higher	0.6		57.0	24.6	15.0	3.3	475

^a Results not statistically reliable, n<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether an association exists between the outcome indicator and the disaggregate variables. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c No trips includes households that use a main drinking water source located in their dwelling, yard, or plot or that use bottled water as their main drinking water source.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.5.1: Coverage of the Types of WASH Interventions Promoted by Feed the Future in the Uganda ZOI During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Type of WASH intervention	All households (%)	Resilience Focus Zone	
		Non-RFZ (%)	RFZ (%)
Participated in training or counseling on hygiene practices	30.8	30.3	32.9
Received training on how to make soap for home use	4.8	4.9	4.2
Received support to make home water or sanitation improvements	26.1	24.5	34.2
Received support to have piped water source installed on the household's premises	0.7	0.7	0.5
Received support to improve piped water source already installed on the household's premises	0.6	0.5	0.8
Received support to have a pit latrine built or installed for home use	18.5	16.6	28.1
Received support to have sanitation facility other than a pit latrine built or installed for home use	12.4	12.8	10.1
Received support to improve an existing sanitation facility for home use	17.8	17.8	18.2
Heard of any program or group working to build or improve water systems for communal use in your area	18.1	17.7	20.0
Accessed an improved community water point	23.7	24.1	21.6
Received information from an extension worker about water services they provide in the area	5.1	4.9	5.9
Saw advertisements or received marketing materials on sanitation services	11.5	11.0	14.1
Used a water or sanitation service provider	19.1	16.6	31.9
Number of households (n)	2,995	1,342	1,653
Noticed an increase in the amount of money the service provider collects	30.1	29.5	31.8
Noticed the service provider has improved the services they provide	55.6	54.1	59.7
Number of households that use a water or sanitation service provider (n)	741	221	520
Has found it easier to remain clean and comfortable during their menstrual period	51.8	52.7	46.6
Number of women of reproductive age (15-49 years) (n)^a	3,208	1,394	1,814

WASH=water, sanitation, and hygiene; RFZ=Resilience Focus Zone; ZOI=Zone of Influence

^a Number of women of reproductive age (15-49 years) who are de jure household members

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 3.5.2: Percent Distribution of Program Participants in the Uganda ZOI who Participated in Specific Types of WASH Interventions of the Types Promoted by Feed the Future During the 12 Months Preceding the Survey, by Sex and Age, Feed the Future Phase Two ZOI and RFZ Midline Survey

Type of WASH intervention	Participant sex		Participant age		Number of participants (n) ^a
	Male (%)	Female (%)	<30 years (%)	30+ years (%)	
Participated in training or counseling on hygiene practices	32.9	67.1	30.1	69.9	1,213
Received support to access soap for their home	23.0	77.0	46.7	53.3	146
Received support to make home sanitation improvements	39.0	61.0	26.2	73.8	759
Saw advertisements or received marketing materials on sanitation services	51.2	48.8	29.3	70.7	457
Accessed an improved community water point	40.2	59.8	46.1	53.9	1,239
Has found it easier to remain clean and comfortable during their menstrual period ^b			67.7	32.3	1,586

WASH=water, sanitation, and hygiene; RFZ=Resilience Focus Zone; ZOI=Zone of Influence

^a Number of de jure household members who participated in specific water, sanitation, or hygiene interventions

^b Asked of only women of reproductive age (15-49 years of age)

Notes:

Estimates are sample-weighted; numbers of observations are unweighted.

Cells are greyed out when information on participants in the household was not collected.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 4.1.1: Comparison of Poverty Indicators at the \$1.90 per Person per Day (2011 PPP) Threshold in the Uganda ZOI and RFZ, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Geographic area and indicator	Baseline (2019)			Midline (2022)			Diff.	Sig.
	Estimate	95% CI	n ^a	Estimate	95% CI	n ^a		
ZOI								
Prevalence of poverty (%) ^b	41.1	37.1, 45.3	2,530	37.1	34.5, 39.7	2,995	-4.0	uc
Depth of poverty of the poor (mean %) ^c	33.8	32.2, 35.4	1,105	35.6		1,108	1.8	na
Prevalence of near-poor (%) ^d	13.7	12.1, 15.3	2,530	11.8		2,995	-1.9	na
RFZ								
Prevalence of poverty (%) ^b	37.9	25.8, 51.7	435	40.0	37.1, 42.9	1,653	2.1	uc
Depth of poverty of the poor (mean %) ^c	41.3	36.1, 46.5	213	41.6		610	0.3	na
Prevalence of near-poor (%) ^d	8.8	6.3, 12.2	435	12.5		1,653	3.7	na

CI=confidence interval, na=not applicable, PPP=purchasing power parity, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households—not the number of household members—even though the indicators measure the percentage of individuals rather than the percentage of households

^b Statistical tests of difference could not be conducted. An asterisk (*) indicates that the baseline and midline estimate 95% CIs are not overlapping, while "uc" indicates that the baseline and midline CIs overlap. The prevalence of poverty, sometimes referred to as the poverty incidence or poverty headcount ratio, presented in this table is the percentage of individuals living below the \$1.90 per person per day (2011 PPP) threshold. The prevalence of poverty was calculated by first converting the \$1.90 per person per day (2011 PPP) poverty line into local currency units (LCU) by multiplying the \$1.90 per person per day (2011 PPP) poverty line by the 2011 PPP conversion rate of the ZOI Survey country, where LCU 2011 PPP=2,927.65 UGX. The resulting figure was then adjusted for the cumulative price inflation between 2011 and the ZOI Survey year by multiplying by the ratio of the ZOI Survey country's Consumer Price Index (CPI) for the survey year and month to the 2011 CPI, where survey month and year CPI LCU=138.45 and 2011 CPI LCU=85.10. Finally, after applying the household member sampling weight, the sum of the number of households in which the per capita daily consumption in LCU for the year and month of the ZOI Survey was less than the poverty line in LCU for the year and month of the ZOI Survey was divided by the sum of the number of households with consumption data.

^c The depth of poverty of the poor measures, on average, how far the consumption of the poor is below the \$1.90 per person per day (2011 PPP) poverty threshold. The depth of poverty of the poor was calculated by first subtracting the per capita daily consumption expenditure in LCU for the year and month of the ZOI Survey of each poor household in the sample from the \$1.90 per day poverty line in LCU for the year and month of the ZOI Survey. The figure was then divided by the \$1.90 per person per day (2011 PPP) poverty line in LCU for the year and month of the ZOI Survey. Finally, after applying the household member sampling weight, the value for each poor household was summed across all poor households and then divided by the number of sampled poor households with consumption data. Due to the survey-to-survey imputation method used to calculate a consumption aggregate for each household at midline, reliable standard error estimates are not available for the depth of poverty of the poor; therefore, statistical tests of difference were not performed and only the difference between baseline and midline is presented.

^d The percent of people who are near-poor presented in this table is the percentage of individuals living at or above the \$1.90 per person per day (2011 PPP) poverty threshold but below 125 percent of that threshold, or \$2.38 per day. The percent of people who are near-poor was calculated by first converting the \$1.90 per person per day (2011 PPP) poverty line into LCU by multiplying the \$1.90 per person per day (2011 PPP) poverty line by the 2011 PPP conversion rate of the ZOI Survey country, where LCU 2011 PPP=539.70. The resulting figure was then adjusted for the cumulative price inflation between 2011 and the ZOI Survey year by multiplying by the ratio of the ZOI Survey country's CPI for the survey year and month to the 2011 CPI, where survey month and year CPI LCU=118.90 and 2011 CPI LCU=92.87. Finally, after applying the household member sampling weight, the sum of the number of households in which the per capita daily consumption in LCU for the year and month of the ZOI Survey was equal to or greater than the poverty line in LCU for the year and month of the ZOI Survey but below 125 percent of the poverty line in LCU for the year and month of the ZOI Survey was divided by the sum of the number of households with consumption data. Due to the survey-to-survey imputation method used to calculate a consumption aggregate for each household at midline, reliable standard error estimates are not available for the prevalence of near-poor; therefore, statistical tests of difference were not performed and only the difference between baseline and midline is presented.

Notes:

Estimates are sample-weighted; numbers of observations are unweighted.

Indicator estimates are calculated using de jure household members.

At baseline, per capita daily consumption was calculated by first determining the monetary value of daily household consumption expenditures per usual household member for specific categories of items (food consumption over the past 7 days; non-food expenditures over the past 7 days, 1 month, 3 months, and 12 months; housing; and durable goods) and then aggregating the values across all categories, as detailed in the Guide to Feed the Future Statistics. Per capita consumption values were then used to estimate the baseline poverty indicator values.

At midline, per capita daily consumption was calculated using a survey-to-survey imputation method developed by the World Bank. The approach required first creating a model to predict per capita consumption using data from the phase two ZOI Baseline Survey, and then using the baseline model's coefficients and phase two ZOI and RFZ Midline Survey data on the model's predictor variables to impute per capita consumption for each household at midline. Imputed per capita consumption values were then used to estimate the midline poverty indicator values.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 4.1.2: Poverty Indicators at the \$2.15 per Person per Day (2017 PPP) and \$1.25 per Person per Day (2005 PPP) Thresholds in the Uganda ZOI and RFZ, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Geographic area, poverty threshold, and indicator	Baseline (2019)			Midline (2022)		
	Estimate	95% CI	n ^a	Estimate	95% CI	n ^a
ZOI						
\$2.15/day (2017 PPP) poverty threshold						
Prevalence of poverty (%) ^b	48.0	43.6, 52.4	2,531	42.3	39.6, 45.0	2,995
Depth of poverty of the poor (mean %) ^c	34.3	32.4, 36.1	1,139	37.3		1,267
Prevalence of near-poor (%) ^d	14.2	12.7, 15.9	2,531	12.0		2,995
\$1.25 poverty threshold (2005 PPP)						
Prevalence of poverty (%) ^e				35.7	33.1, 38.3	2,995
Depth of poverty of the poor (mean %) ^f				35.2		1,067
Prevalence of near-poor (%) ^g				11.8		2,995
RFZ						
\$2.15/day (2017 PPP) poverty threshold						
Prevalence of poverty (%) ^b	46.5	33.2, 60.2	435	44.4	41.4, 47.4	1,653
Depth of poverty of the poor (mean %) ^c	42.4	34.8, 50.0	219	42.9		698
Prevalence of near-poor (%) ^d	11.8	9.0, 15.2	435	9.6		1,653
\$1.25 poverty threshold (2005 PPP)						
Prevalence of poverty (%) ^e				38.8	35.9, 41.7	1,653
Depth of poverty of the poor (mean %) ^f				41.2		587
Prevalence of near-poor (%) ^g				9.6		1,653

CI=confidence interval, PPP=purchasing power parity, RFZ=Resilience Focus Zone, ZOI=Zone of Influence, na=not applicable

^a Number of households—not the number of household members—even though the indicators measure the percentage of individuals rather than the percentage of households

^b The prevalence of poverty was calculated by first converting the \$2.15 per person per day (2017 PPP) poverty line into local currency units (LCU) by multiplying the \$2.15 per person per day (2017 PPP) poverty line by the 2017 PPP conversion rate of the ZOI Survey country, where LCU 2017 PPP=3,243.07 UGX. The resulting figure was then adjusted for the cumulative price inflation between 2017 and the ZOI Survey year by multiplying by the ratio of the ZOI Survey country's Consumer Price Index (CPI) for the survey year and month to the 2017 CPI, where survey month and year CPI LCU=138.48 and 2017 CPI LCU=112.10. Finally, after applying the household member sampling weight, the sum of the number of households in which the per capita daily consumption in LCU for the year and month of the ZOI Survey was less than the poverty line in LCU for the year and month of the ZOI Survey was divided by the sum of the number of households with consumption data.

^c The depth of poverty of the poor measures, on average, how far the consumption of the poor is below the \$2.15 per person per day (2017 PPP) poverty threshold. The depth of poverty of the poor was calculated by first subtracting the per capita daily consumption expenditure in LCU for the year and month of the ZOI Survey of each poor household in the sample from the \$2.15 per person per day (2017 PPP) poverty line in LCU for the year and month of the ZOI Survey. The figure was then divided by the \$2.15 per person per day (2017 PPP) poverty line in LCU for the year and month of the ZOI Survey. Finally, after applying the household member sampling weight, the value for each poor household was summed across all poor households and then divided by the number of sampled poor households with consumption data.

^d The percent of people who are near-poor is the percentage of individuals living at or above the \$2.15 per person per day (2017 PPP) poverty threshold but below 125 percent of that threshold, or \$2.69 per day. The calculation of the percent of people who are near-poor is similar way to the calculation of the prevalence of poverty, but in the final step, after applying the household member sampling weight, the sum of the number of households in which the per capita daily consumption in LCU for the year and month of the ZOI Survey was equal to or greater than the poverty line in LCU for the year and month of the ZOI Survey but below 125 percent of the poverty line in LCU for the year and month of the ZOI Survey was divided by the sum of the number of households with consumption data.

^e The prevalence of poverty is the percentage of individuals living below the \$1.25 (2005 PPP) per person per day poverty threshold. The prevalence of poverty was calculated by first converting the \$1.25 per person per day (2005 PPP) poverty line into LCU by multiplying the \$1.25 per person per day (2005 PPP) poverty line by the 2005 PPP conversion rate of the ZOI Survey country, where LCU 2005 PPP=2,850.33 UGX. The resulting figure was then adjusted for the cumulative price inflation between 2005 and the ZOI Survey year by multiplying by the ratio of the ZOI Survey country's CPI for the survey year and month to the 2005 CPI, where survey month and year CPI LCU=138.48 and 2005 CPI LCU=45.22. Finally, after applying the household member sampling weight, the sum of the number of households in which the per capita daily consumption in LCU for the year and month of the ZOI Survey was less than the poverty line in LCU for the year and month of the ZOI Survey was divided by the sum of the number of households with consumption data.

^f The depth of poverty of the poor measures, on average, how far the consumption of the poor is below the \$1.25 per person per day (2005 PPP) poverty threshold. The depth of poverty of the poor was calculated by first subtracting the per capita daily consumption expenditure in LCU for the year and month of the ZOI Survey of each poor household in the sample from the \$1.25 per person per day (2005 PPP) poverty line in LCU for the year and month of the ZOI Survey. The figure was then divided by the \$1.25 per person per day (2005 PPP) poverty line in LCU for the year and month of the ZOI Survey. Finally, after applying the household member sampling weight, the value for each poor household was summed across all poor households and then divided by the number of sampled poor households with consumption data.

^g The percent of people who are near-poor is the percentage of individuals living at or above the \$1.25 per person per day (2005 PPP) poverty threshold but below 125 percent of that threshold, or \$1.56 per day. The calculation of the percent of people who are near-poor is similar to the calculation of the prevalence of poverty, but in the final step, after applying the household member sampling weight, the sum of the number of households in which the per capita daily consumption in LCU for the year and month of the ZOI Survey was equal to or greater than the poverty line in LCU for the year and month of the ZOI Survey but below 125 percent of the poverty line in LCU for the year and month of the ZOI Survey was divided by the sum of the number of households with consumption data.

Notes:

Estimates are sample-weighted; numbers of observations are unweighted.

Indicator estimates are calculated using de jure household members.

At baseline, per capita daily consumption was calculated by first determining the monetary value of daily household consumption expenditures per usual household member for specific categories of items (food consumption over the past 7 days; non-food expenditures over the past 7 days, 1 month, 3 months, and 12 months; housing; and durable goods) and then aggregating the values across all categories, as detailed in the Guide to Feed the Future Statistics. Per capita consumption values were then used to estimate the baseline prevalence of poverty values.

At midline, per capita daily consumption was calculated using a survey-to-survey imputation method developed by the World Bank. The approach required first creating a model to predict per capita consumption using data from the phase two ZOI Baseline Survey, and then using the baseline model's coefficients and phase two ZOI Midline Survey data on the model's predictor variables to impute per capita consumption for each household at midline. Imputed per capita consumption values were then used to estimate the midline poverty indicator values.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 4.2.1: Comparison of the Percent of Households in the Uganda ZOI that Have or Own Selected Assets, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Asset	Baseline (2019)		Midline (2022)		Sig. ^a
	%	95% CI	%	95% CI	
Electricity (from any source)	34.6	31.1, 38.3	34.7	30.5, 39.1	n/s
Electricity (from public source main grid)	11.4	8.1, 15.7	8.3	5.3, 12.7	n/s
Radio	48.6	45.8, 51.4	43.8	40.2, 47.4	*
Television	10.1	8.1, 12.5	14.4	11.0, 18.6	n/s
Mobile phone	73.7	71.0, 76.3	80.3	77.3, 83.0	**
Non-mobile telephone	1.0	0.6, 1.8	1.1	0.5, 2.5	n/s
Computer	2.0	1.3, 3.0	1.8	1.2, 2.5	n/s
Refrigerator	1.7	1.1, 2.6	1.8	1.1, 2.9	n/s
Cassette/CD/DVD player	5.0	4.0, 6.2	5.7	4.3, 7.5	n/s
Table	70.6	67.6, 73.5	73.0	70.2, 75.5	n/s
Chair	81.9	79.4, 84.2	84.2	81.0, 86.9	n/s
Sofa set	22.8	19.3, 26.7	25.0	21.4, 29.1	n/s
Bed	80.5	77.3, 83.4	80.4	77.2, 83.3	n/s
Cupboard	20.3	17.9, 23.0	20.6	17.7, 23.8	n/s
Clock	6.3	5.3, 7.5	6.3	4.9, 8.2	n/s
Watch	16.3	14.1, 18.9	21.2	18.0, 24.7	*
Bank account	23.2	20.4, 26.1	51.2	45.8, 56.6	***
Bicycle	24.1	21.0, 27.4	24.5	20.7, 28.7	n/s
Motorcycle or motor scooter	8.5	7.2, 9.9	9.3	7.6, 11.3	n/s
Animal-drawn cart	1.0	0.5, 1.9	0.7	0.4, 1.4	n/s
Car or truck	1.7	1.1, 2.5	2.2	1.5, 3.2	n/s
Boat with a motor	0.0	na	0.3	0.1, 0.7	na
Boat without a motor	0.2	0.0, 0.5	0.6	0.3, 1.2	n/s
Hand tools for farm work	77.4	73.5, 80.9	79.9	76.0, 83.4	n/s
Mechanical plow	0.1	0.0, 0.3	2.6	1.7, 3.9	***
Broad bed maker	0.1	0.0, 0.3	1.4	0.8, 2.5	***
Knapsack chem, sprayer	9.4	7.7, 11.5	12.4	10.1, 15.0	n/s
Modern beehive	4.2	3.2, 5.5	2.8	2.0, 3.8	*
Granary	6.2	4.5, 8.5	6.3	4.7, 8.4	n/s
Modern silo	0.5	0.2, 1.0	1.6	1.0, 2.5	**
Grain bags	59.2	55.4, 62.9	55.6	50.3, 60.8	n/s
Tarpaulins	29.8	26.8, 32.9	27.5	24.6, 30.7	n/s
Number of households (n)	2,540		2,995		

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 4.2.2: Percent Distribution of Households in the Uganda ZOI by Asset-based Wealth Quintile, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)						Midline (2022)					
	Lowest	Second	Middle	Fourth	Highest	<i>n^{a,b}</i>	Lowest	Second	Middle	Fourth	Highest	<i>n^{a,b}</i>
All households	19.8	20.3	19.5	18.8	21.6	2,540	22.4	22.1	19.5	18.5	17.6	2,995
Gendered household type												
Male and female adults	19.4	20.3	21.0	19.8	19.5	1,836	19.8	21.2	20.6	19.1	19.3	2,401
Female adults only	23.6	19.9	16.5	18.2	21.8	450	30.4	28.0	14.9	17.0	9.8	399
Male adults only	16.0	20.9	14.1	12.2	36.8	253	40.0	21.0	14.2	13.4	11.5	194
Children only	^	^	^	^	^		^	^	^	^	^	
Shock exposure severity												
Did not experience any shocks	8.4	14.3	5.2	6.1	66.0	68	^	^	^	^	^	19
Low	16.2	18.2	19.3	19.0	27.3	830	24.5	20.2	18.9	17.8	18.6	1,019
Moderate	22.3	20.8	17.2	19.8	19.8	848	23.0	23.3	16.8	17.3	19.6	981
High	22.1	22.4	23.9	18.3	13.3	777	19.6	23.0	22.6	20.3	14.5	972
Region												
East	22.0	29.9	21.0	13.4	13.7	761	23.3	27.9	19.8	14.8	14.3	345
Karamoja	73.5	5.7	3.5	3.7	13.6	218	83.8	6.3	4.6	3.1	2.3	744
North	46.2	15.8	12.0	10.4	15.7	602	45.8	25.0	9.6	13.4	6.2	416
Southwest	3.3	19.4	23.4	25.8	28.1	959	4.6	20.9	24.9	24.2	25.3	1,490
Resilience Focus Zone												
Non-RFZ	17.3	22.2	20.7	18.9	20.9	2,104	16.9	23.6	20.8	20.0	18.8	1,342
RFZ	33.5	10.2	12.9	17.9	25.6	436	50.8	14.6	12.5	10.6	11.5	1,653
Household education												
No education	61.7	17.2	9.0	5.1	7.0	100	63.1	25.3	4.8	6.8	0.0	195
Less than primary year 3	43.0	33.5	14.0	6.7	2.8	106	63.6	16.5	14.2	4.2	1.4	204
Completed primary year 3	28.2	27.2	24.4	13.2	6.9	714	35.3	31.9	17.6	10.2	5.0	795
Completed primary	15.5	20.2	22.2	22.2	20.0	1,120	15.5	21.8	23.3	22.8	16.6	1,322
Completed 'A' level or higher	5.7	7.2	8.1	23.8	55.2	500	6.4	9.2	15.2	23.6	45.5	479

^a Results not statistically reliable, n<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 4.2.3: Comparison of the Percent of Households in the Uganda ZOI in the Lowest Wealth Quintile According to the Comparative Wealth Index, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^c	Sig. ^d
	%	95% CI	n ^{a,b}	%	95% CI	n ^{a,b}			
All households	39.8	36.2, 43.4	2,540	38.1	33.9, 42.6	2,995	-1.7	0.57	n/s
Gendered household type									
Male and female adults	39.5	35.6, 43.5	1,836	34.2	30.3, 38.3	2,401	-5.3	0.07	n/s
Female adults only	42.5	35.9, 49.4	450	53.7	46.9, 60.4	399	11.2	0.02	*
Male adults only	36.9	27.8, 47.0	253	57.5	42.9, 70.9	194	20.6	0.02	*
Children only	^	^	1	^	^	1	na	na	na
Shock exposure severity									
Did not experience any shocks	22.7	12.4, 37.9	68	^	^	19	na	na	na
Low	34.5	29.5, 39.8	886	40.2	33.6, 47.3	1,019	5.7	0.19	n/s
Moderate	43.4	38.0, 48.9	766	38.9	33.1, 45.1	981	-4.5	0.28	n/s
High	43.8	39.3, 48.5	808	35.4	30.9, 40.2	972	-8.4	0.01	*
Region									
East	51.2	43.5, 58.8	761	45.4	34.3, 56.9	345	-5.8	0.40	n/s
Karamoja	79.2	57.1, 91.6	218	88.5	81.6, 93.1	744	9.3	0.30	n/s
North	62.0	51.2, 71.7	602	65.5	53.9, 75.6	416	3.5	0.64	n/s
Southwest	22.4	19.1, 26.0	959	17.8	14.5, 21.7	1,490	-4.6	0.07	n/s
Resilience Focus Zone									
Non-RFZ	39.1	35.5, 42.8	2,104	33.6	28.8, 38.8	1,342	-5.5	0.09	n/s
RFZ	43.6	28.5, 60.1	436	61.1	55.8, 66.3	1,653	17.5	0.046	*
Household education									
No education	78.9	67.2, 87.2	100	83.0	70.7, 90.8	195	4.1	0.57	n/s
Less than primary year 3	75.3	64.5, 83.6	106	77.8	67.3, 85.7	204	2.5	0.71	n/s
Completed primary year 3	54.8	49.4, 60.1	714	58.4	51.7, 64.8	795	3.6	0.41	n/s
Completed primary	35.4	31.2, 39.8	1,120	30.9	27.3, 34.7	1,322	-4.5	0.12	n/s
Completed 'A' level or higher	13.0	10.0, 16.6	500	12.0	8.3, 17.1	479	-1.0	0.74	n/s

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, disaggregate sample sizes may not total to the aggregated sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^d Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 4.2.4: Percent Distribution of Households in the Uganda ZOI by Wealth Quintile According to the Comparative Wealth Index, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)						Midline (2022)					
	Lowest	Second	Middle	Fourth	Highest	<i>n^{a,b}</i>	Lowest	Second	Middle	Fourth	Highest	<i>n^{a,b}</i>
All households	39.8	32.6	13.4	5.3	8.9	2,540	38.1	15.6	24.1	17.6	4.6	2,995
Gendered household type												
Male and female adults	39.5	34.4	13.3	5.0	7.8	1,836	34.2	16.3	25.5	18.7	5.3	2,401
Female adults only	42.5	30.3	13.8	5.5	7.9	450	53.7	12.9	19.5	11.5	2.4	399
Male adults only	36.9	24.1	13.4	6.8	18.8	253	57.5	10.9	16.0	15.3	0.2	194
Children only	^	^	^	^	^		^	^	^	^	^	
Shock exposure severity												
Did not experience any shocks	22.7	7.7	18.5	16.1	35.0	68	^	^	^	^	^	19
Low	34.5	32.6	13.5	6.5	12.9	886	40.2	14.6	21.2	19.1	5.0	1,019
Moderate	43.4	29.7	14.9	5.1	6.8	766	38.9	14.5	23.9	17.5	5.1	981
High	43.8	37.8	11.3	2.8	4.3	808	35.4	17.7	27.1	16.1	3.7	972
Region												
East	51.2	31.1	8.0	2.8	6.9	761	45.4	15.7	20.9	15.2	2.9	345
Karamoja	79.2	5.7	4.6	4.0	6.6	218	88.5	4.0	4.3	2.0	1.1	744
North	62.0	19.0	9.1	4.1	5.8	602	65.5	9.6	15.7	7.0	2.2	416
Southwest	22.4	41.3	18.3	6.9	11.2	959	17.8	19.3	31.5	24.6	6.7	1,490
Resilience Focus Zone												
Non-RFZ	39.1	34.0	13.4	5.2	8.4	2,104	33.6	16.7	26.0	18.9	4.8	1,342
RFZ	43.6	25.1	13.5	5.9	11.9	436	61.1	10.1	14.5	10.8	3.5	1,653
Household education												
No education	78.9	13.0	6.7	0.3	1.1	100	83.0	7.8	9.2	0.0	0.0	195
Less than primary year 3	75.3	21.9	2.4	0.0	0.4	106	77.8	13.0	7.6	1.7	0.0	204
Completed primary year 3	54.8	35.0	7.2	1.2	1.7	714	58.4	17.1	17.3	6.7	0.5	795
Completed primary	35.4	37.1	15.8	5.5	6.3	1,120	30.9	17.4	29.8	18.4	3.5	1,322
Completed 'A' level or higher	13.0	23.4	20.8	12.9	29.8	500	12.0	10.3	24.3	37.7	15.7	479

^ Results not statistically reliable, *n*<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.1.1: Comparison of the Prevalence of Moderate and Severe Food Insecurity in the Uganda ZOI, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Characteristic	Baseline (2019)			Midline (2022)			Diff.	Sig. ^c
	%	95% CI	n ^{a,b}	%	95% CI	n ^{a,b}		
All households	70.7	63.5, 77.9	2,533	84.6	80.6, 88.5	2,981	13.9	s
Gendered household type								
Male and female adults	71.1	64.5, 77.7	1,829	84.2	80.1, 88.3	2,390	13.1	s
Female adults only	73.1	63.2, 83.0	450	87.4	83.5, 91.3	397	14.3	s
Male adults only	63.3	53.6, 73.0	253	82.3	73.0, 91.6	193	19.0	s
Children only	^	^		^	^		na	na
Wealth quintile								
Highest (wealthiest)	47.2	39.9, 54.5	570	65.1	59.0, 71.2	475	17.9	s
Fourth	60.4	53.6, 67.2	407	80.7	76.3, 85.1	439	20.3	s
Middle	75.9	71.4, 80.4	447	87.6	83.4, 91.8	482	11.7	s
Second	82.9	79.3, 86.5	483	90.5	88.4, 92.6	557	7.6	s
Lowest (poorest)	88.4	83.3, 93.5	626	94.5	91.8, 97.2	1,028	6.1	s
Shock exposure severity								
Did not experience any shocks	35.4	24.4, 46.4	68	^	^	19	na	na
Low	59.0	52.8, 65.2	883	76.8	71.3, 82.3	1,014	17.7	s
Moderate	74.5	67.5, 81.5	762	85.5	80.3, 90.7	975	11.0	s
High	83.1	78.4, 87.8	806	91.4	88.9, 93.9	970	8.3	s
Region								
East	82.2	78.9, 85.5	758	88.6	87.4, 89.8	345	6.4	s
Karamoja	87.8	83.7, 91.9	218	95.9	90.6, 100.0	742	8.1	s
North	80.1	76.2, 84.0	601	86.2	79.9, 92.5	409	6.1	n/s
Southwest	60.5	54.2, 66.8	956	80.3	77.1, 83.5	1,485	19.8	s
Resilience Focus Zone								
Non-RFZ	70.7	63.1, 78.3	2,091	84.0	79.9, 88.1	1,336	13.3	s
RFZ	70.6	54.9, 86.3	434	87.6	77.4, 97.8	1,645	17.0	s
Household education								
No education	89.7	82.4, 97.0	102	89.3	78.5, 100.0	195	-0.4	n/s
Less than primary year 3	83.9	77.1, 90.7	105	94.8	88.5, 100.0	203	10.9	s
Completed primary year 3	80.0	71.8, 88.2	714	92.1	89.5, 94.7	794	12.1	s
Completed primary	69.5	63.1, 75.9	1,115	84.8	81.5, 88.1	1,316	15.3	s
Completed 'A' level or higher	52.8	45.0, 60.6	497	69.4	61.7, 77.1	473	16.6	s

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households—not the number of household members—even though the prevalence of food insecurity is calculated using household member weights

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, disaggregate sample sizes may not total to the aggregated sample size.

^c Whether a statistically significant difference exists between the baseline and midline estimates was determined by comparing the difference of midline and baseline point estimates to the midline margin of error (MoE) at the 95 percent level. If the difference of midline and baseline point estimates is greater than the midline MoE, it is considered statistically significant. If not, it is considered not statistically significant. Differences found to be statistically significant are indicated by s=significant; n/s=not significant and na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.1.2: Comparison of the Prevalence of Moderate and the Prevalence of Severe Food Insecurity in the Uganda ZOI, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Characteristic	Baseline (2019)					Midline (2022)						
	Moderate		Severe		n ^{a,b}	Moderate			Severe			n ^{a,b}
	%	95% CI	%	95% CI		%	95% CI	Sig. ^c	%	95% CI	Sig. ^c	
All households	42.3	39.2, 45.4	28.4	22.4, 34.3	2,533	47.9	45.3, 50.6	s	36.6	31.1, 42.2	s	2,981
Gendered household type												
Male and female adults	43.9	40.8, 47.0	27.2	21.4, 33.0	1,829	47.9	44.8, 51.0	s	35.6	29.9, 41.3	s	2,391
Female adults only	39.4	34.8, 44.0	33.6	25.7, 41.5	450	47.5	43.8, 51.2	s	41.8	36.2, 47.4	s	396
Male adults only	36.5	31.1, 41.9	26.8	19.8, 33.8	253	49.2	44.0, 54.4	s	39.5	28.2, 50.8	s	193
Children only	^	^	^	^	1	^	^	na	^	^	na	1
Wealth quintile												
Highest (wealthiest)	35.4	29.7, 41.1	11.8	8.7, 14.9	570	47.4	43.3, 51.5	s	17.7	11.8, 23.6	n/s	475
Fourth	39.6	36.2, 43.0	20.9	16.5, 25.3	407	49.0	45.6, 52.4	s	31.7	26.7, 36.7	s	439
Middle	46.8	43.1, 50.5	29.1	22.5, 35.7	447	52.1	48.4, 55.8	s	35.5	31.1, 39.9	s	482
Second	46.1	40.8, 51.4	36.8	29.5, 44.1	483	50.0	47.4, 52.6	s	40.4	36.6, 44.2	n/s	557
Lowest (poorest)	44.3	41.5, 47.1	44.0	38.2, 49.8	626	41.7	37.6, 45.8	n/s	52.8	46.4, 59.2	s	1,028
Shock exposure severity												
Did not experience any shocks	20.6	12.5, 28.7	14.8	8.8, 20.8	68	^	^	na	^	^	na	19
Low	39.8	35.2, 44.4	19.3	15.2, 23.4	883	45.9	43.5, 48.3	s	30.9	24.6, 37.2	s	1,014
Moderate	44.5	41.5, 47.5	30.0	23.7, 36.3	762	48.8	45.6, 52.0	s	36.7	29.9, 43.5	n/s	975
High	44.9	41.3, 48.5	38.2	31.8, 44.6	806	49.4	45.6, 53.2	s	42.0	37.1, 46.9	n/s	970
Region												
East	41.9	39.2, 44.6	40.3	35.9, 44.7	758	49.0	46.6, 51.4	s	39.6	37.0, 42.2	n/s	345
Karamoja	44.5	40.9, 48.1	43.3	38.7, 47.9	218	35.9	33.2, 38.6	s	60.0	54.9, 65.1	s	742
North	49.3	47.5, 51.1	30.7	27.5, 33.9	601	46.7	44.2, 49.2	s	39.6	34.0, 45.2	s	409
Southwest	39.5	34.9, 44.1	20.9	18.8, 23.0	957	49.7	47.3, 52.1	s	30.6	29.1, 32.1	s	1,485
Resilience Focus Zone												
Non-RFZ	42.6	39.1, 46.1	28.1	21.9, 34.3	2,091	48.8	46.8, 50.8	s	35.1	30.4, 39.8	s	1,336
RFZ	41	36.6, 45.4	29.7	17.3, 42.1	434	43.2	34.5, 51.9	n/s	44.4	26.0, 62.8	n/s	1,645

Characteristic	Baseline (2019)					Midline (2022)						
	Moderate		Severe		n ^{a,b}	Moderate			Severe			n ^{a,b}
	%	95% CI	%	95% CI		%	95% CI	Sig. ^c	%	95% CI	Sig. ^c	
Household education												
No education	42.0	34.8, 49.2	47.7	38.8, 56.6	102	37.6	30.9, 44.3	n/s	51.8	37.3, 66.3	n/s	195
Less than primary year 3	42.9	37.5, 48.3	41.0	30.9, 51.1	105	42.6	34.0, 51.2	n/s	52.2	39.1, 65.3	n/s	203
Completed primary year 3	43.7	39.6, 47.8	36.3	28.2, 44.4	714	50.6	47.6, 53.6	s	41.5	36.8, 46.2	s	794
Completed primary	43.0	39.9, 46.1	26.5	21.8, 31.2	1,115	48.4	45.2, 51.6	s	36.4	32.3, 40.5	s	1,316
Completed 'A' level or higher	38.5	32.5, 44.5	14.3	9.5, 19.1	497	45.6	42.3, 48.9	s	23.9	17.6, 30.2	s	473

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households—not the number of household members—even though the prevalence of food insecurity is calculated using household member weights

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, disaggregate sample sizes may not total to the aggregated sample size.

^c Whether a statistically significant difference exists between the baseline and midline estimates was determined by comparing the difference of midline and baseline point estimates to the midline margin of error (MoE) at the 95 percent level. If the difference of midline and baseline point estimates is greater than the midline MoE, it is considered statistically significant. If not, it is considered not statistically significant. Differences found to be statistically significant are indicated by s=significant; n/s=not significant and na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted numbers of households.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.2.1: Comparison of the Percent of Households in the Uganda ZOI Exposed to Each Shock or Stress During the 12 Months Preceding the Survey, in Total and by Region, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Shock or stress	Total			Region											
	Est.	95% CI	Sig. ^a	East			Karamoja			North			Southwest		
	Est.	95% CI	Sig. ^a	Est.	95% CI	Sig. ^a	Est.	95% CI	Sig. ^a	Est.	95% CI	Sig. ^a	Est.	95% CI	Sig. ^a
Experienced (%)															
Too much rain															
Baseline	50.5	45.9, 55.1		73.9	67.3, 79.7		66.3	51.3, 78.6		40.3	33.2, 47.8		42.9	35.6, 50.5	
Midline	65.0	60.7, 69.1	***	81.4	72.8, 87.7	n/s	37.8	29.4, 47.1	**	50.4	43.7, 57.2	*	66.5	59.6, 72.7	***
Too little rain															
Baseline	69.8	65.4, 73.8		77.8	71.1, 83.3		78.3	72.0, 83.6		74.9	66.0, 82.1		63.5	56.5, 70.0	
Midline	79.0	75.6, 82.1	***	75.9	66.5, 83.3	n/s	83.3	77.5, 87.8	n/s	86.9	78.4, 92.4	*	77.2	72.9, 80.9	***
Hail or frost															
Baseline	30.8	26.0, 36.1		45.0	37.8, 52.3		10.1	6.7, 14.8		25.8	20.3, 32.3		28.8	20.8, 38.4	
Midline	49.8	43.8, 55.9	***	64.0	48.7, 76.9	*	11.4	7.2, 17.7	n/s	24.9	19.3, 31.4	n/s	57.6	48.8, 65.9	***
Erosion of land															
Baseline	18.4	15.0, 22.4		25.4	18.0, 34.6		9.9	6.0, 15.9		3.5	1.5, 7.9		22.0	16.8, 28.4	
Midline	35.3	30.5, 40.4	***	41.5	30.0, 54.1	*	6.3	4.0, 9.8	n/s	15.1	9.4, 23.4	**	43.7	36.3, 51.4	***
Disease affecting crops															
Baseline	45.4	41.0, 49.8		53.1	45.2, 60.7		49.4	37.5, 61.3		40.0	29.6, 51.3		43.7	37.4, 50.3	
Midline	47.2	42.6, 51.9	n/s	60.0	48.6, 70.3	n/s	57.6	45.8, 68.7	n/s	46.9	37.7, 56.4	n/s	39.9	34.0, 46.1	n/s
Pests affecting crops															
Baseline	36.4	32.0, 40.9		52.5	46.3, 58.6		52.4	42.5, 62.2		41.9	29.7, 55.2		25.8	20.3, 32.1	
Midline	39.9	33.9, 46.2	n/s	63.3	53.1, 72.4	n/s	63.9	52.6, 73.9	n/s	19.2	12.5, 28.3	**	32.3	23.1, 43.1	n/s
Weeds affecting crops															
Baseline	30.5	26.8, 34.4		29.3	22.5, 37.2		66.9	53.6, 77.9		36.7	27.4, 47.0		24.9	20.5, 29.8	
Midline	43.0	39.1, 47.0	***	48.1	37.6, 58.8	**	54.2	44.7, 63.4	n/s	27.5	19.4, 37.3	n/s	44.1	39.6, 48.8	***
Disease affecting livestock															
Baseline	29.1	26.3, 32.1		35.5	30.0, 41.4		39.4	25.8, 54.9		40.4	32.6, 48.8		21.0	17.6, 24.9	
Midline	45.5	41.6, 49.4	***	57.7	47.1, 67.7	***	30.0	24.3, 36.5	n/s	46.6	38.3, 55.1	n/s	41.8	37.2, 46.6	***
Animals stolen															
Baseline	16.7	14.4, 19.2		19.5	15.2, 24.6		17.1	8.1, 32.8		15.9	11.6, 21.3		15.8	12.6, 19.6	
Midline	26.5	23.4, 29.9	***	33.3	27.2, 40.0	**	31.9	27.0, 37.3	*	22.0	16.5, 28.7	n/s	24.1	19.5, 29.4	**
Increased prices of agricultural or livestock inputs															
Baseline	42.6	39.7, 45.6		54.7	47.6, 61.7		18.1	11.3, 27.7		44.1	38.0, 50.5		39.5	35.5, 43.6	
Midline	65.3	62.0, 68.4	***	64.4	56.0, 72.0	n/s	51.4	46.0, 56.8	***	56.1	48.5, 63.3	*	70.8	66.4, 74.9	***
Unable to sell crops, livestock, etc., at fair price															
Baseline	26.2	23.7, 28.9		31.5	24.7, 39.2		6.1	3.8, 9.6		25.2	20.0, 31.3		26.4	23.4, 29.6	
Midline	43.4	39.3, 47.6	***	49.3	39.2, 59.4	**	30.2	25.2, 35.6	***	37.7	29.5, 46.6	*	44.7	38.9, 50.6	***

Shock or stress	Region														
	Total			East			Karamoja			North			Southwest		
	Est.	95% CI	Sig. ^a	Est.	95% CI	Sig. ^a	Est.	95% CI	Sig. ^a	Est.	95% CI	Sig. ^a	Est.	95% CI	Sig. ^a
Loss of land															
Baseline	5.2	4.0, 6.7		6.3	4.2, 9.5		7.0	2.1, 21.0		6.8	3.6, 12.3		3.9	2.6, 5.7	
Midline	19.2	16.1, 22.6	***	21.0	15.3, 28.2	***	19.2	13.8, 26.0	*	8.0	5.1, 12.2	n/s	22.1	17.2, 27.8	***
Assets stolen or destroyed															
Baseline	19.3	16.7, 22.2		20.0	16.0, 24.8		20.6	12.8, 31.4		11.6	7.9, 16.7		21.9	17.8, 26.6	
Midline	36.8	33.9, 39.8	***	39.0	34.1, 44.1	***	46.5	39.2, 54.0	***	34.8	25.2, 45.7	***	34.9	31.2, 38.9	***
Increase in food prices															
Baseline	74.2	71.0, 77.1		81.4	77.6, 84.7		85.7	79.6, 90.3		78.5	70.7, 84.7		68.3	63.2, 73.0	
Midline	95.4	93.8, 96.6	***	95.0	91.4, 97.1	***	96.6	92.9, 98.4	**	94.4	90.6, 96.7	***	95.8	93.2, 97.4	***
Delay in food assistance															
Baseline	10.6	8.4, 13.3		14.9	11.8, 18.7		6.6	3.8, 11.3		9.6	5.3, 16.8		9.6	6.5, 14.2	
Midline	37.1	32.3, 42.1	***	47.6	38.4, 57.0	***	61.5	53.8, 68.6	***	4.9	1.7, 13.3	n/s	39.2	31.8, 47.2	***
Unemployment of youth															
Baseline	23.1	20.5, 26.0		37.6	32.0, 43.5		6.3	3.1, 12.6		28.1	21.5, 35.8		16.9	13.7, 20.7	
Midline	55.3	51.6, 58.9	***	70.3	61.0, 78.2	***	56.3	48.2, 64.1	***	36.6	31.4, 42.1	n/s	54.5	49.2, 59.6	***
Human disease outbreak															
Baseline	20.8	18.6, 23.1		23.4	19.6, 27.6		44.9	38.6, 51.3		25.4	20.9, 30.5		15.4	12.5, 18.8	
Midline	42.3	38.3, 46.3	***	59.4	52.1, 66.2	***	52.3	45.2, 59.3	n/s	19.5	13.4, 27.4	n/s	40.5	34.9, 46.4	***
Severe illness in family															
Baseline	44.8	41.3, 48.4		42.4	37.4, 47.7		36.9	31.6, 42.5		66.0	59.7, 71.7		38.5	32.8, 44.6	
Midline	63.6	59.5, 67.4	***	69.4	56.6, 79.8	***	73.2	66.1, 79.2	***	67.5	62.7, 72.0	n/s	58.0	52.9, 63.0	***
Death in household															
Baseline	13.9	12.0, 16.0		15.7	12.2, 20.0		6.2	3.1, 11.9		14.6	11.3, 18.7		13.6	10.9, 17.0	
Midline	18.7	15.7, 22.2	*	18.7	12.8, 26.5	n/s	21.9	18.0, 26.4	***	16.6	13.7, 19.9	n/s	19.0	14.3, 24.9	n/s
No shocks experienced (%)															
Baseline	2.6	1.7, 4.0		0.0	na		0.4	0.1, 3.1		1.3	0.5, 3.2		4.4	2.7, 7.1	
Midline	0.4	0.2, 0.8	***	0.0	na	na	1.0	0.2, 5.4	n/s	0.6	0.2, 2.0	n/s	0.4	0.2, 1.0	***
Mean number of shocks experienced															
Baseline	6.1	5.8, 6.3		7.4	6.9, 7.9		6.3	5.6, 6.9		6.3	5.6, 7.0		5.4	5.1, 5.8	
Midline	9.1	8.7, 9.4	***	10.6	9.8, 11.4	***	8.9	8.2, 9.5	***	7.2	6.6, 7.7	n/s	9.1	8.6, 9.5	***
Number of households (n)															
Baseline	2,529			756			217			599			957		
Midline	2,993			345			742			416			1,490		

CI=confidence interval, Est.=estimate, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether an association exists between the outcome indicator and the disaggregate variables. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.2.2: Comparison of the Percent Distribution of Households in the Uganda ZOI by Perceived Severity on Household Food Consumption of Each Shock Experienced During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Shock or stress	Not severe			Somewhat severe			Severe			Extremely severe			n ^{b,c}
	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	
Too much rain													
Baseline	8.0	6.2, 10.3		16.4	14.3, 18.7		36.5	32.3, 40.9		39.1	35.3, 43.1		1,343
Midline	2.3	1.6, 3.3	***	13.9	11.1, 17.3	n/s	46.5	41.9, 51.1	**	37.4	33.2, 41.8	n/s	1,661
Too little rain													
Baseline	5.5	4.4, 6.8		15.9	13.5, 18.6		32.2	27.7, 37.2		46.4	41.5, 51.4		1,777
Midline	2.4	1.7, 3.5	***	14.5	11.6, 18.0	n/s	46.9	42.4, 51.4	***	36.2	31.6, 41.0	**	2,425
Hail or frost													
Baseline	9.1	6.2, 13.0		15.8	12.1, 20.4		31.4	25.8, 37.7		43.7	38.4, 49.1		776
Midline	3.6	2.3, 5.5	**	16.0	11.8, 21.3	n/s	41.1	34.5, 47.9	*	39.4	32.4, 46.8	n/s	1,129
Erosion of land													
Baseline	10.6	7.5, 14.6		17.8	14.4, 21.7		33.4	27.0, 40.5		38.3	30.0, 47.3		429
Midline	2.9	1.6, 5.1	***	18.9	13.7, 25.5	n/s	44.1	38.8, 49.6	*	34.1	28.1, 40.5	n/s	802
Disease affecting crops													
Baseline	8.7	6.7, 11.2		18.6	15.3, 22.3		39.1	34.5, 44.0		33.6	28.2, 39.4		1,131
Midline	4.8	3.0, 7.6	*	23.4	20.0, 27.2	n/s	44.1	39.6, 48.8	n/s	27.7	23.4, 32.5	n/s	1,480
Pests affecting crops													
Baseline	7.9	6.2, 10.1		23.9	19.8, 28.6		35.9	31.1, 41.0		32.2	27.1, 37.8		985
Midline	6.4	3.9, 10.2	n/s	17.7	14.1, 22.0	*	42.5	38.9, 46.3	*	33.3	28.2, 38.9	n/s	1,284
Weeds affecting crops													
Baseline	19.9	15.3, 25.4		26.6	23.1, 30.5		38.0	32.7, 43.5		15.5	12.2, 19.6		794
Midline	9.4	6.6, 13.1	***	25.9	22.0, 30.3	n/s	47.7	43.1, 52.4	**	17.0	12.7, 22.3	n/s	1,377
Disease affecting livestock													
Baseline	18.5	15.2, 22.2		26.4	22.5, 30.8		29.9	25.9, 34.3		25.2	20.9, 30.0		783
Midline	3.7	2.5, 5.4	***	17.1	14.0, 20.7	***	43.2	38.6, 47.9	***	36.1	31.5, 40.9	**	1,257

Shock or stress	Not severe			Somewhat severe			Severe			Extremely severe			n ^{b,c}
	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	
Animals stolen													
Baseline	16.2	12.5, 20.6		23.5	19.1, 28.6		31.3	26.5, 36.6		29.0	23.4, 35.3		420
Midline	3.3	2.0, 5.3	***	14.2	10.4, 19.1	**	49.3	44.8, 53.8	***	33.2	28.3, 38.5	n/s	805
Increased prices of agricultural or livestock inputs													
Baseline	10.3	8.1, 13.1		25.6	21.9, 29.7		37.3	33.1, 41.7		26.8	22.2, 32.0		1,052
Midline	2.6	1.6, 4.0	***	19.9	16.4, 23.9	*	50.5	45.8, 55.3	***	27.0	23.2, 31.2	n/s	1,826
Unable to sell crops, livestock, etc., at fair price													
Baseline	11.4	8.5, 15.2		23.4	19.5, 27.7		39.5	34.0, 45.4		25.6	19.9, 32.3		633
Midline	2.7	1.6, 4.5	***	18.9	15.2, 23.2	n/s	52.8	47.9, 57.7	***	25.5	21.9, 29.6	n/s	1,207
Loss of land													
Baseline	7.1	3.1, 15.6		16.9	10.4, 26.3		34.4	24.7, 45.7		41.5	31.2, 52.7		141
Midline	2.3	1.0, 5.3	n/s	12.0	8.4, 16.8	n/s	51.6	44.1, 59.1	*	34.1	27.1, 41.8	n/s	571
Assets stolen or destroyed													
Baseline	14.6	11.1, 18.8		23.4	18.7, 28.9		28.4	23.5, 33.9		33.6	27.5, 40.4		464
Midline	4.6	2.9, 7.1	***	18.0	14.8, 21.8	n/s	53.7	48.2, 59.1	***	23.6	19.4, 28.5	*	1,063
Increase in food prices													
Baseline	3.2	2.4, 4.4		16.1	13.3, 19.4		38.9	34.7, 43.2		41.8	37.2, 46.5		1,903
Midline	1.0	0.6, 1.7	***	7.5	5.7, 9.6	***	30.2	26.9, 33.7	**	61.3	56.6, 65.8	***	2,851
Delay in food assistance													
Baseline	4.4	2.0, 9.1		15.3	10.2, 22.3		38.8	29.1, 49.6		41.6	33.1, 50.6		266
Midline	0.6	0.3, 1.4	*	11.4	8.3, 15.4	n/s	52.5	48.4, 56.5	*	35.5	31.7, 39.6	n/s	1,093
Unemployment of youth													
Baseline	9.6	7.0, 13.0		15.9	12.5, 20.0		37.3	32.4, 42.5		37.1	30.9, 43.9		638
Midline	1.9	1.2, 3.0	***	15.1	12.3, 18.5	n/s	47.6	43.9, 51.3	**	38.4	32.1, 38.6	n/s	1,529
Human disease outbreak													
Baseline	12.6	9.5, 16.4		23.3	18.5, 28.8		31.2	26.1, 36.8		33.0	27.4, 39.1		561
Midline	3.3	2.2, 5.1	***	13.8	11.0, 17.3	**	50.4	45.1, 55.6	***	32.5	27.2, 38.2	n/s	1,292

Shock or stress	Not severe			Somewhat severe			Severe			Extremely severe			n ^{b,c}
	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	
Severe illness in family													
Baseline	13.5	11.0, 16.5		20.8	17.5, 24.4		30.4	26.6, 34.4		35.4	31.2, 39.8		1,151
Midline	2.9	1.9, 4.3	***	14.2	11.3, 17.6	**	40.3	36.1, 44.6	***	42.7	38.4, 47.0	*	1,940
Death in household													
Baseline	12.5	8.5, 17.9		11.6	7.8, 16.8		25.8	19.8, 32.8		50.2	42.9, 57.5		323
Midline	5.0	2.9, 8.6	**	6.9	4.6, 10.3	n/s	27.9	22.1, 34.5	n/s	60.2	52.9, 67.0	n/s	551

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05,

** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Number of households that reported experiencing the shock or stress during the 12 months preceding the survey

^c Records missing information for severity have been excluded from the estimates. The unweighted sample size reflects this loss in observations.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.2.3: Comparison of Mean Shock Exposure Index Scores in the Uganda ZOI, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)			Midline (2022)			Sig. ^c
	Mean	95% CI	n ^{a,b}	Mean	95% CI	n ^{a,b}	
All households	18.0	17.2, 18.9	2,523	28.6	27.2, 29.8	2,991	***
Gendered household type							
Male and female adults	19.0	18.1, 19.9	1,822	29.5	28.1, 30.9	2,397	***
Female adults only	16.3	15.0, 17.5	447	24.6	22.7, 26.4	399	***
Male adults only	13.8	12.1, 15.6	253	24.6	22.3, 26.9	194	***
Children only	^	^		^	^		na
Wealth quintile							
Highest (wealthiest)	14.3	13.1, 15.5	567	27.2	25.5, 29.0	475	***
Fourth	18.2	16.6, 19.9	406	29.4	27.2, 31.7	442	***
Middle	19.6	18.3, 20.9	446	29.5	27.2, 31.8	486	***
Second	19.3	18.0, 20.5	481	29.3	27.5, 31.1	557	***
Lowest (poorest)	19.1	17.6, 20.5	623	27.3	25.5, 29.0	1,031	***
Region							
East	21.4	19.9, 22.9	755	34.7	31.3, 38.0	345	***
Karamoja	17.9	15.4, 20.4	215	28.5	26.3, 30.7	740	***
North	18.4	16.2, 20.7	596	20.3	18.9, 21.6	416	n/s
Southwest	16.4	15.2, 17.7	957	28.5	26.8, 30.2	1,490	***
Resilience Focus Zone							
Non-RFZ	18.0	17.0, 18.9	2,090	28.8	27.3, 30.3	1,342	***
RFZ	18.3	16.9, 19.7	433	27.1	25.8, 28.4	1,649	***
Household education							
No education	16.1	14.4, 17.8	102	24.7	21.8, 27.6	194	***
Less than primary year 3	14.9	12.7, 17.0	105	24.8	22.7, 26.9	203	***
Completed primary year 3	19.9	19.0, 21.2	714	28.2	26.2, 30.0	797	***
Completed primary	17.9	16.9, 18.9	1,107	29.8	28.3, 31.3	1,322	***
Completed 'A' level or higher	16.3	14.9, 17.6	495	27.1	25.1, 29.0	475	***

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households that experienced at least one shock or stress during the 12 months preceding the survey for which the shock exposure index score could be calculated

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.2.4: COVID-19 Knowledge and its Effects on Households in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Midline Survey

COVID-19 effect	%
Someone in household has had or been suspected of having COVID-19	15.6
Household member has died of COVID-19	3.6
Household lost income or was unable to work due to COVID-19	72.6
Household spent savings or sold assets due to COVID-19	68.8
Household had difficulty obtaining or purchasing food or goods due to COVID-19	88.2
Household members had worry or concern due to COVID-19	96.3
Household members had difficulty moving freely or socializing due to COVID-19	94.0
Household has been affected by COVID-19 in any other way	6.6
Number of households (n)	2,993

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.3.1: Comparison of the Mean Ability to Recover from Shocks and Stresses Index Scores in the Uganda ZOI, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^c	Sig. ^d
	Mean	95% CI	n ^{a,b}	Mean	95% CI	n ^{a,b}			
All households	4.4	4.3, 4.5	2,455	3.9	3.7, 4.0	2,972	-0.5	<0.001	***
Gendered household type									
Male and female adults	4.5	4.4, 4.6	1,794	3.9	3.7, 4.0	2,385	-0.6	<0.001	***
Female adults only	4.2	4.0, 4.4	431	3.7	3.5, 4.0	397	-0.5	0.003	**
Male adults only	4.4	4.2, 4.6	229	3.9	3.6, 4.2	189	-0.5	0.002	**
Children only	^	^		^	^		na	na	na
Wealth quintile									
Highest (wealthiest)	4.8	4.6, 4.9	519	4.3	4.0, 4.5	470	-0.5	0.001	**
Fourth	4.5	4.4, 4.7	403	4.1	3.9, 4.3	440	-0.4	0.001	**
Middle	4.3	4.1, 4.4	441	3.8	3.6, 4.0	484	-0.5	<0.001	***
Second	4.1	4.0, 4.3	474	3.6	3.4, 3.8	554	-0.5	<0.001	***
Lowest (poorest)	4.3	4.2, 4.5	618	3.6	3.4, 3.8	1,024	-0.7	<0.001	***
Region									
East	4.1	4.0, 4.3	754	3.5	3.3, 3.8	345	-0.6	<0.001	***
Karamoja	5.1	4.9, 5.3	214	3.5	3.2, 3.7	731	-1.6	<0.001	***
North	4.6	4.5, 4.8	585	4.0	3.7, 4.3	413	-0.6	<0.001	***
Southwest	4.4	4.2, 4.5	902	4.0	3.8, 4.2	1,483	-0.4	0.006	**
Resilience Focus Zone									
Non-RFZ	4.4	4.3, 4.5	2,025	3.9	3.7, 4.1	1,334	-0.5	<0.001	***
RFZ	4.7	4.5, 5.0	430	3.6	3.5, 3.8	1,638	-1.1	<0.001	***
Household education									
No education	4.2	3.8, 4.5	98	3.6	3.3, 3.9	188	-0.6	0.03	*
Less than primary year 3	4.2	3.9, 4.5	104	3.4	3.1, 3.7	203	-0.8	<0.001	***
Completed primary year 3	4.2	4.1, 4.4	704	3.8	3.6, 4.0	795	-0.4	<0.001	***
Completed primary	4.5	4.3, 4.6	1,082	3.9	3.7, 4.1	1,314	-0.6	<0.001	***
Completed 'A' level or higher	4.7	4.5, 4.8	467	4.0	3.7, 4.2	472	-0.7	<0.001	***

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households that experienced at least one shock or stress during the 12 months preceding the survey for which the shock exposure index score could be calculated

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^d Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.3.2: Comparison of the Percent Distribution of Households in the Uganda ZOI by Their Perceived Ability to Meet Their Food Needs at the Time of the Survey, Compared to Before Shock Exposure, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)							Midline (2022)							n ^{a,b}		
	Better		Same		Worse		n ^{a,b}	Better			Same			Worse			
	%	95% CI	%	95% CI	%	95% CI		%	95% CI	Sig. ^c	%	95% CI	Sig. ^c	%		95% CI	Sig. ^c
All households	35.5	32.1, 39.0	40.1	37.7, 42.5	24.4	21.8, 27.3	2,461	21.5	18.2, 25.1	***	33.4	29.5, 37.5	**	45.2	40.5, 49.9	***	2,974
Gendered household type																	
Male and female adults	36.7	33.2, 40.3	39.7	37.2, 42.4	23.6	20.8, 26.7	1,799	21.9	18.5, 25.6	***	32.8	28.4, 37.5	*	45.4	40.2, 50.6	***	2,387
Female adults only	31.8	25.8, 38.3	38.5	33.6, 43.7	29.7	24.2, 35.9	432	18.3	12.5, 25.9	**	36.4	29.5, 44.0	n/s	45.3	37.2, 53.5	**	397
Male adults only	32.9	26.6, 39.9	46.3	39.4, 53.2	20.9	16.2, 26.4	229	23.2	13.7, 36.5	n/s	34.5	24.8, 45.7	n/s	42.3	34.2, 50.8	***	189
Children only	^	^	^	^	^	^	1	^	^	na	^	^	na	^	^	na	1
Wealth quintile																	
Highest (wealthiest)	47.1	40.4, 54.0	39.0	33.5, 44.7	13.9	9.9, 19.3	519	29.4	21.7, 38.5	**	36.5	29.6, 43.9	n/s	34.1	25.9, 43.4	***	470
Fourth	37.6	31.3, 44.3	42.8	37.0, 48.7	19.7	15.1, 25.2	404	23.9	18.3, 30.6	**	37.3	30.8, 44.3	n/s	38.8	31.8, 46.3	***	440
Middle	30.3	25.2, 36.0	39.7	35.1, 44.5	30.0	26.4, 33.8	442	18.5	14.4, 23.5	**	33.6	26.8, 41.0	n/s	47.9	40.0, 55.9	***	484
Second	28.8	23.8, 34.4	40.4	36.2, 44.8	30.8	26.3, 35.7	477	16.0	12.3, 20.6	***	32.2	26.1, 39.0	*	51.7	45.3, 58.1	***	554
Lowest (poorest)	33.5	28.0, 39.5	38.8	33.7, 44.1	27.7	22.9, 33.1	619	21.2	17.1, 25.9	***	28.6	23.9, 33.9	**	50.2	44.5, 56.0	***	1,026
Region																	
East	27.2	22.6, 32.4	42.4	37.9, 47.1	30.3	26.5, 34.5	755	12.6	8.1, 19.2	***	31.2	22.2, 42.0	*	56.1	47.2, 64.7	***	345
Karamoja	54.4	44.7, 63.7	40.2	31.4, 49.8	5.4	2.3, 12.0	216	18.9	13.6, 25.7	***	26.6	19.4, 35.3	*	54.5	45.7, 62.9	***	733
North	41.8	34.2, 49.8	36.0	29.2, 43.3	22.2	17.9, 27.3	588	31.7	24.4, 40.0	n/s	26.7	20.9, 33.4	n/s	41.6	33.4, 50.3	***	413
Southwest	34.6	29.5, 40.0	40.7	37.7, 43.7	24.8	20.6, 29.4	902	22.5	17.6, 28.4	**	37.6	32.0, 43.7	n/s	39.9	32.7, 47.4	***	1,483
Resilience Focus Zone																	
Non-RFZ	33.2	29.6, 36.9	40.4	37.8, 43.1	26.4	23.6, 29.4	2,029	21.1	17.4, 25.3	***	34.8	30.3, 39.7	*	44.1	38.7, 49.6	***	1,334
RFZ	47.4	40.4, 54.6	38.3	33.0, 44.0	14.2	10.3, 19.4	432	23.2	19.4, 27.6	***	26.0	21.8, 30.6	***	50.8	45.9, 55.7	***	1,640
Household education																	
No education	26.8	17.5, 38.9	42.9	31.1, 55.5	30.3	20.5, 42.2	98	16.7	9.5, 27.8	n/s	36.3	25.3, 49.1	n/s	46.9	34.6, 59.7	n/s	189
Less than primary year 3	32.6	22.6, 44.4	39.5	31.6, 48.1	27.9	20.1, 37.4	104	13.3	7.1, 23.7	**	35.2	25.6, 46.2	n/s	51.5	41.5, 61.3	***	203
Completed primary year 3	30.5	25.7, 35.8	39.1	35.1, 43.2	30.4	26.3, 34.8	705	20.7	16.6, 25.5	**	31.7	26.0, 38.0	*	47.6	40.9, 54.3	***	796
Completed primary	36.1	32.1, 40.4	42.3	39.0, 45.6	21.6	18.4, 25.2	1,087	22.9	18.7, 27.7	***	33.5	28.5, 39.0	**	43.6	37.6, 49.7	***	1,314
Completed 'A' level or higher	43.9	37.0, 51.0	35.8	30.4, 41.7	20.3	15.8, 25.7	467	21.0	15.4, 28.1	***	34.5	28.8, 40.8	n/s	44.4	37.3, 51.7	***	472

^ Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households that experienced at least one shock or stress during the 12 months preceding the survey

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.3.3: Comparison of the Percent Distribution of Households in the Uganda ZOI by Their Perceived Ability to Meet Their Food Needs During the Year After the Survey, Compared to Before Shock Exposure, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI Baseline and Midline Surveys

Household characteristic	Baseline (2019)							Midline (2022)							n ^{a,b}		
	Better		Same		Worse		n ^{a,b}	Better			Same			Worse			
	%	95% CI	%	95% CI	%	95% CI		%	95% CI	Sig. ^c	%	95% CI	Sig. ^c	%		95% CI	Sig. ^c
All households	47.9	44.8, 50.9	34.0	31.6, 36.5	18.1	16.1, 20.3	2,461	37.2	32.8, 41.9	***	32.9	29.8, 36.2	n/s	29.9	26.3, 33.7	***	2,974
Gendered household type																	
Male and female adults	48.9	45.3, 52.4	34.0	30.9, 37.2	17.1	14.9, 19.6	1,799	37.9	33.4, 42.6	***	31.8	28.5, 35.4	n/s	30.3	26.5, 34.3	***	2,387
Female adults only	44.1	36.9, 51.4	32.8	27.6, 38.4	23.2	18.4, 28.8	432	32.5	24.9, 41.1	*	39.7	33.4, 46.4	n/s	27.8	20.7, 36.2	n/s	397
Male adults only	47.3	40.7, 54.1	36.8	29.6, 44.6	15.9	11.5, 21.6	229	38.7	27.1, 51.8	n/s	32.4	23.0, 43.5	n/s	28.9	21.4, 37.7	**	189
Children only	^	^	^	^	^	^	1	^	^	na	^	^	na	^	^	na	1
Wealth quintile																	
Highest (wealthiest)	59.3	54.0, 64.3	30.3	25.5, 35.6	10.4	7.4, 14.5	519	49.5	39.9, 59.1	n/s	31.5	23.4, 40.9	n/s	19.0	13.0, 27.0	*	470
Fourth	50.3	44.2, 56.5	35.7	29.8, 42.1	14.0	10.2, 18.9	404	43.1	34.1, 52.5	n/s	35.6	28.9, 42.9	n/s	21.3	16.3, 27.5	*	440
Middle	45.3	40.0, 50.7	34.6	30.1, 39.5	20.1	16.6, 24.1	442	37.5	30.7, 44.8	n/s	34.2	28.2, 40.6	n/s	28.4	22.6, 35.0	*	484
Second	39.2	34.0, 44.8	35.4	30.8, 40.2	25.4	20.7, 30.7	477	30.7	24.9, 37.3	*	31.6	26.4, 37.2	n/s	37.7	30.6, 45.3	**	554
Lowest (poorest)	45.3	40.4, 50.3	34.2	30.2, 38.5	20.4	16.9, 24.5	619	29.0	23.5, 35.3	***	32.1	26.3, 38.4	n/s	38.9	32.9, 45.3	***	1,026
Region																	
East	38.5	32.5, 44.9	37.3	32.0, 42.9	24.2	19.8, 29.2	755	25.3	18.7, 33.3	**	36.9	30.4, 43.9	n/s	37.8	30.1, 46.1	**	345
Karamoja	58.8	47.5, 69.2	36.3	27.8, 45.8	4.9	2.7, 9.0	216	21.3	15.4, 28.7	***	36.4	30.0, 43.4	n/s	42.2	35.7, 49.0	***	733
North	59.4	53.0, 65.4	24.7	19.4, 30.8	16.0	12.7, 19.9	588	50.3	40.9, 59.7	n/s	19.3	13.9, 26.1	n/s	30.4	22.2, 40.1	**	413
Southwest	46.3	42.0, 50.6	36.0	32.8, 39.4	17.7	14.7, 21.1	902	40.8	33.9, 48.1	n/s	35.1	30.6, 39.9	n/s	24.1	19.3, 29.7	*	1,483
Resilience Focus Zone																	
Non-RFZ	47.3	44.1, 50.6	33.5	30.8, 36.3	19.2	17.1, 21.5	2,029	38.8	33.8, 44.2	**	32.8	29.3, 36.5	n/s	28.4	24.3, 32.8	***	1,334
RFZ	50.6	42.2, 59.0	36.8	32.2, 41.6	12.6	7.3, 20.6	432	28.9	24.5, 33.8	***	33.5	29.3, 37.9	n/s	37.6	33.7, 41.6	***	1,640
Household education																	
No education	45.9	32.0, 60.5	29.5	18.0, 44.4	24.6	15.8, 36.1	98	28.5	19.4, 39.8	n/s	36.8	26.7, 48.3	n/s	34.7	23.3, 48.1	n/s	189
Less than primary year 3	43.0	33.6, 52.9	34.0	24.5, 45.0	23.0	14.4, 34.6	104	24.5	14.7, 37.9	*	33.1	23.7, 44.1	n/s	42.4	32.9, 52.5	**	203
Completed primary year 3	42.0	37.2, 47.0	36.6	33.0, 40.5	21.3	18.4, 24.6	705	35.9	29.3, 43.0	n/s	31.2	26.5, 36.3	n/s	32.9	27.7, 38.6	***	796
Completed primary	48.1	44.3, 52.0	35.0	31.8, 38.4	16.9	13.8, 20.5	1,087	37.0	32.0, 42.3	***	34.5	30.6, 38.6	n/s	28.5	23.8, 33.8	***	1,314
Completed 'A' level or higher	58.1	52.1, 63.9	28.0	22.9, 33.7	13.9	10.5, 18.2	467	44.2	35.6, 53.2	*	30.3	25.3, 35.9	n/s	25.5	19.7, 32.3	**	472

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households that experienced at least one shock or stress during the 12 months preceding the survey

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.4.1: Comparison of the Mean Social Capital Index Scores in the Uganda ZOI, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^c	Sig. ^d
	Mean	95% CI	n ^{a,b}	Mean	95% CI	n ^{a,b}			
All households	61.9	59.4, 64.5	2,529	60.1	57.9, 62.3	2,992	-1.8	0.28	n/s
Gendered household type									
Male and female adults	63.7	61.0, 66.4	1,827	61.0	58.6, 63.3	2,398	-2.7	0.14	n/s
Female adults only	56.2	51.7, 60.8	448	56.0	50.8, 61.2	399	-0.2	0.95	n/s
Male adults only	59.5	53.0, 66.1	253	56.7	50.1, 63.3	194	-2.8	0.55	n/s
Children only	^	^		^	^		na	na	na
Wealth quintile									
Highest (wealthiest)	65.1	60.7, 69.6	567	66.7	63.0, 70.4	475	1.6	0.59	n/s
Fourth	62.3	57.5, 67.2	407	63.6	59.4, 67.8	442	1.3	0.69	n/s
Middle	58.5	54.9, 62.2	447	61.6	57.4, 65.8	486	3.1	0.27	n/s
Second	63.4	59.9, 66.9	484	58.1	54.1, 62.2	557	-5.3	0.050	n/s
Lowest (poorest)	59.9	54.9, 65.0	624	52.5	49.3, 55.7	1,032	-7.4	0.02	*
Shock exposure severity									
Did not experience any shocks	41.3	29.9, 52.6	68	^	^	19	na	na	na
Low	59.1	54.9, 63.3	884	56.8	52.9, 60.7	1,019	-2.3	0.43	n/s
Moderate	65.4	61.7, 69.1	764	63.0	59.6, 66.3	981	-2.4	0.33	n/s
High	63.4	60.4, 66.5	807	60.5	57.3, 63.7	971	-2.9	0.19	n/s
Region									
East	62.7	58.6, 66.7	756	59.7	54.5, 64.9	345	-3.0	0.37	n/s
Karamoja	45.2	31.8, 58.6	217	55.2	48.1, 62.2	741	10.0	0.19	n/s
North	68.4	64.2, 72.5	599	50.4	45.0, 55.8	416	-18.0	<0.001	***
Southwest	60.9	57.1, 64.7	957	64.3	61.3, 67.3	1,490	3.4	0.18	n/s
Resilience Focus Zone									
Non-RFZ	62.6	59.9, 65.3	2,094	60.8	58.2, 63.4	1,342	-1.8	0.35	n/s
RFZ	58.6	50.9, 66.3	435	56.3	52.8, 59.9	1,650	-2.3	0.60	n/s
Household education									
No education	50.3	41.4, 59.1	102	50.7	42.4, 59.0	195	0.4	0.94	n/s
Less than primary year 3	57.1	49.5, 64.6	105	54.8	46.3, 63.2	203	-2.3	0.69	n/s
Completed primary year 3	62.3	59.3, 65.4	715	58.1	53.8, 62.3	798	-4.2	0.11	n/s
Completed primary	61.3	57.8, 64.8	1,112	61.7	59.1, 64.2	1,321	0.4	0.87	n/s
Completed 'A' level or higher	65.9	62.0, 69.9	495	61.5	56.8, 66.1	475	-4.4	0.15	n/s

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^d Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.4.2: Comparison of the Mean Bonding Social Capital Scores in the Uganda ZOI, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^c	Sig. ^d
	Mean	95% CI	n ^{a,b}	Mean	95% CI	n ^{a,b}			
All households	65.1	62.5, 67.7	2,529	60.6	58.4, 62.7	2,992	-4.5	0.009	**
Gendered household type									
Male and female adults	67.3	64.5, 70.1	1,827	61.5	59.2, 63.9	2,398	-5.8	0.002	**
Female adults only	59.5	54.5, 64.4	448	56.0	50.3, 61.8	399	-3.5	0.37	n/s
Male adults only	59.8	53.1, 66.5	253	57.8	51.7, 63.9	194	-2.0	0.66	n/s
Children only	^	^	1	^	^	1	na	na	na
Wealth quintile									
Highest (wealthiest)	65.5	61.0, 70.0	567	66.2	61.9, 70.5	475	0.7	0.81	n/s
Fourth	63.8	58.4, 69.3	407	63.4	59.0, 67.9	442	-0.4	0.91	n/s
Middle	62.7	58.5, 66.8	447	61.9	57.4, 66.5	486	-0.8	0.81	n/s
Second	68.4	64.6, 72.2	484	59.4	54.9, 64.0	557	-9.0	0.003	**
Lowest (poorest)	64.9	60.4, 69.4	624	53.7	50.4, 57.0	1,032	-11.2	<0.001	***
Shock exposure severity									
Did not experience any shocks	42.4	29.9, 54.9	68	^	^	19	na	na	na
Low	62.5	58.1, 66.8	884	59.4	55.2, 63.5	1,019	-3.1	0.31	n/s
Moderate	68.7	65.0, 72.4	764	64.1	60.5, 67.7	981	-4.6	0.08	n/s
High	66.4	63.4, 69.5	807	58.5	55.0, 62.0	971	-7.9	<0.001	***
Region									
East	67.2	63.0, 71.3	756	57.2	52.7, 61.7	345	-10.0	0.002	**
Karamoja	50.8	40.7, 60.9	217	57.2	50.7, 63.6	741	6.4	0.29	n/s
North	71.4	67.5, 75.2	599	53.6	48.0, 59.2	416	-17.8	<0.001	***
Southwest	63.3	59.1, 67.5	957	65.0	61.9, 68.1	1,490	1.7	0.53	n/s
Resilience Focus Zone									
Non-RFZ	65.9	63.0, 68.9	2,094	61.1	58.7, 63.6	1,342	-4.8	0.01	*
RFZ	60.6	55.2, 66.1	435	57.7	54.2, 61.2	1,650	-2.9	0.37	n/s
Household education									
No education	54.0	44.6, 63.4	102	47.7	37.7, 57.7	195	-6.3	0.37	n/s
Less than primary year 3	63.7	55.7, 71.6	105	56.3	47.3, 65.4	203	-7.4	0.23	n/s
Completed primary year 3	66.3	63.1, 69.5	715	60.0	56.2, 63.8	798	-6.3	0.01	*
Completed primary	64.1	60.6, 67.7	1,112	61.8	59.1, 64.5	1,321	-2.3	0.31	n/s
Completed 'A' level or higher	67.8	63.7, 71.9	495	61.2	56.2, 66.1	475	-6.6	0.04	*

^ Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^d Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.4.3: Comparison of the Mean Bridging Social Capital Scores in the Uganda ZOI, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^c	Sig. ^d
	Mean	95% CI	n ^{a,b}	Mean	95% CI	n ^{a,b}			
All households	58.8	55.9, 61.6	2,529	59.6	56.7, 62.4	2,992	0.8	0.69	n/s
Gendered household type									
Male and female adults	60.1	57.2, 63.1	1,827	60.4	57.4, 63.5	2,398	0.3	0.89	n/s
Female adults only	53.0	48.1, 57.9	448	56.0	49.9, 62.1	399	3.0	0.46	n/s
Male adults only	59.3	51.3, 67.3	253	55.6	46.8, 64.4	194	-3.7	0.54	n/s
Children only	^	^		^	^		na	na	na
Wealth quintile									
Highest (wealthiest)	64.8	59.8, 69.8	567	67.2	63.2, 71.2	475	2.4	0.46	n/s
Fourth	60.8	56.0, 65.7	407	63.8	58.5, 69.1	442	3.0	0.41	n/s
Middle	54.4	50.1, 58.7	447	61.3	56.5, 66.1	486	6.9	0.04	*
Second	58.4	54.1, 62.6	484	56.8	52.2, 61.4	557	-1.6	0.63	n/s
Lowest (poorest)	55.0	49.0, 61.0	624	51.3	46.9, 55.7	1,032	-3.7	0.33	n/s
Shock exposure severity									
Did not experience any shocks	40.2	28.6, 51.7	68	^	^	19	na	na	na
Low	55.8	51.1, 60.5	884	54.3	49.9, 58.8	1,019	-1.5	0.66	n/s
Moderate	62.1	58.1, 66.2	764	61.8	58.1, 65.6	981	-0.3	0.91	n/s
High	60.4	56.7, 64.1	807	62.5	58.1, 66.9	971	2.1	0.47	n/s
Region									
East	58.2	53.7, 62.6	756	62.2	55.0, 69.3	345	4.0	0.34	n/s
Karamoja	39.6	22.4, 56.7	217	53.2	44.3, 62.0	741	13.6	0.16	n/s
North	65.3	60.1, 70.6	599	47.2	40.3, 54.1	416	-18.1	<0.001	***
Southwest	58.5	54.4, 62.6	957	63.5	59.9, 67.1	1,490	5.0	0.07	n/s
Resilience Focus Zone									
Non-RFZ	59.2	56.4, 62.0	2,094	60.5	57.2, 63.8	1,342	1.3	0.56	n/s
RFZ	56.6	45.5, 67.6	435	55.0	50.6, 59.4	1,650	-1.6	0.79	n/s
Household education									
No education	46.5	36.7, 56.3	102	53.7	43.4, 64.0	195	7.2	0.32	n/s
Less than primary year 3	50.5	41.1, 59.8	105	53.2	43.0, 63.3	203	2.7	0.70	n/s
Completed primary year 3	58.3	54.7, 61.9	715	56.1	50.7, 61.5	798	-2.2	0.50	n/s
Completed primary	58.5	54.5, 62.5	1,112	61.5	58.5, 64.6	1,321	3.0	0.23	n/s
Completed 'A' level or higher	64.1	59.7, 68.5	495	61.7	56.2, 67.3	475	-2.4	0.51	n/s

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^d Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.4.4: Comparison of the Percent of Households in the Uganda ZOI that Believe Local Government Will Help the Community Cope with Future Shocks and Stresses, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^c	Sig. ^d
	%	95% CI	n ^{a,b}	%	95% CI	n ^{a,b}			
All households	61.1	56.7, 65.3	2,294	59.0	53.9, 64.0	2,794	-2.1	0.55	n/s
Gendered household type									
Male and female adults	60.0	55.3, 64.6	1,656	58.1	52.8, 63.3	2,239	-1.9	0.59	n/s
Female adults only	63.8	56.9, 70.2	405	65.5	56.6, 73.4	373	1.7	0.76	n/s
Male adults only	63.6	55.1, 71.4	232	55.8	44.9, 66.2	181	-7.8	0.26	n/s
Children only	^	^	1	^	^	1	na	na	na
Wealth quintile									
Highest (wealthiest)	66.0	58.5, 72.8	513	58.3	47.3, 68.7	443	-7.7	0.25	n/s
Fourth	63.2	56.6, 69.3	369	61.6	53.7, 68.9	414	-1.6	0.76	n/s
Middle	65.5	57.6, 72.6	405	56.7	49.3, 63.8	453	-8.8	0.10	n/s
Second	57.6	49.9, 64.9	448	62.3	55.6, 68.6	527	4.7	0.34	n/s
Lowest (poorest)	52.9	47.1, 58.7	559	56.0	50.4, 61.4	957	3.1	0.45	n/s
Shock exposure severity									
Did not experience any shocks	79.5	66.0, 88.6	60	^	^	17	na	na	na
Low	64.0	58.8, 68.9	792	56.5	50.8, 62.0	938	-7.5	0.050	n/s
Moderate	60.2	54.2, 65.9	695	63.1	56.7, 69.1	913	2.9	0.50	n/s
High	57.5	51.4, 63.3	741	57.3	49.8, 64.6	924	-0.2	0.98	n/s
Region									
East	58.2	50.8, 65.1	688	48.6	40.7, 56.5	318	-9.6	0.08	n/s
Karamoja	56.1	40.2, 70.8	203	79.2	68.2, 87.1	708	23.1	0.02	*
North	56.0	50.0, 61.7	530	38.2	30.9, 46.1	370	-17.8	<0.001	***
Southwest	64.7	57.4, 71.4	873	67.5	58.7, 75.3	1,398	2.8	0.60	n/s
Resilience Focus Zone									
Non-RFZ	60.4	56.0, 64.6	1,889	58.5	52.6, 64.2	1,271	-1.9	0.62	n/s
RFZ	64.7	49.3, 77.6	405	61.7	55.5, 67.5	1,523	-3.0	0.70	n/s
Household education									
No education	59.1	44.4, 72.4	93	69.5	55.2, 80.8	182	10.4	0.29	n/s
Less than primary year 3	49.2	36.0, 62.6	92	74.5	62.3, 83.8	191	25.3	0.005	**
Completed primary year 3	60.9	55.1, 66.3	651	60.6	55.5, 65.5	754	-0.3	0.95	n/s
Completed primary	63.2	58.0, 68.1	1,016	57.0	51.3, 62.6	1,227	-6.2	0.11	n/s
Completed 'A' level or higher	58.8	51.5, 65.8	442	57.0	47.0, 66.5	440	-1.8	0.77	n/s

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^b Number of households that reported, "Yes, the government would provide help" or "No, the government would not provide help." Estimates exclude households that reported local government will not need to help the community cope with shocks or stresses in the future.

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^e Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.4.5: Percent of Households in the Uganda ZOI Participating in Group-based Savings, Micro-finance, or Lending Programs, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^c	Sig. ^d
	%	95% CI	n ^{a,b}	%	95% CI	n ^{a,b}			
All households	59.8	56.1, 63.3	2,178	62.9	59.3, 66.4	2,624	3.1	0.22	n/s
Gendered household type									
Male and female adults	60.8	56.8, 64.7	1,734	65.2	61.3, 68.9	2,231	4.4	0.12	n/s
Female adults only	55.8	49.4, 62.0	444	49.9	43.3, 56.4	393	-5.9	0.20	n/s
Male adults only									
Children only									
Wealth quintile									
Highest (wealthiest)	65.7	60.7, 70.3	442	72.8	63.4, 80.5	411	7.1	0.16	n/s
Fourth	70.1	63.8, 75.6	362	73.7	68.0, 78.8	398	3.6	0.36	n/s
Middle	59.5	51.3, 67.1	392	68.5	61.6, 74.7	428	9.0	0.09	n/s
Second	54.5	47.8, 60.9	423	57.0	50.9, 62.9	482	2.5	0.57	n/s
Lowest (poorest)	49.6	42.6, 56.6	559	46.0	40.1, 52.1	905	-3.6	0.45	n/s
Shock exposure severity									
Did not experience any shocks	52.0	37.8, 65.9	42	^	^	12	na	na	na
Low	57.2	52.7, 61.6	720	60.9	55.2, 66.4	877	3.7	0.30	n/s
Moderate	60.1	54.5, 65.5	685	65.9	61.1, 70.4	863	5.8	0.12	n/s
High	62.3	57.0, 67.3	725	61.8	55.2, 68.0	869	-0.5	0.92	n/s
Region									
East	37.2	32.0, 42.6	648	44.5	31.3, 58.4	289	7.3	0.33	n/s
Karamoja	60.7	42.6, 76.3	196	36.7	28.5, 45.7	689	-24.0	0.02	*
North	60.3	54.8, 65.6	531	68.2	63.6, 72.4	347	7.9	0.03	*
Southwest	68.9	63.2, 74.1	803	73.4	69.6, 76.8	1,299	4.5	0.18	n/s
Resilience Focus Zone									
Non-RFZ	58.9	54.8, 62.9	1,794	64.7	60.6, 68.6	1,165	5.8	0.049	*
RFZ	64.3	56.1, 71.7	384	53.9	48.9, 58.9	1,459	-10.4	0.03	*
Household education									
No education	33.2	19.3, 50.8	91	30.5	16.7, 48.9	177	-2.7	0.82	n/s
Less than primary year 3	40.5	27.2, 55.4	84	47.7	36.4, 59.2	184	7.2	0.45	n/s
Completed primary year 3	55.5	50.0, 60.8	626	57.3	50.2, 64.2	690	1.8	0.68	n/s
Completed primary	62.8	58.1, 67.3	978	66.4	62.4, 70.2	1,170	3.6	0.24	n/s
Completed 'A' level or higher	67.4	61.7, 72.7	399	70.4	62.3, 77.3	403	3.0	0.54	n/s

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households with a primary adult female decision-maker

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^d Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.5.1: Knowledge, Perception, and Concern about Impacts of Climate Change on Daily Life among Households in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Midline Survey

	All households (%)	Sex of household respondent		Age of household respondent	
		Male (%)	Female (%)	15-29 years (%)	30+ years (%)
Knowledge, perception, or concern					
Knowledge about climate change					
A lot	16.1	16.2	16.0	11.3	17.7
Moderate	30.8	29.1	32.3	23.7	33.0
A little	42.3	43.3	41.4	50.8	39.6
None	10.9	11.5	10.3	14.2	9.8
Perception of weather changes					
Perceive that weather has become more problematic	88.0	88.1	87.9	82.8	89.6
Concern about the impacts of climate change on day-to-day life and income					
Very concerned	31.2	32.6	30.0	23.1	33.8
Moderately concerned	31.1	32.7	29.7	30.9	31.2
Somewhat concerned	32.5	30.2	34.7	38.4	30.7
Not concerned	5.1	4.5	5.7	7.7	4.3
Number of households (n)	2,995	1,409	1,586	766	2,229

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.5.2: Percent of Households in the Uganda ZOI that Received Information to Help Prepare for or Manage Weather-related Challenges During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Information on weather-related challenges	%
Received information	26.7
Number of households (n)	2,995
Information provided by (person or group)^a	
Friends or neighbors	35.7
Family	7.4
Non-governmental organization	9.0
Community group	8.1
Community leaders	12.6
Government officials	29.8
Agrodealers or agrovets	7.9
Cooperative	1.4
Other	14.0
Information provided by (mechanism)^a	
Word of mouth	38.7
Community meeting	11.4
Training	3.5
Television	6.9
Radio	65.3
SMS/messaging application	0.5
Voice calls (call center, etc.)	0.3
Mobile application	0.3
Website	0.1
Email	0.0
Other	0.7
Number of households that received information (n)	739

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Multiple responses are allowed, so percentages may sum to more than 100 percent.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.5.3: Percent of Households that Grew Crops in the Uganda ZOI that Received Training or Support or Adopted Practices to Help Manage Weather-related Challenges Related to Crop Production During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Crop-specific information	Received training or support (%)	Adopted practice (%)
Any topic	5.4	25.2
Topic^a		
Interpretation and use of weather-related information for crop production	4.0	
New crops or varieties	4.4	15.2
Planting practices	4.7	19.9
Crop, pest, or disease management	3.6	12.1
Soil management	3.5	16.2
Water management	2.0	3.5
Post-harvest management	1.4	4.5
Other	0.2	0.4
Number of households that grew crops (n)	2,579	2,579

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Multiple responses are allowed, so percentages may sum to more than 100 percent.

Note: Estimates are sample-weighted; numbers of observations are unweighted. Source:

2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.5.4: Percent of Households that Raised Livestock or Farm Animals in the Uganda ZOI that Received Training or Support or Adopted Practices to Manage Weather-related Challenges Related to Raising Livestock During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Livestock-specific information	Received training or support (%)	Adopted practice (%)
Any topic	4.8	11.5
Topic^a		
Interpretation and use of weather-related information for animal production	2.8	
New animals or breeds	2.5	3.1
Animal management	2.9	7.5
Feeding practices	2.7	5.4
Fodder storage	1.8	2.8
Water points	1.2	2.9
Animal health	3.3	8.5
Other	0.1	0.5
Number of households that raised livestock or farm animals (n)	1,831	1,831

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Multiple responses are allowed, so percentages may sum to more than 100 percent.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.5.5: Percent of Households in the Uganda ZOI that Faced Barriers to Adopting Crop or Farm Animal Practices to Help Manage the Impacts of Weather-related Problems During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Barrier	%
Faced barriers adopting new crop or livestock practices	23.7
Number of agricultural households (n)	2,674
Barriers faced^a	
Land access	44.8
Money for inputs	53.1
Money for labor	28.1
Knowledge	18.9
Labor availability	10.0
Infrastructure or materials	5.5
Markets for buying inputs or selling crops	7.4
Time	6.7
Other	7.7
Number of agricultural households that faced barriers (n)	536

RFZ=Resilience Focus Zone; ZOI=Zone of Influence

^a Multiple responses are allowed, so percentages may sum to more than 100 percent. Note:

Estimates are sample-weighted; numbers of observations are unweighted. Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.5.6: Percent of Households in the Uganda ZOI that Participated in a Group or Committee Working to Strengthen the Community's Ability to Manage Weather-related Problems During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Group participation	%
Participated in a group or committee	7.9
Number of households (n)	2,995
Percent of households that participated in a group that:	
Created a formal plan	51.5
Took action	44.0
Number of households with at least one member participating in a group (n)^a	197
Percent distribution of group participants by sex	
Male	46.8
Female	53.2
Percent distribution of group participants by age category	
<30 years old	22.9
30+ years old	77.1
Number of group participants (n)^b	225

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households that have at least one de jure household member who participated in the survey

^b De jure household members only

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.5.7: Percent of Households in the Uganda ZOI that Obtained Insurance, by Type and Support Received to Obtain Insurance During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Insurance details	%
Obtained insurance	0.9
Number of households (n)	2,995
Type of insurance obtained^a	
Health	^
Life	^
Crop	^
Livestock or other animals	^
Received support in obtaining the insurance	^
Number of households that obtained insurance (n)	23
Obtained insurance to protect against weather-related problems	^
Number of households that obtained crop or livestock insurance (n)	4
Percent distribution of individuals who received support to obtain insurance by sex	
Male	^
Female	^
Percent distribution of individuals who received support to obtain insurance by age category	
<30 years old	^
30+ years old	^
Number of individuals who received support to obtain insurance (n)^b	20

^a Results not statistically reliable, n<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Multiple responses are allowed, so percentages may sum to more than 100 percent.

^b De jure household members only

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.5.8: Percent of Households in the Uganda ZOI that Received Payments for Weather-related Problems that Affected Their Crop or Livestock Production During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Payment details	%
Received payment	0.7
Number of agricultural households (n)^a	2,674
Type of payment received^b	
Insurance payout	20.5
Cash transfer	23.1
Voucher	3.1
In-kind payment	54.8
Payment provider^b	
Government	^
Community group	^
Formal bank	^
Insurance company	^
Non-governmental organization	^
Other	^
Number of agricultural households that received a payment (n)	29
Percent distribution of payment recipients by sex	
Male	80.7
Female	19.3
Percent distribution of payment recipients by age category	
<30 years old	27.0
30+ years old	73.0
Number of payment recipients (n)^c	38

^ Results not statistically reliable, n<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Household that cultivated crops or raised livestock or other farm animals in the 12 months preceding the survey

^b Multiple responses are allowed, so percentages may sum to more than 100 percent.

^c De jure household members only

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.5.9: Percent of Households in the Uganda ZOI that Participated in a Group or Committee Working to Monitor or Influence Land Use During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Group participation	%
Participated in a land use group or committee	3.1
Number of households (n)	2,995
Percent of households that participated in a group that:	
Created a formal plan	44.4
Took action	38.9
Number of households with at least one group participant (n)^a	83
Percent distribution of group participants by sex	
Male	63.9
Female	36.1
Percent distribution of group participants by age category	
<30 years old	29.0
30+ years old	71.0
Number of group participants (n)^b	89

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households that have at least one de jure household member who participated in a land use group or committee

^b De jure household members only

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.5.10: Percent of Households in the Uganda ZOI that Participated in a Training or Received Support to Help Diversify Their Sources of Income During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Participation	%
Participated in a training or received support on income diversification	9.6
Number of households (n)	2,995
Participated in training or received support on:	
New crops or products	32.0
Planting trees for cash or credit	8.7
Diversification into the livestock sector	24.7
Wage labor on other farms	6.6
Non-farm wage labor	12.0
Salaried off-farm agricultural employment	3.8
Salaried off-farm non-agricultural employment	4.7
Petty trade	27.5
Agricultural-related self-employment	10.2
Non-agricultural-related self-employment	14.1
Migration or remittances	0.0
Renting out farming equipment	0.7
Renting out land	5.8
Other	9.2
Number of households with at least one member participating in training or receiving support on income diversification (n)^a	247
Percent distribution of individuals who received income diversification support by sex	
Male	48.5
Female	51.5
Percent distribution of individuals who received income diversification support by age category	
<30 years old	29.8
30+ years old	70.2
Number of individuals who received income diversification training or support (n)^b	285

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households that have at least one de jure household member who participated in training or received support on income diversification

^b De jure household members only

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.6.1: Coverage of the Types of Resilience Interventions Promoted by Feed the Future in the Uganda ZOI During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Type of resilience intervention	All households (%)	Resilience Focus Zone	
		Non-RFZ (%)	RFZ (%)
Participated in financial training	15.5	16.9	8.7
Received support to access financial services	22.2	23.7	14.6
Received a cash transfer or voucher	12.7	14.1	5.5
Received a conditional cash transfer or voucher	3.4	4.0	0.7
Received an unconditional cash transfer or voucher	9.3	10.1	4.9
Received food or food vouchers	1.9	1.5	3.9
Participated in a local development group or committee	17.1	17.7	14.2
Participated in a disaster risk reduction training or group	4.8	5.2	2.4
Contributed to the development of a disaster risk reduction plan	4.0	4.5	1.7
Participated in a program or group working to improve local governance	3.1	3.5	1.6
Participated in reading or writing training	4.4	4.1	5.7
Participated in vocational training	11.7	12.6	7.3
Participated in an empowerment group for girls or young women	7.6	8.2	4.3
Number of households (n)	2,995	1,342	1,653

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.6.2: Percent Distribution of Household Members in the Uganda ZOI who Participated in Specific Types of Resilience Interventions of the Types Promoted by Feed the Future During the 12 Months Preceding the Survey and by Sex and Age, Feed the Future Phase Two ZOI and RFZ Midline Survey

Type of resilience intervention	Sex of participant		Age of participant		Number of participants (n) ^a
	Male (%)	Female (%)	<30 years (%)	30+ years (%)	
Participated in financial training	45.7	54.3	23.8	76.2	439
Received support to access financial services	44.6	55.4	19.6	80.4	677
Received a cash transfer or voucher ^b	41.8	58.2	20.8	79.2	341
Received food or food vouchers	37.5	62.5	53.7	46.3	91
Participated in a local development group or committee	48.5	51.5	21.1	78.9	529
Participated in a disaster risk reduction training or group	52.5	47.5	30.7	69.3	124
Contributed to the development of a disaster risk reduction plan	64.5	35.5	31.6	68.4	90
Participated in a program or group working to improve local governance	64.7	35.3	34.3	65.7	77
Participated in reading or writing training	44.1	55.9	60.3	39.7	186
Participated in vocational training	38.2	61.8	61.9	38.1	318
Participated in an empowerment group for girls or young women	8.3	91.7	35.6	64.4	191

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a De jure household members only

^b Question only asked who received the cash or voucher, regardless of whether it was conditional or not.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.7.1: Comparison of the Percent of Households in the Uganda RFZ Exposed to Each Shock or Stress During the 12 Months Preceding the Survey, in Total, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

	Baseline (2019)		Midline (2022)		
Shock or stress	Est.	95% CI	Est.	95% CI	Sig. ^a
Experienced (%)					
Too much rain	48.0	36.8, 59.5	44.1	38.2, 50.1	n/s
Too little rain	76.8	66.9, 84.4	85.1	81.5, 88.1	n/s
Hail or frost	21.4	15.6, 28.7	27.2	22.4, 32.5	n/s
Erosion of land	16.1	11.0, 23.0	16.6	13.4, 20.5	n/s
Disease affecting crops	48.4	41.1, 55.7	52.2	45.3, 59.0	n/s
Pests affecting crops	34.1	22.4, 48.1	46.5	39.9, 53.3	n/s
Weeds affecting crops	43.1	34.2, 52.4	49.3	43.8, 54.8	n/s
Disease affecting livestock	33.8	26.5, 41.9	37.8	33.3, 42.4	n/s
Animals stolen	20.1	14.8, 26.6	28.6	25.0, 32.4	*
Increased prices of agricultural or livestock inputs	32.3	27.4, 37.5	57.1	53.0, 61.2	***
Unable to sell crops, livestock, etc., at fair price	17.9	14.8, 21.5	37.7	33.4, 42.3	***
Loss of land	6.2	3.1, 11.8	19.0	15.2, 23.6	***
Assets stolen or destroyed	29.6	23.0, 37.1	36.9	31.5, 42.7	n/s
Increase in food prices	74.9	69.5, 79.7	95.7	93.6, 97.1	***
Delay in food assistance	5.2	3.0, 8.9	40.5	34.6, 46.6	***
Unemployment of youth	19.8	15.9, 24.3	48.7	43.9, 53.5	***
Human disease outbreak	25.2	20.4, 30.8	45.3	40.5, 50.2	***
Severe illness in family	50.0	44.1, 55.9	69.1	64.3, 73.6	***
Malaria	41.4	36.0, 47.1	62.0	57.2, 66.6	***
Tuberculosis	4.0	2.0, 8.1	8.2	6.1, 11.1	*
Other	18.9	13.5, 25.9	24.8	20.8, 29.2	n/s
Death in household	12.0	9.2, 15.5	18.8	16.0, 21.9	**
No shocks experienced (%)	0.8	0.2, 3.0	0.6	0.1, 2.8	n/s
Mean number of shocks experienced	6.1	5.6, 6.6	8.6	8.2, 9.0	***
Number of households (n) ^{b,c}	435		1,651		

CI=confidence interval, Est.=estimate, RFZ=Resilience Focus Zone, ZOI = Zone of Influence

^a Significance tests were performed to determine whether an association exists between the outcome indicator and the disaggregate variables. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Number of households that reported experiencing the shock or stress during the 12 months preceding the survey

^c Records missing information for severity have been excluded from the estimates. The unweighted sample size reflects this loss in observations.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.7.2: Comparison of the Percent Distribution of Households in the Uganda RFZ by Perceived Severity of Each Shock Experienced During the 12 Months Preceding the Survey on Household Food Consumption, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Shock or stress	Not severe			Somewhat severe			Severe			Extremely severe			n ^{b,c}
	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	
Too much rain													
Baseline	5.1	2.8, 9.0		19.2	15.3, 23.9		35.1	26.2, 45.1		40.6	30.7, 51.3		235
Midline	5.3	3.9, 7.3	n/s	16.8	13.6, 20.5	n/s	39.0	33.5, 44.7	n/s	38.9	33.0, 45.3	n/s	748
Too little rain													
Baseline	6.2	3.7, 10.2		11.0	7.1, 16.6		31.3	25.3, 38.0		51.5	43.3, 59.6		340
Midline	2.5	1.2, 5.1	*	11.4	9.0, 14.3	n/s	42.8	38.7, 47.0	**	43.4	39.3, 47.6	n/s	1,395
Hail or frost													
Baseline	10.5	5.7, 18.6		14.1	7.9, 24.0		31.8	20.0, 46.5		43.5	29.7, 58.4		95
Midline	6.1	4.0, 9.2	n/s	15.9	11.6, 21.5	n/s	37.1	31.5, 43.0	n/s	40.9	34.9, 47.2	n/s	458
Erosion of land													
Baseline	13.0	6.2, 25.2		18.9	10.6, 31.4		35.0	21.3, 51.8		33.0	24.7, 42.6		64
Midline	7.0	4.2, 11.5	n/s	19.0	13.4, 26.2	n/s	41.2	34.8, 47.9	n/s	32.8	26.1, 40.3	n/s	284
Disease affecting crops													
Baseline	7.6	5.2, 11.0		18.6	10.9, 30.0		36.4	28.9, 44.6		37.4	28.7, 47.0		203
Midline	4.1	2.0, 8.4	n/s	18.3	15.4, 21.6	n/s	45.2	40.3, 50.2	n/s	32.4	27.7, 37.4	n/s	871
Pests affecting crops													
Baseline	7.9	4.9, 12.4		32.4	22.8, 43.7		28.3	23.1, 34.0		31.4	20.8, 44.5		163
Midline	2.5	1.4, 4.3	**	10.9	8.5, 14.0	***	43.1	36.9, 49.5	***	43.5	36.6, 50.7	n/s	774
Weeds affecting crops													
Baseline	14.5	9.1, 22.3		29.4	22.1, 38.1		36.4	28.5, 45.0		19.7	13.1, 28.7		203
Midline	6.6	4.7, 9.2	*	23.7	20.2, 27.6	n/s	42.6	38.8, 46.6	n/s	27.1	22.5, 32.2	n/s	835
Disease affecting livestock													
Baseline	12.4	6.9, 21.2		29.2	19.6, 41.1		27.4	16.8, 41.3		31.1	23.6, 39.7		165
Midline	3.6	2.3, 5.5	*	19.2	15.5, 23.5	n/s	40.6	35.6, 45.8	n/s	36.6	31.4, 42.1	n/s	651
Animals stolen													
Baseline	13.1	7.0, 23.1		23.3	13.6, 36.9		28.3	15.7, 45.5		35.4	24.0, 48.7		79
Midline	2.2	1.1, 4.3	**	11.5	8.0, 16.2	n/s	42.6	34.8, 50.8	n/s	43.7	36.3, 51.4	n/s	467

Shock or stress	Not severe			Somewhat severe			Severe			Extremely severe			n ^{b,c}
	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	
Unable to access crop or livestock inputs													
Baseline	21.3	15.4, 28.7		20.1	14.6, 27.1		28.0	18.5, 40.0		30.6	21.3, 41.7		122
Midline	2.4	1.6, 3.8	***	19.3	15.6, 23.6	n/s	48.0	43.3, 52.8	**	30.2	25.6, 35.4	n/s	959
Unable to sell crops, livestock, etc., at fair price													
Baseline	13.9	7.3, 24.8		17.2	7.2, 35.6		43.5	29.2, 58.9		25.5	14.2, 41.4		68
Midline	1.6	0.8, 3.0	**	23.0	18.5, 28.1	n/s	48.1	43.8, 52.3	n/s	27.4	22.1, 33.4	n/s	639
Loss of land													
Baseline	^	^		^	^		^	^		^	^		29
Midline	2.1	0.7, 5.8	na	14.1	9.4, 20.6	na	43.3	35.8, 51.1	na	40.5	32.4, 49.1	na	316
Assets stolen or destroyed													
Baseline	14.2	10.0, 19.6		21.6	15.1, 30.0		33.6	22.4, 47.1		30.6	18.9, 45.5		101
Midline	2.9	1.3, 6.3	***	15.6	11.4, 21.0	n/s	44.5	39.1, 50.0	n/s	37.0	31.0, 43.5	n/s	584
Increase in food prices													
Baseline	3.2	1.6, 6.3		10.7	5.5, 20.0		34.6	28.6, 41.0		51.5	44.8, 58.2		341
Midline	0.9	0.4, 2.1	n/s	6.1	4.2, 8.7	n/s	32.1	27.7, 36.9	n/s	60.9	55.1, 66.4	*	1,571
Delay in food assistance													
Baseline	^	^		^	^		^	^		^	^		24
Midline	2.1	0.9, 4.7	na	18.9	13.7, 25.5	na	46.2	40.1, 52.5	na	32.8	24.7, 42.2	na	639
Unemployment of youth													
Baseline	12.8	6.1, 24.9		14.2	11.4, 17.5		51.3	46.3, 56.2		21.8	14.1, 32.0		77
Midline	2.6	1.5, 4.5	*	17.1	12.9, 22.3	n/s	46.7	41.4, 52.1	n/s	33.6	28.8, 38.8	*	788
Human disease outbreak													
Baseline	14.7	7.9, 25.7		28.6	20.3, 38.6		31.5	22.4, 42.2		25.2	18.2, 33.8		137
Midline	2.4	1.4, 4.0	**	18.4	14.5, 23.2	n/s	51.6	46.6, 56.6	***	27.5	21.8, 34.1	n/s	732
Severe illness in family													
Baseline	13.9	7.7, 24.0		19.1	14.5, 24.8		26.4	19.3, 35.1		40.5	35.6, 45.6		213
Midline	3.4	2.2, 5.1	*	17.7	14.0, 22.1	n/s	40.3	37.0, 43.6	**	38.6	33.0, 44.6	n/s	1,125

Shock or stress	Not severe			Somewhat severe			Severe			Extremely severe			n ^{b,c}
	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	
Death in household													
Baseline	10.0	3.0, 28.9		0.8	0.1, 7.4		26.5	12.1, 48.4		62.7	44.4, 77.9		43
Midline	8.7	4.4, 16.4	n/s	8.0	4.7, 13.4	**	28.2	21.4, 36.1	n/s	55.1	46.4, 63.5	n/s	312

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05,

** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Number of households that reported experiencing the shock or stress during the 12 months preceding the survey

^c Records missing information for severity have been excluded from the estimates. The unweighted sample size reflects this loss in observations.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.7.3: Comparison of Mean Shock Exposure Index Scores in the Uganda RFZ, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)			Midline (2022)			Sig. ^c
	Mean	95% CI	n ^{a,b}	Mean	95% CI	n ^{a,b}	
All households	18.3	16.8, 19.8	433	27.1	25.8, 28.5	1,649	***
Gendered household type							
Male and female adults	18.9	17.3, 20.5	309	27.8	26.5, 29.1	1,321	***
Female adults only	16.7	15.3, 18.2	86	25.0	23.1, 26.9	217	***
Male adults only	17.2	13.6, 20.9	37	23.0	19.3, 26.8	110	*
Children only	^	^	1	^	^	1	na
Wealth quintile							
Highest (wealthiest)	16.4	13.2, 19.5	90	24.9	21.7, 28.8	202	***
Fourth	17.9	15.3, 20.6	59	27.2	24.5, 29.8	183	***
Middle	16.2	14.9, 17.6	42	27.5	25.6, 29.4	214	***
Second	22.0	17.9, 26.1	39	26.1	24.2, 28.1	252	n/s
Lowest (poorest)	19.7	18.1, 21.2	203	27.8	26.2, 29.5	798	***
Region							
Karamoja	17.9	15.4, 20.4	215	28.5	26.3, 30.7	740	***
North	24.8	24.7, 24.9	50	19.3	17.4, 21.2	129	***
Southwest	17.8	15.6, 20.1	168	26.9	25.1, 28.8	780	***
Household education							
No education	18.1	15.9, 20.2	57	28.7	25.1, 32.3	168	***
Less than primary year 3	16.4	14.0, 18.8	40	26.3	24.4, 28.2	171	***
Completed primary year 3	20.5	18.7, 22.3	111	27.7	26.1, 29.3	445	***
Completed primary	18.3	16.7, 19.9	154	27.0	25.4, 28.7	635	***
Completed 'A' level or higher	15.3	10.5, 20.1	71	25.9	23.4, 28.5	230	***

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households for which a shock exposure index score could be calculated

^b Records missing information for the severity variables have been excluded from the estimates. The unweighted sample size reflects this loss in observations.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.7.4: Household Resilience Capacity Index in the Uganda RFZ, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)			Midline (2022)			Diff.
	Estimate	95% CI	n ^a	Estimate	95% CI	n ^a	
Household resilience capacity index	54.8	50.3, 59.3	435	39.3	37.2, 41.4	1,650	-15.5
Gendered household type							
Male and female adults	56.1	50.8, 61.4	310	40.2	38.0, 42.4	1,322	-15.9
Female adults only	51.0	43.7, 58.3	87	34.0	31.0, 37.1	217	-17.0
Male adults only	53.4	48.7, 58.1	37	39.1	34.8, 43.4	110	-14.3
Children only	^	^	1	^	^	1	na
Wealth quintile							
Highest (wealthiest)	67.6	64.2, 70.9	90	57.8	54.4, 61.3	202	-9.8
Fourth	64.9	62.0, 67.8	59	50.4	47.6, 53.3	183	-14.5
Middle	57.9	50.6, 65.1	42	47.9	45.4, 50.4	214	-10.0
Second	52.5	47.1, 58.0	40	42.2	40.4, 43.9	252	-10.3
Lowest (poorest)	39.1	34.7, 43.6	204	29.8	27.3, 32.4	799	-9.3
Shock exposure severity							
Did not experience any shocks	^	^	3	^	^	11	na
Low	56.8	51.3, 62.3	138	38.2	35.1, 41.2	550	-18.6
Moderate	51.9	46.8, 57.1	154	40.1	38.1, 42.2	564	-11.8
High	55.2	50.7, 59.7	138	40.1	37.2, 42.9	523	-15.1
Region							
Karamoja	45.9	37.0, 54.7	217	32.2	28.6, 35.7	741	-13.7
North	36.9	32.2, 41.6	50	33.0	29.3, 36.7	129	-3.9
Southwest	61.9	58.4, 65.5	168	48.3	45.3, 51.2	780	-13.6
Household education							
No education	37.9	28.6, 47.1	57	25.1	22.1, 28.1	169	-12.8
Less than primary year 3	44.0	35.9, 52.1	40	30.3	27.3, 33.2	171	-13.7
Completed primary year 3	52.6	48.8, 56.3	111	36.2	33.6, 38.9	446	-16.4
Completed primary	57.4	52.6, 62.2	156	43.6	41.4, 45.8	634	-13.8
Completed 'A' level or higher	67.1	62.8, 71.4	71	50.7	47.6, 53.7	230	-16.4

^ Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.7.5: Comparison of Absorptive, Adaptive, and Transformative Capacity Indices for Households in the Uganda RFZ 2019-2022, Feed the Future Phase Two ZOI Baseline and Midline Surveys

Index and component	Baseline (2019)					Midline (2022)					Diff.	p-value ^c	Sig. ^d
	Original scale ^a	Original estimate	Rescaled estimate ^b	Rescaled 95% CI	n	Original estimate	Rescaled estimate ^b	Rescaled 95% CI	n				
Absorptive capacity index			59.4	55.8, 62.9	435		40.4	38.2, 42.6	1,650				
Availability of informal safety nets	0-4	2.7	68.1	63.4, 72.8	435	2.0	49.5	45.8, 53.1	1,650	-18.7	<0.001	***	
Bonding social capital	0-2	1.2	60.6	55.5, 65.7	435	1.2	57.7	54.0, 61.4	1,650	-2.9	0.36	n/s	
Access to cash savings	0-1	0.7	73.3	66.2, 80.4	435	0.5	47.9	43.3, 52.5	1,650	-25.4	<0.001	***	
Access to remittances	0-1	0.1	5.6	2.9, 8.4	435	0.1	5.0	3.6, 6.4	1,650	-0.6	0.70	n/s	
Asset ownership	0-29	9.3	37.1	34.9, 39.2	435	8.2	32.8	31.1, 34.6	1,650	-4.2	0.003	**	
Shock preparedness and mitigation	0-3	1.1	37.1	32.5, 41.8	435	0.6	21.6	19.0, 24.1	1,650	-15.6	<0.001	***	
Availability of/access to humanitarian assistance	0-1	0.3	25.9	18.6, 33.2	435	0.4	35.3	28.9, 41.7	1,650	9.4	0.060	n/s	
Adaptive capacity index			46.5	42.7, 50.3	435		41.0	39.3, 42.7	1,650				
Aspirations/confidence to adapt/locus of control index	0-16	10.8	56.9	54.6, 59.1	435	10.8	56.9	55.1, 58.6	1,650	0.0	1.00	n/s	
Bridging social capital	0-2	1.1	56.6	45.5, 67.6	435	1.1	55.0	50.4, 59.6	1,650	-1.6	0.80	n/s	
Linking social capital	0-4	1.8	45.8	42.7, 49.0	435	1.2	30.0	26.3, 33.7	1,650	-15.9	<0.001	***	
Social network index	0-6	2.6	61.1	56.8, 65.5	435	1.8	41.5	37.1, 46.0	1,650	-19.6	<0.001	***	
Education/training	0-3	1.8	59.5	51.7, 67.3	435	1.5	49.7	45.3, 54.1	1,650	-9.8	0.03	*	
Livelihood diversification	0-18	3.2	27.1	23.4, 30.9	435	3.3	28.9	26.8, 31.0	1,650	1.8	0.41	n/s	
Exposure to information	0-11	4.1	37.6	31.7, 43.6	435	3.9	35.5	31.9, 39.0	1,650	-2.2	0.54	n/s	
Asset ownership	0-29	9.3	37.1	34.9, 39.2	435	8.2	32.8	31.1, 34.6	1,650	-4.2	0.003	**	
Availability of financial services	0-2	0.7	34.3	26.5, 42.2	435	0.2	11.8	7.5, 16.1	1,650	-22.5	<0.001	***	
Transformative capacity index			50.1	43.3, 57.0	435		43.0	39.7, 46.4	1,650				
Availability of/access to formal safety nets	0-5	0.8	19.0	14.7, 23.4	435	0.7	18.1	14.9, 21.3	1,650	-0.9	0.74	n/s	
Availability of markets	0-3	1.6	52.9	38.3, 67.5	435	1.7	57.0	50.3, 63.6	1,650	4.0	0.62	n/s	
Availability of/access to communal natural resources	0-4	1.1	27.4	21.7, 33.1	435	1.1	27.1	23.4, 30.9	1,650	-0.3	0.94	n/s	
Availability of/access to basic services	0-5	2.2	44.6	37.0, 52.1	435	1.6	31.1	26.6, 35.6	1,650	-13.5	0.003	**	
Availability of/access to infrastructure	0-4	1.5	51.5	42.0, 60.9	435	1.7	57.8	52.4, 63.2	1,650	6.4	0.250	n/s	
Availability of/access to agricultural services	0-2	0.8	39.9	32.7, 47.1	435	0.6	29.7	24.4, 35.0	1,650	-10.2	0.03	*	
Availability of/access to livestock services	0-2	1.1	54.3	44.0, 64.6	435	1.0	49.6	44.5, 54.8	1,650	-4.7	0.42	n/s	
Bridging social capital	0-2	1.1	56.6	45.5, 67.6	435	1.1	55.0	50.4, 59.6	1,650	-1.6	0.80	n/s	
Linking social capital	0-4	1.8	45.8	42.7, 49.0	435	1.2	30.0	26.3, 33.7	1,650	-15.9	<0.001	***	
Collective action	0-10	1.1	12.7	9.5, 16.0	435	0.6	6.4	5.1, 7.7	1,650	-6.3	<0.001	***	
Social cohesion	0-4	1.5	37.0	33.0, 41.0	435	1.2	29.8	26.7, 32.9	1,650	-7.3	0.005	**	
Household-level gender index ^e	0-4	2.3	58.7	54.8, 62.6	265	0.6	15.3	12.9, 17.8	542	-43.4	<0.001	***	
Gender equitable decision-making index ^f	0-8	4.9	61.0	58.2, 63.8	384	5.3	66.0	63.8, 68.2	1,458	5.1	0.006	**	

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a The original scale refers the range of possible scores; the maximum score in the ZOI Survey data may be less than the maximum possible score.

^b Scale: 0-100

^c Significance tests were performed to determine whether differences exist between baseline and midline estimates for the component indicators. Significance tests were not performed for the overall absorptive, adaptive, or transformative index estimates due to concerns about the comparability of baseline and midline index estimates created using factor analysis with large differences in sample size and important differences in the sampling design approaches.

^d Differences found to be statistically significant are indicated by level: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; n/s=not significant; na=not applicable.

^e The household-level gender index is not included in the calculation of the transformative capacity index because all 4 pieces of the component indicator are available for only 265 households in 2019 and 542 households in 2022 in which the resilience module respondent is married and has a child under 18 years of age.

^f The gender equitable decision-making index is not included in the calculation of the transformative capacity index because the component indicator is available for only primary adult female decision-makers; however, the results are presented for the 384 female respondents in 2019 and 1,458 female respondents in 2022 who are de jure household members who completed the Women's Empowerment in Agriculture module.

Note: See Appendix A5, Criteria used to calculate the resilience capacity indices, for information detailing how each component indicator was calculated.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.7.6: Comparison of the Index of Community Resilience for the Uganda RFZ, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Component	Baseline (2019)					Midline (2022)					p- value ^b Sig. ^c	
	Original scale	Original estimate	Rescaled estimate ^a	Rescaled 95% CI	n	Original estimate	Rescaled estimate ^b	Rescaled 95% CI	n	Diff.		
Index of community resilience			32.4	28.2, 36.5	435		27.2	25.0, 29.3				
Natural resource management groups	0-4	0.3	8.8	5.5, 12.1	435	0.2	7.3	4.8, 9.7	1,650	-1.6	0.45	n/s
Community disaster risk reduction index	0-2	0.4	24.2	20.2, 28.2	435	0.4	22.5	17.9, 27.1	1,650	-1.7	0.58	n/s
Social protection index	0-7 ^d		48.6	40.2, 57.0	435		30.9	27.4, 34.4	1,650	-17.6		
Managing and maintaining public goods	0-6	0.8	15.0	11.5, 18.5	435	0.4	7.7	6.2, 9.2	1,650	-7.3	<0.001	***
Conflict mitigation	0-2	1.6	81.3	75.8, 86.7	435	1.5	74.6	70.6, 78.6	1,650	-6.7	0.052	n/s

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Scale: 0-100

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^c Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^d The original scale listed is not the range of possible scores, but rather the number of variables that were combined using factor analysis to generate the index.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.7.7: Comparison of the Reduced Coping Strategies Index for Households in the Uganda RFZ, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Household characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^b	Sig. ^c
	Estimate	95% CI	n ^a	Estimate	95% CI	n ^a			
Overall rCSI	8.5	6.7, 10.3	435	16.1	14.7, 17.4	1,629	7.6	<0.001	***
Gendered household type									
Male and female adults	8.5	6.8, 10.1	310	16.2	14.7, 17.7	1,302	7.7	<0.001	***
Female adults only	9.0	6.0, 12.1	87	17.5	15.2, 19.8	216	8.5	<0.001	***
Male adults only	7.4	4.4, 10.3	37	11.3	9.3, 13.3	110	3.9	0.03	*
Children only	^	^	1	^	^	1	na	na	na
Region									
Karamoja	9.5	7.1, 11.9	218	19.0	17.0, 21.0	735	9.5	<0.001	***
North	9.8	4.7, 14.9	50	6.1	4.5, 7.7	126	-3.7	0.13	n/s
Southwest	7.8	4.9, 10.7	167	14.4	12.2, 16.7	768	6.6	<0.001	***
Household education									
No education	13.6	9.3, 18.0	57	22.5	19.8, 25.2	169	8.9	<0.001	***
Less than primary year 3	6.9	4.7, 9.1	40	19.5	16.9, 22.0	170	12.6	<0.001	***
Completed primary year 3	10.9	8.9, 13.0	110	17.1	15.2, 19.0	441	6.2	<0.001	***
Completed primary	7.6	5.4, 9.9	157	14.9	13.4, 16.4	624	7.3	<0.001	***
Completed 'A' level or higher	4.2	1.3, 7.1	71	9.9	8.4, 11.5	225	5.7	<0.001	***
Wealth quintile									
Highest (wealthiest)	4.4	3.6, 5.1	90	8.3	6.4, 10.2	197	3.9	<0.001	***
Fourth	6.9	1.1, 12.8	59	13.0	10.8, 15.3	183	6.1	0.06	n/s
Middle	9.5	8.1, 10.9	42	14.1	12.0, 16.3	211	4.6	<0.001	***
Second	11.8	5.4, 18.2	39	16.3	13.8, 18.8	251	4.5	0.19	n/s
Lowest (poorest)	11.2	8.8, 13.6	205	18.9	17.1, 20.7	787	7.7	<0.001	***
Shock exposure severity									
Did not experience any shocks	^	^	3	^	^	11	na	na	na
Low	4.7	2.9, 6.6	138	13.5	11.8, 15.2	545	8.8	<0.001	***
Moderate	8.8	5.9, 11.7	153	16.3	14.6, 18.0	553	7.5	<0.001	***
High	12.3	9.6, 15.1	138	18.3	16.0, 20.6	517	6.0	0.001	**

^a Results not statistically reliable, n<30

CI=confidence interval, rCSI=Reduced Coping Strategies Index, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^c Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Disaggregates based on individual household members (i.e., gendered household type, household education, wealth quintile, and poverty status) are calculated using de jure household members.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.7.8: Comparison of Coping Strategies included in the Reduced Coping Strategies Index Used by Households in the Uganda RFZ During the 7 Days Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Coping strategy	Practiced coping strategy						
	Baseline (2019)		Midline (2022)		Diff.	p-value ^a	Sig. ^b
	(%)	95% CI	(%)	95% CI			
Ate less expensive or preferred foods	66.0	58.9, 72.4	84.9	81.4, 87.9	18.9	<0.001	***
Limited portions at mealtime	33.1	27.9, 38.9	77.7	72.7, 82.1	44.6	<0.001	***
Reduced the number of meals per day	27.2	22.9, 32.0	75.1	69.7, 79.9	47.9	<0.001	***
Borrowed food or money from friends and relatives	34.9	29.8, 40.3	59.9	55.6, 63.9	25.0	<0.001	***
Limited adult intake ^c	16.4	12.6, 21.1	41.0	36.4, 45.9	24.6	<0.001	***
Number of households (n)	435		1,629				

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^b Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^c The numerator includes only households with at least one child under 5 years of age in the household roster that reported limiting adult food intake.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 5.7.9: Comparison of the Mean Number of Days Each Coping Strategy included in the Reduced Coping Strategies Index Was Used During the 7 Days Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Coping strategy	Mean number of days practiced								
	Baseline (2019)			Midline (2022)			Diff.	p-value ^b	Sig. ^c
	Estimate	95% CI	n ^a	Estimate	95% CI	n ^a			
Ate less expensive or preferred foods	3.9	3.6, 4.1	292	3.9	3.7, 4.1	1,378	0.0	0.88	n/s
Limited portions at mealtime	3.9	3.4, 4.3	149	3.6	3.4, 3.9	1,265	-0.3	0.39	n/s
Reduced the number of meals per day	4.2	3.7, 4.7	127	3.9	3.7, 4.1	1,216	-0.3	0.31	n/s
Borrowed food or money from friends and relatives	2.6	2.2, 2.9	156	2.5	2.4, 2.7	957	-0.1	0.76	n/s
Limited adult intake ^d	3.6	2.8, 4.3	76	3.2	2.9, 3.5	659	-0.4	0.39	n/s

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Includes only households in the Uganda RFZ that used the coping strategy during the week preceding the survey

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^c Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^d Includes only households that have at least one child under 5 years of age in the household roster

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.2.1: Comparison of 5DE Scores and 5DE Components in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Indicator	Baseline (2019)			Midline (2022)			Diff.	p-value ^b	Sig. ^c
	Est.	95% CI	n ^a	Est.	95% CI	n ^a			
Phase two ZOI									
5DE score	0.92	0.91, 0.93	1,938	0.90	0.88, 0.92	2,335	-0.02	0.07	n/s
18-29 years	0.88	0.86, 0.90	537	0.85	0.81, 0.88	640	-0.03	0.05	n/s
30 years or older	0.93	0.93, 0.94	1,401	0.92	0.90, 0.94	1,695	-0.01	0.16	n/s
Women achieving empowerment (%)	75.8	72.9, 78.5	1,938	72.2	68.0, 76.1	2,335	-3.6	0.16	n/s
18-29 years	65.6	60.3, 70.5	537	60.6	54.0, 66.9	640	-5.0	0.24	n/s
30 years or older	79.5	76.5, 82.2	1,401	76.3	72.0, 80.2	1,695	-3.2	0.21	n/s
Average adequacy score of disempowered women ^d	0.67	0.66, 0.68	533	0.64	0.63, 0.66	711	-0.03	0.002	**
18-29 years	0.65	0.64, 0.67	198	0.61	0.58, 0.63	258	-0.04	0.002	**
30 years or older	0.68	0.67, 0.69	335	0.66	0.65, 0.68	453	-0.02	0.046	*
Resilience Focus Zone									
5DE score	0.93	0.91, 0.96	342	0.88	0.85, 0.93	1,277	-0.05	0.02	*
18-29 years	0.91	0.87, 0.94	97	0.85	0.81, 0.88	370	-0.06	<0.001	***
30 years or older	0.94	0.92, 0.97	245	0.89	0.87, 0.92	907	-0.05	0.004	**
Women achieving empowerment (%)	79.8	72.6, 85.5	342	67.1	60.7, 72.9	1,277	-12.7	0.005	**
18-29 years	75.6	66.2, 83.1	97	59.7	51.6, 67.3	370	-15.9	0.86	n/s
30 years or older	81.4	73.7, 87.2	245	70.1	63.6, 75.9	907	-11.3	0.87	n/s
Average adequacy score of disempowered women ^d	0.67	0.65, 0.69	82	0.63	0.61, 0.65	422	-0.04	0.013	*
18-29 years	^	^	26	0.64	0.59, 0.64	151	na	na	na
30 years or older	0.69	0.66, 0.71	56	0.64	0.62, 0.66	271	-0.05	0.006	**

^a Results not statistically reliable, n<30

5DE=Five Domains of Empowerment, CI=confidence interval, Est.=estimate, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of primary adult female decision-makers who are de jure household members

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^c Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^d Also referred to as the proportion of weighted indicators in which disempowered individuals have adequate achievements. The range of the average adequacy score is 0-1.

Notes:

Estimates are sample-weighted; numbers of observations are unweighted.

Women are not included in the estimates if they do not participate in agriculture and therefore do not have input into productive decisions or if they are missing a value for any of the other five Abbreviated Women's Empowerment in Agriculture Index indicators.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.2.2: Comparison of Empowerment among Women in the Uganda ZOI, in Total and by Selected Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Characteristic	Baseline (2019)			Midline (2022)			Sig. ^c
	%	95% CI	n ^{a,b}	%	95% CI	n ^{a,b}	
All women	75.8	72.9, 78.5	1,938	72.2	68.0, 76.1	2,335	n/s
Wealth quintile							
Highest (wealthiest)	84.5	77.8, 89.5	293	83.1	77.2, 87.8	352	n/s
Fourth	78.8	74.0, 82.8	339	80.8	75.6, 85.1	367	n/s
Middle	76.0	70.1, 81.1	373	74.0	66.5, 80.3	391	n/s
Second	74.9	69.7, 79.5	403	65.3	57.6, 72.4	435	*
Lowest (poorest)	67.0	60.6, 72.8	530	60.9	53.3, 67.9	790	n/s
Shock exposure severity							
Did not experience any shocks	^	^	20	^	^	7	na
Low	74.5	70.2, 78.5	576	68.2	62.9, 73.0	698	n/s
Moderate	76.5	71.5, 80.8	633	78.5	74.1, 82.3	784	n/s
High	76.7	72.8, 80.2	701	69.6	62.1, 76.2	843	n/s
Region							
East	57.2	52.0, 62.3	565	56.7	40.3, 71.8	242	n/s
Karamoja	75.8	63.7, 84.9	170	53.2	40.4, 65.6	560	**
North	73.0	67.6, 77.8	488	73.6	66.6, 79.6	310	n/s
Southwest	84.7	80.6, 88.0	715	80.3	76.1, 84.0	1,223	n/s
Resilience Focus Zone							
Non-RFZ	75.1	71.8, 78.2	1,596	73.1	68.3, 77.4	1,058	n/s
RFZ	79.8	72.6, 85.5	342	67.1	60.7, 72.9	1,277	**

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of primary adult female decision-makers who are de jure household members

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Notes:

Estimates are sample-weighted; numbers of observations are unweighted.

Women are not included in the estimates if they do not participate in household agricultural production and therefore do not have input into productive decisions or if they are missing a value for any of the other five Five Domains of Empowerment indicators.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.3.1: Comparison of Adequate Achievement in Each A-WEAI indicator among Women with Data for All Six A-WEAI indicators in the Uganda ZOI, in Total and by Age, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

A-WEAI indicator and women's age (years)	Baseline (2019)		Midline (2022)		Sig. ^a
	%	95% CI	%	95% CI	
Input in productive decisions	97.7	96.9, 98.3	98.9	98.2, 99.3	**
18-29 years	95.8	93.7, 97.2	97.1	94.8, 98.3	n/s
30+ years	98.4	97.4, 99.0	99.6	99.0, 99.8	**
Ownership of assets	98.1	97.2, 98.7	94.0	92.1, 95.4	***
18-29 years	97.5	95.6, 98.6	87.9	83.8, 91.1	***
30+ years	98.3	97.4, 98.9	96.1	94.3, 97.3	**
Access to and decisions on credit	60.4	57.2, 63.5	64.3	60.2, 68.1	n/s
18-29 years	49.8	44.3, 55.2	57.7	52.1, 63.0	*
30+ years	64.2	60.8, 67.4	66.6	62.3, 70.6	n/s
Control over income	97.1	96.0, 97.9	98.6	97.7, 99.2	*
18-29 years	95.6	93.1, 97.2	96.4	93.0, 98.1	n/s
30+ years	97.7	96.4, 98.5	99.4	98.8, 99.7	**
Group membership	83.0	80.3, 85.3	77.8	73.7, 81.5	*
18-29 years	73.9	69.4, 78.0	65.3	58.4, 71.6	*
30+ years	86.2	83.6, 88.4	82.2	77.9, 85.9	n/s
Workload	67.0	63.5, 70.3	55.1	51.4, 58.8	***
18-29 years	67.1	61.8, 72.0	53.6	47.5, 59.7	**
30+ years	67.0	63.4, 70.4	55.6	51.8, 59.4	***
Number of all women (n)^b	1,938		2,335		
Number of women 18-29 years (n)^b	537		640		
Number of women 30 years or older (n)^b	1,401		1,695		

A-WEAI=Abbreviated Women's Empowerment in Agriculture Index, CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Number of primary adult female decision-makers who are de jure household members

Notes:

Estimates are sample-weighted; numbers of observations are unweighted.

Estimates are equivalent to uncensored headcounts.

Women are not included in the estimates if they do not participate in household agricultural production and therefore do not have input into productive decisions or if they are missing a value for any of the other five A-WEAI indicators.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.3.2: Comparison of Adequate Achievement in Each A-WEAI indicator among Disempowered Women in the Uganda ZOI, in Total and by Age, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

A-WEAI indicator and women's age (years)	Baseline (2019)		Midline (2022)		Sig. ^a
	%	95% CI	%	95% CI	
Input in productive decisions	91.3	88.7, 93.3	96.5	94.0, 98.0	***
18-29 years	89.2	84.5, 92.6	92.8	86.8, 96.2	n/s
30+ years	92.5	88.2, 95.3	98.7	96.6, 99.5	**
Ownership of assets	96.3	94.0, 97.7	85.8	81.1, 89.5	***
18-29 years	94.9	89.9, 97.5	75.8	67.5, 82.5	***
30+ years	97.1	94.9, 98.4	91.7	87.6, 94.5	**
Access to and decisions on credit	12.1	9.2, 15.7	17.1	12.9, 22.5	n/s
18-29 years	13.0	9.3, 17.9	21.0	14.6, 29.3	n/s
30+ years	11.5	8.1, 16.1	14.9	10.5, 20.7	n/s
Control over income	89.4	86.1, 92.0	95.6	92.0, 97.6	**
18-29 years	87.2	81.3, 91.4	90.8	82.3, 95.5	n/s
30+ years	90.8	86.8, 93.7	98.3	96.3, 99.2	***
Group membership	44.6	39.7, 49.6	34.8	29.0, 41.2	*
18-29 years	39.5	33.0, 46.4	30.2	21.8, 40.2	n/s
30+ years	47.7	41.5, 54.0	37.5	29.8, 46.0	n/s
Workload	42.6	37.0, 48.4	31.9	26.6, 37.7	**
18-29 years	43.6	35.6, 52.0	32.4	24.7, 41.1	n/s
30+ years	42.0	36.3, 47.9	31.6	24.7, 39.5	*
Number of all women (n)^b	533		711		
Number of women 18-29 years (n)^b	198		258		
Number of women 30 years or older (n)^b	335		453		

A-WEAI=Abbreviated Women's Empowerment in Agriculture Index, CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Number of primary adult female decision-makers who are de jure household members

Notes:

Estimates are sample-weighted; numbers of observations are unweighted.

Women are not included in the estimates if they do not participate in household agricultural production and therefore do not have input into productive decisions or if they are missing a value for any of the other five A-WEAI indicators.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.3.3: Comparison of Adequate Achievement in Each A-WEAI indicator among All Women in the Uganda ZOI, in Total and by Age, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

A-WEAI indicator and women's age (years)	Baseline (2019)			Midline (2022)			Sig. ^b
	%	95% CI	n ^a	%	95% CI	n ^a	
Input in productive decisions	97.7	96.9, 98.3	1,942	98.9	98.2, 99.3	2,337	**
18-29 years	95.8	93.7, 97.2	538	97.1	94.8, 98.3	640	n/s
30+ years	98.4	97.4, 99.0	1,404	99.6	99.0, 99.8	1,697	**
Ownership of assets	97.4	96.4, 98.1	2,186	92.4	90.6, 93.9	2,624	***
18-29 years	96.5	94.4, 97.8	642	85.9	81.9, 89.1	750	***
30+ years	97.7	96.7, 98.4	1,544	94.8	93.0, 96.1	1,874	***
Access to and decisions on credit	58.2	55.0, 61.3	2,184	62.3	58.1, 66.4	2,621	n/s
18-29 years	47.9	42.8, 53.0	642	56.0	50.3, 61.6	750	*
30+ years	62.2	58.8, 65.5	1,542	64.6	60.2, 68.8	1,871	n/s
Control over income	95.3	94.0, 96.2	2,177	95.5	94.2, 96.6	2,617	n/s
18-29 years	93.8	91.3, 95.5	641	93.8	90.8, 95.8	747	n/s
30+ years	95.8	94.5, 96.9	1,536	96.2	94.5, 97.4	1,870	n/s
Group membership	81.1	78.6, 83.4	2,186	75.6	71.4, 79.3	2,624	*
18-29 years	71.8	67.1, 76.1	642	62.9	55.9, 69.3	750	*
30+ years	84.7	82.3, 86.9	1,544	80.2	76.0, 83.9	1,874	n/s
Workload	67.3	64.1, 70.3	2,185	55.8	52.3, 59.3	2,624	***
18-29 years	66.3	61.5, 70.8	641	54.1	48.0, 60.2	750	**
30+ years	67.6	64.2, 70.8	1,544	56.4	52.8, 60.0	1,874	***

A-WEAI=Abbreviated Women's Empowerment in Agriculture Index, CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Number of primary adult female decision-makers who are de jure household members

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.4.1: Comparison of Women's Participation in Economic Activities in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Economic activity	Baseline (2019)		Midline (2022)		Sig. ^a
	%	95% CI	%	95% CI	
Any economic activity^b	96.1	94.9, 97.0	95.1	93.0, 96.6	n/s
Any agriculture economic activity^c	91.2	88.7, 93.1	91.3	88.1, 93.7	n/s
Food crop farming	89.5	86.7, 91.7	88.7	84.8, 91.7	n/s
Cash crop farming	46.7	42.9, 50.6	41.8	37.0, 46.8	n/s
Livestock raising	51.5	48.0, 55.0	61.0	56.1, 65.7	**
Fishing or fishpond culture	0.9	0.5, 1.8	0.9	0.5, 1.5	n/s
Non-farm economic activities	29.0	26.0, 32.2	28.7	25.7, 31.9	n/s
Wage or salaried employment	25.6	22.2, 29.3	25.1	21.6, 29.1	n/s
Number of women (n)^d	2,186		2,624		

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Economic activities include food crop farming, cash crop farming, livestock raising, fishing or fishpond culture, non-farm economic activities, and wage and salary employment.

^c Agriculture economic activities include food crop farming, cash crop farming, livestock raising, and fishing or fishpond culture.

^d Number of primary adult female decision-makers who are de jure household members

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.4.2: Comparison of Women's Input in Decision-making on Economic Activities in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Economic activity	Baseline (2019)			Midline (2022)			Sig. ^b
	%	95% CI	n ^a	%	95% CI	n ^a	
Any economic activity	98.5	97.8, 98.9	2,075	99.0	98.1, 99.4	2,464	n/s
Any agriculture economic activity	97.6	96.9, 98.2	1,940	99.0	98.3, 99.4	2,332	**
Food crop farming	96.4	95.3, 97.2	1,900	98.4	97.6, 99.0	2,266	***
Cash crop farming	92.7	90.1, 94.7	964	95.6	93.7, 96.9	930	*
Livestock raising	90.5	86.9, 93.1	1,079	95.8	94.3, 96.9	1,308	**
Fishing or fishpond culture	^	^	18	^	^	17	na
Non-farm economic activities	96.6	92.5, 98.5	690	97.9	95.7, 98.9	766	n/s
Wage or salaried employment	98.8	97.3, 99.5	544	99.2	97.8, 99.7	602	n/s

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Number of primary adult female decision-makers who are de jure household members and participated in the economic activity

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.4.3: Comparison of Women's Ownership of Productive Resources in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Productive resource	Baseline (2019)		Midline (2022)		Sig. ^a
	%	95% CI	%	95% CI	
At least 1 large or 2 small assets^b	97.4	96.4, 98.1	92.4	90.6, 93.9	***
Agricultural land	79.0	75.4, 82.1	58.2	54.3, 61.9	***
Large livestock	20.1	17.8, 22.6	16.3	13.6, 19.5	n/s
Small livestock	45.9	41.9, 50.0	39.8	36.2, 43.5	*
Poultry, ducks, turkeys, and pigeons	50.6	47.1, 54.2	47.0	42.9, 51.2	n/s
Fishpond or fishing equipment	0.3	0.1, 0.6	0.4	0.2, 1.0	n/s
Hand tools for farming	89.0	86.0, 91.4	85.0	82.0, 87.7	*
Non-mechanized farm equipment	5.2	4.1, 6.7	2.5	1.7, 3.5	***
Mechanized farm equipment	1.8	1.1, 2.8	0.1	0.0, 0.4	***
Non-farm business equipment	15.5	12.1, 19.6	6.4	4.7, 8.6	***
House or other structures	29.3	25.2, 33.8	25.7	20.2, 32.1	n/s
Large consumer durables	24.5	20.4, 29.0	20.1	16.8, 23.7	n/s
Small consumer durables	55.6	51.3, 59.8	56.3	50.6, 61.8	n/s
Cell phone	61.6	57.8, 65.3	50.1	45.3, 54.9	***
Non-agricultural land	7.9	6.3, 9.9	7.4	6.0, 9.0	n/s
Means of transportation	24.9	21.6, 28.5	10.5	8.3, 13.3	***
Number of women (n)^c	2,186		2,624		

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Small assets include chickens, hand tools for farming, non-mechanized farming equipment, and small consumer durables. Large assets include fishpond or fishing equipment, mechanized farm equipment, agricultural land, non-agricultural land, a house or other structures, large consumer durables, non-farm business equipment, means of transportation, large or small livestock, and cell phones.

^c Number of primary adult female decision-makers who are de jure household members

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.4.4: Comparison of Women's Credit Access in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Characteristic	Baseline (2019)			Midline (2022)			Sig. ^b
	%	95% CI	n ^a	%	95% CI	n ^a	
Household received a loan							
Any loan	62.1	58.9, 65.2	2,185	68.9	64.6, 72.9	2,621	*
In-kind loan	7.1	5.0, 10.1	2,185	6.2	4.5, 8.5	2,621	n/s
Cash loan	59.8	56.4, 63.1	2,185	68.1	63.8, 72.1	2,621	**
Source of loan							
Non-governmental organization	5.0	3.8, 6.6	2,182	5.9	4.2, 8.3	2,614	n/s
Informal lender	2.6	1.7, 4.0	2,184	4.3	3.0, 6.0	2,615	n/s
Formal lender	4.4	3.4, 5.8	2,182	6.0	4.4, 8.1	2,614	n/s
Friends or relatives	19.7	16.8, 23.0	2,172	33.5	29.9, 37.3	2,614	***
Group-based micro-finance or lending	33.2	29.5, 37.2	2,181	37.3	32.3, 42.5	2,617	n/s
Informal credit or savings groups	18.6	14.9, 22.9	2,184	21.6	16.9, 27.3	2,618	n/s
Religious group	2.1	1.4, 3.0	2,180	2.5	1.8, 3.6	2,617	n/s
Local merchant or trader	9.1	6.4, 12.7	2,181	9.8	7.4, 12.8	2,614	n/s
Other	0.8	0.5, 1.3	2,178	0.9	0.5, 1.6	2,617	n/s
Woman contributed to credit decision							
Any decision	93.6	91.4, 95.3	1,287	90.4	87.4, 92.8	1,754	n/s
On whether to borrow	91.0	88.3, 93.1	1,287	86.3	83.1, 89.0	1,754	*
On how to use loan	91.5	88.8, 93.6	1,287	88.0	84.7, 90.6	1,754	n/s

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Number of primary adult female decision-makers who are de jure household members for 'Household received a loan' and 'Source of loan' sections; number of primary adult female decision-makers who are de jure household members whose household received a loan for 'Women contributed to credit decision' section

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.4.5: Comparison of Women's Input in Decision-making on Use of Income and Household Expenditures in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Economic activity	Baseline (2019)			Midline (2022)			Sig. ^b
	%	95% CI	n ^a	%	95% CI	n ^a	
Any economic activity or household expenditures^c	98.3	97.5, 98.9	2,186	99.0	98.4, 99.4	2,624	n/s
Any economic activity^d	97.3	96.3, 98.1	2,062	98.8	97.9, 99.4	2,447	*
Any agriculture economic activity^e	96.1	94.9, 97.1	1,907	98.0	97.0, 98.6	2,306	**
Food crop farming	94.5	93.0, 95.7	1,846	97.4	96.2, 98.3	2,228	***
Cash crop farming	90.8	87.5, 93.3	958	95.7	93.3, 97.2	918	**
Livestock raising	87.5	83.0, 90.9	1,054	94.3	92.6, 95.7	1,285	**
Fishing or fishpond culture	^	^	17	^	^	17	na
Non-farm economic activities	96.9	94.4, 98.3	690	98.0	96.2, 98.9	766	n/s
Wage or salaried employment	97.6	95.4, 98.8	544	99.6	98.7, 99.9	603	*
Minor household expenditures ^f	96.3	95.2, 97.1	2,152	97.6	96.6, 98.3	2,581	*
Major household expenditures ^g	87.8	85.4, 89.9	1,457	88.4	85.3, 90.9	1,524	n/s

^ Results not statistically reliable, n<30

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Number of primary adult female decision-makers who are de jure household members and participated in the economic activity

^b Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^c Activities include food crop farming, cash crop farming, livestock raising, fishing or fishpond culture, non-farm economic activities, wage and salary employment, minor household expenditures, and major household expenditures.

^d Economic activities include food crop farming, cash crop farming, livestock raising, fishing or fishpond culture, non-farm economic activities, and wage and salary employment.

^e Agriculture economic activities include food crop farming, cash crop farming, livestock raising, and fishing or fishpond culture.

^f Minor household expenditures include food for daily consumption or other household needs.

^g Major household expenditures include large appliances such as a refrigerator.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.4.6: Comparison of Group Membership among Women in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Type of group	Baseline (2019)		Midline (2022)		Sig. ^a
	%	95% CI	%	95% CI	
Any group	81.1	78.6, 83.4	75.6	71.4, 79.3	*
Agricultural producers' group	8.8	6.9, 11.1	10.5	8.6, 12.6	n/s
Water users' group	3.3	2.3, 4.6	4.3	2.9, 6.2	n/s
Forest users' group	0.5	0.3, 0.8	1.0	0.5, 1.9	n/s
Credit or micro-finance group	49.0	45.6, 52.5	50.0	45.8, 54.2	n/s
Mutual help or insurance group	36.8	32.1, 41.6	41.0	36.2, 46.1	n/s
Trade and business association	4.2	3.1, 5.7	3.0	2.1, 4.4	n/s
Civic or charitable group	8.1	6.0, 10.7	4.5	3.0, 6.8	*
Local government	12.8	10.4, 15.6	10.1	8.3, 12.3	n/s
Religious group	40.2	36.7, 43.9	33.7	30.3, 37.1	**
Other women's group	39.3	35.4, 43.4	9.0	6.2, 12.9	***
Other group	0.5	0.2, 1.0	0.7	0.3, 1.8	n/s
Number of women (n)^b	2,186		2,624		

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Number of primary adult female decision-makers who are de jure household members

Notes:

Estimates are sample-weighted; numbers of observations are unweighted.

Estimates include all interviewed individuals, even those who reported that no group exists or that they are unaware of the existence of a group in their community. These individuals, who report that none of the groups exist or who are unaware of any groups, are counted as having inadequate achievement of this empowerment indicator.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 6.4.7: Comparison of Women's Time Allocation in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Activity	Performed activity					Mean time devoted (hours: minutes)				
	Baseline (2019)		Midline (2022)		Sig. ^a	Baseline (2019)		Midline (2022)		Sig. ^a
	%	95% CI	%	95% CI		Mean	95% CI	Mean	95% CI	
Work activities	97.5	96.5, 98.2	98.9	98.2, 99.3	**	8:46	8:31, 9:02	9:35	9:22, 9:49	***
Sleeping and resting	100.0	na	100.0	na	na	10:43	10:31, 10:55	9:32	9:19, 9:46	***
Eating and drinking	98.8	97.9, 99.3	99.1	98.5, 99.5	n/s	1:32	1:28, 1:36	1:43	1:38, 1:47	***
Personal care	85.1	80.7, 88.6	90.3	88.3, 92.1	*	0:48	0:44, 0:51	0:56	0:53, 0:59	***
School and homework ^b	1.9	1.2, 3.0	1.7	1.1, 2.6	n/s	0:02	0:01, 0:04	0:02	0:00, 0:03	n/s
Work as employed ^b	6.0	4.9, 7.3	5.2	3.8, 7.0	n/s	0:22	0:16, 0:26	0:17	0:11, 0:23	n/s
Own business work ^b	15.6	13.2, 18.2	18.9	15.6, 22.7	n/s	0:50	0:40, 1:01	0:54	0:41, 1:06	n/s
Farming, livestock, fishing ^b	61.8	57.3, 66.1	58.2	53.5, 62.8	n/s	2:49	2:35, 3:02	2:38	2:23, 2:52	n/s
Shopping, getting services ^b	9.3	7.6, 11.3	10.5	8.0, 13.6	n/s	0:10	0:07, 0:12	0:10	0:07, 0:13	n/s
Weaving, sewing, textile care ^b	2.0	1.4, 2.9	0.9	0.5, 1.4	**	0:02	0:01, 0:03	0:01	0:00, 0:01	*
Cooking ^b	89.7	87.7, 91.5	92.2	90.6, 93.6	*	2:29	2:22, 2:36	2:56	2:45, 3:07	***
Domestic work (including fetching water) ^b	70.2	66.5, 73.7	78.3	74.1, 82.0	**	1:21	1:14, 1:28	1:52	1:42, 2:01	***
Care for children, adults, elderly ^b	37.7	35.1, 40.5	40.5	36.6, 44.6	n/s	0:29	0:26, 0:34	0:33	0:29, 0:37	n/s
Commuting (for work or school) ^b	16.3	13.2, 19.9	16.7	13.7, 20.1	n/s	0:13	0:10, 0:16	0:14	0:11, 0:17	n/s
Travel (not for work or school)	35.1	31.1, 39.3	12.5	10.3, 15.2	***	0:35	0:29, 0:40	0:14	0:11, 0:17	***
Watching TV, listening to radio, reading	10.3	8.5, 12.4	14.2	11.4, 17.5	*	0:11	0:08, 0:13	0:14	0:10, 0:18	n/s
Exercising	1.2	0.6, 2.1	1.0	0.5, 1.8	n/s	0:01	0:00, 0:01	0:01	0:01, 0:02	n/s
Social activities and hobbies	32.9	29.9, 36.1	41.1	36.2, 46.1	**	0:41	0:37, 0:46	0:59	0:50, 1:08	***
Religious activities	30.2	27.2, 33.4	43.0	39.1, 46.9	***	0:38	0:32, 0:44	0:44	0:38, 0:50	n/s
Other	4.2	3.1, 5.6	0.4	0.2, 1.0	***	0:05	0:03, 0:07	0:00	0:00, 0:00	***
Number of women (n)^c	2,185		2,624			2,185		2,624		

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05,

** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Activities considered to be work in the workload indicator calculation

^c Number of primary adult female decision-makers who are de jure household members

Notes:

Estimates are sample-weighted; numbers of observations are unweighted.

Women who did not partake in an activity are included with a time of 0 hours for that activity when estimating the mean time devoted to the activity.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.1: Crops Cultivated in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Crop	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Bean																			
Baseline	80.5	77.5, 83.2		74.4		84.6		79.3		80.9		86.8		25.7		66.2		90.1	
Midline	73.0	68.2, 77.4	**	69.5	n/s	75.2	**	70.3	*	74.0	*	66.7	**	53.1	**	63.6	n/s	83.5	*
Cassava																			
Baseline	51.0	45.7, 56.2		54.4		48.6		48.6		51.7		57.1		6.6		67.4		44.6	
Midline	41.3	34.9, 48.1	*	45.1	*	39.0	*	46.5	n/s	39.4	**	55.2	n/s	4.7	n/s	59.8	n/s	32.8	n/s
Coffee																			
Baseline	13.7	10.2, 18.2		16.5		11.8		9.6		15.1		19.1		0.0		0.2		19.0	
Midline	11.3	8.4, 15.2	n/s	12.0	n/s	11.0	n/s	4.8	*	13.8	n/s	20.3	n/s	0.0	na	0.5	n/s	14.8	n/s
Maize																			
Baseline	61.6	56.8, 66.1		69.9		55.9		64.1		60.7		90.7		51.0		67.0		45.7	
Midline	59.2	54.3, 63.9	n/s	64.8	n/s	55.7	n/s	64.5	n/s	57.2	n/s	75.9	**	57.6	n/s	75.3	n/s	44.8	n/s
Matooke																			
Baseline	28.5	24.3, 33.1		29.9		27.6		26.2		29.3		39.3		0.4		2.2		39.0	
Midline	26.9	22.0, 32.5	n/s	28.4	n/s	26.0	n/s	18.2	*	30.2	n/s	47.6	n/s	0.5	n/s	1.4	n/s	35.2	n/s
Millet																			
Baseline	27.9	23.0, 33.3		21.0		32.6		20.7		30.3		28.1		13.8		23.8		31.3	
Midline	18.7	14.1, 24.5	*	12.8	*	22.4	*	16.2	n/s	19.7	**	14.8	n/s	4.3	n/s	14.0	*	25.1	n/s
Sorghum																			
Baseline	20.3	15.7, 25.8		20.2		20.4		17.0		21.4		23.5		86.9		20.9		11.5	
Midline	17.1	13.5, 21.4	n/s	16.7	n/s	17.4	n/s	17.3	n/s	17.0	n/s	15.5	n/s	81.4	n/s	10.4	*	10.2	n/s
Sweet potato																			
Baseline	46.5	40.8, 52.3		42.6		49.1		42.1		47.9		44.7		15.7		50.6		48.6	
Midline	32.6	27.9, 37.6	***	27.2	***	35.9	**	32.4	*	32.6	***	28.4	*	5.5	n/s	31.8	**	39.2	n/s
Groundnuts^b																			
Baseline	19.0	15.5, 23.1		15.0		21.7		14.1		20.6		12.5		18.9		17.2		23.1	
Midline	8.8	6.6, 11.7	***	6.8	**	10.1	***	6.1	***	9.9	***	8.8	n/s	3.9	**	4.9	**	11.6	**
Other																			
Baseline	34.5	29.9, 39.3		43.2		28.5		36.1		33.9		33.8		36.2		61.4		21.5	
Midline	37.7	32.0, 43.7	n/s	48.9	n/s	30.8	n/s	43.5	n/s	35.6	n/s	33.3	n/s	28.3	n/s	69.6	n/s	25.7	n/s
Total number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; n/s=not significant; na=not applicable.

^b Groundnut responses were recoded from the "Other" open response option.

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

Note: Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.2a: Availability of Seeds in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Seed availability	Total			Sex				Age (years)				Region							
				Male		Female		15-29		30+		East		Karamoja		North		Southwest	
	(%)	95% CI	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a
Purchased seeds for crops cultivated^b																			
Bought all seeds																			
Baseline	51.9	48.1, 55.7		55.1		49.7		57.1		50.2		65.6		60.4		40.3		49.9	
Midline	57.3	52.2, 62.3	n/s	61.9	n/s	54.4	n/s	63.5	n/s	55.0	n/s	52.2	n/s	72.1	n/s	59.7	***	55.7	n/s
Did not buy any seeds																			
Baseline	23.7	20.8, 26.9		18.4		27.3		18.6		25.4		18.2		15.3		19.3		29.5	
Midline	20.1	16.9, 23.8	n/s	16.6	n/s	22.3	n/s	15.8	n/s	21.7	n/s	24.4	n/s	15.1	n/s	12.8	n/s	22.7	n/s
Bought some, received some																			
Baseline	24.4	21.9, 27.0		26.4		23.0		24.3		24.4		16.2		24.3		40.4		20.6	
Midline	22.6	19.8, 25.6	n/s	21.5	*	23.2	n/s	20.7	n/s	23.3	n/s	23.3	n/s	12.8	n/s	27.5	***	21.6	n/s
Reported availability of seeds in the community^b																			
Baseline	80.7	78.1, 83.1		78.0		82.6		79.5		81.1		73.4		60.5		82.1		85.8	
Midline	73.3	68.7, 77.4	**	71.7	*	74.3	**	75.6	n/s	72.5	**	71.0	n/s	60.7	n/s	77.7	n/s	74.3	**
Total number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Sources of purchased seeds^d																			
Agriculture dealer with voucher																			
Baseline	0.5	0.2, 1.2		0.1		0.8		0.1		0.6		0.0		0.0		0.1		1.0	
Midline	0.3	0.1, 0.7	n/s	0.1	n/s	0.4	n/s	0.6	n/s	0.1	n/s	0.0	na	0.1	n/s	0.2	n/s	0.4	n/s
Breeder																			
Baseline	1.4	0.8, 2.5		1.6		1.2		1.5		1.3		0.7		1.2		0.6		2.2	
Midline	1.4	0.8, 2.4	n/s	2.0	n/s	1.1	n/s	2.4	n/s	1.0	n/s	4.4	*	2.4	n/s	0.9	n/s	0.3	*
Friend or relative																			
Baseline	20.1	17.1, 23.5		20.9		19.5		20.0		20.2		12.2		15.2		26.9		21.5	
Midline	18.3	15.1, 21.9	n/s	16.7	n/s	19.3	n/s	19.8	n/s	17.7	n/s	24.1	*	20.8	n/s	18.9	n/s	15.1	n/s
Market (non-agriculture dealer)																			
Baseline	41.5	36.9, 46.3		41.2		41.8		38.6		42.6		35.1		90.8		47.4		35.7	
Midline	40.9	34.8, 47.4	n/s	41.3	n/s	40.7	n/s	42.5	n/s	40.3	n/s	17.7	**	62.9	***	61.1	n/s	35.2	n/s
Retail outlet (seed grower or vendor)																			
Baseline	15.7	13.3, 18.4		17.9		14.0		16.0		15.6		27.6		1.7		13.5		11.8	
Midline	12.7	10.1, 16.0	n/s	12.3	*	13.0	n/s	9.7	*	14.0	n/s	23.3	n/s	6.6	n/s	8.7	*	11.9	n/s
Retailer or agriculture dealer with cash																			
Baseline	28.6	25.1, 32.3		28.3		28.8		28.1		28.7		31.0		7.7		22.9		33.0	
Midline	35.5	30.6, 40.7	*	37.3	*	34.3	n/s	35.3	n/s	35.6	*	42.0	n/s	8.3	n/s	34.2	n/s	38.6	n/s

Seed availability	Total			Sex				Age (years)				Region							
				Male		Female		15-29		30+		East		Karamoja		North		Southwest	
	(%)	95% CI	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a
Wholesaler seed grower or vendor																			
Baseline	6.9	5.2, 9.2		7.6		6.4		9.0		6.2		4.6		0.0		5.8		9.8	
Midline	8.1	6.3, 10.4	n/s	10.0	n/s	6.8	n/s	9.3	n/s	7.6	n/s	8.4	n/s	5.6	**	7.2	n/s	8.9	n/s
Other																			
Baseline	1.1	0.6, 1.9		0.3		1.6		0.0		1.5		0.7		0.0		1.1		1.5	
Midline	1.2	0.8, 1.9	n/s	1.3	n/s	1.2	n/s	1.2	*	1.3	n/s	1.8	n/s	0.0	na	0.8	n/s	1.5	n/s
Number of farmers who purchased some/all seeds (n)^c																			
Baseline	1,893			878		1,015		502		1,391		602		157		548		586	
Midline	2,497			1,026		1,471		745		1,752		226		718		441		1,112	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^aSignificance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^bIncludes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^cRecords missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^dIncludes farmers who are de jure household members and purchased all or some seeds in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.2b: Reasons for Not Purchasing Seeds in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Reason for not purchasing seeds	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Did not trust quality																			
Baseline	0.3	0.1, 0.9		0.3		0.3		0.0		0.4		0.4		0.0		0.5		0.2	
Midline	1.2	0.6, 2.8	n/s	1.5	n/s	1.1	n/s	0.9	n/s	1.3	n/s	2.2	n/s	3.2	n/s	0.6	n/s	0.9	n/s
Fake seeds or packaging issues																			
Baseline	0.7	0.3, 1.4		0.9		0.5		0.7		0.6		0.0		0.0		1.2		0.6	
Midline	0.2	0.0, 1.2	n/s	0.6	n/s	0.0	n/s	0.8	n/s	0.0	n/s	0.8	n/s	0.4	n/s	0.0	n/s	0.0	n/s
High cost																			
Baseline	7.5	5.5, 10.1		8.8		6.6		9.6		6.9		9.8		8.7		8.8		5.8	
Midline	13.6	10.6, 17.3	**	18.7	**	11.0	n/s	16.6	n/s	12.8	**	20.8	*	24.4	n/s	17.9	*	7.4	n/s
Lack of information about type to buy																			
Baseline	0.3	0.1, 1.0		0.0		0.5		0.0		0.4		0.0		0.0		0.4		0.4	
Midline	0.8	0.3, 2.2	n/s	0.3	*	1.1	n/s	1.0	n/s	0.7	n/s	2.4	n/s	2.5	*	0.1	n/s	0.3	n/s
Lack of information about where to buy																			
Baseline	0.4	0.1, 1.0		0.0		0.6		0.4		0.4		0.0		0.5		0.5		0.4	
Midline	0.9	0.4, 1.8	n/s	0.8	n/s	0.9	n/s	0.4	n/s	1.0	n/s	2.4	n/s	4.4	n/s	0.5	n/s	0.0	n/s
Lack of or limited availability																			
Baseline	1.3	0.7, 2.2		2.1		0.8		1.7		1.1		0.0		1.0		2.6		0.9	
Midline	0.4	0.1, 1.0	*	0.6	n/s	0.2	n/s	1.0	n/s	0.2	*	0.0	na	6.8	n/s	0.0	**	0.0	*
Own saved seed, seed from friend or relative (not purchased)																			
Baseline	92.0	89.7, 93.9		90.6		92.9		89.8		92.7		88.8		77.9		90.6		95.2	
Midline	88.6	84.6, 91.6	n/s	84.8	n/s	90.6	n/s	88.2	n/s	88.7	n/s	85.3	n/s	53.3	*	90.3	n/s	93.1	n/s
Received from aid distribution																			
Baseline	3.2	2.2, 4.6		4.7		2.3		2.6		3.4		5.7		15.0		1.6		2.3	
Midline	1.6	1.0, 2.6	*	2.5	n/s	1.1	n/s	1.1	n/s	1.7	n/s	1.5	*	10.1	n/s	1.3	n/s	0.8	n/s
Other																			
Baseline	1.3	0.6, 2.4		1.8		1.0		3.1		0.7		0.5		0.0		2.6		0.8	
Midline	4.0	2.4, 6.8	*	3.1	n/s	4.5	**	4.3	n/s	4.0	**	0.0	n/s	2.2	n/s	9.9	n/s	3.5	*
Number of farmers who did not purchase any/some seeds (n)^b																			
Baseline	1,156			479		677		279		877		261		75		406		414	
Midline	1,396			527		869		364		1,032		145		239		282		730	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

Note: Includes farmers who are de jure household members. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.3a: Fertilizer Use in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Fertilizer use	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Applied fertilizer to crops^b																			
Baseline	18.3	15.2, 21.8		22.9		15.2		18.9		18.1		38.9		3.5		4.9		16.2	
Midline	20.5	16.6, 25.1	n/s	24.0	n/s	18.3	n/s	17.7	n/s	21.6	n/s	35.8	n/s	4.8	n/s	8.3	n/s	22.8	n/s
Total number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Reported availability of fertilizer in the community^d																			
Baseline	73.8	66.8, 79.7		73.4		74.2		82.1		70.9		78.8		0.0		52.0		70.5	
Midline	71.8	62.8, 79.4	n/s	66.7	n/s	76.0	n/s	70.7	n/s	72.2	n/s	66.0	n/s	95.3	n/s	69.1	n/s	75.2	n/s
Type of fertilizer applied^d																			
Inorganic																			
Baseline	37.4	30.6, 44.8		36.6		38.2		33.5		38.8		41.8		^		39.1		31.5	
Midline	42.8	34.6, 51.4	n/s	46.5	n/s	39.8	n/s	40.0	n/s	43.7	n/s	45.6	n/s	23.4	na	62.2	n/s	38.3	n/s
Organic																			
Baseline	42.2	34.2, 50.5		39.4		45.0		43.5		41.7		32.4		^		33.3		55.1	
Midline	45.0	35.3, 55.0	n/s	36.2	n/s	52.2	n/s	45.1	n/s	44.9	n/s	35.5	n/s	72.8	na	27.6	n/s	53.2	n/s
Both																			
Baseline	16.8	13.0, 21.4		20.3		13.2		14.7		17.5		18.9		^		25.7		13.4	
Midline	12.1	7.9, 18.0	n/s	17.2	n/s	7.9	n/s	14.9	n/s	11.2	n/s	18.9	n/s	3.8	na	10.2	n/s	8.3	n/s
Don't know																			
Baseline	3.7	0.9, 13.4		3.7		3.6		8.3		2.1		7.0		^		2.0		0.0	
Midline	0.1	0.01, 1.02	n/s	0.1	n/s	0.1	n/s	0.0	n/s	0.2	n/s	0.0	n/s	0.0	na	0.0	n/s	0.2	n/s
Purchased fertilizer^d																			
Baseline	74.0	66.5, 80.2		79.6		68.1		77.7		72.6		86.2		^		80.5		60.2	
Midline	65.2	53.0, 75.7	n/s	77.3	n/s	55.4	n/s	62.9	n/s	65.9	n/s	77.8	n/s	60.2	na	87.6	n/s	53.3	n/s
Number of farmers who applied fertilizer (n)^c																			
Baseline	468			249		219		121		347		278		10		30		150	
Midline	504			242		262		114		390		112		34		40		318	
Sources of fertilizer, if farmer purchased fertilizer^e																			
Bought from agriculture dealer with voucher																			
Baseline	1.3	0.4, 3.7		1.2		1.3		0.3		1.6		0.6		^		^		2.6	
Midline	1.3	0.4, 4.1	n/s	1.7	n/s	1.0	n/s	0.0	n/s	1.7	n/s	1.0	n/s	^	na	0.0	na	2.1	n/s
Bought from friend or relative																			
Baseline	11.2	7.4, 16.5		14.0		7.8		9.6		11.8		7.7		^		^		14.6	
Midline	7.6	4.5, 12.7	n/s	5.2	*	10.5	n/s	11.3	n/s	6.6	n/s	6.3	n/s	^	na	8.0	na	9.2	n/s

Fertilizer use	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Bought from retail outlet																			
Baseline	23.1	17.2, 30.2		19.8		27.0		25.0		22.4		28.1		^		^		16.1	
Midline	16.0	10.2, 24.3	n/s	13.8	n/s	18.5	n/s	13.5	n/s	16.8	n/s	19.7	n/s	^	na	11.7	na	14.1	n/s
Bought from retailer or agriculture dealer																			
Baseline	38.4	31.8, 45.4		37.6		39.3		37.5		38.7		36.4		^		^		39.6	
Midline	53.2	45.6, 60.5	**	59.3	**	46.2	n/s	60.6	n/s	51.0	*	51.0	n/s	^	na	82.9	na	47.5	n/s
Bought from wholesaler																			
Baseline	7.8	4.9, 12.4		7.5		8.3		10.1		7.0		3.4		^		^		16.1	
Midline	11.5	6.2, 20.4	n/s	12.8	n/s	10.1	n/s	10.7	n/s	11.8	n/s	14.0	n/s	^	na	0.0	na	12.7	n/s
Bought in market (non-agriculture dealer)																			
Baseline	18.8	12.5, 27.1		19.6		17.7		17.0		19.4		22.9		^		^		12.5	
Midline	13.7	9.0, 20.3	n/s	14.4	n/s	13.0	n/s	9.9	n/s	14.8		12.9	n/s	^	na	13.2	na	14.7	n/s
Received from aid distribution																			
Baseline	0.2	0.0, 1.7		0.4		0.0		0.0		0.3		0.4		^		^		0.0	
Midline	0.6	0.1, 2.7	n/s	0.3	n/s	1.0	n/s	0.0	na	0.8	n/s	0.0	n/s	^	na	0.0	na	1.3	n/s
Used compost or manure from own animals																			
Baseline	16.2	10.7, 23.8		17.6		14.4		19.0		15.1		18.1		^		^		16.2	
Midline	9.1	5.7, 14.3	n/s	8.3	n/s	10.0	n/s	6.3	n/s	9.9	n/s	10.3	n/s	^	na	0.7	na	9.0	n/s
Other																			
Baseline	0.4	0.1, 1.6		0.4		0.3		0.5		0.3		0.3		^		^		0.0	
Midline	2.8	1.1, 6.9	n/s	2.7	n/s	2.9	n/s	1.8	n/s	3.0	n/s	0.0	n/s	^	na	0.0	na	6.2	*
Number of farmers who purchased fertilizer (n)^c																			
Baseline	354			201		153		94		260		240		0		24		90	
Midline	331			181		150		74		257		89		18		33		191	

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes farmers who are de jure household members and applied fertilizer in the 12 months preceding the survey

^e Includes all farmers who are de jure household members and purchased fertilizer in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.3b: Reasons for Not Using Fertilizer in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Reason for not using fertilizer ^b	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Beliefs and misconceptions																			
Baseline	6.9	5.4, 8.7		6.7		7.0		4.5		7.6		7.5		7.1		9.0		5.5	
Midline	8.9	7.3, 10.8	n/s	9.0	n/s	8.8	n/s	9.4	**	8.7	n/s	3.4	n/s	21.0	**	12.2	n/s	6.3	n/s
Did not trust quality																			
Baseline	2.3	1.6, 3.4		2.2		2.3		1.3		2.6		1.8		0.8		2.0		2.8	
Midline	3.1	1.7, 5.4	n/s	3.3	n/s	2.9	n/s	2.1	n/s	3.4	n/s	3.6	n/s	1.8	n/s	1.8	n/s	3.9	n/s
Fake fertilizer or packaging																			
Baseline	0.8	0.5, 1.4		0.9		0.8		0.8		0.8		0.9		0.5		0.5		1.0	
Midline	2.0	1.1, 3.6	n/s	2.3	n/s	1.8	n/s	2.2	n/s	1.9	n/s	9.4	**	1.8	n/s	0.2	n/s	0.5	n/s
High cost																			
Baseline	72.2	68.7, 75.5		71.9		72.4		70.6		72.8		82.1		57.6		66.2		73.8	
Midline	77.7	74.6, 80.5	*	79.0	*	76.9	n/s	76.6	n/s	78.1	*	89.2	n/s	35.4	*	82.3	***	79.9	n/s
Lack of information about type to buy																			
Baseline	6.1	4.6, 8.0		6.3		5.9		7.3		5.7		0.5		10.0		7.1		7.0	
Midline	7.0	5.4, 9.2	n/s	5.9	n/s	7.7	n/s	6.5	n/s	7.3	n/s	3.6	**	18.9	n/s	4.6	n/s	7.2	n/s
Lack of information about where to buy																			
Baseline	5.6	4.3, 7.2		5.7		5.5		6.9		5.2		0.5		12.0		6.3		6.3	
Midline	6.2	4.6, 8.3	n/s	6.0	n/s	6.3	n/s	8.2	n/s	5.4	n/s	6.9	**	19.3	n/s	3.1	n/s	5.0	n/s
Lack of knowledge about value added																			
Baseline	8.0	6.2, 10.3		7.4		8.4		7.7		8.1		1.4		27.0		11.0		6.5	
Midline	13.0	10.7, 15.8	**	14.9	***	11.9	n/s	14.4	**	12.5	*	2.4	n/s	46.6	**	17.0	n/s	7.4	n/s
Limited availability																			
Baseline	12.6	10.0, 15.7		15.5		10.7		11.9		12.8		2.6		17.7		19.9		11.5	
Midline	11.7	9.4, 14.6	n/s	12.2	n/s	11.5	n/s	12.4	n/s	11.5	n/s	9.8	*	33.0	n/s	12.5	n/s	7.5	n/s
Other																			
Baseline	1.5	0.9, 2.2		1.2		1.6		1.6		1.4		2.9		0.6		1.3		1.1	
Midline	10.4	8.4, 12.8	***	10.9	***	10.1	***	12.0	***	9.7	***	6.8	n/s	1.9	n/s	16.2	***	10.0	***
Number of farmers who did not apply fertilizer (n)^c																			
Baseline	1,965			819		1,146		504		1,461		467		182		642		674	
Midline	2,683			1,030		1,653		802		1,881		188		823		504		1,168	
Reasons for not purchasing fertilizer^d																			
Did not trust quality																			
Baseline	0.9	0.1, 10.2		0.0		1.5		^		1.2		3.4		^		^		0.0	
Midline	1.4	0.3, 7.2	n/s	0.4	n/s	1.9	n/s	4.1	na	0.6	n/s	^	na	^	na	^	na	0.4	n/s
Fake fertilizer or packaging																			
Baseline	0.0	na		0.0		0.0		^		0.0		0.0		^		^		0.0	
Midline	5.2	1.5, 16.9	n/s	9.6	n/s	3.4	n/s	10.9	na	3.3	n/s	^	na	^	na	^	na	0.5	n/s

Reason for not using fertilizer ^b	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male		Female		15-29		30+		East		Karamoja		North		Southwest	
				(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a		
High cost																			
Baseline	43.0	31.0, 55.7		46.9		40.4		^		40.7		50.5		^		^		41.2	
Midline	29.7	16.8, 46.9	n/s	36.4	n/s	26.9	n/s	33.9	na	28.3	n/s	^	na	^	na	^	na	23.9	n/s
Lack of information about type to buy																			
Baseline	1.9	0.3, 9.7		0.0		3.1		^		2.4		0.0		^		^		2.9	
Midline	1.1	0.2, 6.0	n/s	0.0	n/s	1.6	n/s	0.5	na	1.4	n/s	^	na	^	na	^	na	1.3	n/s
Lack of information about where to buy																			
Baseline	0.7	0.1, 7.9		0.0		1.2		^		1.0		0.0		^		^		1.2	
Midline	3.5	0.7, 15.4	n/s	4.3	n/s	3.2	n/s	4.5	na	3.2	n/s	^	na	^	na	^	na	0.3	n/s
Limited availability																			
Baseline	3.6	1.1, 11.4		1.1		5.3		^		4.6		1.6		^		^		4.9	
Midline	5.2	1.4, 17.9	n/s	8.4	n/s	3.9	n/s	6.0	na	4.9	n/s	^	na	^	na	^	na	3.8	n/s
Preferred to use compost or manure from own animals																			
Baseline	41.6	31.0, 52.9		32.3		47.6		^		40.6		38.0		^		^		41.5	
Midline	62.9	44.5, 78.2	*	60.7	*	63.8	n/s	51.4	na	66.7	*	^	na	^	na	^	na	67.9	*
Received from aid distribution or free of charge																			
Baseline	7.3	2.8, 17.8		11.9		4.3		^		8.2		9.9		^		^		5.5	
Midline	7.6	3.1, 17.3	n/s	3.3	n/s	9.3	n/s	12.2	na	6.0	n/s	^	na	^	na	^	na	7.2	n/s
Other																			
Baseline	5.4	1.4, 18.4		8.9		3.2		^		7.0		1.6		^		^		7.7	
Midline	4.6	1.8, 11.1	n/s	4.9	n/s	4.5	n/s	8.7	na	3.2	n/s	^	na	^	na	^	na	4.7	n/s
Number of farmers who applied but did not purchase fertilizer (n) ^c																			
Baseline	114			48		66		27		87		38		10		6		60	
Midline	173			61		112		40		133		23		16		7		127	

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes farmers who are de jure household members and did not apply any fertilizer in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes farmers who are de jure household members and did not purchase fertilizer in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.4a: Pesticide Use in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Pesticide use	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male		Female		15-29		30+		East		Karamoja		North		Southwest	
				(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a		
Applied pesticides ^b																			
Baseline	33.4	28.9, 38.2		43.8		26.3		33.6		33.4		45.9		15.8		41.0		25.4	
Midline	36.3	31.1, 42.0	n/s	46.2	n/s	30.2	n/s	34.0	n/s	37.2	n/s	53.7	n/s	6.1	*	42.1	n/s	31.5	n/s
Total number of farmers (n) ^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Sources of pesticides applied ^d																			
Bought from agriculture dealer with voucher																			
Baseline	1.6	0.9, 2.7		1.1		2.1		1.3		1.6		0.7		^		3.0		1.2	
Midline	1.4	0.6, 3.3	n/s	1.9	n/s	1.0	n/s	0.0	n/s	1.9	n/s	1.4	n/s	0.0	na	1.9	n/s	1.2	n/s
Bought from friend or relative																			
Baseline	5.8	3.5, 9.4		6.8		4.7		9.8		4.4		10.8		^		3.9		2.9	
Midline	4.1	2.5, 6.7	n/s	2.0	*	6.0	n/s	4.9	n/s	3.8	n/s	8.1	n/s	8.0	n/s	5.2	n/s	0.4	*
Bought from retail outlet																			
Baseline	23.3	18.5, 28.9		21.7		25.0		23.6		23.1		28.2		^		12.0		29.3	
Midline	18.8	13.6, 25.3	n/s	20.4	n/s	17.3	n/s	19.7	n/s	18.5	n/s	25.9	n/s	13.5	n/s	9.5	n/s	19.9	n/s
Bought from retailer or agriculture dealer																			
Baseline	38.5	33.5, 43.7		40.1		36.6		38.6		38.4		34.1		^		46.5		38.0	
Midline	59.6	52.6, 66.3	***	59.9	***	59.4	***	64.2	**	58.1	***	54.9	**	20.3	n/s	66.4	*	59.8	**
Bought from wholesaler																			
Baseline	5.0	3.1, 7.8		3.9		6.2		5.6		4.8		2.7		^		1.8		9.8	
Midline	10.3	6.4, 16.2	*	10.7	*	10.0	n/s	9.9	n/s	10.5	n/s	15.8	**	5.7	n/s	2.1	n/s	12.0	n/s
Bought in market (non-agriculture dealer)																			
Baseline	27.4	21.6, 34.1		30.5		23.9		25.7		27.9		27.9		^		35.5		20.6	
Midline	16.8	11.6, 23.7	*	18.2	*	15.5	n/s	15.6	n/s	17.3	*	18.0	n/s	32.1	n/s	23.2	n/s	11.4	n/s
Made at home																			
Baseline	3.2	2.0, 5.1		2.6		3.8		1.1		3.9		2.1		^		3.1		3.5	
Midline	0.2	0.1, 0.5	***	0.3	*	0.0	**	0.1	n/s	0.2	***	0.0	*	4.4	n/s	0.2	n/s	0.1	*
Received from aid distribution																			
Baseline	2.7	1.6, 4.6		3.2		2.1		1.9		3.0		1.8		^		0.8		1.7	
Midline	0.4	0.2, 1.1	**	0.6	*	0.3	*	0.0	n/s	0.5	**	0.6	n/s	15.4	*	0.0	n/s	0.1	n/s
Other																			
Baseline	0.7	0.3, 1.7		0.5		0.9		0.9		0.6		0.7		^		0.7		0.5	
Midline	2.0	0.8, 5.2	n/s	1.5	n/s	2.5	n/s	1.9	n/s	2.1	n/s	3.7	n/s	2.0	n/s	1.1	n/s	1.5	n/s

Pesticide use	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male		Female		15-29		30+		East		Karamoja		North		Southwest	
				(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a
Number of farmers who applied pesticides (n) ^c																			
Baseline	834			464		370		209		625		334		27		253		220	
Midline	813			417		396		208		605		155		44		166		448	

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes farmers who are de jure household members and applied pesticides in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.4b: Reasons for Not Applying Pesticides in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

pesticides	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male	Female			15-29		30+		East		Karamoja		North		Southwest	
				(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a		
Beliefs and misconceptions																			
Baseline	6.0	4.7, 7.6		6.6		5.7		4.2		6.6		3.9		19.8		2.9		6.3	
Midline	6.9	5.6, 8.7	n/s	10.7	*	5.1	n/s	6.8	n/s	7.0	n/s	5.4	n/s	13.7	n/s	7.8	*	5.4	n/s
Did not trust quality																			
Baseline	2.1	1.5, 3.1		2.5		1.9		1.4		2.4		2.7		0.7		2.4		2.0	
Midline	1.4	0.9, 2.3	n/s	2.1	n/s	1.1	n/s	1.5	n/s	1.4	n/s	1.7	n/s	1.7	n/s	0.3	**	1.7	n/s
High cost																			
Baseline	65.7	61.9, 69.3		63.0		67.1		63.1		66.5		82.3		54.6		54.9		65.2	
Midline	69.5	65.3, 73.4	n/s	70.1	n/s	69.3	n/s	69.8	n/s	69.4	n/s	78.4	n/s	43.0	n/s	72.4	**	72.1	n/s
Lack of information about type to buy																			
Baseline	7.2	5.7, 9.1		6.5		7.6		7.2		7.3		3.0		11.4		6.6		8.5	
Midline	8.8	6.9, 11.1	n/s	8.8	n/s	8.7	n/s	10.6	n/s	8.1	n/s	10.2	n/s	22.0	n/s	6.0	n/s	6.5	n/s
Lack of information about where to buy																			
Baseline	6.3	4.6, 8.6		6.9		6.0		6.8		6.1		2.8		10.8		5.1		7.4	
Midline	6.4	4.6, 8.9	n/s	8.4	n/s	5.5	n/s	9.1	n/s	5.4	n/s	11.3	*	22.1	n/s	3.6	n/s	2.6	**
Lack of knowledge on value added																			
Baseline	7.5	5.7, 9.8		6.3		8.1		7.2		7.6		3.0		16.3		8.8		7.6	
Midline	12.5	10.5, 14.9	**	13.4	***	12.1	*	12.9	*	12.4	**	5.0	n/s	42.1	***	12.6	n/s	7.8	n/s
Not available or lack of market for agricultural inputs																			
Baseline	9.1	7.1, 11.5		12.2		7.4		7.7		9.5		2.9		20.5		16.3		7.1	
Midline	6.5	4.8, 8.8	n/s	7.0	*	6.3	n/s	8.9	n/s	5.6	*	11.7	*	22.1	n/s	3.4	***	2.8	n/s
Pesticides not needed																			
Baseline	21.1	18.2, 24.3		22.0		20.6		24.0		20.1		8.5		13.0		29.2		23.4	
Midline	19.0	15.8, 22.6	n/s	16.4	n/s	20.2	n/s	21.2	n/s	18.1	n/s	10.8	n/s	6.6	n/s	34.4	n/s	17.8	n/s
Other																			
Baseline	1.1	0.6, 2.0		1.0		1.1		0.7		1.2		2.2		1.4		0.5		0.9	
Midline	4.1	2.7, 6.3	**	6.0	***	3.2	n/s	4.2	*	4.1	**	0.7	n/s	0.4	n/s	7.1	***	4.7	*
Number of farmers who did not apply pesticide (n) ^b																			
Baseline	1,599			604		995		416		1,183		411		165		419		604	
Midline	2,374			855		1,519		708		1,666		145		813		378		1,038	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

Note: Includes farmers who are de jure household members and did not apply pesticides in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.5: Agricultural Equipment Use in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Equipment	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Drip irrigation																			
Baseline	0.6	0.3, 1.1		0.5		0.6		0.7		0.5		1.3		0.5		0.2		0.4	
Midline	1.0	0.6, 1.7	n/s	1.5	*	0.7	n/s	1.0	n/s	1.1	n/s	2.3	n/s	0.2	n/s	0.9	n/s	0.7	n/s
Hoe																			
Baseline	98.8	98.2, 99.3		98.7		98.9		97.7		99.2		97.6		99.4		99.5		99.1	
Midline	97.9	97.1, 98.6	n/s	97.5	n/s	98.2	n/s	97.5	n/s	98.1	*	96.2	n/s	96.0	n/s	99.1	n/s	98.4	n/s
Mechanized dryers																			
Baseline	0.1	0.0, 0.2		0.1		0.0		0.0		0.1		0.2		0.0		0.0		0.0	
Midline	0.4	0.2, 1.0	*	0.7	n/s	0.3	n/s	0.6	n/s	0.4	n/s	0.4	n/s	0.0	na	0.5	n/s	0.5	n/s
Plow																			
Baseline	29.2	25.5, 33.2		34.6		25.5		25.5		30.4		13.8		52.8		50.5		24.0	
Midline	20.9	16.0, 26.9	*	27.5	n/s	16.8	*	24.1	n/s	19.7	**	38.0	**	25.3	*	45.5	n/s	1.2	***
Power or motorized tiller																			
Baseline	0.0	na		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
Midline	0.4	0.2, 0.8	**	0.5	n/s	0.4	*	0.6	n/s	0.4	*	1.0	n/s	0.6	n/s	0.3	n/s	0.2	n/s
Rotator for irrigation																			
Baseline	0.1	0.0, 0.4		0.2		0.1		0.2		0.1		0.4		0.0		0.2		0.0	
Midline	0.8	0.4, 1.5	*	1.1	n/s	0.6	*	1.2	*	0.6	n/s	2.0	n/s	0.0	na	0.4	n/s	0.6	*
Solar dryers or drying racks																			
Baseline	0.1	0.0, 0.4		0.0		0.2		0.0		0.1		0.0		0.0		0.0		0.2	
Midline	0.8	0.5, 1.5	**	0.9	*	0.8	*	0.4	n/s	1.0	*	0.8	n/s	0.2	n/s	1.1	n/s	0.8	n/s
Tractor																			
Baseline	0.9	0.4, 1.9		1.1		0.7		0.5		1.0		0.7		6.0		0.9		0.4	
Midline	4.0	2.1, 7.6	*	4.6	*	3.6	*	3.5	*	4.2	*	2.2	n/s	6.6	n/s	3.9	n/s	4.4	n/s
Walking tractor																			
Baseline	0.1	0.0, 0.3		0.1		0.0		0.0		0.1		0.0		0.5		0.2		0.0	
Midline	0.8	0.4, 1.6	**	1.3	*	0.5	*	0.3	n/s	1.0	*	0.8	n/s	0.5	n/s	0.9	n/s	0.8	n/s
Shellers^b																			
Baseline																			
Midline	2.0	1.3, 3.0		2.4		1.8		2.5		1.8		2.2		0.1		1.5		2.5	
Threshers^b																			
Baseline																			
Midline	1.8	1.2, 2.7		2.4		1.4		2.0		1.7		1.0		0.1		2.1		2.2	
Mills^b																			
Baseline																			
Midline	11	3.2, 14.6		12.4		10.1		8.8		11.8		18		0.4		3.3		13.6	

Equipment	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Chippers^b																			
Baseline																			
Midline	0.5	0.2, 0.9		0.9		0.2		0.7		0.4		0.8		0.3		0.3		0.4	
Other																			
Baseline	5.5	4.1, 7.3		5.7		5.4		4.1		5.9		5.8		3.8		0.7		7.9	
Midline	5.6	3.8, 8.3	n/s	5.0	n/s	6.0	n/s	2.6	n/s	6.8	n/s	4.4	n/s	0.3	n/s	0.0	n/s	9.8	n/s
None																			
Baseline	0.0	0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0	
Midline	0.0	0.0	na	0.0	na	0.0	na	0.0	na	0.0	na	0.0	na	0.0	na	0.0	na	0.0	na
Number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b These response options were not included in the baseline questionnaire.

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

Note: Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.6a: Information Sources for Planting and Selling Crops Used by Farmers in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Information on planting and selling crops	Total			Sex				Age (years)				Region							
				Male		Female		15-29		30+		East		Karamoja		North		Southwest	
	(%)	95% CI	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a
Received information on planting and selling crops^b																			
Baseline	26.2	23.4, 29.1		28.9		24.3		27.4		25.8		23.6		24.3		21.4		29.9	
Midline	17.1	14.8, 19.6	***	20.6	**	14.9	***	18.7	**	16.5	***	21.4	n/s	11.1	**	17.8	n/s	15.9	***
Total number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Sources of information^d																			
Agriculture extension worker																			
Baseline	7.3	5.4, 9.9		8.9		6.0		3.3		8.7		3.4		10.8		7.5		8.5	
Midline	7.1	4.0, 12.5	n/s	5.9	n/s	8.2	n/s	9.9	n/s	6.0	n/s	5.0	n/s	10.3	n/s	11.3	n/s	5.7	n/s
Agro-input dealer																			
Baseline	5.6	3.7, 8.4		8.1		3.6		3.0		6.5		8.1		0.0		8.2		4.2	
Midline	6.2	4.0, 9.7	n/s	10.6	n/s	2.5	n/s	7.9	n/s	5.5	n/s	7.3	n/s	0.9	n/s	11.2	n/s	3.6	n/s
Farmer's group or association																			
Baseline	10.5	7.4, 14.7		11.3		9.8		6.8		11.8		4.0		7.6		16.2		11.3	
Midline	9.0	6.2, 12.8	n/s	10.6	n/s	7.6	n/s	11.7	n/s	7.8	n/s	5.1	n/s	7.1	n/s	18.2	n/s	6.4	n/s
Friend or neighbor																			
Baseline	23.0	18.5, 28.1		18.5		26.5		26.6		21.7		13.8		47.5		23.4		24.3	
Midline	20.1	15.9, 25.0	n/s	20.4	n/s	19.8	n/s	33.2	n/s	14.5	*	12.2	n/s	10.0	***	43.3	*	13.0	*
Other farmer																			
Baseline	10.3	8.0, 13.2		9.2		11.2		9.7		10.5		18.9		8.4		7.6		8.0	
Midline	9.9	5.8, 16.4	n/s	12.5	n/s	7.6	n/s	13.3	n/s	8.5	n/s	1.7	***	3.1	n/s	29.1	n/s	4.8	n/s
Trader or market stakeholder																			
Baseline	4.1	2.5, 6.6		5.7		2.8		4.0		4.1		5.2		0.0		4.0		4.0	
Midline	2.9	1.4, 5.6	n/s	1.8	n/s	3.8	n/s	3.9	n/s	2.4	n/s	1.4	n/s	0.0	na	5.1	n/s	2.8	n/s
Village agent																			
Baseline	4.1	2.5, 6.6		5.0		3.4		4.0		4.2		8.7		2.4		3.0		2.9	
Midline	7.3	4.2, 12.3	n/s	8.1	n/s	6.6	n/s	7.0	n/s	7.4	n/s	17.8	n/s	9.0	n/s	5.9	n/s	2.0	n/s
Other																			
Baseline	2.0	1.2, 3.2		1.6		2.3		2.0		2.0		4.4		1.6		3.0		0.7	
Midline	1.4	0.7, 3.1	n/s	1.2	n/s	1.7	n/s	0.7	n/s	1.8	n/s	0.0	***	3.1	n/s	1.5	n/s	2.0	n/s
Number of farmers who received any information (n)^c																			
Baseline	629			305		324		179		450		183		42		154		250	
Midline	570			275		295		172		398		63		96		135		276	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes all farmers who are de jure household members and received information on planting and selling crops in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.6b: Methods of Getting Information for Planting and Selling Crops Used by Farmers in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Information on planting and selling crops	Total			Sex				Age (years)				Region							
				Male		Female		15-29		30+		East		Karamoja		North		Southwest	
	(%)	95% CI	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a
Received information on planting and selling crops^b																			
Baseline	26.2	23.4, 29.1		28.9		24.3		27.4		25.8		23.6		24.3		21.4		29.9	
Midline	17.1	14.8, 19.6	***	20.6	**	14.9	***	18.7	**	16.5	***	21.4	n/s	11.1	**	17.8	n/s	15.9	***
Total number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Methods of getting information^d																			
Face-to-face meeting																			
Baseline	19.1	14.8, 24.3		14.9		22.6		14.8		20.7		13.9		31.4		19.6		19.9	
Midline	31.9	25.9, 38.7	**	29.6	**	34.0	*	35.1	**	30.6	*	33.7	*	34.7	n/s	56.3	**	17.5	n/s
Internet																			
Baseline	0.2	0.1, 1.0		0.3		0.2		0.5		0.2		0.6		0.0		0.0		0.2	
Midline	0.3	0.0, 2.3	n/s	0.7	n/s	0.0	n/s	0.0	n/s	0.5	n/s	0.0	n/s	0.0	na	0.0	na	0.7	n/s
Mobile phone messaging																			
Baseline	0.4	0.1, 1.5		0.2		0.6		0.7		0.3		0.9		0.0		0.0		0.4	
Midline	0.6	0.1, 2.3	n/s	0.1	n/s	0.9	n/s	1.7	n/s	0.1	n/s	1.5	n/s	0.0	na	0.5	n/s	0.1	n/s
Newspaper																			
Baseline	0.8	0.3, 2.3		0.9		0.8		0.3		1.0		0.5		0.0		1.1		0.9	
Midline	0.0	na	n/s	0.0	n/s	0.0	n/s	0.0	n/s	0.0	n/s	0.0	n/s	0.0	na	0.0	n/s	0.0	n/s
Phone call																			
Baseline	0.6	0.2, 1.8		0.2		0.9		0.0		0.8		0.0		0.0		0.5		0.9	
Midline	6.8	3.6, 12.5	**	5.4	*	8.0	*	5.8	n/s	7.3	**	12.1	n/s	6.3	*	6.1	*	4.4	n/s
Radio program																			
Baseline	48.1	43.4, 52.8		48.7		47.6		47.8		48.2		42.2		21.5		43.7		54.2	
Midline	51.8	44.3, 59.2	n/s	53.3	n/s	50.6	n/s	45.2	n/s	54.6	n/s	55.6	n/s	33.2	n/s	39.1	n/s	58.8	n/s
Television																			
Baseline	3.6	1.9, 6.5		4.1		3.2		5.6		2.8		1.3		0.0		0.3		5.8	
Midline	4.9	2.7, 8.7	n/s	6.6	n/s	3.4	n/s	3.3	n/s	5.6	n/s	3.5	n/s	0.0	na	1.4	n/s	8.2	n/s
Other																			
Baseline	2.0	1.2, 3.2		1.6		2.3		2.0		2.0		4.4		1.6		3.0		0.7	
Midline	1.4	0.7, 3.1	n/s	1.2	n/s	1.7	n/s	0.7	n/s	1.8	n/s	0.0	***	3.1	n/s	1.5	n/s	2.0	n/s
Number of farmers who received any information (n)^c																			
Baseline	629			305		324		179		450		183		42		154		250	
Midline	570			275		295		172		398		63		96		135		276	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes all farmers who are de jure household members and received information on planting and selling crops in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.6c: Types of Information Used for Planting and Selling Crops Used by Farmers in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Information on planting and selling crops	Total			Sex				Age (years)				Region							
				Male		Female		15-29		30+		East		Karamoja		North		Southwest	
	(%)	95% CI	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a
Received information on planting and selling crops^b																			
Baseline	26.2	23.4, 29.1		28.9		24.3		27.4		25.8		23.6		24.3		21.4		29.9	
Midline	17.1	14.8, 19.6	***	20.6	**	14.9	***	18.7	**	16.5	***	21.4	n/s	11.1	**	17.8	n/s	15.9	***
Total number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Types of information used^d																			
Agro-input dealer																			
Baseline	6.7	4.3, 10.3		9.4		4.4		4.5		7.4		7.0		2.4		15.1		4.0	
Midline	15.9	10.4, 23.6	*	14.7	n/s	16.9	**	18.2	*	14.9	n/s	28.0	*	7.1	n/s	23.7	n/s	6.0	n/s
Availability of equipment and machinery																			
Baseline	1.6	0.7, 3.4		1.3		1.8		1.8		1.5		0.0		0.1		1.4		2.3	
Midline	5.3	2.8, 10.1	*	4.4	n/s	6.2	*	12.7	*	2.3	n/s	8.5	n/s	2.8	n/s	12.4	*	0.1	*
Availability of seeds																			
Baseline	26.5	22.7, 30.6		26.7		26.3		27.4		26.2		30.3		20.3		33.7		23.0	
Midline	25.6	20.6, 31.4	n/s	25.6	n/s	25.7	n/s	33.0	n/s	22.5	n/s	27.5	n/s	22.0	n/s	37.6	n/s	18.6	n/s
Crop disease and health issues																			
Baseline	21.4	17.3, 26.1		25.3		18.2		15.6		23.4		15.2		39.7		28.0		20.0	
Midline	25.4	20.0, 31.6	n/s	28.5	n/s	22.7	n/s	28.6	n/s	24.0	n/s	32.6	**	16.5	n/s	35.0	n/s	17.3	n/s
Crop rotation																			
Baseline	44.9	37.7, 52.3		41.9		47.3		36.8		47.7		24.9		22.1		32.1		59.0	
Midline	36.7	30.1, 44.0	n/s	38.5	n/s	35.3	n/s	43.8	n/s	33.7	**	55.2	**	24.0	n/s	51.1	*	20.3	***
Information on weather																			
Baseline	11.5	8.8, 14.9		12.1		11.0		8.8		12.5		12.2		49.5		11.9		7.9	
Midline	15.1	11.3, 19.8	n/s	16.7	n/s	13.6	n/s	17.1	n/s	14.2	n/s	11.9	n/s	12.7	**	16.9	n/s	16.1	*
Prices of inputs																			
Baseline	22.3	18.5, 26.5		22.3		22.2		24.2		21.6		23.3		53.0		26.2		17.9	
Midline	27.3	21.2, 34.4	n/s	30.9	n/s	24.2	n/s	33.1	n/s	24.9	n/s	17.0	n/s	14.5	***	52.0	*	21.1	n/s
Prices of outputs																			
Baseline	18.6	14.7, 23.4		21.2		16.5		20.4		18.0		17.3		62.8		25.7		12.9	
Midline	16.5	12.2, 21.9	n/s	18.3	n/s	14.9	n/s	19.9	n/s	15.1	n/s	13.1	n/s	4.6	***	36.1	n/s	9.2	n/s
Type of crops																			
Baseline	34.9	30.4, 39.7		35.1		34.8		35.4		34.7		28.0		49.7		44.5		33.0	
Midline	50.7	43.3, 58.1	***	48.1	*	52.9	***	56.4	*	48.3	**	65.1	***	28.4	n/s	50.3	n/s	45.6	*

Information on planting and selling crops	Total			Sex				Age (years)				Region							
				Male		Female		15-29		30+		East		Karamoja		North		Southwest	
	(%)	95% CI	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a
Use of fertilizer																			
Baseline	26.2	21.9, 31.1		32.1		21.4		25.6		26.4		39.0		8.2		13.9		27.1	
Midline	38.0	31.3, 45.1	**	36.1	n/s	39.6	***	35.2	n/s	39.2	*	54.8	n/s	8.5	n/s	34.2	**	34.3	n/s
Other																			
Baseline	2.2	1.2, 3.8		2.5		1.9		1.3		2.5		3.1		0.0		4.4		1.2	
Midline	3.5	2.0, 6.2	n/s	3.4	n/s	3.6	n/s	1.8	n/s	4.2	n/s	0.9	n/s	1.6	n/s	0.0	n/s	7.1	*
Number of farmers who received any information (n)^c																			
Baseline	629			305		324		179		450		183		42		154		250	
Midline	570			275		295		172		398		63		96		135		276	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes all farmers who are de jure household members and received information on planting and selling crops in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.7a: Reasons for Growing Crops and Methods Used for Collecting Crops in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Purpose and harvesting method	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Reason for growing crops																			
Grown for home consumption only																			
Baseline	43.1	39.3, 46.7		31.9		50.5		46.4		41.9		52.1		55.8		27.2		44.9	
Midline	42.3	37.9, 46.8	n/s	32.6	n/s	48.3	n/s	43.6	n/s	41.8	n/s	46.5	n/s	87.7	***	25.4	n/s	41.0	n/s
Grown for market only																			
Baseline	2.1	1.4, 3.0		3.4		1.2		3.1		1.7		1.2		0.0		3.2		2.1	
Midline	1.8	1.2, 2.7	n/s	3.0	n/s	1.1	n/s	3.7	n/s	1.2	n/s	1.5	n/s	0.3	n/s	4.0	n/s	1.2	n/s
Grown for both home consumption and market																			
Baseline	54.8	51.3, 58.4		64.5		48.3		50.5		56.3		46.5		44.2		69.6		53.0	
Midline	55.8	51.5, 60.0	n/s	64.2	n/s	50.5	n/s	52.8	n/s	56.9	n/s	52.0	n/s	11.9	***	70.6	n/s	57.6	n/s
Other																			
Baseline	0.1	0.0, 0.4		0.1		0.0		0.1		0.0		0.2		0.0		0.0		0.0	
Midline	0.1	0.0, 0.5	n/s	0.2	n/s	0.0	na	0.0	n/s	0.2	n/s	0.0	n/s	0.2	n/s	0.0	na	0.2	n/s
Methods of collecting crops from garden or field^b																			
Baskets																			
Baseline	21.0	17.6, 24.7		17.1		23.5		18.4		21.8		7.7		0.6		22.2		29.0	
Midline	25.9	21.7, 30.6	n/s	21.3	n/s	28.7	n/s	23.2	n/s	26.9	n/s	18.1	**	5.6	**	21.3	n/s	34.7	n/s
Buckets																			
Baseline	21.4	18.5, 24.6		23.5		20.0		23.0		20.9		19.1		26.7		30.8		17.4	
Midline	22.2	18.3, 26.6	n/s	23.3	n/s	21.4	n/s	24.8	n/s	21.2	n/s	25.3	n/s	12.5	**	37.5	n/s	15.1	n/s
Drums																			
Baseline	1.3	0.8, 2.0		1.3		1.2		1.0		1.3		0.6		0.4		0.9		1.9	
Midline	1.6	1.0, 2.4	n/s	1.9	n/s	1.3	n/s	1.8	n/s	1.5	n/s	2.1	n/s	3.0	n/s	1.0	n/s	1.4	n/s
Own garment or clothes																			
Baseline	0.8	0.5, 1.3		1.3		0.5		0.6		0.9		1.4		0.0		1.9		0.1	
Midline	16.6	13.2, 20.7	***	14.2	***	18.1	***	15.3	***	17.1	***	17.1	***	43.1	***	9.1	*	15.6	***
Tarpaulins																			
Baseline	17.6	15.0, 20.5		19.1		16.5		15.0		18.4		16.4		6.5		15.2		20.5	
Midline	21.2	18.1, 24.7	n/s	20.8	n/s	21.5	*	18.7	n/s	22.2	n/s	21.9	n/s	14.3	n/s	13.3	n/s	26.0	n/s
Wooden boxes																			
Baseline	8.7	6.8, 11.0		8.7		8.7		7.9		8.9		4.9		45.3		6.9		7.6	
Midline	1.3	0.8, 2.0	***	2.1	***	0.8	***	1.0	***	1.4	***	1.2	**	0.8	***	1.2	***	1.4	***
Woven, single-layer bags																			
Baseline	84.4	81.7, 86.7		84.6		84.2		85.5		84.0		89.5		78.8		80.9		84.1	
Midline	82.2	79.2, 84.8	n/s	82.7	n/s	81.8	n/s	83.2	n/s	81.8	n/s	82.3	*	56.8	**	90.0	*	82.6	n/s
Other																			
Baseline	4.7	3.4, 6.5		5.9		3.9		5.7		4.4		2.4		4.3		10.1		3.3	
Midline	13.6	10.8, 17.1	***	14.2	**	13.3	***	12.8	**	14.0	***	7.5	*	2.7	n/s	16.3	n/s	16.7	***

Purpose and harvesting method	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male		Female		15-29		30+		East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
				(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a								
Total number of farmers (n) ^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Multiple responses allowed. Percentages may add up to more than 100.

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

Note: Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.7b: Methods of Transporting Harvested Crops in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Transportation method	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Methods of transporting harvested crops from garden or field to drying or storage location ^b																			
By bicycle																			
Baseline	16.9	14.1, 20.2		25.0		11.4		16.8		16.9		13.0		6.2		41.6		7.9	
Midline	19.4	15.0, 24.7	n/s	25.7	n/s	15.5	n/s	22.4	n/s	18.3	n/s	21.3	n/s	4.3	n/s	44.6	n/s	9.0	n/s
By motorbike or boda boda																			
Baseline	9.7	8.1, 11.7		10.6		9.1		9.3		9.9		9.3		3.7		11.0		10.0	
Midline	13.2	10.0, 17.2	n/s	15.3	n/s	11.9	n/s	12.4	n/s	13.5	n/s	25.3	**	2.3	n/s	12.4	n/s	10.4	n/s
By wheelbarrow or cart																			
Baseline	1.7	1.0, 2.8		2.4		1.1		1.5		1.7		2.5		0.9		2.6		0.8	
Midline	1.1	0.7, 1.9	n/s	2.2	n/s	0.5	n/s	1.7	n/s	0.9	n/s	1.8	n/s	0.6	n/s	1.7	n/s	0.7	n/s
On a cow-driven cart																			
Baseline	0.1	0.0, 0.2		0.1		0.1		0.0		0.1		0.2		0.4		0.0		0.0	
Midline	0.1	0.0, 0.5	n/s	0.0	n/s	0.1	n/s	0.0	na	0.1	n/s	0.0	n/s	0.1	n/s	0.3	n/s	0.0	na
On a donkey or mule																			
Baseline	0.3	0.1, 0.7		0.4		0.3		0.2		0.3		0.6		3.0		0.0		0.0	
Midline	0.3	0.1, 0.9	n/s	0.5	n/s	0.2	n/s	0.5	n/s	0.3	n/s	1.4	n/s	0.0	n/s	0.2	n/s	0.0	na
On a truck																			
Baseline	1.3	0.8, 2.0		2.0		0.8		0.3		1.6		1.0		2.6		1.5		1.2	
Midline	1.7	0.7, 4.3	n/s	2.4	n/s	1.3	n/s	1.1	n/s	1.9	n/s	1.2	n/s	0.3	n/s	1.9	n/s	2.1	n/s
On a tractor																			
Baseline	0.3	0.1, 1.3		0.3		0.3		0.5		0.3		0.0		4.0		0.1		0.2	
Midline	0.0	na	n/s	0.0	n/s	0.0	n/s	0.0	n/s	0.0	n/s	0.0	na	0.0	n/s	0.0	n/s	0.0	n/s
On public transportation																			
Baseline	0.4	0.2, 1.0		0.8		0.1		0.2		0.5		0.1		0.0		0.6		0.5	
Midline	0.6	0.2, 1.6	n/s	0.7	n/s	0.6	n/s	0.0	n/s	0.9	n/s	0.4	n/s	0.0	na	1.7	n/s	0.3	n/s
Walked or carried crops on head																			
Baseline	87.9	85.5, 89.9		84.5		90.2		85.7		88.6		89.6		90.1		82.1		89.6	
Midline	85.8	82.3, 88.7	n/s	83.1	n/s	87.5	n/s	85.5	n/s	85.9	n/s	79.5	*	89.0	n/s	83.0	n/s	89.2	n/s
Other																			
Baseline	1.2	0.8, 1.9		0.6		1.6		1.5		1.1		0.8		2.3		0.4		1.7	
Midline	4.3	2.9, 6.3	***	4.2	***	4.3	*	3.2	n/s	4.7	***	3.5	n/s	8.2	*	1.9	n/s	5.1	*
Total number of farmers (n) ^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^aSignificance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^bMultiple responses allowed. Percentages may add up to more than 100.

^cRecords missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

Note: Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.8a: Methods of Drying and Storing Harvested Crops Used in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Drying method	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Dry their harvested crops ^b																			
Baseline	96.8	95.7, 97.7		96.9		96.8		96.1		97.1		98.2		97.4		97.0		96.0	
Midline	94.0	92.2, 95.4	**	93.3	*	94.4	*	94.8	n/s	93.6	**	93.7	*	91.6	*	96.6	n/s	93.2	*
Total number of farmers (n) ^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Methods of drying crops ^d																			
Bare ground (dirt)																			
Baseline	57.7	53.2, 62.1		56.2		58.8		58.7		57.4		57.9		57.0		76.7		48.4	
Midline	62.4	55.5, 68.7	n/s	64.9	n/s	60.8	n/s	66.7	n/s	60.7	n/s	55.8	n/s	39.9	n/s	79.2	n/s	60.4	n/s
Drying yard with cemented ground																			
Baseline	0.3	0.2, 0.6		0.4		0.3		0.3		0.3		0.2		0.0		0.5		0.3	
Midline	0.4	0.1, 1.3	n/s	0.4	n/s	0.3	n/s	0.3	n/s	0.4	n/s	0.9	n/s	0.0	na	0.0	n/s	0.4	n/s
Ground covered with straw																			
Baseline	0.9	0.5, 1.8		0.7		1.1		1.5		0.7		0.8		0.0		2.1		0.4	
Midline	1.6	0.7, 3.7	n/s	1.2	n/s	1.8	n/s	2.4	n/s	1.2	n/s	1.6	n/s	0.8	n/s	0.3	n/s	2.3	n/s
Ground plastered with cow dung																			
Baseline	5.3	4.0, 7.0		7.2		4.0		5.0		5.4		8.1		47.8		4.1		0.0	
Midline	5.2	4.0, 6.7	n/s	5.8	n/s	4.9	n/s	6.6	n/s	4.7	n/s	3.3	*	45.0	n/s	3.5	n/s	0.3	n/s
Left to dry on plant																			
Baseline	7.5	5.7, 9.8		7.2		7.7		6.1		8.0		2.3		7.7		5.8		11.0	
Midline	7.5	5.2, 10.8	n/s	7.7	n/s	7.4	n/s	4.6	n/s	8.6	n/s	1.9	n/s	3.1	n/s	5.0	n/s	11.8	n/s
Raised platform																			
Baseline	5.1	3.4, 7.7		5.3		5.0		5.6		5.0		12.0		2.8		1.5		3.7	
Midline	4.7	2.9, 7.7	n/s	6.7	n/s	3.5	n/s	3.3	n/s	5.3	n/s	15.3	n/s	2.3	n/s	0.9	n/s	2.7	n/s
Tarpaulin																			
Baseline	48.3	44.4, 52.3		48.1		48.5		46.1		49.1		44.8		17.1		39.5		57.8	
Midline	43.0	38.0, 48.1	n/s	43.5	n/s	42.7	n/s	40.0	n/s	44.1	n/s	52.0	n/s	17.8	n/s	44.4	n/s	42.8	***
Other																			
Baseline	1.0	0.6, 1.6		1.3		0.8		1.1		0.9		2.2		3.0		0.4		0.5	
Midline	4.9	2.3, 10.1	*	3.6	n/s	5.6	*	3.2	n/s	5.5	n/s	4.8	n/s	4.7	n/s	0.1	n/s	7.3	n/s
Number of farmers who dry their harvested crops (n) ^c																			
Baseline	2,352			1,036		1,316		598		1,754		729		188		649		786	
Midline	2,953			1,178		1,775		850		2,103		281		775		510		1,387	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes farmers who are de jure household members and who dried their harvested crops in the 12 months preceding the survey

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.8b: Storage of Harvested Crops in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Storage method	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Store their harvested crops^b																			
Baseline	78.3	75.8, 80.5		82.8		75.2		78.0		78.3		81.6		89.8		92.3		68.6	
Midline	89.6	87.4, 91.4	***	87.6	*	90.8	***	91.5	***	88.9	***	87.7	n/s	89.1	n/s	94.5	n/s	88.0	***
Total number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Containers or materials used to store harvested crops^d																			
Buckets																			
Baseline	7.9	6.0, 10.3		7.2		8.4		6.9		8.3		5.8		12.6		8.2		8.3	
Midline	9.3	7.2, 12.0	n/s	7.6	n/s	10.4	n/s	12.6	*	8.1	n/s	15.3	*	8.4	n/s	5.6	n/s	8.9	n/s
Drums																			
Baseline	1.4	0.9, 2.3		0.7		1.9		1.1		1.5		0.6		0.0		1.5		2.0	
Midline	2.5	1.4, 4.3	n/s	2.2	n/s	2.6	n/s	1.3	n/s	2.9	n/s	2.1	n/s	4.3	n/s	0.1	*	3.5	n/s
Hermetic bags																			
Baseline	15.6	11.8, 20.2		11.2		18.9		10.8		17.2		7.0		0.4		3.8		30.3	
Midline	9.9	7.3, 13.3	*	12.4	n/s	8.5	**	12.8	n/s	8.8	**	20.7	**	6.7	**	13.6	*	4.2	***
Own garment or clothes																			
Baseline	2.4	1.6, 3.5		3.3		1.7		2.8		2.2		2.6		6.5		1.6		2.2	
Midline	8.9	7.1, 11.2	***	7.5	**	9.7	***	7.9	**	9.3	***	10.2	***	21.1	**	1.1	n/s	10.4	***
Tarpaulins																			
Baseline	7.3	5.9, 9.0		7.7		7.0		6.2		7.7		9.6		0.0		4.9		8.5	
Midline	8.9	6.9, 11.4	n/s	9.2	n/s	8.7	n/s	7.7	n/s	9.4	n/s	15.8	n/s	10.3	***	1.9	*	9.5	n/s
Woven, single-layer bags																			
Baseline	84.1	80.3, 87.3		87.3		81.7		86.4		83.3		88.0		75.0		97.4		74.4	
Midline	82.7	79.3, 85.7	n/s	81.4	*	83.6	n/s	80.9	n/s	83.5	n/s	86.3	n/s	62.4	n/s	75.8	***	88.4	***
Woven, two- or three-layer bags																			
Baseline	5.5	3.7, 7.9		4.6		6.2		4.8		5.7		3.4		1.1		7.5		5.9	
Midline	13.4	10.6, 17.0	***	15.9	***	12.0	*	12.0	**	14.0	***	31.6	***	5.5	**	6.9	n/s	10.8	n/s
Nothing, piled on ground																			
Baseline	12.7	10.4, 15.3		12.3		12.9		10.7		13.3		18.0		26.0		10.5		9.2	
Midline	9.5	7.6, 11.9	n/s	8.8	n/s	10.0	n/s	8.9	n/s	9.8	n/s	13.1	n/s	4.1	***	10.9	n/s	8.3	n/s
Other																			
Baseline	1.3	0.8, 2.1		1.7		0.9		1.0		1.4		1.3		6.1		0.4		1.2	
Midline	2.4	1.5, 4.0	n/s	2.3	n/s	2.5	*	2.4	n/s	2.4	n/s	1.4	n/s	2.0	n/s	2.1	n/s	3.1	n/s

Storage method	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Locations used to store harvested crops^d																			
Cribs																			
Baseline	2.8	1.8, 4.3		3.4		2.3		0.5		3.6		0.8		1.9		4.9		2.7	
Midline	3.5	2.6, 4.8	n/s	3.2	n/s	3.7	n/s	5.1	***	2.9	n/s	1.8	n/s	7.1	n/s	3.7	n/s	3.5	n/s
Granaries																			
Baseline	8.8	6.8, 11.3		10.3		7.7		5.5		9.9		4.2		71.8		7.8		3.6	
Midline	8.1	5.8, 11.1	n/s	9.2	n/s	7.4	n/s	9.2	n/s	7.6	n/s	6.3	n/s	58.7	n/s	4.6	n/s	1.9	n/s
Other constructed stores																			
Baseline	8.4	6.7, 10.5		9.8		7.3		9.2		8.1		7.3		4.9		4.4		12.1	
Midline	9.3	7.4, 11.6	n/s	6.6	n/s	10.9	*	4.9	*	11.0	n/s	4.9	n/s	2.1	n/s	4.7	n/s	14.7	n/s
Residential house																			
Baseline	86.4	83.2, 89.1		85.1		87.4		88.5		85.8		93.5		45.4		93.8		83.1	
Midline	86.4	83.5, 88.8	n/s	88.2	n/s	85.2	n/s	89.0	n/s	85.4	n/s	92.9	n/s	41.1	n/s	96.5	n/s	86.2	n/s
Storage silos																			
Baseline	1.5	0.8, 2.7		1.2		1.7		0.3		1.9		0.1		4.1		1.2		2.2	
Midline	1.2	0.7, 1.9	n/s	1.1	n/s	1.2	n/s	0.9	n/s	1.2	n/s	0.0	n/s	1.7	n/s	1.3	n/s	1.4	n/s
Warehouses																			
Baseline	1.5	0.9, 2.5		1.4		1.6		1.9		1.4		1.4		1.6		0.7		2.1	
Midline	1.8	1.2, 2.7	n/s	1.8	n/s	1.8	n/s	1.4	n/s	1.9	n/s	1.1	n/s	2.2	n/s	1.9	n/s	1.9	n/s
Stored harvest elsewhere																			
Baseline	0.3	0.1, 0.7		0.2		0.3		0.1		0.3		0.6		0.0		0.0		0.3	
Midline	1.2	0.7, 1.9	**	1.2	*	1.1	*	0.9	n/s	1.2	*	0.6	n/s	0.7	n/s	1.0	*	1.6	*
Number of farmers who store their harvested crops (n)^c																			
Baseline	1,976			910		1,066		500		1,476		611		175		620		570	
Midline	2,803			1,103		1,700		821		1,982		260		764		499		1,280	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes farmers who are de jure household members and who stored their harvested crops in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.9a: Points of Sale and Means of Transport to Market for Selling Harvested Crops in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Sale of crops	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Sell their crops^b																			
Baseline	57.0	53.3, 60.7		68.1		49.5		53.6		58.1		47.9		44.2		72.8		55.1	
Midline	57.7	53.2, 62.1	n/s	67.4	n/s	51.7	n/s	56.4	n/s	58.2	n/s	53.5	n/s	12.3	***	74.6	n/s	59.0	n/s
Total number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Points of sale of harvested crops^d																			
Outside the house or on roadside																			
Baseline	13.6	10.9, 16.8		13.4		13.7		10.4		14.5		9.9		9.9		16.3		13.7	
Midline	17.6	13.8, 22.3	n/s	12.3	n/s	21.9	*	16.6	n/s	18.0	n/s	14.1	n/s	22.2	n/s	14.0	n/s	21.0	n/s
At the local market																			
Baseline	48.4	42.5, 54.3		48.9		48.0		48.4		48.4		50.2		83.4		52.1		42.3	
Midline	41.5	34.7, 48.7	n/s	44.8	n/s	38.8	n/s	47.8	n/s	39.2	*	54.2	n/s	68.2	n/s	47.3	n/s	32.3	n/s
To a trader																			
Baseline	40.8	35.0, 46.9		41.0		40.6		40.9		40.8		30.1		5.6		42.8		47.1	
Midline	44.5	38.5, 50.6	n/s	47.1	n/s	42.4	n/s	46.5	n/s	43.7	n/s	53.7	*	12.9	n/s	43.3	n/s	42.8	n/s
To a retailer																			
Baseline	18.3	15.3, 21.8		18.4		18.3		13.3		19.9		35.3		9.0		8.0		18.5	
Midline	16.9	13.4, 21.1	n/s	19.9	n/s	14.5	n/s	15.3	n/s	17.5	n/s	29.1	n/s	8.7	n/s	20.2	**	10.6	**
To a wholesaler																			
Baseline	11.2	8.9, 14.0		11.7		10.8		11.9		11.0		9.2		13.7		15.9		8.9	
Midline	9.5	7.4, 12.2	n/s	10.6	n/s	8.7	n/s	8.9	n/s	9.8	n/s	8.9	n/s	0.8	n/s	8.7	*	10.6	n/s
Other																			
Baseline	1.7	1.1, 2.7		1.5		1.8		1.7		1.7		2.6		2.0		0.6		2.0	
Midline	4.6	3.2, 6.6	**	5.3	**	4.1	n/s	4.0	n/s	4.9	**	4.3	n/s	4.3	n/s	7.2	***	3.2	n/s
Transportation of harvested crops to point of sale^d																			
Walked or carried crops on head																			
Baseline	45.0	40.1, 50.0		39.1		50.6		46.3		44.6		58.7		77.4		36.8		41.7	
Midline	39.6	33.9, 45.5	n/s	34.9	n/s	43.3	n/s	38.6	n/s	39.9	n/s	48.6	n/s	81.0	n/s	29.6	n/s	40.8	n/s
By wheelbarrow or cart																			
Baseline	0.7	0.3, 1.3		0.8		0.5		0.0		0.8		0.3		1.0		1.2		0.4	
Midline	0.4	0.1, 1.1	n/s	0.0	*	0.6	n/s	0.4	n/s	0.3	n/s	0.0	n/s	0.0	n/s	0.4	n/s	0.5	n/s
By bicycle																			
Baseline	24.9	20.7, 29.5		28.9		21.1		25.8		24.6		17.0		12.9		44.0		17.0	
Midline	21.2	16.0, 27.4	n/s	27.8	n/s	15.8	n/s	29.2	n/s	18.3	n/s	21.7	n/s	5.3	n/s	42.1	n/s	8.7	*

Sale of crops	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
On a cow-driven cart																			
Baseline	0.0	0.0, 0.3		0.0		0.1		0.0		0.1		0.0		0.0		0.1		0.0	
Midline	0.0	na	n/s	0.0	na	0.0	n/s	0.0	na	0.0	n/s	0.0	na	0.0	na	0.0	n/s	0.0	na
On a donkey or mule																			
Baseline	0.2	0.0, 1.0		0.3		0.1		0.1		0.2		0.9		0.0		0.0		0.0	
Midline	0.1	0.0, 0.9	n/s	0.0	n/s	0.2	n/s	0.0	n/s	0.2	n/s	0.6	n/s	0.0	na	0.0	na	0.0	na
By motorbike or boda boda																			
Baseline	37.0	32.1, 42.3		37.6		36.5		34.0		38.0		30.5		15.9		37.6		41.3	
Midline	33.1	27.1, 39.8	n/s	38.1	n/s	29.1	n/s	38.5	n/s	31.2	n/s	39.8	n/s	12.6	n/s	42.0	n/s	25.9	*
On public transportation																			
Baseline	2.9	1.7, 4.7		3.5		2.3		2.2		3.1		1.3		2.1		5.6		1.9	
Midline	1.7	1.0, 2.7	n/s	2.1	n/s	1.3	n/s	1.9	n/s	1.6	n/s	4.2	n/s	0.0	n/s	1.6	n/s	0.8	n/s
On a tractor																			
Baseline	0.1	0.0, 0.4		0.1		0.2		0.0		0.2		0.5		1.0		0.0		0.0	
Midline	0.2	0.1, 0.8	n/s	0.4	n/s	0.0	n/s	0.0	na	0.3	n/s	0.0	n/s	0.0	n/s	0.0	n/s	0.4	n/s
On a truck																			
Baseline	4.5	3.3, 6.1		5.4		3.6		2.6		5.1		5.8		2.7		4.3		4.2	
Midline	1.8	1.0, 3.2	**	2.3	**	1.4	*	2.0	n/s	1.8	**	0.6	*	0.0	n/s	1.8	n/s	2.3	n/s
Other																			
Baseline	2.0	1.0, 3.6		1.5		2.3		3.1		1.6		2.5		2.1		0.7		2.5	
Midline	23.9	19.6, 28.9	***	19.8	***	27.2	***	19.7	***	25.4	***	16.1	***	4.3	n/s	17.3	***	31.5	***
Number of farmers who sell their crops (n)^c																			
Baseline	1,410			732		678		344		1,066		360		86		492		472	
Midline	1,613			734		879		420		1,193		162		121		380		950	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^aSignificance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes farmers who cultivate crops for only market production or for both home consumption and market production. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.9b: Major Problems Faced in Selling Harvested Crops in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Major problem faced	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Sell their crops ^b																			
Baseline	57.0	53.3, 60.7		68.1		49.5		53.6		58.1		47.9		44.2		72.8		55.1	
Midline	57.7	53.2, 62.1	n/s	67.4	n/s	51.7	n/s	56.4	n/s	58.2	n/s	53.5	n/s	12.3	***	74.6	n/s	59.0	n/s
Total number of farmers (n) ^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Major problems faced in selling crops ^d																			
Access to markets																			
Baseline	45.4	40.9, 50.0		48.4		42.6		42.1		46.4		56.4		34.8		44.8		41.9	
Midline	40.5	36.5, 44.6	n/s	46.1	n/s	35.9	n/s	39.2	n/s	41.0	n/s	63.8	n/s	38.7	n/s	34.7	*	35.2	n/s
High costs of agriculture inputs																			
Baseline	51.1	46.8, 55.5		55.9		46.7		53.6		50.4		59.5		21.7		59.8		44.4	
Midline	60.2	54.2, 65.9	*	63.1	n/s	57.7	**	51.8	n/s	63.3	**	75.0	n/s	31.7	n/s	41.1	**	67.4	***
High transaction cost																			
Baseline	32.0	27.8, 36.6		32.6		31.5		29.8		32.7		29.8		15.1		31.7		34.7	
Midline	26.7	23.3, 30.4	n/s	31.4	n/s	22.9	*	27.1	n/s	26.5	n/s	59.8	***	22.7	n/s	20.8	n/s	17.8	***
High transportation cost																			
Baseline	47.6	42.5, 52.7		50.2		45.1		50.9		46.5		51.1		19.2		49.6		47.2	
Midline	47.4	42.1, 52.7	n/s	54.8	n/s	41.4	n/s	44.2	n/s	48.6	n/s	78.7	***	42.0	*	43.3	n/s	38.1	n/s
Lack of grading																			
Baseline	24.1	20.0, 28.8		27.0		21.4		25.2		23.8		32.9		12.1		26.1		20.1	
Midline	25.6	21.5, 30.3	n/s	26.8	n/s	24.7	n/s	27.9	n/s	24.8	n/s	42.2	n/s	19.9	n/s	21.8	n/s	21.9	n/s
Lack of information on agriculture inputs																			
Baseline	37.3	32.9, 42.0		39.9		35.0		42.1		35.9		43.5		9.2		41.9		34.2	
Midline	46.5	41.3, 51.8	**	50.2	*	43.5	*	43.3	n/s	47.7	**	53.5	n/s	22.8	n/s	42.5	n/s	47.2	**
Lack of information on markets																			
Baseline	35.2	30.9, 39.7		40.1		30.6		37.4		34.5		51.9		8.4		34.2		30.9	
Midline	42.5	37.4, 47.7	*	47.1	n/s	38.7	*	42.0	n/s	42.7	*	56.1	n/s	27.1	n/s	36.2	n/s	41.7	**
Low price																			
Baseline	84.6	82.0, 86.9		89.0		80.5		83.7		84.9		84.1		91.3		93.5		78.6	
Midline	79.4	74.6, 83.6	*	83.6	n/s	76.1	n/s	79.9	n/s	79.3	*	81.6	n/s	77.1	n/s	88.1	n/s	73.3	*
Low quality																			
Baseline	45.0	39.6, 50.6		47.6		42.7		42.8		45.7		51.9		13.9		37.2		49.7	
Midline	48.0	43.2, 52.9	n/s	49.5	n/s	46.8	n/s	47.4	n/s	48.3	n/s	60.6	n/s	23.9	n/s	41.5	n/s	48.1	n/s
No value added																			
Baseline	35.9	31.1, 41.1		37.3		34.7		31.9		37.2		41.4		7.1		28.9		40.5	
Midline	27.7	23.0, 32.9	*	30.7	n/s	25.3	*	29.6	n/s	27.0	**	42.1	n/s	21.1	*	23.9	n/s	24.8	**

Major problem faced	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male	Sig. ^a	Female	Sig. ^a	15-29	Sig. ^a	30+	Sig. ^a	East	Sig. ^a	Karamoja	Sig. ^a	North	Sig. ^a	Southwest	Sig. ^a
				(%)		(%)		(%)		(%)		(%)		(%)		(%)		(%)	
Processing																			
Baseline	22.0	17.8, 26.8		26.5		17.8		22.8		21.8		37.7		2.9		24.8		15.1	
Midline	28.9	25.1, 32.9	*	31.5	n/s	26.7	**	28.0	n/s	29.2	*	43.9	n/s	17.3	n/s	24.6	n/s	26.2	**
Storage																			
Baseline	41.7	36.8, 46.7		46.1		37.6		49.3		39.3		51.2		11.0		54.2		32.1	
Midline	44.5	40.4, 48.8	n/s	47.1	n/s	42.5	n/s	43.9	n/s	44.8	n/s	64.6	*	16.7	n/s	42.5	n/s	39.1	n/s
Other																			
Baseline	8.1	6.6, 9.8		10.0		6.3		8.5		7.9		13.5		5.2		6.9		6.8	
Midline	6.3	4.3, 9.1	n/s	6.4	n/s	6.2	n/s	5.8	n/s	6.5	n/s	3.2	***	4.4	n/s	6.2	n/s	7.7	n/s
Number of farmers who sell their crops (n)^c																			
Baseline	1,410			732		678		344		1,066		360		86		492		472	
Midline	1,613			734		879		420		1,193		162		121		380		950	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes farmers who cultivate crops for only market production or for both home consumption and market production. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.10a: Use and Sources of Agricultural Credit in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Use and source of credit	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Took agricultural credit (cash or in-kind) ^b																			
Baseline	24.0	21.0, 27.3		23.7		24.2		21.5		24.8		12.1		42.4		20.8		29.5	
Midline	22.4	19.4, 25.8	n/s	23.5	n/s	21.7	n/s	20.2	n/s	23.2	n/s	12.6	n/s	15.4	***	20.4	n/s	28.6	n/s
Total number of farmers (n) ^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Sources of agricultural credit ^d																			
Commercial bank																			
Baseline	5.9	3.9, 8.9		9.8		3.3		0.9		7.3		7.1		2.1		4.6		6.7	
Midline	4.1	2.4, 7.1	n/s	5.8	n/s	2.9	n/s	2.0	n/s	4.8	n/s	9.8	n/s	1.3	n/s	1.2	n/s	4.3	n/s
Farmers' group																			
Baseline	14.0	10.1, 19.2		11.1		16.0		14.0		14.1		5.4		0.0		23.4		14.7	
Midline	4.7	2.8, 7.9	***	4.6	n/s	4.8	**	4.3	n/s	4.9	**	4.1	n/s	0.0	na	10.8	n/s	3.2	**
Government scheme																			
Baseline	0.4	0.1, 1.2		1.0		0.0		0.4		0.4		0.0		2.4		0.8		0.0	
Midline	0.3	0.1, 1.7	n/s	0.7	n/s	0.1	n/s	0.2	n/s	0.4	n/s	0.0	na	0.6	n/s	0.0	n/s	0.5	n/s
Input supplier																			
Baseline	3.2	1.9, 5.4		3.9		2.7		3.9		3.0		13.0		0.0		0.8		2.5	
Midline	2.1	1.1, 4.1	n/s	1.8	n/s	2.3	n/s	0.8	n/s	2.6	n/s	0.0	**	0.6	n/s	2.4	n/s	2.5	n/s
Micro-finance institution																			
Baseline	2.8	1.7, 4.7		4.0		2.1		1.4		3.3		1.8		0.0		5.7		2.5	
Midline	4.4	2.5, 7.5	n/s	5.8	n/s	3.5	n/s	4.9	n/s	4.2	n/s	8.9	n/s	0.0	na	1.2	n/s	5.1	n/s
Private money lender																			
Baseline	4.2	2.5, 7.1		6.6		2.7		6.7		3.5		10.1		2.8		1.8		4.1	
Midline	2.4	1.3, 4.4	n/s	2.7	n/s	2.2	n/s	4.6	n/s	1.7	n/s	5.9	n/s	0.0	n/s	1.2	n/s	2.4	n/s
Relative or friend																			
Baseline	17.9	14.1, 22.4		23.5		14.1		27.7		15.0		31.7		20.2		6.0		18.8	
Midline	14.6	11.4, 18.5	n/s	11.4	**	16.7	n/s	24.9	n/s	11.2	n/s	4.3	***	33.0	n/s	8.5	n/s	16.8	n/s
Savings and credit cooperative organization																			
Baseline	14.4	10.5, 19.4		13.8		14.8		10.5		15.5		18.1		0.0		9.0		17.6	
Midline	10.6	7.0, 15.8	n/s	15.4	n/s	7.3	*	8.4	n/s	11.3	n/s	14.1	n/s	1.1	n/s	0.8	**	14.2	n/s
Trader																			
Baseline	8.9	5.7, 13.6		6.2		10.7		9.6		8.7		9.9		0.0		1.0		12.8	
Midline	6.2	4.1, 9.3	n/s	7.8	n/s	5.1	n/s	6.7	n/s	6.0	n/s	17.8	n/s	2.4	n/s	1.5	n/s	6.1	n/s
Village savings and loan association																			
Baseline	43.7	36.4, 51.2		37.6		47.7		39.5		44.9		28.6		88.5		55.0		36.1	
Midline	58.2	51.5, 64.6	**	55.0	*	60.4	*	49.1	n/s	61.2	**	44.4	n/s	70.6	n/s	76.1	n/s	53.4	**

Use and source of credit	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Other																			
Baseline	0.7	0.2, 1.8		0.4		0.8		0.5		0.7		0.9		0.0		1.5		0.4	
Midline	2.3	1.3, 4.1	*	1.6	n/s	2.8	n/s	3.7	n/s	1.9	n/s	0.0	n/s	0.4	n/s	1.2	n/s	3.3	**
Number of farmers who took agricultural credit (n)^c																			
Baseline	542			234		308		122		420		92		74		133		243	
Midline	767			322		445		203		564		37		134		114		482	

CI=confidence intervals, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes farmers who are de jure household members and took agricultural credit in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.10b: Reasons for Taking Agricultural Credit in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Use and reason for taking agricultural credit	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male	Sig. ^a	Female	Sig. ^a	15-29	Sig. ^a	30+	Sig. ^a	East	Sig. ^a	Karamoja	Sig. ^a	North	Sig. ^a	Southwest	Sig. ^a
				(%)		(%)		(%)		(%)		(%)		(%)		(%)		(%)	
Took agricultural credit (cash or in-kind) ^b																			
Baseline	24.0	21.0, 27.3		23.7		24.2		21.5		24.8		12.1		42.4		20.8		29.5	
Midline	22.4	19.4, 25.8	n/s	23.5	n/s	21.7	n/s	20.2	n/s	23.2	n/s	12.6	n/s	15.4	***	20.4	n/s	28.6	n/s
Total number of farmers (n) ^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Reasons for taking agricultural credit ^d																			
Expand or purchase land																			
Baseline	13.7	10.1, 18.4		13.5		13.8		8.3		15.3		9.1		2.5		4.9		19.3	
Midline	16.0	12.1, 20.8	n/s	16.9	n/s	15.3	n/s	19.1	*	14.9	n/s	17.4	n/s	7.3	n/s	16.2	n/s	16.4	n/s
Hire labor																			
Baseline	47.7	41.5, 53.9		50.5		45.7		48.3		47.4		38.4		93.2		69.2		35.2	
Midline	46.2	37.7, 54.9	n/s	48.6	n/s	44.6	n/s	52.6	n/s	44.1	n/s	42.3	n/s	75.9	*	76.3	n/s	33.8	n/s
Improve land																			
Baseline	5.3	3.6, 7.8		7.4		3.9		5.6		5.2		7.8		0.0		5.4		5.6	
Midline	9.3	6.0, 14.1	n/s	9.3	n/s	9.3	n/s	8.0	n/s	9.7	n/s	33.5	**	0.0	na	2.4	n/s	8.1	n/s
Improve marketing of crops																			
Baseline	1.5	0.5, 4.0		3.1		0.3		1.5		1.4		2.9		0.0		0.8		1.6	
Midline	1.6	0.7, 3.4	n/s	2.0	n/s	1.3	n/s	1.3	n/s	1.6	n/s	5.3	n/s	0.0	na	0.0	n/s	1.6	n/s
Crop failure																			
Baseline	8.1	4.5, 14.0		7.5		8.4		7.2		8.3		3.0		12.4		3.1		10.1	
Midline	1.7	0.8, 3.5	**	0.2	***	2.8	n/s	1.9	n/s	1.7	*	2.0	n/s	1.2	**	0.1	n/s	2.3	*
Mitigate uncertainty due to weather																			
Baseline	1.7	0.9, 3.1		2.8		0.9		0.8		1.9		0.0		5.1		2.6		1.2	
Midline	0.3	0.1, 1.7	*	0.7	n/s	0.1	n/s	0.0	n/s	0.4	*	2.3	n/s	0.0	**	0.0	n/s	0.1	n/s
Mitigate uncertainty in market prices																			
Baseline	0.9	0.4, 2.3		1.8		0.4		0.4		1.1		0.0		5.4		1.1		0.4	
Midline	0.5	0.1, 2.0	n/s	0.0	n/s	0.8	n/s	1.7	n/s	0.0	n/s	2.9	n/s	2.3	n/s	0.0	n/s	0.0	n/s
Purchase agriculture inputs																			
Baseline	61.0	54.9, 66.7		62.5		59.9		65.0		59.8		81.8		35.6		41.2		67.3	
Midline	55.6	48.5, 62.4	n/s	52.4	n/s	57.7	n/s	47.9	*	58.0	n/s	64.3	**	26.6	n/s	52.7	n/s	57.6	n/s
Purchase or repair machinery																			
Baseline	1.8	0.8, 3.9		3.0		1.0		2.0		1.7		0.0		11.5		0.0		1.3	
Midline	0.7	0.2, 2.4	n/s	0.2	*	1.1	n/s	0.3	n/s	0.9	n/s	5.5	n/s	0.0	n/s	0.1	n/s	0.1	n/s

Use and reason for taking agricultural credit	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male		Female		15-29		30+		East		Karamoja		North		Southwest	
				(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a	(%)	Sig. ^a		
Other																			
Baseline	1.4	0.6, 3.3		1.6		1.3		1.2		1.5		1.5		0.0		3.0		1.1	
Midline	12.1	7.7, 18.4	***	13.3	***	11.3	**	12.0	*	12.1	***	2.5	n/s	2.8	n/s	6.3	n/s	16.7	***
Number of farmers who took agricultural credit (n) ^c																			
Baseline	542			234		308		122		420		92		74		133		243	
Midline	767			322		445		203		564		37		134		114		482	

CI=confidence intervals, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes farmers who are de jure household members and took agricultural credit in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.10c: Reasons for Not Taking Agricultural Credit and Membership in Farmers' Groups in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Use and reason for not taking agricultural credit	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Took agricultural credit (cash or in-kind)^b																			
Baseline	24.0	21.0, 27.3		23.7		24.2		21.5		24.8		12.1		42.4		20.8		29.5	
Midline	22.4	19.4, 25.8	n/s	23.5	n/s	21.7	n/s	20.2	n/s	23.2	n/s	12.6	n/s	15.4	***	20.4	n/s	28.6	n/s
Total number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Reasons for not taking any agricultural credit^d																			
Credit not needed																			
Baseline	55.5	51.4, 59.6		51.8		58.0		57.7		54.7		43.5		60.7		48.1		66.5	
Midline	53.9	48.9, 58.7	n/s	49.1	n/s	56.7	n/s	56.3	n/s	52.9	n/s	43.7	n/s	32.8	***	66.4	**	56.4	*
Denied credit																			
Baseline	2.1	1.4, 3.0		2.7		1.6		1.8		2.1		2.1		3.3		1.7		2.1	
Midline	4.5	3.0, 6.6	*	5.0	n/s	4.1	*	4.3	n/s	4.6	n/s	6.8	**	7.5	n/s	3.5	n/s	3.2	n/s
Lack of trust in lenders																			
Baseline	1.9	1.4, 2.6		2.5		1.5		1.3		2.1		1.6		2.7		2.3		1.8	
Midline	3.3	2.3, 4.6	*	4.8	*	2.3	n/s	3.7	n/s	3.1	n/s	9.2	***	0.9	n/s	2.0	n/s	1.5	n/s
No collateral																			
Baseline	22.0	18.6, 25.8		25.0		19.9		22.8		21.7		37.0		16.4		23.6		12.4	
Midline	20.2	17.3, 23.4	n/s	23.8	n/s	18.0	n/s	20.6	n/s	20.0	n/s	36.1	n/s	32.4	n/s	25.6	n/s	6.8	*
Not available in community																			
Baseline	7.1	5.6, 9.0		9.1		5.7		10.0		6.0		9.4		4.6		10.7		3.8	
Midline	8.0	6.4, 10.0	n/s	12.2	n/s	5.5	n/s	7.2	n/s	8.3	n/s	11.5	n/s	11.8	n/s	6.8	n/s	6.2	n/s
Not aware of services																			
Baseline	5.9	4.5, 7.7		6.1		5.7		5.8		5.9		10.0		8.5		6.5		2.8	
Midline	10.3	8.0, 13.1	**	12.0	**	9.2	*	9.0	n/s	10.8	**	11.8	n/s	5.7	n/s	7.7	n/s	11.8	***
Terms and conditions not favorable																			
Baseline	22.7	19.6, 26.1		23.1		22.4		16.6		24.8		19.9		27.1		26.8		21.7	
Midline	29.5	26.3, 32.8	**	30.2	*	29.0	*	29.8	***	29.3	n/s	27.5	n/s	33.6	n/s	34.0	n/s	27.2	n/s
Other																			
Baseline	2.0	1.3, 3.2		2.3		1.8		0.6		2.5		2.6		1.0		1.8		1.9	
Midline	7.1	5.7, 8.9	***	8.8	***	6.1	***	5.8	***	7.6	***	7.3	*	2.6	n/s	8.1	**	7.4	***
Number of farmers who did not take agricultural credit (n)^c																			
Baseline	1,891			834		1,057		503		1,388		653		118		539		581	
Midline	2,420			950		1,470		713		1,707		263		723		430		1,004	

Use and reason for not taking agricultural credit	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Have active membership in a farmers' group^b																			
Baseline	19.2	16.4, 22.3		22.4		17.0		15.0		20.6		10.7		25.8		30.0		17.4	
Midline	16.9	13.8, 20.6	n/s	19.0	n/s	15.6	n/s	15.6	n/s	17.4	n/s	15.4	n/s	9.2	n/s	26.6	n/s	14.2	n/s
Total number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Believe that an active membership in a farmers' group is useful^e																			
Baseline	88.5	84.5, 91.5		89.9		87.2		84.7		89.4		80.2		95.1		92.8		86.3	
Midline	92.1	88.9, 94.4	n/s	92.0	n/s	92.2	n/s	92.1	n/s	92.1	n/s	93.4	*	90.1	n/s	90.4	n/s	93.2	n/s
Number of farmers who are active members of a farmers' group (n)^c																			
Baseline	479			253		226		103		376		87		42		211		139	
Midline	431			199		232		101		330		44		82		133		172	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes farmers who are de jure household members and did not take agricultural credit in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

^e Includes farmers who are de jure household members and have an active membership in a farmers' group

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.11: Participation in Additional Activities for Cash or Food in the Uganda ZOI, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Additional work	Total			Sex				Age (years)				Region							
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Participated in additional activities for cash or food^b																			
Baseline	45.6	42.9, 48.4		57.6		37.4		48.0		44.8		44.3		68.6		53.2		40.1	
Midline	41.5	38.6, 44.5	*	50.1	**	36.2	n/s	43.5	n/s	40.8	n/s	48.4	n/s	30.3	***	42.9	*	39.9	n/s
Total number of farmers (n)^c																			
Baseline	2,433			1,068		1,365		625		1,808		745		192		672		824	
Midline	3,187			1,272		1,915		916		2,271		300		857		544		1,486	
Additional activities^d																			
Agriculture trader																			
Baseline	12.5	10.1, 15.4		14.2		10.7		11.4		12.9		14.5		0.7		10.7		14.7	
Midline	12.0	9.2, 15.4	n/s	15.1	n/s	9.3	n/s	15.8	n/s	10.5	n/s	18.5	n/s	0.8	n/s	15.2	n/s	8.6	*
Driver																			
Baseline	3.7	2.4, 5.6		6.5		0.8		8.4		2.0		3.3		1.9		6.3		2.6	
Midline	3.7	2.3, 6.0	n/s	5.9	n/s	1.9	n/s	2.8	*	4.1	n/s	8.0	n/s	0.4	n/s	0.9	*	3.5	n/s
Fish business																			
Baseline	1.4	0.7, 2.9		1.7		1.1		1.4		1.4		1.1		0.0		1.1		2.0	
Midline	4.1	2.4, 6.9	*	3.6	n/s	4.5	*	6.8	*	3.0	n/s	6.7	*	0.7	n/s	4.0	n/s	3.2	n/s
Government service																			
Baseline	5.4	3.9, 7.5		6.1		4.7		0.9		7.0		4.5		8.9		3.3		6.7	
Midline	6.3	3.9, 10.2	n/s	7.1	n/s	5.6	n/s	1.2	n/s	8.4	n/s	6.1	n/s	7.2	n/s	5.3	n/s	6.9	n/s
Homestead gardening																			
Baseline	23.3	19.5, 27.5		22.2		24.4		24.1		23.0		18.8		28.1		18.0		28.3	
Midline	35.3	29.3, 41.9	**	35.9	**	34.8	*	32.4	n/s	36.5	**	36.2	*	43.4	*	28.5	n/s	37.5	n/s
Input supplier																			
Baseline	3.4	2.1, 5.3		3.8		3.0		3.9		3.2		2.0		0.0		1.1		6.3	
Midline	4.4	3.0, 6.3	n/s	4.7	n/s	4.0	n/s	4.0	n/s	4.5	n/s	2.8	n/s	0.3	n/s	1.5	n/s	7.1	n/s
Livestock rearing																			
Baseline	41.2	36.9, 45.6		46.7		35.4		36.4		42.9		34.3		55.5		56.5		32.5	
Midline	35.8	30.6, 41.3	n/s	37.3	*	34.5	n/s	34.9	n/s	36.2	n/s	48.2	*	20.8	***	42.6	*	28.0	n/s
Mobile mechanic																			
Baseline	1.6	1.0, 2.5		2.9		0.3		3.0		1.1		0.2		0.9		4.7		0.5	
Midline	2.8	1.8, 4.5	n/s	4.4	n/s	1.4	n/s	2.4	n/s	3.0	*	3.2	n/s	0.0	n/s	4.0	n/s	2.3	*

Additional work	Total			Sex		Age (years)				Region									
	(%)	95% CI	Sig. ^a	Male (%)	Sig. ^a	Female (%)	Sig. ^a	15-29 (%)	Sig. ^a	30+ (%)	Sig. ^a	East (%)	Sig. ^a	Karamoja (%)	Sig. ^a	North (%)	Sig. ^a	Southwest (%)	Sig. ^a
Off-farm day labor																			
Baseline	32.8	28.7, 37.1		35.9		29.5		36.4		31.5		37.7		46.7		25.7		32.2	
Midline	42.0	37.4, 46.6	**	49.5	***	35.5	n/s	42.8	n/s	41.6	**	43.4	n/s	58.9	n/s	33.0	n/s	43.7	*
Paid domestic work																			
Baseline	8.4	6.0, 11.7		10.5		6.2		8.8		8.3		17.9		3.0		6.9		5.3	
Midline	11.7	8.8, 15.4	n/s	12.7	n/s	10.8	*	10.8	n/s	12.0	n/s	18.7	n/s	30.5	***	3.4	n/s	10.1	n/s
Private service																			
Baseline	14.1	11.6, 17.2		18.1		10.0		16.0		13.5		6.6		5.5		7.7		23.9	
Midline	22.5	17.4, 28.6	**	23.7	n/s	21.4	**	20.5	n/s	23.3	**	16.3	*	2.0	n/s	10.9	n/s	34.2	n/s
Small business																			
Baseline	37.2	33.4, 41.1		32.2		42.4		31.8		39.1		35.6		53.8		35.7		35.9	
Midline	40.9	36.4, 45.5	n/s	37.5	n/s	43.8	n/s	43.7	*	39.8	n/s	57.7	***	37.9	n/s	35.7	n/s	35.6	n/s
Other																			
Baseline	5.5	3.8, 7.8		6.4		4.5		3.7		6.1		7.3		2.8		5.5		5.0	
Midline	11.0	8.4, 14.3	**	13.4	**	9.0	*	12.9	**	10.3	n/s	7.5	n/s	5.1	n/s	17.5	**	10.2	n/s
Number of farmers who participated in additional activities (n) ^c																			
Baseline	1,148			616		532		310		838		335		130		351		332	
Midline	1,258			604		654		371		887		148		268		237		605	

CI=confidence interval, RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Associations found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Includes all farmers who are de jure household members and cultivated crops in the 12 months preceding the survey

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^d Includes all farmers who are de jure household members and participated in activities in addition to crop farming for cash or food in the 12 months preceding the survey. Multiple responses allowed. Percentages may add up to more than 100.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.12.1: Coverage of the Types of Interventions Related to Cultivating Crops Promoted by Feed the Future in the Uganda ZOI During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Type of crop cultivation intervention	All crop farmers (%)	Resilience Focus Zone	
		Non-RFZ (%)	RFZ (%)
Used agricultural credit to rent or buy mechanized farm equipment	4.5	4.3	5.7
Participated in training on the following topics:			
Growing iron-rich beans	6.2	5.9	7.4
Growing orange-fleshed sweet potatoes	7.0	6.5	9.3
Growing green-leafy vegetables	17.6	17.0	20.9
Growing soy beans	7.7	8.2	5.0
Growing seed oil crops	3.4	2.8	6.3
How to evaluate input quality	4.8	5.4	2.0
Pest or disease management practices	10.7	11.5	6.8
Post harvest management practices	8.3	8.8	5.8
Crop-related record keeping	3.8	4.0	3.0
Accessing information on market prices or set prices for farm products	7.5	8.2	3.9
Advertising or promoting farm products	4.2	4.6	1.9
Finding new places to sell farm products or finding new buyers	4.3	4.9	1.4
Number of crop farmers (n)^a	3,258	1,403	1,855
Services provided by farmers' groups			
Trainings	48.3	46.9	60.7
Mechanized equipment	3.8	4.2	0.3
Crop inputs	32.3	32.5	30.7
Processing equipment	6.1	6.3	3.9
Storage or storage equipment	4.9	5.3	1.1
Crop transport to market	5.3	5.1	7.6
Marketing	21.4	21.8	18.5
Access to markets for selling crops	10.7	10.2	15.6
Crop aggregation	1.0	0.7	3.6
Financial support	34.3	37.0	11.9
Crop insurance	0.6	0.6	0.4
Number of crop farmers who were active members of a farmers' group (n)^b	431	237	194
Price of crop inputs purchased through farmers' group			
More than outside	17.0	^	^
About the same as outside	66.4	^	^
Less than outside	16.7	^	^
Number of farmers who purchased crop inputs through a farmers' group (n)	48	25	23

^a Results not statistically reliable, n<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of farmers who are de jure household members and who were responsible for cultivating crops during the 12 months preceding the survey

^b Number of farmers who are de jure household members and who were responsible for cultivating crops during the 12 months preceding the survey and who were active members of a farmers' group during that time

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.12.2: Percent Distribution of Household Members in the Uganda ZOI who Engaged in Specific Types of Interventions Related to Cultivating Crops Promoted by Feed the Future During the 12 Months Preceding the Survey by Sex and Age, Feed the Future Phase Two ZOI and RFZ Midline Survey

Type of crop cultivation intervention	Sex of participants		Age of participants		Number of participants (n) ^a
	Male (%)	Female (%)	<30 years (%)	30+ years (%)	
Used agricultural credit to rent or buy mechanized farm equipment	44.4	55.6	17.8	82.2	166
Participated in training on the following topics:					
Growing iron-rich beans	39.9	60.1	23.5	76.5	203
Growing orange-fleshed sweet potatoes	33.3	66.7	26.4	73.6	245
Growing green-leafy vegetables	35.9	64.1	25.9	74.1	604
Growing soy beans	39.3	60.7	19.9	80.1	201
Growing seed oil crops	45.3	54.7	26.5	73.5	147
How to evaluate input quality	43.9	56.1	27.5	72.5	110
Pest or disease management practices	40.5	59.5	22.7	77.3	279
Post harvest management practices	40.6	59.4	25.9	74.1	228
Crop-related record keeping	51.5	48.5	32.8	67.2	105
Accessing information on market prices or set prices for farm products	36.9	63.1	29.1	70.9	178
Advertising or promoting farm products	31.2	68.8	34.3	65.7	92
Finding new places to sell farm products or finding new buyers	42.0	58.0	32.0	68.0	97
Services provided by farmers' groups					
Trainings	43.1	56.9	20.8	79.2	226
Mechanized equipment	^	^	^	^	11
Crop inputs	49.9	50.1	22.6	77.4	138
Processing equipment	^	^	^	^	20
Storage or storage equipment	^	^	^	^	14
Crop transport to market	^	^	^	^	23
Marketing	51.3	48.7	39.6	60.4	76
Markets for selling crops	42.9	57.1	16.1	83.9	48
Crop aggregation	^	^	^	^	9
Financial support	40.4	59.6	29.4	70.6	119
Crop insurance	^	^	^	^	2

^ Results not statistically reliable, n<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of de jure household members who participated in specific types of agriculture programming of the types promoted by Feed the Future during the 12 months preceding the survey

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.12.3: Coverage of the Types of Interventions Related to Raising Livestock Promoted by Feed the Future in the Uganda ZOI During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Type of livestock intervention	All livestock farming households (%)	Resilience Focus Zone	
		Non-RFZ (%)	RFZ (%)
Participated in training on livestock farming	9.7	9.7	9.9
Participated in training on improving health and productivity of livestock	6.7	6.6	7.5
Participated in training related to commercialization of livestock	1.4	1.5	0.9
Received support to access livestock vaccines and medication	0.9	0.9	0.9
Receive any veterinary service from a trained veterinary officer	3.4	3.4	3.5
Number of households that raised livestock or other farm animals (n)^a	2,172	1,050	1,122
Received training on milk processing and safety	0.9	0.8	1.1
Number of households that own milk-producing farm animals (n)^b	1,244	655	589

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of households that owned or raised livestock during the 12 months preceding the survey

^b Number of households that owned milk cows, other cattle, horses, donkeys, mules, goats, or sheep at the time of the survey

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 7.12.4: Percent Distribution of Household Members in the Uganda ZOI who Engaged in Specific Types of Interventions Related to Raising Livestock Promoted by Feed the Future During the 12 Months Preceding the Survey by Sex and Age, Feed the Future Phase Two ZOI and RFZ Midline Survey

Type of livestock intervention	Sex of participants		Age of participants		Number of participants (n) ^a
	Male (%)	Female (%)	<30 years (%)	30+ years (%)	
Participated in training on livestock farming	64.3	35.7	24.5	75.5	261
Participated in training on improving health and productivity of livestock	67.7	32.3	30.5	69.5	192
Participated in training related to commercialization of livestock	68.6	31.4	38.0	62.0	33
Received support to access livestock vaccines and medication	^	^	^	^	28
Receive any veterinary service from a trained veterinary officer	72.0	28.0	29.8	70.2	91
Received training on milk processing and safety	^	^	^	^	18

^ Results not statistically reliable, n<30

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of de jure household members who participated in specific types of livestock programming of the types promoted by Feed the Future during the 12 months preceding the survey

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 8.1.1: Comparison of the Percent of Women of Reproductive Age in the Uganda ZOI Consuming a Diet of Minimum Diversity, in Total and by Selected Woman and Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^c	Sig. ^d
	%	95% CI	n ^{a,b}	%	95% CI	n ^{a,b}			
All women of reproductive age	29.2	26.5, 32.2	2,318	47.1	43.3, 51.0	3,085	17.9	<0.001	***
Women's age									
15-19	32.5	27.2, 38.2	486	47.9	41.9, 54.0	672	15.4	<0.001	***
20-24	25.1	20.4, 30.5	401	52.8	45.8, 59.7	580	27.7	<0.001	***
25-29	25.7	20.3, 32.0	405	49.1	41.5, 56.7	525	23.4	<0.001	***
30-34	31.9	25.4, 39.1	304	38.5	31.6, 45.9	416	6.6	0.19	n/s
35-39	32.0	25.5, 39.3	279	41.7	34.1, 49.6	363	9.7	0.07	n/s
40-44	27.7	21.1, 35.3	224	46.7	37.8, 55.8	284	19.0	0.001	**
45-49	30.1	24.1, 36.9	219	49.5	41.2, 57.8	245	19.4	<0.001	***
Gendered household type									
Male and female adults	29.2	26.1, 32.5	1,939	47.1	43.0, 51.3	2,740	17.9	<0.001	***
Female adults only	29.2	24.0, 35.0	367	46.8	38.9, 55.0	335	17.6	<0.001	***
Male adults only	^	^	12	^	^	10	na	na	na
Children only	—	—	0	—	—	0	na	na	na
Region									
East	25.5	19.9, 32.0	691	59.2	52.4, 65.6	333	33.7	<0.001	***
Karamoja	11.0	4.2, 26.2	197	34.5	22.8, 48.4	870	23.5	0.007	**
North	34.6	28.9, 40.7	569	31.0	22.8, 40.7	418	-3.6	0.51	n/s
Southwest	30.5	26.7, 34.5	861	48.4	43.3, 53.5	1,464	17.9	<0.001	***
Resilience Focus Zone									
Non-RFZ	30.1	27.0, 33.4	1,907	48.1	43.8, 52.5	1,346	18.0	<0.001	***
RFZ	24.5	18.7, 31.5	402	41.7	35.5, 48.2	1,739	17.2	<0.001	***
Wealth quintile									
Highest (wealthiest)	42.3	37.2, 47.6	525	59.3	52.7, 65.5	526	17.0	<0.001	***
Fourth	34.9	29.2, 41.1	395	52.5	45.0, 60.0	475	17.6	<0.001	***
Middle	23.6	19.4, 28.5	421	51.6	44.9, 58.3	512	28.0	<0.001	***
Second	21.5	17.2, 26.5	439	39.2	33.7, 44.9	524	17.7	<0.001	***
Lowest (poorest)	21.8	16.3, 28.4	538	32.3	26.0, 39.4	1,048	10.5	0.02	*
Shock exposure severity									
Did not experience any shocks	17.0	7.2, 34.8	39	^	^	13	na	na	na
Low	27.8	24.0, 32.0	762	47.5	42.7, 52.5	1,036	19.7	<0.001	***
Moderate	31.4	27.0, 36.2	696	44.8	39.6, 50.2	951	13.4	<0.001	***
High	29.4	24.7, 34.6	804	48.5	43.0, 54.0	1,081	19.1	<0.001	***

^ Results not statistically reliable, n<30; — Results not available, n=0

CI=confidence interval, RFZ= Resilience Focus Zone, ZOI=Zone of Influence

^a Number of women of reproductive age (15-49 years of age) who are de facto household members

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, the sum of disaggregate sample sizes may not equal the overall sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^d Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 8.1.2: Comparison of the Percent of Women of Reproductive Age in the Uganda ZOI who Consumed Foods in Each Food Group During the 24 Hours Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Food group	Baseline (2019)		Midline (2022)		Sig. ^a
	%	95% CI	%	95% CI	
Grains, roots, and tubers	95.7	94.3, 96.8	95.5	93.9, 96.6	n/s
Legumes and beans	63.0	59.1, 66.8	78.2	74.6, 81.5	***
Nuts and seeds	26.8	23.4, 30.5	27.4	23.8, 31.4	n/s
Dairy products	20.4	17.3, 23.9	23.7	19.2, 28.9	n/s
Meat and organ meats	22.8	19.5, 26.6	32.8	29.3, 36.6	***
Eggs	5.3	4.2, 6.7	11.7	9.6, 14.1	***
Vitamin A-rich dark green leafy vegetables	49.1	45.8, 52.4	54.7	50.5, 58.9	*
Other vitamin A-rich vegetables and fruits	29.8	26.4, 33.5	43.3	38.6, 48.2	***
Other fruits	51.9	48.3, 55.5	70.5	66.9, 73.8	***
Other vegetables	19.0	16.7, 21.6	31.7	28.0, 35.6	***
Number of women of reproductive age (n)^b	2,318		3,085		

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Number of women of reproductive age (15-49 years of age) who are de facto household members

Note: Estimates are sample-weighted; numbers of observations are unweighted

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 8.1.3: Differences in Dietary Diversity and Consumption of Vitamin A-rich Fruits and Vegetables among Women of Reproductive Age With and Without the Red Palm Question, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Information on dietary diversity	Baseline (2019)			Midline (2022)		
	With red palm (%)	Without red palm (%)	Diff.	With red palm (%)	Without red palm (%)	Diff.
Ate other vitamin A-rich fruits or vegetables	29.8	28.1	1.7	43.3	35.3	8.0
Women's food group score: Number of food groups consumed						
0	0.3	0.3	0.0	0.1	0.1	0.0
1	2.3	2.3	0.0	1.1	1.1	0.0
2	16.2	16.4	-0.2	8.5	9.0	-0.5
3	26.7	27.0	-0.3	18.8	21.0	-1.7
4	25.3	25.5	-0.2	24.4	25.4	-0.8
5	16.1	15.6	0.6	16.2	17.4	1.3
6	7.4	7.2	0.1	7.3	11.1	1.2
7	3.5	3.4	0.1	3.5	6.9	0.5
8	1.5	1.5	0.0	2.9	2.9	0.0
9	0.6	0.6	0.0	2.5	2.5	0.0
10	0.1	0.1	0.0	3.2	3.2	0.0
Achieved MDD	29.2	28.5	0.7	47.1	43.9	3.2
Number of women of reproductive age (n)	2,318			3,085		

MDD=minimum dietary diversity, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

Notes:

The Diff. column shows the percentage point difference in estimates calculated with the red palm question (V429) minus the estimates calculated without the red palm question (V429).

Estimates are presented for de facto household members.

Estimates are sample weighted; the numbers of women are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 8.2.1: Coverage of the Types of Women's Health and Nutrition Interventions Promoted by Feed the Future in the Uganda ZOI During the 12 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Type of women's health and nutrition intervention	All women (%)	Resilience Focus Zone	
		Non-RFZ (%)	RFZ (%)
Received counseling on maternal nutrition from a health worker in the 4 months preceding the survey	22.0	21.1	26.7
Received counseling on the healthy timing and spacing of pregnancies in the 4 months preceding the survey	35.0	35.4	33.3
Number of women of reproductive age 15-49 years (n)^a	3,208	1,394	1,814
Received an antenatal check-up from a health worker	83.5	82.3	88.7
Number of women 15-49 years of age who were pregnant in the past 12 months (n)^b	531	199	332
Mean number of antenatal check-ups received while pregnant	4.9	4.8	5.0
Received iron and folic acid while pregnant	92.9	91.3	97.1
Number of women 15-49 years of age who gave birth in the past 12 months (n)^c	132	40	92

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of women of reproductive age who are de jure household members

^b Number of women of reproductive age who were pregnant during the 12 months preceding the survey and are de jure household members

^c Number of women of reproductive age who gave birth during the 12 months preceding the survey and are de jure household members

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 8.2.2: Percent Distribution of Women of Reproductive Age who Participated in Women's Health and Nutrition Interventions of the Type that Feed the Future Promoted in the Uganda ZOI During the 12 Months Preceding the Survey by Age, Feed the Future Phase Two ZOI and RFZ Midline Survey

	Woman's age (years)		Number of participants (n)
	15-19 years (%)	20-49 years (%)	
Women's health and nutrition intervention			
Received counseling on maternal nutrition from a health worker ^a	15.0	85.0	792
Received counseling on the healthy timing and spacing of pregnancies ^a	15.6	84.4	1,094
Received an antenatal check-up from a health worker ^b	14.8	85.2	455
Mean number of antenatal check-ups received while pregnant ^c	4.2	5.1	132
Received iron and folic acid while pregnant ^c	18.1	81.9	126

RFZ = Resilience Focus Zone, ZOI=Zone of Influence

^a Number of women of reproductive age who are de jure household members

^b Number of women of reproductive age who were pregnant during the 12 months preceding the survey and are de jure household members

^c Number of women of reproductive age who gave birth during the 12 months preceding the survey and are de jure household members

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 8.3.1: Comparison of Prevalence of Exclusive Breastfeeding among Children 0-5 Months of Age in the Uganda ZOI, in Total and by Selected Child and Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^c	Sig. ^d
	%	95% CI	n ^{a,b}	%	95% CI	n ^{a,b}			
All children 0-5 months of age	49.2	39.0, 59.4	161	41.9	31.4, 53.3	188	-7.3	0.35	n/s
Child's sex									
Male	38.8	26.7, 52.4	87	32.9	20.2, 48.6	94	-5.9	0.55	n/s
Female	60.3	47.7, 71.6	74	51.1	39.0, 63.1	94	-9.2	0.30	n/s
Region									
East	33.2	20.9, 48.4	55	^	^	22	na	na	na
Karamoja	^	^	12	75.9	57.5, 88.0	64	na	na	na
North	49.3	23.1, 75.9	33	^	^	15	na	na	na
Southwest	53.6	38.9, 67.7	61	45.0	28.7, 62.4	87	-8.6	0.46	n/s
Resilience Focus Zone									
Non-RFZ	47.9	38.1, 58.0	131	33.1	22.2, 46.3	68	-14.8	0.07	n/s
RFZ	^	^	28	76.0	64.7, 84.6	120	na	na	na
Wealth quintile									
Highest (wealthiest)	46.9	27.9, 66.8	34	^	^	27	na	na	na
Fourth	^	^	22	48.7	23.5, 74.6	30	na	na	na
Middle	37.5	19.6, 59.5	31	^	^	23	na	na	na
Second	37.1	20.6, 57.3	31	^	^	27	na	na	na
Lowest (poorest)	51.1	32.3, 69.5	43	55.2	36.1, 72.9	81	4.1	0.77	n/s
Shock exposure severity									
Did not experience any shocks	^	^	4	^	^	2	na	na	na
Low	67.0	47.0, 82.3	36	53.5	31.8, 73.9	64	-13.5	0.36	n/s
Moderate	44.0	30.4, 58.6	64	42.0	26.6, 59.1	52	-2.0	0.86	n/s
High	48.5	33.6, 63.7	53	32.9	21.5, 46.8	70	-15.6	0.13	n/s

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^b Number of children 0-5 months of age who are de facto household members

^c Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, disaggregate sample sizes may not total to the aggregated sample size.

^d Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^e Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 8.3.2: Comparison of the Percent of Children 6-23 Months of Age in the Uganda ZOI who Consumed a Minimum Acceptable Diet, in Total and by Selected Child and Household Characteristics, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Characteristic	Baseline (2019)			Midline (2022)			Diff.	p-value ^c	Sig. ^d
	%	95% CI	n ^{a,b}	%	95% CI	n ^{a,b}			
All children 6-23 months of age	11.1	8.4, 14.5	555	27.2	21.7, 33.4	669	16.1	<0.001	***
Child's sex									
Male	10.0	6.7, 14.8	285	32.7	25.2, 41.3	353	22.7	<0.001	***
Female	12.2	8.3, 17.6	270	20.2	13.8, 28.7	316	8.0	0.07	n/s
Child's age									
6-11 months	11.9	7.3, 18.7	180	26.0	17.6, 36.5	200	14.1	0.01	*
12-17 months	9.0	5.6, 14.3	214	32.6	23.3, 43.4	238	23.6	<0.001	***
18-23 months	13.0	7.5, 21.8	161	22.7	13.7, 35.1	231	9.7	0.14	n/s
Child's breastfeeding status									
Breastfeeding	11.5	8.7, 15.0	447	30.1	23.6, 37.4	528	18.6	<0.001	***
Not breastfeeding	9.6	4.3, 20.1	108	15.6	8.1, 27.9	141	6.0	0.34	n/s
Gendered household type									
Male and female adults	11.3	8.6, 14.7	496	26.6	21.2, 32.8	624	15.3	<0.001	***
Female adults only	10.7	4.5, 23.5	55	32.4	13.7, 59.2	40	21.7	0.10	n/s
Male adults only	^	^	4	^	^	5	na	na	na
Children only	—	—	0	—	—	0	na	na	na
Region									
East	9.4	6.3, 13.7	176	24.0	16.0, 34.3	89	14.6	0.005	**
Karamoja	4.9	1.5, 15.0	68	11.4	6.6, 18.8	181	6.5	0.12	n/s
North	8.6	3.8, 18.6	124	15.3	8.9, 25.2	94	6.7	0.21	n/s
Southwest	14.1	9.7, 19.9	187	37.1	28.0, 47.3	305	23.0	<0.001	***
Resilience Focus Zone									
Non-RFZ	10.9	8.0, 14.7	446	29.4	23.1, 36.5	293	18.5	<0.001	***
RFZ	12.5	7.3, 20.5	107	16.0	12.2, 20.8	376	3.5	0.37	n/s
Wealth quintile									
Highest (wealthiest)	21.9	13.6, 33.1	96	47.1	33.3, 61.3	94	25.2	0.005	**
Fourth	10.7	5.5, 19.6	80	36.8	25.6, 49.6	91	26.1	<0.001	***
Middle	13.4	8.1, 21.3	97	28.6	15.9, 46.0	109	15.2	0.07	n/s
Second	5.9	2.4, 13.4	117	24.2	15.0, 36.6	125	18.3	0.003	**
Lowest (poorest)	6.6	3.6, 11.8	165	10.4	6.3, 16.6	250	3.8	0.25	n/s
Shock exposure severity									
Did not experience any shocks	^	^	9	^	^	2	na	na	na
Low	12.4	7.2, 20.6	181	23.6	16.6, 32.6	227	11.2	0.03	*
Moderate	9.7	5.8, 15.9	164	29.9	21.9, 39.3	212	20.2	<0.001	***
High	11.8	7.7, 17.8	196	27.0	18.5, 37.7	225	15.2	0.006	**

^ Results not statistically reliable, n<30; — Results not available, n=0

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of children 6-23 months of age who are de facto household members

^b Records missing information for the disaggregate variables have been excluded from the disaggregated estimates. The unweighted sample size reflects this loss in observations; therefore, disaggregate sample sizes may not total to the aggregated sample size.

^c Significance tests were performed to determine whether a difference exists between the baseline and midline estimates.

^d Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 8.3.3: Comparison of the Percent of Children 6-23 Months of Age in the Uganda ZOI Achieving Minimum Feeding Frequency and Dietary Diversity, and Consuming Foods from Each of the Food Groups in the Minimum Acceptable Diet Indicator, in Total and by Breastfeeding Status and Age, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Breastfeeding status and indicator or food group	All children			Children's age								
				6-11 months			12-17 months			18-23 months		
	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a
All children 6-23 months of age												
Achieving minimum meal frequency												
Baseline	34.1	28.2, 40.5		42.0	33.7, 50.9		33.1	25.2, 42.1		26.6	17.1, 38.7	
Midline	47.9	40.6, 55.4	**	54.8	43.5, 65.5	n/s	46.5	36.5, 56.7	n/s	43.5	33.8, 53.8	*
Achieving minimum dietary diversity												
Baseline	25.4	21.1, 30.2		18.7	13.1, 26.0		30.2	23.5, 38.0		26.5	19.2, 35.3	
Midline	47.9	42.2, 53.8	***	39.9	30.9, 49.6	***	61.0	50.7, 70.5	***	41.5	31.4, 52.4	*
Consuming:												
Grains, roots, and tubers												
Baseline	91.3	88.3, 93.5		83.0	76.1, 88.3		94.0	90.3, 96.4		96.7	91.9, 98.7	
Midline	93.4	90.0, 95.8	n/s	90.9	82.4, 95.6	n/s	94.3	89.3, 97.1	n/s	94.7	90.7, 97.0	n/s
Legumes and nuts												
Baseline	60.2	54.4, 65.7		46.9	37.9, 56.1		65.7	57.3, 73.2		67.6	56.9, 76.8	
Midline	73.4	68.2, 78.0	***	63.5	54.0, 72.1	*	77.1	67.4, 84.6	n/s	78.3	69.4, 85.1	n/s
Dairy products												
Baseline	30.3	25.0, 36.3		31.3	23.3, 40.6		27.8	20.7, 36.2		32.7	23.6, 43.2	
Midline	28.5	21.1, 37.2	n/s	31.2	21.7, 42.6	n/s	32.4	22.9, 43.7	n/s	22.1	15.0, 31.4	n/s
Meats and organ meats												
Baseline	11.8	8.9, 15.6		8.7	5.0, 14.7		13.9	9.2, 20.5		12.5	7.7, 19.7	
Midline	30.9	24.5, 38.0	***	24.8	18.8, 32.0	***	31.3	21.5, 43.1	**	35.6	24.0, 49.2	**
Eggs												
Baseline	6.1	4.0, 9.3		8.3	4.0, 16.6		6.5	3.8, 11.0		3.2	1.3, 7.6	
Midline	11.5	7.9, 16.4	*	10.8	6.6, 17.2	n/s	13.5	8.1, 21.6	n/s	10.0	5.2, 18.4	n/s
Breast milk												
Baseline	79.4	75.4, 82.9		97.3	93.7, 98.9		84.2	77.8, 89.0		53.3	43.4, 63.0	
Midline	80.0	75.3, 83.9	n/s	98.1	93.4, 99.4	n/s	82.8	75.7, 88.1	n/s	61.3	51.0, 70.7	n/s
Vitamin A-rich fruits and vegetables												
Baseline	45.4	39.3, 51.6		29.1	21.2, 38.4		49.4	40.8, 58.1		58.1	49.4, 66.3	
Midline	57.6	51.8, 63.2	**	43.3	34.9, 52.1	*	69.1	58.7, 77.9	**	58.2	46.8, 68.9	n/s
Other fruits and vegetables												
Baseline	41.6	36.6, 46.8		31.4	23.8, 40.2		47.9	40.2, 55.8		44.6	35.6, 54.0	
Midline	64.1	58.7, 69.2	***	53.8	45.0, 62.4	***	70.8	61.9, 78.3	***	66.2	55.2, 75.7	**

Breastfeeding status and indicator or food group	Children's age											
	All children			6-11 months			12-17 months			18-23 months		
	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a
Number of children 6-23 months of age (n)^c												
Baseline	555			180			214			161		
Midline	669			200			238			231		
Breastfed children 6-23 months of age												
Achieving minimum meal frequency												
Baseline	36.1	30.8, 41.8		40.9	32.5, 49.8		32.3	24.3, 41.6		34.3	22.6, 48.1	
Midline	48.3	39.8, 56.9	*	55.7	44.3, 66.6	*	41.7	31.0, 53.2	n/s	47.1	33.2, 61.5	n/s
Achieving minimum dietary diversity												
Baseline	28.5	23.7, 33.8		19.2	13.4, 26.6		33.4	26.1, 41.5		37.3	26.5, 49.5	
Midline	51.8	45.7, 57.9	***	40.6	31.5, 50.3	***	65.1	54.2, 74.7	***	49.1	35.0, 63.3	n/s
Consuming:												
Grains, roots, and tubers												
Baseline	90.1	86.6, 92.8		83.0	75.8, 88.3		93.3	88.9, 96.1		97.9	91.5, 99.5	
Midline	92.6	88.4, 95.3	n/s	90.8	82.1, 95.5	n/s	94.2	88.7, 97.1	n/s	92.8	86.6, 96.2	n/s
Legumes and nuts												
Baseline	58.9	52.7, 64.8		47.7	38.8, 56.9		65.0	55.6, 73.3		68.6	58.2, 77.3	
Midline	72.9	66.9, 78.2	***	62.8	53.1, 71.6	*	77.2	67.0, 84.9	n/s	81.1	72.0, 87.8	*
Dairy products												
Baseline	29.5	23.8, 35.9		29.8	21.5, 39.7		27.7	20.6, 36.0		32.6	21.5, 46.0	
Midline	27.4	19.8, 36.5	n/s	31.7	22.1, 43.1	n/s	28.0	18.0, 40.8	n/s	20.4	12.1, 32.3	n/s
Meats and organ meats												
Baseline	10.3	7.3, 14.4		8.5	4.8, 14.8		12.2	7.3, 19.8		9.9	5.4, 17.5	
Midline	30.2	23.3, 38.2	***	25.1	18.9, 32.4	***	30.6	20.8, 42.5	**	36.8	20.0, 57.5	*
Eggs												
Baseline	6.4	4.0, 10.0		7.6	3.4, 16.1		6.9	3.9, 12.0		2.8	1.0, 7.6	
Midline	11.4	7.4, 17.1	n/s	11.0	6.7, 17.5	n/s	13.2	7.6, 21.9	n/s	9.5	3.8, 21.9	n/s
Vitamin A-rich fruits and vegetables												
Baseline	42.7	36.0, 49.7		28.5	20.8, 37.8		48.6	39.8, 57.5		58.8	45.2, 71.1	
Midline	56.8	50.5, 62.9	**	43.1	34.4, 52.2	*	72.4	61.9, 80.9	***	54.3	39.3, 68.6	n/s
Other fruits and vegetables												
Baseline	40.5	35.0, 46.3		30.7	22.7, 40.1		46.1	37.7, 54.6		48.5	35.7, 61.5	
Midline	60.5	54.6, 66.2	***	53.1	44.2, 61.8	***	70.4	60.9, 78.4	***	57.2	42.6, 70.7	n/s
Number of breastfed children 6-23 months of age (n)^d												
Baseline	447			175			180			92		
Midline	528			194			200			134		

Breastfeeding status and indicator or food group	All children			Children's age								
				6-11 months			12-17 months			18-23 months		
	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a
Non-breastfed children 6-23 months of age												
Achieving minimum meal frequency												
Baseline	26.4	13.8, 44.5		^	^		37.3	16.8, 63.6		17.8	7.5, 36.6	
Midline	46.4	34.6, 58.6	*	^	^	na	69.4	44.5, 86.5	n/s	37.7	25.3, 52.1	*
Achieving minimum milk feeding frequency												
Baseline	20.8	9.8, 38.8		^	^		20.3	7.1, 46.0		17.0	6.9, 36.2	
Midline	32.1	19.8, 47.5	n/s	^	^	na	52.3	28.5, 75.1	*	24.1	13.6, 38.9	n/s
Achieving minimum dietary diversity												
Baseline	13.4	6.3, 26.3		^	^		13.6	4.7, 33.4		14.2	5.2, 33.1	
Midline	32.5	21.5, 45.9	*	^	^	na	41.5	19.6, 67.4	n/s	29.6	18.5, 43.9	n/s
Consuming:												
Grains, roots, and tubers												
Baseline	95.7	87.2, 98.6		^	^		97.9	82.9, 99.8		95.3	86.6, 98.5	
Midline	97.0	88.0, 99.3	n/s	^	^	na	95.0	65.3, 99.5	n/s	97.7	90.2, 99.5	n/s
Legumes and nuts												
Baseline	65.3	51.0, 77.3		^	^		69.6	51.5, 83.2		66.6	46.9, 81.8	
Midline	75.5	61.6, 85.5	n/s	^	^	na	77.1	49.8, 91.9	n/s	73.7	56.9, 85.6	n/s
Dairy products												
Baseline	33.7	20.2, 50.5		^	^		28.4	12.4, 52.8		32.8	18.4, 51.3	
Midline	33.0	20.6, 48.4	n/s	^	^	na	53.7	29.5, 76.2	n/s	24.9	14.3, 39.7	n/s
Meats and organ meats												
Baseline	17.6	9.2, 30.9		^	^		22.7	10.4, 42.7		15.5	6.3, 33.5	
Midline	33.4	22.8, 46.1	n/s	^	^	na	34.5	16.8, 57.9	n/s	33.9	22.2, 47.9	n/s
Eggs												
Baseline	5.2	1.9, 13.6		^	^		4.1	0.4, 29.2		3.7	0.7, 16.4	
Midline	11.9	6.1, 21.8	n/s	^	^	na	15.2	5.0, 37.9	n/s	10.9	4.4, 24.6	n/s
Vitamin A-rich fruits and vegetables												
Baseline	55.8	43.6, 67.4		^	^		53.7	35.8, 70.6		57.3	43.9, 69.7	
Midline	60.9	48.8, 71.7	n/s	^	^	na	53.6	30.0, 75.7	n/s	64.4	52.1, 75.1	n/s
Other fruits and vegetables												
Baseline	46.1	32.0, 60.7		^	^		57.9	34.8, 78.0		40.1	24.5, 58.1	
Midline	78.5	64.5, 88.0	***	^	^	na	72.9	46.4, 89.3	n/s	80.5	66.8, 89.5	***

Breastfeeding status and indicator or food group	All children			Children's age								
				6-11 months			12-17 months			18-23 months		
	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a	%	95% CI	Sig. ^a
Number of non-breastfed children 6-23 months of age (n)^e												
Baseline	108			5			34			69		
Midline	141			6			38			97		

^a Results not statistically reliable, n<30

CI=confidence interval, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Significance tests were performed to determine whether a difference exists between the baseline and midline estimates. Differences found to be statistically significant are indicated by level: * p<0.05, ** p<0.01, *** p<0.001; n/s=not significant; na=not applicable.

^b Number of children 6-23 months of age who are de facto household members

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 8.3.4: Differences in Dietary Diversity and Consumption of Vitamin A-rich Fruits and Vegetables among Children 6-23 Months of Age With and Without the Red Palm Question, Feed the Future Phase Two ZOI and RFZ Baseline and Midline Surveys

Information on dietary diversity	Baseline (2019)			Midline (2022)		
	With red palm (%)	Without red palm (%)	Diff.	With red palm (%)	Without red palm (%)	Diff.
Ate vitamin A-rich fruits or vegetables	45.4	45.3	0.1	57.6	51.3	6.3
Children's food group score: Number of food groups consumed						
0	0.0	0.0	0.0	0.3	0.3	0.0
1	5.7	5.8	-0.1	1.5	1.5	0.0
2	14.3	14.2	0.1	8.9	9.0	-0.1
3	23.8	23.8	0.0	13.1	15.0	-1.9
4	30.8	30.8	0.0	28.3	28.9	-0.6
5	16.9	16.9	0.0	27.5	26.3	1.2
6	6.7	6.7	0.0	14.4	12.9	1.5
7	1.7	1.7	0.0	4.5	4.5	0.0
8	0.0	0.0	0.0	1.6	1.6	0.0
Achieved MDD	25.4	25.4	0.0	47.9	45.2	2.7
Number of children 6-23 months of age (n)	555			669		

MDD=minimum dietary diversity, RFZ=Resilience Focus Zone, ZOI=Zone of Influence

Notes:

The Diff. column shows the percentage point difference in estimates calculated with the red palm question (V559) minus estimates calculated without the red palm question (V559).

Estimates are presented for de facto household members.

Estimates are sample weighted; the numbers of children are unweighted.

Sources: 2019 Feed the Future Uganda P2-ZOI Baseline Survey; 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 8.4.1: Coverage of the Types of Children's Health and Nutrition Interventions Promoted by Feed the Future in the Uganda ZOI During the 4 or 6 Months Preceding the Survey, Feed the Future Phase Two ZOI and RFZ Midline Survey

Type of children's health and nutrition intervention	All children (%)	Resilience Focus Zone	
		Non-RFZ (%)	RFZ (%)
Child's caregiver received counseling on breastfeeding from a health worker in the 4 months preceding the survey	42.7	39.5	57.8
Child's caregiver received counseling on infant and young child feeding from a health worker in the 4 months preceding the survey	43.3	42.1	49.0
Number of children under 2 years (n)^a	883	375	508
Child's caregiver received counseling on growth monitoring from a health worker in the 4 months preceding the survey	37.3	34.5	49.3
Child's weight and/or height were checked by a health worker in the 4 months preceding the survey	45.7	41.8	62.7
Attended a Child Health Day in the 6 months preceding the survey	54.3	53.9	56.0
Number of children under 5 years (n)^b	2,444	979	1,465
Received treatment for malnutrition in the 4 months preceding the survey	71.0	66.8	83.5
Received ready-to-use therapeutic food in the 4 months preceding the survey	49.6	40.8	75.8
Number of children under 5 years diagnosed as malnourished (n)^c	258	76	182
Received deworming medication in the 6 months preceding the survey	81.4	79.5	89.9
Number of children 12-59 months (n)^d	2,045	817	1,228
Received vitamin A supplementation in the 6 months preceding the survey	82.4	81.2	87.5
Number of children 6-59 months (n)^e	2,250	908	1,342

RFZ = Resilience Focus Zone; ZOI=Zone of Influence

^a Number of children under 2 years of age who are de jure household members

^b Number of children under 5 years of age who are de jure household members

^c Number of children under 5 years of age who were diagnosed as malnourished and who are de jure household members

^d Number of children 12-59 months of age who are de jure household members

^e Number of children 6-59 months of age who are de jure household members

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

Table 8.4.2: Percent Distribution of Beneficiaries of Children's Health and Nutrition Interventions of the Type Promoted by Feed the Future in the Uganda ZOI During the 4 or 6 Months Preceding the Survey by Sex and Age, Feed the Future Phase Two ZOI and RFZ Midline Survey

Type of children's health and nutrition intervention	Sex of child		Age of child		Number of child beneficiaries (n)
	Male (%)	Female (%)	<2 years (%)	2+ years (%)	
Child's caregiver received counseling on breastfeeding ^a	56.0	44.0			428
Child's caregiver received counseling on infant and young child feeding ^a	55.9	44.1			398
Child's caregiver received counseling on growth monitoring ^b	51.7	48.3	43.0	57.0	1,048
Child's weight and/or height were checked by a health worker ^b	53.5	46.5	54.5	45.5	1,323
Attended a Child Health Day ^b	51.4	48.6	41.8	58.2	1,388
Received treatment for malnutrition ^c	57.2	42.8	35.4	64.6	196
Received ready-to-use therapeutic food ^c	54.4	45.6	36.8	63.2	158
Received deworming medication ^d	51.4	48.6	24.5	75.5	1,750
Received vitamin A supplementation ^e	52.4	47.6	33.7	66.3	1,886

RFZ=Resilience Focus Zone, ZOI=Zone of Influence

^a Number of children under 2 years of age who are de jure household members

^b Number of children under 5 years of age who are de jure household members

^c Number of children under 5 years of age who were diagnosed as malnourished in the 4 months preceding the survey and who are de jure household members

^d Number of children 12-59 months of age who are de jure household members

^e Number of children 6-59 months of age who are de jure household members

Note: Estimates are sample-weighted; numbers of observations are unweighted.

Source: 2022 Feed the Future Uganda P2-ZOI and RFZ Midline Survey

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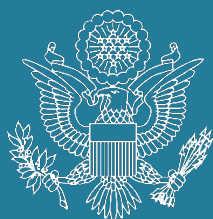
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